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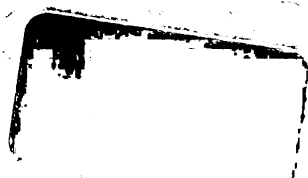
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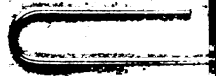
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REPORT



PROVINCIAL INSTRUCTOR IN ROAD-MAKING ONTARIO

• 1896.

(PUBLISHED BY THE ONTARIO DEPARTMENT OF AGRICULTURE.)

PRINTED BY ORDER OF
THE LEGISLATIVE ASSEMBLY OF ONTARIO



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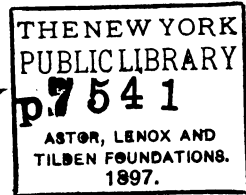
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TABLE OF CONTENTS.

Letter of Transmission	PAGE 5
Introductory.....	7
The Road Problem	10

PART I.—COUNTRY ROADS.

Statute Labor	18
Statute Labor Commuted	15
Statute Labor Abolished	15
Existing Roads.....	16
Road Metal	18
Gravel	18
Broken Stone	19
Boulders	21
Placing Metal on the Roads	22
Rolling Country Roads	23
Drainage of Macadam Roads.....	23
Dimensions of Country Roads	25
Crowning the Road.....	26
Hills	27
Embankments and Cuttings	28
Location of Roads	28
Repair and Maintenance	29
Road Machinery	30
Bridges and Culverts.....	31
Wagon Tires.....	34
Instruction for Pathmasters	36
Snow Obstructions.....	37
Convict Labor	38
Artistic Treatment of Roads.....	39
Development of Roads in Ontario	42

PART II.—TOWN STREETS.

Street Improvement	47
Choice of Pavements	49
Macadamized Town Streets	52
Form of Specification for Macadam Streets	55

APPENDIX.

Report on Town Streets.....	59
Report on Township Roads ...	67
Cement—Concrete Sidewalks	75
Road Laws	76

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REPORT
OF THE
PROVINCIAL INSTRUCTOR IN ROAD-MAKING.

To the Honorable JOHN DRYDEN,

Minister of Agriculture.

SIR:—I herewith beg to submit to you my report for the year 1896 on Road and Street Improvement in Ontario.

The scope of this report is necessarily confined largely to a general review of the situation rather than a detailed statement of the work in which I have been engaged; this for the reason that, my duties being of an educational nature, and the office having been created under an Order-in-Council bearing the recent date, April 15th, 1896, the period has been too brief to anticipate any radical effect upon methods which have held for nearly a century; for the further reason that, while much literature relating to road improvement in England, France and the United States has been laid before the people of Ontario by the 'public' press, little or none of a distinctly local character has been available, and a discussion of the subject in a manner applicable solely to this Province will therefore be of value and interest.

I have the honor to be,

Sir,

Your obedient servant,

A. W. CAMPBELL,

Provincial Instructor in Road-Making

DEPARTMENT OF AGRICULTURE,

TORONTO, Ontario, 23rd March, 1897.



PROVINCIAL INSTRUCTOR IN ROAD-MAKING.

INTRODUCTORY.

The improvement of roads in Ontario has not kept pace with the development of the Province in population, wealth, civilization nor commerce. This lack is seriously felt by all classes of society, and has, within the past three years, created an almost revolutionary movement in favor of better roads. Were economic methods instituted, millions of dollars would be added to agricultural values, and the mercantile and banking interests would be placed on a more stable basis. This movement is not confined to Ontario alone, of the Provinces of the Dominion, but is already beginning to take definite form in the others, especially Quebec, Nova Scotia and New Brunswick.

During my tenure of office, I have found the towns even more in need of reform in matters relating to street improvement, than are the rural communities, and they have been quite as ready to receive instruction. It should be understood throughout this report that streets of towns are quite distinct in very many particulars from country roads, and that the curbing, boulevarding, width of macadamized driveway and other features advocated for town streets are not applicable to township highways.

PROCEDURE.

In the performance of my duties, the procedure has been, upon receipt of a request for my services from a township, county, town or city council, to appoint a date upon which to visit the municipality. Accompanied usually by members of the council, the Board of Works, or a deputation, I drive over the roads or streets, and point out defects in their construction and maintenance. I further examine the material available for road-making, the system of expenditure, and the administration of road laws. Having made this investigation, and informed myself as to the lines along which reform is most needed, I deliver a public address on the subject, directing my remarks, however, more particularly to the members of the council. Subsequently I present to the council a written report upon the condition of the roads and streets examined, the material available, etc., etc., with recommendations as to the best methods and systems to be adopted. These local reports I find necessary in the great majority of instances. Oral instruction merely, particularly on technical points, is misconstrued, overlooked or forgotten to such an extent as to render a written statement of the greatest value in securing substantial results. They form also a record for succeeding councils. Two of these reports are appended to this volume, to show the ground covered by them—one for a town, and the other for a township municipality.

Municipalities visited in this way up to the end of 1896 were :

City of Stratford.

TOWNS AND VILLAGES.

Cobourg,	Orangeville,	Barrie,
Arnprior,	Brighton,	Port Hope,
Ingersoll,	Carleton Place,	Cornwall,
Winchester,	Iroquois,	Morrisburg,
Pembroke,	Galt,	Newmarket,
Seaforth,	Clinton,	Woodstock,
Berlin,	Paisley,	Paris.

TOWNSHIPS.

Beckwith,	Hamilton,	York,
Matilda,	Mountain,	Winchester,
Williamsburg,	Finch,	Osnabruck,
Roxborough,	Cornwall,	Kenyon,
Charlottenburg,	Lochiel,	Lancaster,
Pembroke,	Tuckersmith,	Stanley,
North Norwich,	Cavan,	East Nissouri,
Elderslie,	Greenock,	East Zorra,
West Zorra,	Puslinch,	Hallowell.

INTEREST TAKEN.

In consulting with municipal councils, and in publicly discussing the question of road reform, the interest displayed has been most gratifying, and promises, in due time, the most satisfactory results. My attention has been directed to a number of instances of actual road construction in which improved methods have already been adopted with beneficial effect, as a result of the distribution of literature on the subject.

Public sentiment has been awakened with regard to the matter, and in every district of the Province I find the keenest interest taken in the discovery of ways and means whereby the neglected condition of roads and streets may be overcome. In every locality visited, my instruction has been met by an evident intention to carry it into practical use in the improvement of the highways.

MACHINERY EXHIBIT.

At the Industrial Fair, Toronto, in September, 1896, an exhibit of road-making machinery was arranged, and attracted a great amount of attention. It is intended to continue this exhibit in succeeding years, as it affords municipal officers and others interested in road-making an opportunity to learn not only the importance of using modern implements, but to see the different makes in actual operation, side by side, assisting in the selection of a pattern that will be most suited to their requirements.

MODEL STREETS.

Models of properly improved streets were constructed in Ingersoll and in Orangeville under my supervision.

BULLETINS.

Two bulletins were issued during the year, No. 1 being a brief pamphlet intended chiefly for pathmasters, while No. 2 contained a more carefully developed outline of the principles of construction.

STATUTE LABOR.

A consideration of road and street improvement naturally divides itself under two heads :

- (1) The system of expenditure and control.
- (2) The principles of construction.

With regard to the first of these, it is not the intention of this report to urge any alterations from present systems, but to define a proper administration of those already employed. With township roads, under this division, a consideration of statute labor and its reorganization is an important feature. It is almost universally admitted that, as at present administered, this method of collecting a road tax is ineffectual, wasteful and unjust, and that wherever an effort to put it on a business like basis fails, the only course is to replace it with a law which will render justice to all, and will produce better roads. If the endeavor to reorganize the statute labor fails to make it effectual in producing the class of roads required, the process will at least be one of education, and will lead to the substitution of a more desirable system.

STATUTE LABOR ABOLISHED OR COMMUTED.

The power to abolish the statute labor system is now in the hands of the people, and has been taken advantage of in at least one instance, that of Malden township, Essex County. The township of North Monaghan, Peterborough County, submitted a by-law to abolish statute labor in January, 1896, which was carried by a vote of 2 to 1 ; plans are now being framed to establish a new system throughout the entire municipality, statute labor having last year been done away with in only a part of the township. Other townships, in whole or in part, have substituted a system of compulsory commutation. As far as I have been able to learn, these townships are Winchester, Barton, Niagara, Sarawak.

COUNTY ROADS.

The County of Huron some years ago improved a considerable mileage of the leading roads of the county by the issue of debentures, the maintenance of these roads reverting to the townships through which they pass. The county of Hastings constructs and maintains a system of county roads, and the county of Wellington maintains several of the main roads by county grants.

PRINCIPLES OF CONSTRUCTION.

Dealing with the actual principles of construction with regard to township requirements, I have in this report enlarged chiefly upon gravel and broken stone roads, the material for these being generally distributed throughout the province. Drainage of roads, crowning the roads, judiciously locating them, width of roadbed, width of graded portion, treatment of hills, repairs, and machinery are among the principle subjects dwelt upon. With town streets, these details of construction require a more finished treatment to accord with the different surroundings and class of traffic.

TIRES.

The question of tires is a very important one, and calls for special emphasis. Were tires of a suitable width used in all cases, a very great part of the road question would be solved.

FURTHER TOPICS.

In addition, the necessity of giving attention to the appearance of country roads is dealt with, touching upon the relation of fences, trees, etc., to highways. A short historical sketch of some of the early and most important roads is included, dwelling chiefly upon the methods and means used, and the various influences affecting their development.

Much other useful and interesting information might have been included in this report, but it appears advisable to limit the contents to such subjects as seem most applicable to the present condition of the roads and the means available for their improvement.

THE ROAD PROBLEM.

There must be intelligent direction before there can be satisfactory results. It is becoming apparent to the people of the Province that improved highways are greatly needed; and the people are beginning to learn that they are suffering from a sore experience; that their labor has been largely wasted, and, in addition, large appropriations from the general municipal tax funds. The energy which has been, and is now being wasted, would, if properly applied, have each year extended, improved and maintained the highways proportionately with the growth and requirements of the Province in wealth, civilization and commerce. We would, if this energy had been properly guided, be now in possession of a system of roads such as would render a "Good Roads Movement" unnecessary. In their enthusiasm to effect a change there is a great danger of the people grasping at new systems and methods regarding which they are not as yet capable of forming a correct judgment, and which they are not yet qualified to carry into effect. Councillors are influenced by the effect of the agitation on the people, and, wishing to comply with popular demands, are of their own accord hurriedly framing and adopting new plans of raising money with which to carry on the work; such is the effect, but not the intention of the agitation.

The question of road reform is a great economic problem which must be approached slowly, and solved after mature and earnest consideration, based upon sound judgment and the greatest possible measure of experience. The great need of the present crisis, as has been stated, is intelligent direction, in order to avert experiments more costly and less productive of good than those from which the public are endeavoring to escape. There is an absolute necessity of providing municipal councillors, commissioners and the public generally with the fullest possible knowledge of the subject, otherwise there will be failure, discouragement and reaction. The history of roads dates back 1,000 years before the Christian era. In Canada, little better than a century is the sum of our experience. It is apparent that so young a country should look into the matter as presented by the history of roads in other countries to see at least if there is nothing to learn. It is contended that we have an immense mileage of roads, little traffic over them, and a comparatively sparse population, and that therefore the study of roads is something to be postponed until the country is wealthy and able to undertake the construction of highways in an expensive manner. All this is rather but a more potent reason why we should, as a country, conserve our energies and exercise them in the most judicious manner.

This, to a great extent, is more applicable to cities and towns than it is to rural districts. Some of the former to-day are already struggling under a load of debt, thrust upon them by unwise expenditure, by the adoption of wrong methods and by the use of material so unsuitable for street purposes that it has not endured one-half its estimated life. If the amount of money expended by these cities in their speculative efforts were aggregated the amount of waste would be appalling, and more so when it is understood that they are now looking for new material with which to experiment afresh. Climate, local geological conditions and requirements are all very similar throughout wide areas, and yet we have each town and city carrying on its own experiments in systems of taxation and construction, relying almost entirely upon its own experience, and meeting the failures that, with a reliable source of information, could have been avoided. It may be argued that one town can readily obtain information from another on application. This is not the case. Deputation are expensive, inefficient and opposed by the people. Clerks, treasurers and all municipal officers are reluctant or unable to give the necessary labor to correspondence regarding the matter.

In rural municipalities, each with conditions and requirements similar to the others, we have toll road systems, county systems, statute labor systems, commutation of statute labor with direct taxation, direct taxation alone, plans of construction no man can number, describe or understand, embodying the peculiar notion of nearly every ratepayer who, in his turn, has at some time acted as road overseer.

The question is both very broad and very local. No one class of pavement can be clearly specified for all streets, and no one class of road can be adopted in rural districts, but these can all be divided into classes and placed before councils and the people in such a manner as to materially aid in proportioning pavements to requirements, and to the ability of the various communities. There are the country roads which are least travelled; there are those more heavily travelled leading to main highways; there are

the main highways which carry the traffic of extensive areas ; there is another class of roads most largely travelled of all, necessarily rare, which verge finally into the streets of large cities ; there are the streets of villages, of towns, of cities, all of varying requirements, regarding which the most reliable information should be procured. There are materials suitable for all classes of roads, and there are principles involved in the construction of pavements as with any other structure. Every building must be erected to meet certain requirements, and the nature of the structure and material used must be carefully balanced with the use to which it is put—strength, durability, service, in order to meet the requirements of economy, in all of which certain principles of construction must be observed.

There is too great a tendency to localize the question. The actual building of any section of road is a matter of an exceedingly local nature, but in obtaining the information from which to deduce both policy and specific details of construction there should be the most complete records upon which to draw. No city, town, township, county or province can afford to rely solely upon its own efforts, but each should be a source of experience for the other, and all should be able to draw upon the experience of other lands, past and present, to the fullest extent. There is a great deal of past experience on which to draw, but there is a more important present and future experience which should be indexed and made available for reference.

The data for determining the annual expenditure of money and labor on the streets and roads has never been collected with sufficient accuracy to permit a close statement of this element of our national taxation. Enough is known, however, to warrant the assertion that it is millions annually. Of this vast sum the greater part is wasted in ill-contrived and temporary repairs needed to keep the roads in a passable condition. In the present state of our taxation it is difficult to expect the people to impose a special tax to speedily improve country roads generally. It will be found that to attain the end in view the best plan is to assemble as far as possible the forces at work, concentrating the expenditure of money, and diverting to durable work as much as possible of that which is now paid for merely temporary repairs.

It is not to be inferred, nor can it be charged, that councils are less competent to deal with this than with other branches of public affairs. The difficulty is rather that the township councils have had too little to do with highway construction, and there is no person or body sufficiently responsible to the people for the condition of the roads. The most reasonable explanation of existing conditions is to be found in the little attention which has been paid to the effect and value of good roads. This in turn has permitted the administration of our laws to be neglected, and has not stimulated sufficient thought to develop an efficient organization of the ways and means available.

PART I.

COUNTRY ROADS

STATUTE LABOR.

The question of road reform has drawn attention to the statute labor system, the means by which a large percentage of the road mileage of this province has been built and maintained. Farmers' Institutes, Dairymen's Associations, Fruit Growers' and kindred associations, have everywhere discussed this phase of the road movement. It is undoubtedly the case that statute labor has done a great deal toward improving roads in Ontario since the first settlement. It is equally true that the system is entirely unsuited to the conditions found in populous and wealthy communities. It is incompatible with street improvement in cities, towns and villages, and it is urged that, in the long settled townships, the system has outlived its usefulness. It is undeniably the case that in some municipalities, the roads are to-day in a worse condition than they were ten years ago. While the pioneers knew the necessity of good roads and performed their work jealously, there is a sentiment beginning to prevail in certain sections and among the rising generations adverse to performing work on the public highways. Not only so, but those who perform work and those who have the direction of the work, are too frequently inclined to neglect and evade the true spirit of the system as it was exemplified by the early settlers. Where this neglect and evasion exists, not only do the roads interfere with the progress of the immediate locality, and therefore the country as a whole, but injustice arises in various ways between citizen and citizen, and between road division and road division. The citizen who performs his work, is taxed unequally with the ratepayer who is shiftless in its performance, while at the same time he is compelled to endure the inconvenience of a bad road. The section or road division, which does good work on the highways is unjustly treated by the neighboring division which does not do its legitimate work; the further probability is that the division which has failed to perform its work will find it necessary to obtain municipal money grants, thereby placing on the neighboring division a double tax.

A plan whereby the work of road improvement and maintenance may be carried on in a business like way is a subject that many councils are considering. It is well-known that in most municipalities there are no particular plans, other than those which suggest themselves as the work becomes necessary. Many councils are now considering better systems of road improvement and maintenance, but unless these systems are put into operation on business like principles, the result will be inferior work and increased taxation.

A SUPERVISOR.

The direction and oversight of municipal improvements should be placed in the hands of a supervisor who should be a man specially adapted to the work, and whose tenure of office should be as secure as that of the average clerk or treasurer. The supervisor should have charge of all road and bridge improvement, and should be required to report to the council at each meeting; no work should be undertaken until it has been reported upon by him and approved by the council. He would not necessarily be actively engaged during the whole year. He should have charge of all township machinery and material required in the work connected with this department. The many economies that would result will readily suggest themselves to councillors of experience.

PATHMASTERS.

Road divisions or "beats" should be from three to five miles in length. A pathmaster should be a permanent officer, and his division should be such that the most of his travel will lead him over the greatest portion of it. He should not receive a salary, but should, as a slight recompense, be preferred in doing small jobs under the supervisor, where the work is not considered of sufficient importance to be let by contract. He should, in addition, give special attention to all emergency work, such as washouts, broken culverts and bridges. If the time required to oversee statute labor in his division is more than would be needed for his own statute labor, he should be paid for such excess under certificate of the supervisor or council, the object being to secure proper supervision of all work performed; the council or supervisor to determine whether the excess time was actually necessary to oversee the work of the division, in accordance with the local by-law of rules and regulations, which should be framed so as to include this matter.

DUTIES OF SUPERVISOR.

1st. The supervisor should prepare a plan of his district, or township, showing all roads, and the location on them of all culverts, bridges and watercourses, classifying roads according to their requirements.

2nd. He should prepare a list of all culverts and bridges, showing dimensions, material used in construction, their condition and the direction of the watercourses passing through them, with memoranda as to the course of the water and location of the outlet.

3rd. The plans and records should show, as to roads, whether they are of gravel, stone or earth, graded or ungraded, the system of drainage, and nature and extent of traffic upon them. He should carefully study the present and future requirements of traffic on all roads, the class of roads best suited to such traffic, the width and depth of metal, width of graded portion, amount of crown and other details of construction.

4th. He should possess full information as to location, extent and quality of material suitable for road construction, and amount of plank and timber obtainable from ratepayers in the township or district, suitable for repairs.

5th. He should report to the council as early as possible in each year, showing the number and location of culverts and small bridges to be re-built or repaired, with a detailed statement of all material required for this work, and an estimate of the cost. It would also be advantageous to have a probable estimate of material required during the following year, presented each fall so that, if thought advisable, it may be purchased and delivered on the ground during winter months or other most convenient season, so as to utilize as much as possible the labor of ratepayers during the slack season.

6th. His report should specify the condition of all bridges, indicating those which require repairs or reconstruction, together with an estimate of cost, and a statement dealing with such special protection work on streams as he may deem worth the council's consideration. There should also be reference to any needed re-location or deviation of existing roads with a view to doing away with bridges, culverts, expensive grades, cuts or other features which tend to prevent permanent work and economic maintenance.

7th. He should consult with all pathmasters and report to the council, showing the number of days labor in each division, the work to be undertaken, and the amount of money which should be appropriated by the council to properly utilize the statute labor.

8th. He should arrange with divisions desiring to compound statute labor for a term of years, with a view to the construction of permanent and finished work.

9th. He should take stock annually, and report to council on all machinery and implements, showing their condition and where kept.

10th. He should carefully examine all parts of the township where gravel and stone exist, and should by borings and tests, determine the quality and extent, and report thereon to the council. All material, stone, plank, gravel, etc., should be purchased by the supervisor in large quantities, and under instructions from the council, the required amount to be determined by his estimates referred to above. As far as possible the material should be purchased by tender, and due consideration should be given to any ratepayer having material for sale. When purchased it should be delivered and stored at convenient points, and placed in charge of, and used by the pathmaster, subject to the order of the supervisor, and in emergency work.

11th. He should prepare specifications of all work for which the council makes appropriations. Contracts should be awarded to the lowest bidder if proper security is given, but the work should be subject to the approval of the supervisor and all accounts should be certified by him before payment.

A municipality is merely a business corporation, the people its stockholders, the councils its board of directors and the officials the managing staff. If as much attention were paid to the management of a municipality as is devoted to a business enterprise of similar proportions, economy in many directions would follow. No business enterprise could afford to change its board of directors or manager each year, intrusting its affairs to an inexperienced executive.

COMMUTED STATUTE LABOR.

The commutation of statute labor is a system of taxation for road purposes which has been adopted in several municipalities, and is said to operate with a considerable measure of success. The assessment is based upon the statute labor list, the rate being fixed by a local by-law. The usual rate is fifty cents per day. In Barton township the rate is fixed annually according to the total expenditure for the year, and ranges from thirty-five cents to fifty cents per day. It is to be pointed out, nevertheless that, with commutation, pathmasters are still appointed, an amount proportionate to the labor on a beat being given to each pathmaster. Proper supervision of the work is as necessary under a system of money expenditure as with statute labor alone; and it is, therefore, quite as important in the one case as in the other to follow the suggestions for a proper organization of the statute labor system as to reducing the number of pathmasters and appointing a general township supervisor.

STATUTE LABOR ABOLISHED.

The township of Malden, Essex county, is the only township municipality which has yet completely abolished the statute labor system. The change was made in March, 1890. There are about sixty-five miles of road in the township. The total mileage is graded and about two-thirds is metalled with gravel or crushed stone. The township Clerk, Mr. James Honor, in response to an inquiry regarding the change says:—

“We have a perfect system of doing roadwork. Whole concessions are ditched and gravelled. Water is carried until it reaches its proper outlet. No one is allowed to suffer by water being thrown in front of his farm and left there until the next road overseer is appointed to take it off. All work is done in a thorough, systematic and business-like manner. It is let by contract, superintended and received by our council. The change operates successfully and there seems to be no disposition on the part of any ratepayer to return to the old system. The cost of road improvement is paid out of the general township funds, and the amount spent on the roads annually for the past five years was about \$2,000 a year.

“The cause of the change was the corrupt and expensive manner in which statute labor was done. For a number of years previous to 1890 we had commutation of statute labor at fifty cents a day, which worked fairly well for a time until the applicants seemed

to have got all the work done that they required around their farms.] At the same time, our township was divided into seventeen road divisions, and seventeen prominent farmers got this statute labor given them for superintending the work that was not commuted. Each one had his peculiar views about doing road work, selfish motives, cranky notions, and occasionally no ideas at all. Some favored doing the work in the spring and some in the fall; some spread the earth on the road and piled the gravel on top of it; some put on gravel and placed dirt alongside to keep it there.

A township that has good roads, well drained and gravelled, shows a higher state of civilization and taste than one that has not. Consequently a practical man would rather locate in our township than in the neighboring ones and the price of farm land seems to have kept up as compared with other townships."

EXISTING ROADS.

The roads of most townships naturally divide themselves into three classes. The first and least important are those cross roads and little used concession roads which accommodate only a few farms; about three-eighths of the road mileage will fall into this class. In the second and more important groups are those which serve a considerable neighborhood, and comprise, usually, about three-eighths of the road mileage. The third group consists of the main roads which concern the community as a whole and these in some townships merge finally into the streets of large market towns and cities; to this last division will belong nearly the remaining quarter of the road mileage. It is the roads of this last group which, in some instances have, in Ontario, been taken over and constructed as county roads.

It is apparent that each of these three classes have different requirements. Over the first class, the least travelled of all, half a dozen or a dozen vehicles going in one direction may be the average traffic for one day. The neighborhood roads may be called upon to accommodate twenty or thirty vehicles; over the main roads the number of vehicles daily may range from fifty to two hundred. To provide the same strength of construction in each case, would manifestly be either inadequate for the most heavily travelled roads or unnecessarily expensive in the case of the first group mentioned. Those which accommodate the greatest amount of traffic, it is apparent, will need the strongest form of construction, and on them should be concentrated as far as practicable, the money appropriations made by the municipal councils.

To the first group will necessarily belong the dirt roads. These will be gradually gravelled, but it cannot be expected, for a number of years to come, that much expenditure of labor or money can be made on them. The little wear to which they are subjected, however, rendering their maintenance very simple, will make the time required for their improvement comparatively short. These dirt roads are very much abused by the road-makers. With each annual application of statute labor, sod, vegetable mould, and stones, are brought to the centre of the road, to be mixed up into mud with the first fall rains. For six months of the year, a dirt road, (not a clay) is as good a driveway as could be desired for light travel. But in order to extend its usefulness, the greatest care must be taken to see that drains and culverts are placed wherever needed, and that they are always in good working order. It should be well crowned, or rounded up, so as to shed the water freely to the side ditches; but in making this crown, sod and vegetable mould should be carefully excluded. A great deal of injury is done to this kind of road, by running a grading machine along the edges, bringing loose stones and sod to the centre. This stuff should be thrown outward and across the open drain. Under drains should be used judiciously; while a complete system may not be used, tile drains should at least be placed where the water does not leave the side of the road early in the spring, or wherever the ground appears to be continually damp. If open drains are kept in good working order, and if the road is kept properly crowned, its condition in fall and spring will indicate the points at which tile drainage is most needed.

Nearly all roads in Ontario have at one time been mere dirt roads. By a slow process of evolution some have become gravel or broken stone roads. These gravel roads are the highways which will fall into the second and third groups indicated in a preceding section of this article. A dirt road nicely crowned and well drained will make an excellent foundation on which to place a layer of gravel. A driveway which has passed through an apprenticeship as a dirt road, and has, during that time, had due attention given to its drainage requirements, will have indicated the points at which open drains, culverts and underdrains are most needed. With these provided, gravel can be applied to the best advantage.

The object in placing a coating of gravel on a dirt road is not merely to form a hard wearing surface, nor is it to change the consistency of the mud. A coating of gravel or broken stone is placed on a road to form a covering that will not permit water to pass into the earth subsoil, and which will not permit the forming of ruts that will hold water. Unless the dirt road beneath the gravel has been provided with drainage deep enough to secure a good firm foundation, the gravel will sink beneath the wheels, ruts will form and the road will be destroyed.

Very few existing gravel roads show a proper conception of the principles of road-making. The majority are too flat on top to shed the water. The sides of the road form square shoulders which obstruct the water in passing to the surface drains. Open drains are often deep and dangerous, but are not provided with proper outlets. If drains are merely receptacles for water, the subsoil of the roadway absorbs this moisture, is softened, and in the wet weather of fall and spring, traffic forces the gravel covering downward, and the mud to the surface, mixing the gravel and soil to such an extent as to very much destroy the usefulness of the road covering.

As with dirt roads, a great mistake is made by passing a grader along the shoulders of a gravel road, to cut off the shoulders, bringing the loosened material into the centre to round up the roadway. If a portion of the shoulders happens to consist of clean gravel, this may be replaced on the hardened track of the road. But as a rule the shoulders are composed of fine stuff, sand, dirt, loose stones and sod, and is of the greatest injury to the roadway. This stuff is of such a nature as to at once roughen the road, while in wet weather it absorbs water, becomes soft and slushy, and ruins the hard gravel foundation underneath. The material composing these shoulders should be thrown outward across the open drains, and used in levelling the roadside, or it should be otherwise disposed of in such a manner that it will at least not injure the roadway. When these shoulders have been cut off, it will be advisable to pass the road scraper over the centre to take away as much fine stuff, and loose stones as possible; and on this a new layer of gravel or broken stone may be placed to complete the crown of the road.

The drainage usually found on existing roads consists of open ditches on each side of the graded portion, with a depth of about eighteen inches. They are frequently carried through rises of ground past natural watercourses. Little attention is given to the regularity of the grade in the bottom, nor to the amount of fall, as evidenced by the varying depths of stagnant water at wet seasons. The object of these drains was more to procure earth to raise the centre of the road above the water line than to lower that line. The secret of successful drainage is to dispose of water in small quantities before it gains force and headway; and for that reason it should be carried past natural watercourses as seldom as possible.

The best practice does not direct that the old open drains should be deepened for the purpose of draining the subsoil. Deep open drains are expensive, dangerous and unsightly, and the excavated earth generally does more harm than good to the road when used to round it up, especially if piled on top of gravel. When the combined cost of construction and maintenance is considered, a small tile drain laid under the bottom of each open drain is cheaper and more serviceable. It is believed by some that one tile drain placed in the centre of the grade is the better plan. In the case of an old road with a hard surface this is very difficult and expensive, and the roadbed, if metal has been used, is destroyed by so doing. If any obstruction occurs with one central drain, in

order to locate and remove the cause of the difficulty, the roadbed must be broken up, sometimes extensively. Where side drains are constructed, they not only drain the foundation, but they protect it by intercepting the soakage water from the adjacent land. Where open drains exist, as they do on the majority of roads, the cost of excavating for the two underdrains is lessened, and smaller tile may be used. Outlets can easily be had into the tile drains for surface water from the ditches, by the use of catch-basins when necessary, and this is rendered more simple when two tile drains are used.

The success of a drain largely depends upon its outlets. Frequently its usefulness is largely removed if not entirely destroyed by the imperfect outlets on private property, the result of the opposition of the landowner, and the dislike of the municipal council to invoke the aid of the Municipal Act to enforce its construction. A drain without an outlet is useless.

ROAD METAL.

GRAVEL.

The majority of roads in the province which have received a surface covering, are metalled with gravel. It is very plentiful, and where it is to be obtained within easy hauling distance is undoubtedly a most valuable material for use on most country roads, and on many of the little travelled residential streets of towns. In localities where gravel is not within easy hauling distance, however, it will be advisable to consider the use of other metal; since gravel, if not found in the immediate locality, and therefore expensive in transportation, will frequently be found less economical in final cost than a better, though higher priced stone. Gravel may be described as a mass composed of fragments of stone, rounded and worn by the action of air, water and ice, the pebbles representing the hardest bits of the rock from which they were detached. With it, is commonly mingled a variably quantity of boulders, sand, clay and earthy matter.

The rounded, water-worn character of the pebbles composing the gravel, does not permit as perfect a bond as is the case with a road covering of crushed stone, in which the sharp angular sides take a firm, mechanical clasp. Gravel has not very satisfactory wearing qualities, especially when it contains an excessive amount of earthy matter. Where much sand and clay is contained, it should be removed by screening, and the large stones and boulders should be broken. If there is a large proportion of the latter, a method which offers many advantages is to place a rock crusher with a rotary screen attachment in the pit, passing all the material through, thereby removing the earthy substances, and breaking the stone with the one operation. If the number of large stones is not great, it will be the cheaper plan to screen the gravel in the ordinary way or, better, by a rotary screen propelled by steam, the large stones being broken by hand or used in the road foundation.

When being prepared for use on heavily travelled and important roads, care should be taken to remove the dirt from the gravel. Sand and clay, when mixed with gravel afford an easy means of consolidating the road covering in such a way as to be very suitable for traffic in dry weather. But in wet weather, moisture is absorbed, the roadway becomes soft and slushy, ruts form readily, and the bond will be found of a very temporary nature. For little travelled roads, it may be impossible to take the precaution to screen the gravel; and perhaps if the earthy matter is not in excess, screening may be omitted to advantage; it is necessary, however, to emphasize the point that this can be recommended for little travelled roads only.

When gravel is not screened, very much may be accomplished, by care in selecting and taking it from the pit. In drawing gravel to the roads, it is frequently the case that teamsters wish to be in company with each other as much as possible. For this reason, when loading the gravel, while there may be room for only one wagon, in the pit, at the point where the best material is to be found, a dozen teams will drive in, and

the wagons are filled with whatever stuff may be most easily obtained, whether gravel, sand, boulders or clay. Another common cause of earthy matter and sod being mixed with the gravel, is to be found in the practice of tunnelling under an overlying layer of soil which gradually falls into the pit, is mixed with the metal, and is drawn with it to the road. This earth should first be stripped from the surface if it does not form too deep a strata, or care should be taken as it falls to keep it from mixing with the gravel.

In the district formed by the angle of the Ottawa and St. Lawrence Rivers, and a line extending northerly from the vicinity of Kingston, the surface is composed largely of glacial drift or "till," while in some portions of this section, bedrock appears at the surface. Gravel is not very plentiful, but that which does occur is of a hard quality, the waste of trappean rocks, contains little clay and is sometimes comparatively free from sand. It is to be looked for, usually, in narrow ridges and mounds formed in the tunnels and caves of ice, through which were the courses of sub-glacial streams. Occasionally in this district, more especially the easterly portion, lake deposits of the post-glacial age are to be found in stratified layers.

Westerly from Kingston and south of Georgian Bay, the glacial waste is overlaid to a greater extent by lacustrine deposits, of which the chief stratas are the Erie and Saugeen clays. Inter-stratified with these, and in drifts and pockets, gravel is to be found at frequent intervals. The quality, however, is not equal to that of the eastern district, being composed more largely of the waste of the local limestone strata. It is of a soft "cement" variety, and has considerable clay and sand mixed with it. Along the base of the Niagara escarpment from Owen Sound to Brantford, there is a remarkable bed known as Artemesia gravel, having an average width of about twenty three miles, and a considerable depth. This extends easterly also, following the north shore of Lake Ontario, in what is known as the Oak Ridge. While this formation of gravel is comparatively free from earthy matter, it contains a quantity of large stones and boulders which interfere with it as road metal. It is principally composed of the water-worn debris of the Niagara and Guelph limestones, mixed with which are some pebbles and boulders of trappean origin.

In searching for gravel, the clearest indications are usually to be found along the banks of streams, where any extensive strata is apt to be exposed. A post-hole auger affords a convenient means of making tests over the surface of the soil for gravel, but the best implement is generally a simple form of drill. There are cases in which gravel beds may be entered at the level of a stream bed, and water is thereby obtained for washing the metal by natural drainage, affording a cheaper means of freeing it from sand and earthy matter than by screening. Gravel is still being deposited in drifts and bars by the agency of streams; this will be found to partake of the character of the pit gravel of the locality but generally will contain less clay, although sand may easily be in excess. This is usually one of the best sources, as the gravel can be washed by natural drainage. Lake gravel is often a good metal but varies greatly. It is apt to be slatey, an undesirable quality. It will be free from dirt and clay, but contains sufficient sharp sand to secure consolidation, especially if a roller is used. Gravel which retains a perpendicular face in the pit in spring, and shows no trace of slipping when thawing out may generally be assumed to be sufficiently clean and free from clay for use on the road without any treatment other than is necessary to remove stones greater than one inch and a quarter in diameter.

BROKEN STONE.

It is impossible to discuss the wearing qualities of different kinds of stone for macadam roads from the standpoint of names. Granite, limestone, sandstone, are names of different rocks common in this province, but to say that granite is better than limestone, or that limestone is better than sandstone, while true, perhaps, of the best qualities of each, may be quite incorrect, since a good sandstone may be infinitely preferable to a poor limestone or granite. The best stone for a macadam road is that which is hard and tough, not easily affected by the atmosphere, moisture, and the varying conditions of

climate. The choice will generally lie between a cheaper and less durable stone near at hand, and a more costly but better stone from a distance. The city of Cleveland, Ohio, is for example importing trap rock from Poole Island near Bruce Mines, on the north shore of Lake Superior. This rock is very hard, tough, and for qualities of endurance can scarcely be surpassed on the continent.

A great portion of macadam roads in Ontario will be constructed of limestone, since this rock is the most common, quarries being within easy access of almost any part of the province. In quality it ranges from that which is useless to that which is almost equal to trap. Limestone which is tough and closegrained, is an excellent material for roads on which the weight of traffic is not excessive. Some dolomitic limestones, while hard, appear to lack in toughness. Other limestones of a slaty texture have few wearing qualities, are rapidly disintegrated on exposure to the atmosphere, and should be avoided. Some limestones of an open, porous nature, yield readily in this climate to the effects of moisture and frost, merely turning into mud. But the excellent binding qualities of limestone make up largely for a lack of hardness, a sort of cement being formed by the dust, which adds very much to its durability.

Some qualities of granite rank high, but much of it will crumble readily, and wear into sand. It is to be had in many parts of northern Ontario, and could be easily reached by boat or rail. Except when the traffic is unusually heavy on the road to be paved, and the best granite is obtained, the cost of transportation will debar it from use in the greater portion of Ontario. Gneiss, like the granite, occurs in vast quantities in the northern districts, and is of variable quality. That containing much felspar or mica is the least desirable. Sandstone is seldom of service, except when used for very light traffic on country roads, since it also readily crumbles and wears to sand. It is quarried chiefly at the Forks of the Credit.

Limestone is quarried very largely at Queenston, at Hagersville and other points along the Grand river, at Kingston, and the counties east; granite is obtainable in the more northern part of the province. Trap rock occurs in dikes of considerable proportions near Kingston and Gananoque.

In determining the best quality of stone for road purposes, there are four prominent destructive agencies which have to be considered: (1) The crushing action of loads. (2) The grinding action of the wheels. (3) The blows from the shoes of horses. (4) Climatic influences of air, water and frost.

With respect to the first three, a stone may have great hardness and splendid crushing strength, but at the same time be brittle, yielding readily to the grinding effect of wheels, and the blows administered by the hoofs of horses. On the other hand, a stone may be able to resist, in a measure, the second two wearing agencies, those of "abrasion" and "impact," and yet be so soft as to crush readily.

The fourth agency, the decomposing effect of the atmosphere, is one of very great importance. The denser stones, those which absorb the least water, are usually best able to resist the injurious action of frost and moisture. The weight of a stone is usually some indication of durability in this respect, the lighter stones being those of less density, and in consequence are the less desirable.

There are other features which a good rock for road-making should possess. When crushed they should break into a compact form. A stone that, in breaking, takes thin, flakey shapes, will not wear so long as one that breaks into cubical pieces, nor will it consolidate so readily in a roadbed, for a wheel, in passing over the side of a flat stone, will throw it out of place and loosen the stones adjoining.

The tests usually applied in determining the qualities of stone are those which indicate crushing strength; the power to resist impact and abrasion; the density, determined by the weight of the stone; the amount of water absorbed. While elaborate trials may be made, a practical man can judge of the qualities of a stone by applying simple tests; by breaking the stone with a hammer; wearing it on a grindstone; crushing it in a black-

smith's vice ; scratching with an iron nail ; breaking small pieces with the fingers. By such simple means, a general idea of the stone can readily be formed, but no test is conclusive except actual wear on the road, for varying local conditions, lead to varying results.

Boulders.

Broken stone produced from boulders is objected to as road metal on various grounds. The rounded sides will not permit consolidation with the minimum of vacuum. If they have been exposed to the atmosphere the boulders are very apt to be decomposed, are soft and will crumble readily. The mixture of different kinds of rock on the road surface, some hard some soft, permits unequal wear, and produces a rough surface.

While these are defects which certainly are not to be overlooked in the choice of road metal, boulders nevertheless, constitute a very valuable material for the construction of a road, particularly in localities where they are plentiful and gravel or bed rock not readily obtainable.

Boulders are generally the result of glacial action. A large portion of the province is overlaid with "till"; that is, a mixture of boulders, clay and pebbles in varying proportions, brought down by the rivers of ice, and left in a broad irregularly disposed sheet, as the ice melted. In some instances, the retreat of the ice, as the glacial age passed away, was interrupted by pauses and readvances. This created the frontal moraines or ridges of debris pushed out by the glacier and deposited along its front. Subsequently the inland lakes and rivers which have been obliterated by upheavals of the earth's surface, together with lakes and rivers now existing have covered up much of the material and acted upon the glacier waste in many ways. The rain also, washing away the earthy material supporting boulders on the hill tops, has left them exposed, or permitted them to roll to the valleys below.

Frequently these frontal moraines are distinct features of the landscape, as is the case between the southern portion Hastings County and Georgian Bay, where they appear as rough stony hills. In other cases they form merely indications on the surface of the ground, the moraine having been covered with a later deposit of fine stuff. As a rule, the moraines are better sources of road metal than the till, since in the former case there is a less quantity of clay and sand intermingled ; while the boulders taken directly from the till or moraines are generally of a better quality than those at the surface of the ground, which have been exposed to the action of the atmosphere.

The majority of the boulders to be found in Ontario are of Huronian or Laurentian origin ; that is, rocks of gneiss, trap and granite brought down by the glacial rivers from the distant north easterly from Georgian Bay. In some parts, rocks of local origin are to be found, notably in the counties of Glengarry and Wellington, where considerable quantities of limestone are exposed.

In selecting field boulders, care should be taken to discard all rock which shows signs of having "weathered," or decomposed by the action of the atmosphere. Sandstones and granites are peculiarly subject to this disintegration, while soft limestones are very common. Rocks which should be condemned from this cause are those which crumble readily under successive blows of a hammer, or which show iron stains when broken. A little experience will quickly teach a judicious roadman to detect the stone which is unfit for road purposes.

PLACING METAL ON THE ROADS.

In order to obtain a knowledge of the manner in which gravel, broken stone or other road metal should be placed on the roadway, it is evidently necessary to consider the objects to be attained by the use of a surface covering. Experience teaches us that these are :

- (1) To prevent water entering and softening the natural soil beneath.
- (2) To present a hard and durable wearing surface, sufficiently smooth to make travel and the transportation of heavy loads, easy and agreeable.

According to the first of these objects, the covering must be of such a texture as to be as far as possible impenetrable to water, and of such a shape as to shed the water to the side gutters or ditches. According to the second, the road covering must partake of the character of a floor. In summer and in dry weather, a road made of the natural soil, if not sand, is a satisfactory driveway for moderate traffic. It is to make a good wet weather road, that metal is usually applied. If the work done on the road is not successful in producing this, the efforts have either been inadequate or misapplied.

The customary way of putting gravel and broken stone on the road is to merely drop the metal from the wagon, leaving it, as it falls, in a continuation of irregular heaps, to be levelled and consolidated by traffic. This metal is usually placed on the road in June. Vehicles at once turn away from it and find a more serviceable dirt drive along the side of the road or in the gutters. Traffic continues to follow the dirt roadway until such a time, late in the fall, when it has become rutted and muddy. Not until forced to do so by the impassable condition of the dirt track, do vehicles turn to the metalled portion. If this dirt track is between the metal and the ditch, the ruts and holes will interfere with surface drainage ; will prevent the water reaching the ditch quickly from the crown ; will be receptacles for a large amount of water. In soaking away, this water will pass under the gravelled portion, and assist in softening the natural sub-soil which, in a well made road, must be sufficiently dry and firm to support the weight of traffic. The gravel or stone lying loosely on the road, has permitted water to pass through it to the natural soil below, as through a sieve. The first effect of wheels passing over it is to create ruts in the metal itself. These ruts in turn become receptacles to hold water, the gravel in the ruts is quickly forced aside and downward, and the wheels sink as far as the mud will permit them. Winter may freeze the ground, and afford a firm means of travel during the frosty season, but when the weather breaks up in the spring, the roadway, upheaved and honey-combed by the ice which formed in and below it, is many times worse than in the fall.

It is evident that to drop gravel, and especially broken stone, loosely on the road is very far from being an economical system of roadmaking. The summer succeeding the application, the road, after the ruts are worn down, may become a good driveway ; yet the gravel or broken stone has been so churned, and mixed with dirt, that its life and durability have been largely wasted before it becomes sufficiently compact to answer the purpose for which it was intended.

A great deal, by very simple means, can be done towards getting better results from the road covering. The metal when first placed on the road could be nicely spread with a rounded surface, as it is intended to lie when consolidated. The metal, particularly if gravel, will, during the summer, consolidate much better than if left in a high heap. In the fall it will be harder, and, not presenting so formidable an obstacle to travel as a series of irregular mounds through which the horses must wade, vehicles will turn from the temporary dirt tracks earlier in the season, before the subsoil has become very soft. If at this period a man is sent along the road, with a rake to fill up the wheel tracks as they appear in the loose gravel, or broken stone, traffic will, in a short time, very effectually consolidate the road. The expense of a man in raking the metal into the ruts and keeping the road properly rounded up during the fall and spring, will be more than saved in the future repair and re-metalling, while a much better road will be the result.

ROLLING COUNTRY ROADS.

Although much can be done by the use of a rake, to assist traffic in consolidating the metal, as indicated in the preceeding article, "Placing Metal on the Roads," the road, nevertheless, should be made for traffic, not by it. A much better plan is the use of a road roller, the advantages derived from which are several :

(1) A good track is immediately obtained, and vehicles at once take the centre of the road.

(2) A dirt track is not made near the ditch, and, by this means, the side of the road is not cut up and made so uneven as to interfere with surface drainage.

(3) Traffic is not inconvenienced in fall by having to drive through loose gravel or crushed stone.

(4) The gravel or stone is not forced down into the sub-soil by the wheels and feet of the horses ; is not churned and mixed with the earth, and there is in this way a great saving in the amount of metal.

(5) There is a great saving in manual labor, and repairs are more easily and effectually made.

An impediment to the use of heavy rollers in a good many townships, is the insufficient strength of bridges, and culverts ; and, while valid in some instances, the objection is liable to exaggeration in others. Weak wooden bridges and culverts could in many cases be temporarily strengthened sufficiently ; while in others, they could be entirely avoided by first completing the rolling on one side, and then passing around a block or so, to commence work on the other.

It is not probable that many townships councils will find themselves in a position to purchase a roller for the exclusive use of one township. In England, the county councils usually undertake this measure. If this were done in Ontario, the roller could be rented to the townships for use on the main roads. Other townships are so situated that they could rent from towns in the vicinity which own a roller. Or, as suggested in discussing stone crushers, a private citizen could be induced to purchase the roller under proper assurance of employment. When the benefit to be derived from the use of rollers is better understood they will be more generally adopted.

DRAINAGE OF MACADAM ROADS.

A prevailing impression with respect to macadam roads is that an allowance which has had the centre portion raised and levelled, with open trenches at the side, and a layer of gravel loosely dropped on the graded portion, has been macadamized. This is very far removed indeed from the principles laid down by Macadam, the principles which have changed the roads of England from a state of impassibility to a condition of excellence, unsurpassed by any in the world.

Macadam insisted upon the theory that it is the native soil which really supports the weight of traffic ; that while it is preserved in a dry state, it will carry any weight without sinking, and that it does, in fact carry the road and carriage also ; that this native soil must previously be made quite dry, and a covering impenetrable to rain must be placed over it when in this dry state. To the erroneous opinion long acted upon, that, by placing large stones in the road bed and covering them with smaller metal, although the sub-soil was in a soft, wet state, a good road could be made, he attributed the bad condition of most of the roads of Great Britain. The first requisite of a macadam road is drainage.

Subject as we are, to continued autumn rains, followed by a period of severe frost and snow, with the attendant thawing and flooding of spring, it becomes of prime necessity that, in Ontario, drainage must be as perfect as possible. So little is the truth of

this recognized that a drive along not only our country, but many of our city and town roads, would lead one to think that to get the water into the road and to keep it there is the idea of our road-makers.

Underdrainage, except in very rare cases, is entirely ignored. Surface drainage generally consists in heaping up a driveway in the centre of the road allowance, covering this with a loose coating of gravel, and leaving the excavations at the side of the roadway thus formed to hold water, since, although called "drains," they seldom have outlets. The water held in these trenches soaks into and softens the foundation of the roadbed until there is a sort of floating island.

To remedy this state of affairs, gravel is, by our road-makers, continually heaped on in the hope of in some way covering the water and corking up the mud. This is a very expensive and unsatisfactory method of roadmaking. Underdrains of common field tile (provided with outlets) to "lower the water line," and open drains or gutters (also provided with outlets), to carry away surface water, are the practical means of creating a foundation for a road. With a soft, wet, yielding foundation, a road is no more durable than any other structure under similar conditions.

Any thoughtful farmer who knows the effect of underdraining in his fields will at once recognize its usefulness in road-making. In the fall, water will be rapidly removed from the bed of the road and the destructive action of frost lessened. In the spring, the frost will come out of the ground more quickly, and each of those periods in fall and spring are shortened, when, with the foundation and surface thoroughly saturated, the roads are not only impassable on account of the mud, but are injured by traffic more than in all the remaining nine or ten months of the year. One year's statute labor with the annual appropriation, spent in providing proper drains for the roads of the province would do more than can be done in five years with the present method of merely filling the holes which appear, with gravel or crushed stone.

Underdraining and grading should be carried on simultaneously. Unless drained, the traffic during the ensuing autumn and spring will usually leave a graded road as shapeless and rough as a pioneer wagon track can be. If drained, there will be a foundation for the gravel or crushed stone to rest on when applied.

A porous soil, like a sponge, retains in its texture, by attraction, a certain amount of water. When water in excess of this is added, it sinks to the first impenetrable strata, and from there it rises higher and higher until it finds a lateral outlet; just as water poured into a pail will rise higher and higher until it finds an outlet in the side of the pail, or until it flows over the top. Underdraining supplies the necessary outlet for this excess moisture at a proper depth from the surface; it "lowers the water line."

With plastic clays the process is slightly different. Clay will absorb nearly one-half its bulk and weight of water. In drying, it shrinks and is torn apart in different directions. The fissures thus commenced by a tile drain become new drains to lead water to the tile; and so the process of contracting and cracking continues until a net work of fissures is produced, and the stiffest clay is thereby drained.

The injury done to roads by frost is caused entirely by the presence of water. Water expands on freezing, and the more there is under a road, and above frost line, the greater is the injury. In freezing, the particles of soil in immediate contact with the water are first compacted. When room for expansion ceases within the body of the soil itself, owing to its saturated condition, the surface is upheaved. When thawing takes place the sub-soil will be found honey-combed, ready to settle and sink beneath traffic. It is therefore of the utmost importance that the soil should be relieved of all water of saturation as quickly as possible by underdrainage. The impassible condition of most roads in Ontario during the spring, often axle deep with mud, is to be attributed very largely to a wet sub-soil which has been honey-combed by frost.

The extent to which underdrainage should be used, must be controlled by the quality of the soil over which the road is laid. There are instances on sandy, gravelly and porous soils, in which underdrainage is so perfectly provided by nature that artificial

drains are not needed. Clays nearly always are benefited to such an extent as to render the outlay for sub-drainage a most profitable investment. The necessity of drainage is further controlled by the situation of the road. If the ground is high-lying it will be less in need of drainage; but if low-lying, or if it receives the soakage water from up-lying lands, it will usually be impossible to secure a dry sub-soil without the use of tile.

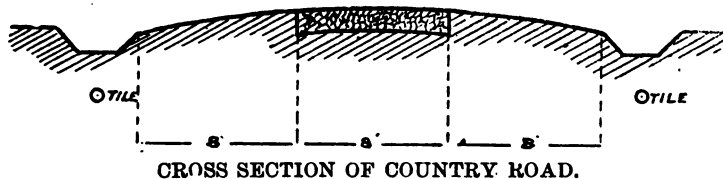
The drain should be placed below frost, otherwise water freezing in the tile will burst and destroy them. If not below frost line the drain will be apt to be stopped up during the spring, the period during which it is most needed. The fall should average one foot in 100; the minimum should never be less than one in 1,000. The tile should be evenly laid; the capacity of the entire drain is regulated by the smallest portion, and to lay tile in an undulating line is equivalent to reducing the bore; when unevenly laid it has a further tendency to catch sand and solid matter, producing stoppages.

There is some dissension among road-makers as to where underdrains should be placed with respect to the roadway. Some prefer one tile drain under the centre of the road, this with the mistaken idea that the water will be caught as it percolates through the road bed. A road bed, if nicely crowned and well compacted under the roller, will shed the surface water to the side gutters without permitting it to pass, to any extent, into the road foundation. It is better practice to place two tile drains, one under each of the surface gutters or ditches where they may if necessary, serve as outlets, catch-basins being provided to permit surface water to enter the tile when natural outlets cannot easily be obtained.

The advantages of the two tile drains, one under each side of the roadway, compared with the plan of having one drain in the centre, are several: The two drains are more effective in securing perfect drainage than one could be; the aftermath of settlements in the centre of the road is avoided; the tile drains on an old road can usually be laid at a greater depth if placed under the old open drains; the expense of placing one drain in the centre of an old road is often greater than that created by placing two drains at the sides; when at the sides they may be more easily used when needed, as outlets for surface drains. The rule, however, of placing tile drains on each side of the roadway need not be rigidly followed, but should be judiciously applied, according to the amount of underdrainage needed, the nature of the soil, the location and the opportunity for obtaining outlets.

DIMENSIONS OF COUNTRY ROADS.

The width of the graded portion on the majority of country roads has already been fixed. Various ideas have prevailed among the different pathmasters regarding this detail, and the width differs accordingly, ranging from ten to forty feet. In the case of old roads which have been coated with gravel or crushed stone, it will seldom be deemed practicable to change the existing width. Twenty-four feet between ditches will generally be found ample, while less than eighteen feet is seldom advisable.



Common field tile underneath open ditches. Gravel or broken-stone roadbed, eight feet wide.

It is not to be understood that the width of the graded roadbed is necessarily the width of the metalled portion. The latter, for the majority of country roads, need not exceed eight feet, but for very largely travelled highways, may increase to twelve or even sixteen feet.

The depth of gravel or stone to be used must vary with the quality of the material, the amount and nature of traffic on the road, and the nature of the sub soil. It is obvious that a dry, compact and stony sub-soil will need less metal than would a plastic clay, difficult of drainage. No definite rule can therefore be laid down other than that from six to twelve inches of well consolidated material will afford a sufficient range to accommodate most circumstances. Under proper conditions, ten inches of metal should accommodate the heaviest traffic to which a gravel or broken stone roadway can be economically subjected.

From the edge of the open drain the graded portion of the roadway should be crowned with a circular rise of one inch to the foot from side to centre. That is, a driveway twenty four feet wide should be one foot higher at the centre than at the side. This amount of crown compared with European practice may appear excessive, but with gravel roads, and roads metalled with the quality of stone commonly available, this crown is no more than sufficient to provide for the wear and settlement, consistent with good surface drainage.

The elevation of the road above the level of the adjacent land, need not be greater than is sufficient to provide against the overflow of storm water, which should always be guarded against. The depth of the open drain must vary according to the amount of fall and the quantity of water to be provided for ; also according to the sub-drainage needed and provided. When tile sub-drains are used, the open drain can usually be very shallow ; in which case the width of the graded roadway can be narrowed, there being no danger of accidents such as are caused by a deep trench at the roadside. The tile drains, as elsewhere stated, should be placed below severe frost, usually a depth of three or four feet.

CROWNING.

A very notable defect of most country roads is the flat or even concave surface ; others present the opposite extreme, and are so rounded up as to be dangerously high in the centre, making it difficult for vehicles to turn out in passing. Roads must be crowned sufficiently to shed the water from the centre to the open drains at the side, otherwise water will stand in the roadway, soak into it, soften and cause rapid wear and decay ; but a crown higher than is necessary to properly drain the surface is also objectionable.

The amount of crown must be sufficient in the newly constructed road to provide for wear and settlement. It should be greater in the softer and rougher varieties of pavement than in those which are hard and smooth ; a gravel road, or a broken stone road, must have a greater crown than a brick pavement. The amount of crown needed on a gravel road or one of broken stone varies with each of these classes according to the quality of gravel or stone, the nature of the subsoil and the care taken by rolling to provide against settlement. The amount of crown for newly built gravel roads has been stated under "Dimensions of Country Roads," as one inch of rise to the foot of width from the edge of the ditch to the centre of the road, and this should be increased in the construction roads on hills according to the steepness of the grade.

The amount of crown should not be more than sufficient to provide for surface drainage. A steeper crown than is necessary tends to confine traffic to the centre of the road, and in turning out, the weight of the load is thrown on one pair of wheels in such a way as to rut the side of the road. The shape of the crown is a matter on which expert road-makers differ, but with the class of material available for roads in Ontario, and the methods and plans of construction, a form as nearly circular as possible will be found most serviceable, and most easily obtained.

HILLS.

Roads located on hills, or along the sides of hills, are generally found difficult to build and maintain, and very often we find a very large proportion of statute labor annually placed on them, to little effect. The difficulty usually arises from the rushes of storm water, from springs underneath the road, and, especially on sidehill roads, from water soaking from higher levels.

The rushes of storm water are often occasioned by the practice of carrying water long distances in open drains, and finally pouring it over the hill by the roadside. If the hill is steep, and a cut has been made, the water is not, and very often at the time of spring floods and freshets, cannot be kept in the open drains, and so is allowed to make a channel of its own down the centre of the road. This condition is the common result of not disposing of water in small quantities along natural watercourses. No water should, as a rule, be allowed to pass over the hills by the roadside, except that which naturally falls on the surface of the slope. Provision should be made for the disposal of water in the drains back of the hill, by carrying it through private property, under the authority of the Drainage Act, if necessary. Property owners, however, should understand the wisdom of permitting drains to be constructed across their lands, when the benefit to be derived is not only better roads but better drainage of their own fields.

The surface drainage on hills should be very pronounced. The crown of the road should be slightly higher than is needed on level ground, a rise of one and one-half inches to the foot from side to centre being advisable for gravel roads. The crown must be sufficient to draw the water to the side gutters, and to do so, it must be sufficient to overcome the tendency of the water to flow directly down the hill, following the line of the wheel tracks. If the water commences to take the latter course, the wheel tracks are quickly deepened to ruts, stones are loosened or protrude, and the road becomes roughened and channeled.

Under-ground currents of water often find outlets on the hillsides. If any of these springy places occur under the roadbed, it will be necessary to tap them at a good depth below the surface with tile drains. In such cases, tile drains will be needed under the open drains at the sides of the road, and the blind drains may then be carried diagonally across the road into the side underdrains. The open drains will sometimes need to be protected with cobble stones, if the hill is long or subject to damaging rushes of water.

Roads passing along the sides of hills are frequently softened and injured by the soakage water from high lying lands. This water should be intercepted before it passes under the road, by a deep drain along the side of the roadway next the hill. Tile should be used, if possible, instead of a deep open drain, and the trench filled with gravel, stone or other porous material, to more readily intercept and absorb the soakage water.

Municipalities will usually find it a measure of economy to bring hill roads to the grade at which they are desired to remain as quickly as possible, and to this end special money grants will be necessary. To leave such work to be performed by statute labor alone is not often satisfactory. The grading done one year is apt to be so inefficient that it is, in a large measure, destroyed by the rushes of water in the ensuing wet seasons. The roadway, moreover, being annually covered or cut for a number of years, is slow to settle, becomes impassable on the slightest provocation, and absorbs labor which is needed by other sections. Municipal grants are needed on main roads, but they are also needed in doing permanent work on the lesser travelled roads, where hills or other unusual circumstances of location render the construction of roads a matter of more than average difficulty. The labor and cost of road building should be equalized as far as practicable among the different sections of the municipality, so that efforts put forth in road improvement will not be retarded by discouraging difficulties.

EMBANKMENTS AND CUTTINGS.

The protection of the sides of cuttings and embankments should be skilfully attended to. It is very common to see these washed away in places after a heavy rain, or after the spring thaw; the sides of the cuttings settle into and fill the open drains, and the water is forced into the road; the sides of embankments wash away leaving dangerous holes in the road. The tendency is to make cuttings and embankments too steep, with a desire to do the least possible amount of work.

The stability of earth slopes is endangered by the action of air and moisture, especially by alternate frost and thaw, and depends on the ease with which water is drained away. A certain amount of moisture increases the strength of the slopes, but too much acts like a lubricant, and reduces the earth to a semi-fluid condition. Clay retains water and becomes pasty. Sand, if in a basin of water-holding earth, becomes a quicksand and is completely unstable. A mixture of sand and clay, the former favoring the access of water, and the latter preventing its escape, is one of the most difficult cases to deal with. There is a certain "angle of repose" at which the tendency of earth to slip is overcome. This angle varies with different kinds of earth, under various conditions of moisture. Wet clay is troublesome, and an angle of about 16 degrees is sometimes needed to secure it. Well drained clay, however, will rest at an angle of 45 degrees, or a slope of one to one. With average gravel and compact earth, a slope of one to one is a safe angle, although first-class gravel will retain an almost vertical face for a considerable time. Sand varies greatly, "water sand" being no better than wet clay. Dry sand usually needs a slope of one and one-half to one. Vegetable earth, also, is apt to be unstable and needs a slope of one and one-half to one. Rules of this description cannot be laid down with sufficient accuracy to be of use, except as they serve to indicate what is to be expected with different soils. The qualities of soils are so variable that it is necessary to learn by observation what slope is needed for a particular piece of earthwork.

The natural form of an earth slope when in permanent repose is a concave curve, with the flattest portion near the bottom. There is a careless tendency to leave the slope rather in the opposite form, with an outward curve. Convex or straight slopes will invariably slip until the natural form is obtained, and in cuttings or embankments approaching ten feet in height, care given to a proper construction in this regard is always profitable.

A dry stone wall at the foot of an embankment or cutting will protect the drain from slipping earth. A coating of sod is one of the best protectors to the slope; and a few inches of vegetable mould over the surface, with a liberal sowing of grass seed, is a measure sometimes adopted.

LOCATION OF ROADS.

A very great number of the roads in this province are faultily located. This has resulted not only in waste and inconvenience in transporting loads, but has very materially increased the difficulty and expense of building and maintaining the roads. The basis of land surveys must be rectilinear; concessions and lots should in the survey be marked off by straight lines. The difficulty has arisen from blindly following these lines in the construction of roads, irrespective of their suitability for the location of a road. While land has been subdivided by straight lines, it was never the intention that these lines should mark unalterably the route of roads.

There is an advantage to the farmer in having all his fields of rectangular shape, as they can be cultivated more easily than when outlined by circular or irregular lines. There is a disadvantage, too, in having an estate cut into separate sections by diverted highways. These are obstacles to the proper alignment of roads in long settled and populous districts, but present little difficulty in new portions of the province. How-

ever, it is usually very much more to the interest of a property owner that the roads leading to his farm should be good and easily maintained than that his farm should be in a compact block with the roads to it impassable during a portion of the year, and even then expensive to build.

It is, of course, desirable that a road between two points should be as direct and short as possible. But a road is not necessarily more direct because it follows absolutely one pointing of the compass. The line followed by a vehicle, leading up the hills and down into the valleys, may be no shorter, nor perhaps as short, as a diverging route following comparatively level ground. Not only may nothing be gained in point of directness by following the line of the original survey, but there is to be considered the greater horse power required to move loads up and down the hills, demanding, too, a greater expenditure of time.

The expense of building a highway is materially increased by steep grades. A moderate divergence of numerous highways in the province would do away with many expensive and unsatisfactory cuts and fills, and with a large number of bridges. The unsuitability of the soil also, if low lying, swampy, or composed of vegetable matter, may render advisable a change of location in favor of a course which will offer a firmer and more easily drained subsoil.

Directness must frequently be sacrificed to obtain an easy grade, and to avoid expensive construction over bad ground, cuttings, fills, bridges and excessive grades. Councils are authorized to alter the location of roads by the Municipal Act, in a manner fair to all parties, and it is advisable that this power be judiciously used whenever circumstances render it practicable. Opposition will no doubt be offered in some cases by the individual property owners affected, but councils representing the general public have a responsibility resting upon them which should not be shirked in a matter so important.

REPAIR AND MAINTENANCE.

Roads should be repaired, not once a year, nor twice, but as soon as signs of wear appear. The statute labor system as usually operated, with one annual overhauling, is the most wasteful plan that could be adopted. Ruts should not be allowed to form in a gravel or stone road when once properly constructed, but material should be kept in place by a constant use of the rake. Settlements and hollows should not be allowed to hold water and create a pitchhole for want of a load of metal. Drains should not be allowed to become obstructed, thereby saturating and softening the whole roadbed, because it is nobody's business to spend a few hours in freeing the water. Culverts should not stand full of water to be burst by the expanding ice because of a neglected outlet. An almost inexhaustible list of these everyday occurrences could be mentioned, which in themselves apparently trifling, become in the aggregate of very great importance. Road-making is made up of details none of which can be overlooked, except at a loss.

The pathmaster should have power to give immediate attention to all emergency work rendered necessary by washouts, etc., either by personal or hired labor. Under the advice of the supervisor, he should be able to send a man over the beat as often as necessary to repair the effect of ordinary wear. Better still, a man should be employed to devote his whole time to a certain mileage of roads, which would include a number of beats, to make repairs as they become necessary. Every farmer, too, should appreciate the value of good roads sufficiently to voluntarily devote time to the roads passing his property, rather than permit them to become bad or impassable because of neglect.

If a general municipal system of employing a man to take care of certain beats is not practicable, a plan followed in at least one township may be of service; in which the residents interested in certain of the main roads have subscribed an amount sufficient in all to keep a man constantly engaged on each road. As a measure of economy, it is certainly advisable to adopt some means to prevent the roads getting out of repair.

ROAD MACHINERY.

Improved and modern road-making machinery is as necessary in the construction and maintenance of highways as are self-binders in the harvest field. Roads cannot be well or economically constructed without suitable tools and implements. If a workman is known by the condition of his tools, a very damaging reflection is certainly cast upon the majority of townships. Not merely are improved machines seldom employed, but the scrapers, plows, wagons, etc., used in the performance of statute labor are rarely on hand as needed, and the utmost carelessness is observed with regard to them.

GRADERS.

Road-grading machines are now very commonly used by townships. Their greatest value is in the shaping of dirt roads. The crowning of the roadway can be done very rapidly by their use, and for this work alone every township requires at least one grader. Some townships using as many as four, feel that they are needed. On gravel roads there is a marked tendency for ruts to form in such a way that the metal is forced upward and out from the wheel tracks, and a road grader can very profitably be used to level the ruts by scraping this gravel back to its place. Another important quality in the maintenance of gravel roads, is the facility with which the shoulders of the roadway can be cut off. These shoulders usually consist of square corners left in the original construction of the road, augmented by dusty material washed down from the centre. They obstruct the flow of water from the gravelled portion to the open drain. When cut off by the grader, the material should be thrown outward and across the ditch and used in levelling the sides of the road.

The stuff from the shoulders and from the ditches should never be drawn to the centre of the road, a practice which has ruined a number of excellent gravel roads. It is a mixture of dust, clay, horse droppings and sod. Placed on a good gravel foundation, it may afford a temporary benefit in summer, but in wet weather it works the destruction of the entire road. The shoulders having been formed, the proper treatment, as stated in the section on "Existing Roads," is to crown the roadway with clean gravel.

ROCK CRUSHERS.

The rock crusher is one of the most important of modern additions to the list of road-making machines. By their use stone can be crushed so much more cheaply than by the old method of hand breaking, that, so far as cost is concerned, stone roads are within the reach of every municipality having suitable rock in the vicinity. In the treatment of gravel a crusher is frequently most valuable, since, if containing many large stones and boulders, it will be possible to place a crusher in the pit and pass all the gravel through. A rotary screen attached to the crusher and driven by the same power will screen the resulting metal, removing sand and clay. Whenever a crusher is employed the rotary screen should be used also to separate the road metal into grades according to size.

The most common type of rock crusher is that in which the breaking is done by the opening and shutting of strong iron jaws arranged so that they can take in large stones, and, by successive openings and shuttings, fracture the stones until the fragments are sufficiently small to drop between the lower margins of the jaws. The fragments are there caught in a chain of pockets, or elevator, and passed up to the rotary screen, which is so designed as to separate the product into grades according to size.

Rock crushers can be readily moved from place to place, and where boulders are plentiful in the fields along the road, may be set up so as to crush stone for short sections of road as required. Municipalities in which boulders are plentiful, will find the property owners willing to provide, at the crusher, all the stone that can be gathered from the fields.

PROVINCIAL INSTRUCTOR IN ROAD-MAKING.

ROLLERS.

Road rollers, the benefit of which are referred to elsewhere, are of designs to be operated by horse or steam. Horse rollers usually consist of one main roll in two sections. The standard diameter is fifty inches, with each section twenty-six inches wide, giving a total rolling width of fifty-two inches. The standard weight is about four tons, but may be increased by loading to six or eight tons. The reversible type, whereby the tongue may be revolved from one side to the other is to be preferred.

Steam rollers generally weigh from ten to twenty tons, the most popular being of ten, twelve or fifteen tons. For country roads a weight of ten tons will be sufficient, especially on gravel roads, and in view of the strain on culverts and bridges. In towns twelve tons, with the ability to load more heavily for surface consolidation, will be satisfactory. If it is proposed to rent it for township work, ten tons will generally be better, and will give excellent results. The lesser weight, while requiring more repeated operation to compact the road metal, is believed by some to do more durable work. Steam rollers, of which illustrations are given, are all very much of the same type.

SELECTION OF MACHINERY.

In all kinds of machinery, the most important points to observe are simplicity of design, quality of material and workmanship, ease of operation, lightness of draft, and adaptability to the work for which it is intended.

OPERATOR.

Graders, crushers and rollers should not be passed around from section to section for everyone and anyone to operate, but capable men should be employed for this purpose and should always accompany them. It is further necessary to see that the operator is a man who knows something of road-making and of machinery, and has good practical judgment. A machine of itself possesses very few of these qualities, and unless the operator does, dissatisfaction and failure will result.

PURCHASE OF MACHINERY.

With regard to the purchase of these machines, it is desirable that every township should possess at least one grading machine, and as many more as circumstances may require. The extent to which broken stone will be needed and the location of the quarry, will suggest the best policy to pursue with respect to the crusher. When one crusher to provide stone for the main roads of a county is sufficient, it may be advisable for the county council to own it, arrangements being made with the various towns and township municipalities, either for the rental of the machine or for the purchase of the crushed metal. Circumstances may render it advisable for a private citizen to provide metal at a fixed price per cord; thus a man owning a threshing machine, and having the steam engine idle for a portion of the year, might be readily induced to consider such a proposition, to the advantage of all parties concerned. If water power is available within reach of a quarry, it will afford a cheap means of obtaining the requisite power.

BRIDGES AND CULVERTS.

Experience has shown durable improvements to be the most economical. One of the most common leakages of municipal funds arises from the maintenance of temporary wooden bridges and culverts. The life of a wooden structure is, at the outside, not more than one-half or one-third that of an iron or stone bridge, or culvert of cement or masonry, while the cost of keeping the former in repair during that period, is very considerable. The price of the timber is constantly rising, and, in many localities, first class material is difficult to obtain at any reasonable price. The cost of iron has been very much reduced of recent years and will probably become still less.

Good judgment is required in estimating the right size of waterway for a particular location. Elaborate rules might be given for determining the probable flow of water from a given area, but these would be of little assistance to the councillor. In this respect, consideration has not only to be given to the amount of the area drained, but also to the amount, slope and general shape of the lands drained, so as to know whether the water will reach the culvert or bridge rapidly and in a large body, or gradually with little force. The waterway must be capable of accommodating not merely the ordinary flow, but the unusual floods. Water checked and permitted to back up is a frequent source of wash-outs, and in the case of culverts permits the water to soak into and soften the roadway. On the other hand a waterway larger than is necessary is needlessly expensive, so that careful consideration as to the size of the culvert or width of span is a matter which must be urged. Error should be on the side of an excessive, rather than insufficient capacity.

When the country was covered with forest, the water was impeded in its flow over the surface of the ground by fallen timber, and rough and broken land; and after reaching the stream, violent rushes were prevented by standing timber, accumulations of floating logs, and other debris. These obstructions were taken away with the removal of the forests and the cultivation of the soil. Drains have been made, and furrows created by the plow, to lead the water quickly to the streams. More recently we find the ravines being stripped of their timber, thus giving an uninterrupted course to the waters swollen by the storms of autumn, and the melted snow and ice, and heavy showers of spring. In the construction of bridges and culverts these considerations must not be overlooked, and for this reason those in charge of township roads should make a systematic study of the volume and behavior of streams passing through every opening. When building or rebuilding any of these structures, ample provision should be made for the free passage of water.

The present practice is generally in opposition to this principle, the tendency being to construct a narrow channel in order to lessen the span of the bridge or culvert. A disposition to use iron, stone or concrete on account of its permanency is exhibited, but with this there is a tendency to construct the channel so as to bring the cost within the limits of a certain amount appropriated for the purpose. From this cause numerous costly structures have been and are being gradually washed away at the times of great water fall.

Permanency in these structures does not consist in the use of durable material only, but also in so designing them as to provide sufficient capacity and power of resistance to any strains likely to arise.

These requirements cannot be measured by the ordinary or the average flows but by the heaviest rushes of water. Economy in this matter does not mean the saving of a few dollars in first cost at the risk of efficiency, but in the expenditure of a sufficient amount to provide for every emergency, thus securing the investment.

BRIDGES.

It is the custom, in erecting iron bridges, to advertise for tenders, and too frequently a committee or council completes the purchase without any advice on the matter, the lowest tender being accepted, irrespective of the quality of material, workmanship or design of the bridge. There is as much variation in the quality of iron bridges as in the quality of timber bridges.

No one pretends to be a capable judge of timber without considerable practical experience, and with iron there is an equal or greater occasion for being able to detect any weakness. Wrought iron should meet the standard of the cold bend test, by which it may be bent, without fracture, until the ends are no farther apart than the thickness of the iron; small bars should not have a less ultimate strength than 55,000 pounds per square inch, nor an elastic limit less than 24,000 pounds.

The sustaining strength of a bridge is dependent upon the weakest portion, just as the strength of a chain is dependent upon the weakest link; and for this reason the matter

of making joints is of very great importance. The tendency is to permit all joints to be riveted, a cheaper method than making "pin" and "screw" joints, but not so strong and durable.

The design of an ordinary highway bridge is chiefly a matter of economy in material, so long as one of the standard forms of beam truss, suspension truss, or bow string is used. No design should be adopted in which the amount and character of every strain cannot be fully determined. But in all these matters respecting the merit of iron, joints, design, it cannot be expected that the average councillor will be able to form, with certainty, a correct conclusion. It is unreasonable to expect it, and the council which undertakes to purchase an iron bridge of any considerable dimensions without the assistance of an expert in iron bridge construction is unreasonable in so doing. To erect a structure which is not positively known to guarantee every probability of safety at the time when the greatest strain is upon it, risking property and endangering life, is little short of crime.

In choosing an iron bridge, it is as well to permit the bridge companies to prepare their own plans and specifications; but these with the stress sheets, should be submitted to an expert builder of iron bridges, and the council should then be guided by his decision. If his services cannot be employed throughout the construction, he should at least be called upon after the erection of the bridge to see that the work has been completed according to the specifications. There are, doubtless, honest bridge companies, but, in competition, when the lowest tender is likely to be accepted, experience has proven the tendency to lower the cost by using low grade material and workmanship.

Stone bridges are the most handsome and durable that can be built. The life of such structures is limited only by the quality of the material and workmanship. Bridges of stone arches built in the time of the Roman Empire are still in existence. Owing to the greater cost, however, they are very rarely met with in Ontario, except in localities where stone is plentiful. With stone that resists decay, with the best quality of cement, with well designed and carefully formed arches and piers, bridges of masonry leave little to be desired in regard to durability. Such arches are monumental, will resist with little repair the wear of centuries, and certainly no form of bridge architecture will accord better with the landscape.

CULVERTS.

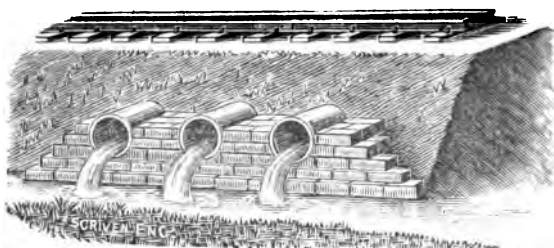
As with bridges, a large portion of the funds of a municipality is wasted on poorly constructed wooden culverts. Wood is only of temporary duration and usually renders poor service while it lasts. For the smaller sizes, cement, earthenware (sewer), iron and box stone culverts should be used. For larger culverts, arches of stone or brick are to be preferred.

Pipe can be safely used up to a diameter of twelve inches. The capacity of the culvert may be increased by laying two parallel lines of pipe. In laying the pipe, the bottom of the trench should be rounded out so that the lower half of the pipe will fit securely, with proper depressions for the "collars" or sockets, and the earth should then be rammed around the pipe solidly, but carefully so as not to fracture it. Pipes are not affected by the expansion of the earth around them, but care must be taken to provide a good fall for water, with provision for taking the water freely from the lower end. Water standing and freezing in the pipe is liable to burst it.

A culvert should be able to discharge water under head without damage to itself, and for this reason it is generally advisable to caulk or close the joints to prevent water from forcing a passage through the joints and along the outside of the pipe. For a similar reason, and to protect the pipe from injury by accidental blows, it is advisable to place a bulk head of brick or stone at each end. The foundation of this parapet wall should be placed below the action of frost. In constructing the bulk-head at the outlet, it should be set so as to provide a rapid fall to avoid the danger of settlement.

For culverts requiring a greater capacity than that offered by three lines of twelve inch pipe, stone masonry should be used throughout. Where large flake stone is plentiful, a very cheap culvert for all small sizes can be built by laying two side walls and covering with flagstone.

For culverts of from five to fifteen feet span, brick or stone masonry should always be employed. The form of the arch is a matter of very great importance. Semicircles, or full centre arches are simple and strong, but in large spans a correspondingly great rise is necessary, which may require an objectionable height. To avoid this, compound curves



may be used, but should always be designed by an engineer for each particular case, and skillfully erected, as certain mathematical principles have to be observed. The number of centres must be uneven, the tangent at the crown must be horizontal and the tangents of the curves at the springing must be vertical.

Cement concrete is being used to some extent in the construction of culverts, and with first-class cement, workmanship and design, will produce excellent results. It is important to place the foundation layers of concrete on a firm and unyielding strata of earth, well below frost and safe from the action of running water. All walls should have a batter on the side next the embankment as a protection against the thrust of freezing earth. Orib work should be firmly braced so that the cement may be thoroughly rammed. Coarse gravel, from which clay and earthy matter has been removed, may be used in forming the concrete, but clean broken stone is much better. The sand should be clean and sharp. The cement should be of some favorably known brand, delivered in barrels or other equally safe receptacles, and should be stored in a tight building, or otherwise properly protected from the atmosphere. Cement which, when opened, indicates by a granular appearance that it has been acted upon and set by the admission of moisture should be rejected.

On roads over which one hundred vehicles pass daily, one may, in almost any town-ship, see the drivers compelled to check their horses to a respectful walk, and then bump over the corners of a box drain or culvert standing above the road. The time and energy wasted, the injury to vehicles and annoyance caused in this way would build, a score of times, good and durable culverts which would not interfere with travel. The wastefulness is plain, the remedy simple, and as such, illustrates the little thought which has been given to the care of highways.

WAGON TIRES.

Vehicles designed for carrying loads of a ton or more should have tires at least four or, better, five inches wide. With expensively built stone roads the effect of narrow tires may be so repaired as to keep the road in a fairly good condition though at very great cost. But broad tires are particularly necessary on roads such as are constructed in this country. Narrow wagon tires are the greatest destroyers of gravel and broken stone

roads built and maintained as they are in Ontario. Even with traffic which is not excessive, our country roads will not be kept in a moderately good condition so long as they are subjected to the strain placed upon them by narrow tired wagons.

There are two aspects of the question of tire widths to be considered.

- (1) The relation of the tire to the amount of horsepower needed to move the load.
- (2) The effect of different widths of tires on the roads.

With respect to the first of these, the tractive force required, the effect of the width of tires varies with the condition of the road. On a smooth, hard, unyielding pavement there is practically no difference. With soft roads, deep with semi-liquid mud, the advantage is slightly in favor of the narrow tire, since the wide feloe has a tendency to carry the mud with it. It is with the intermediate stage, such as exists on a good gravel or dirt country road, that the benefit arises. Through such surfaces the narrow tires cut and the load is, in effect, being constantly drawn up hill, but the broad tires roll smoothly along the top. A wagon having tires wide enough to keep the wheels from cutting into the road will plainly draw more easily than one which plows through the mud instead of running over it. Experiments have shown that the loads which on narrow tires sink to the axles can be drawn without difficulty when broad tires are used. The broader base takes the firmer hold of the road, and the wheel revolves more easily and perfectly. It is admittedly the case that wide tires are an advantage on farms, but there appears to be a persistent effort put forth to prevent their use on the roads.

It is urged against wide tires that they do not roll freely in the ruts made by narrow tires. So long as narrow tires are commonly used this will be the case to some extent; on the other hand, if wide tires were generally used the ruts would not exist. In case, with narrow tires the bottom of the ruts made by the narrow tires are uneven, the narrow rims are constantly grinding against the sides of the ruts, creating the most friction, so that the objectionable difference is not so great as appears on first sight if it exists at all.

It is further contended that the wide tires come in contact with more loose stones than those with a narrow tread. The greater resistance offered in this way is more counter-balanced, however, by the loose stones dropping into the narrow ruts. In the case the wheel goes to the stone, in the other the stone gets in front of the wheel. The regular bottom of the ruts, and the stones in the narrow ruts keep up a constant vibration of the wagon, which transmits a swinging motion to the tongue, galling and injuring the horses and destructive to conveyances.

Broad tires actually improve a road by rolling it down hard, and leaving it smooth. Water runs off without doing injury. Narrow tires cut and grind, burst and plow furrows into the road, leaving ruts and holes to catch and hold water. The difference between the two is on a par with the difference between a pick and a pounder. The one breaks up, the other consolidates.

The narrow tire is a rut producer. With a load of say 2,000 pounds, each wheel supports 500 pounds. Then a narrow, straight line, the width of the tire, supports 500 pounds. With macadam, gravel and dirt roads the narrow tires compressed must have a greater bearing, and so, the width of the tire being fixed, the tire sinks into the road so as to extend the bearing along its circumference. In this way the rut is commenced. Other narrow wheels follow, deepening the rut and turning the earth around it. After a rain water lies in the hollow to assist the work of traction.

Contrary are a benefit rather than a detriment to the road. They do not sink at all so deeply into the road as do the narrow tires. Their broad base runs across the road as well as lengthways. Their broad tread does not sink at all so deeply into the road as do the narrow tires. The result is that the road is broken up readily, and the consequent jolting is avoided.

They do not push loose stones before them, tearing up the road, as do narrow tires, but pass over them, pressing them into the road. Grinding, upheaving and fracturing do not take place as with narrow tires, the road is compacted and compressed, and rendered thereby less pervious to moisture. This means that at all times the road is better, while the cost of maintenance is very greatly reduced. Were the benefit of broad tires better understood in Ontario, our unimproved roads would be very much less cut up, and improved roads would be more cheaply kept in repair.

In descending hills with heavy loads, it is a frequent practice with teamsters to lock one or more wheels. It is evident that the injury resulting to the road, which is very great in any case, is much increased when the load is supported by cutting tires. Hills, even under ordinary traffic, are expensive to maintain, and the width of the tire used with locked wheels becomes a very important consideration. In nearly every part of the province, this is the case to some extent. But there are districts where hills, many of them steep, are of constant occurrence. In such localities the practice of locking wheels is very common, and the necessity for wide tires is of the utmost consequence.

While a width of four or five inches is very satisfactory on farm wagons, the drays and tonnage wagons used for the transportation of excessively heavy freight in towns could reasonably be twice this. England, and all the progressive European countries, have laws regulating the width of tires according to the load vehicles are designed to draw. Sometimes the width is regulated by the size of the axle. In France, a country which presents some of our most perfect models in road-making, tires on market wagons range from three to ten inches, the majority being from four to six. The gauge of the wheels is sometimes set so that the track of the front wheels comes inside the track of the rear wheels. In New York State the turnpike law grants reduced rates of toll to vehicles with broad tires. The Michigan road law provides that users of wide tires are entitled to a rebate of half their road tax.

That wide tires are not more generally used in this country is to some extent the result of prejudice. People are not accustomed to seeing them, and wide tires, therefore, appear strange and awkward. When wide tires are generally used, as they certainly will be, the reverse will be the case, and narrow tires will be looked upon as the oddity, and a very objectionable one. As a means towards overcoming this practice, town and city municipalities could well afford to provide their watering carts, garbage wagons and other vehicles used in corporation work with wide tires.

City and rural municipalities cannot too soon set laws in motion in this regard. Obviously the use of wide tires cannot be made immediately compulsory, without working injustice. Provision could be made, however, that a by-law regulating the matter would come into force after a term of years; fair warning would thereby be given, and little inconvenience would be felt. Before the time for enforcing the by-law had arrived, the use of wide tires would, in anticipation, have gradually become more common. The beneficial effect of wider tires would then be apparent to all, and public opinion would demand and sustain the enactment. The people are competent to judge in this matter, as in others affecting personal and public interests. The great reason that wide tires are not now used is that the public has not had an opportunity to judge of their merits in a practical manner. Further, the users of the road have in the past looked at the matter of draught only. They prefer wide tires on the farms because the fields and lanes are not cut up by them. The roads, however, have been regarded as public property, and the users have not felt a personal interest in their condition.

INSTRUCTION FOR PATHMASTERS.

The council and road supervisor should use every means to interest pathmasters in the subject of road improvement, and to keep them interested. An association of township, or even county pathmasters, should be formed to meet annually for discussion and for the interchange of ideas, and it should be one of the duties of every pathmaster to attend this convention. The council should see that the pathmasters are provided with

printed matter, containing good practical hints on the principles of road-making, and the best means of carrying them into effect. A number of the directions most important to follow may, as a simple measure, accompany each year's road list. The following is a model set of rules, which could be impressed on the pathmaster by this means :

- (1) Carefully plan and lay out the work before calling out the men.
- (2) When preparing plans, keep the work of succeeding years in view.
- (3) Call out for each day only such a number of men and teams as can be properly directed.
- (4) In laying out the work, estimate on a full day's work from every man, and see that it is performed.
- (5) See that all the gravel hauled is clean.
- (6) Every wagon box should hold a quarter of a cord of gravel.
- (7) Specify the number of loads to constitute a day's work.
- (8) Properly grade and crown the road before putting on gravel.
- (9) A fair crown for gravel roads is one inch of rise to each foot of width from the side to the centre.
- (10) Give the roadway on hills a higher crown than on level ground, otherwise water will follow the wheel tracks. One and one-half inches to the foot from side to centre will be sufficient.
- (11) Spread the gravel evenly over the road, and keep it raked or scraped into the wheel and horse tracks until consolidated. A road roller should be used for compacting the gravel.
- (12) On all gravel roads, crown the roadway by cutting off the sides, putting new material in the centre. Do not cover the old gravel foundation with earth from the side.
- (13) Make the width of the road as uniform and the grade as easy as possible.
- (14) Remember that good drainage is absolutely necessary.
- (15) Whenever water stands on the roadway, or by the side of the road, or whenever the ground is seen to remain moist, better drainage is needed.
- (16) See that drains are free from obstruction, that they have a good fall and proper outlets.
- (17) Surface water should be disposed of in small quantities. Great accumulations are hard to handle and are destructive. Obtain outlets into natural water courses as often as possible.
- (18) Instead of having deep ditches to underdrain the road, use tile.
- (19) Give culverts a good fall and free outlet, so that water will not freeze in them.
- (20) Make early arrangements for having on the ground when required, and in good repair, all machinery and implements to be used in the performance of statute labor.

SNOW OBSTRUCTIONS.

To prevent the roads becoming obstructed by snowdrifts during the winter months has become an important question in nearly every portion of the province.

Snowdrifts are caused by the obstruction of the currents of air, those close to the ground, which are carrying the drifting snow. Where the forest covers the ground, the winds have not sufficient strength to carry the snow, so that it lies evenly, and with a uniform depth. Wherever the land has been cleared, currents prevail, and the velocity of these currents depends upon the extent of the clearing. The completeness with which the country is being stripped of trees is largely responsible for the blocked condition of our roads in winter.

Fences along roadways should be either entirely open or very close. The intermediate stage, found in the rail fence, offers an obstruction to the wind but permits the drifting snow to pass through, subside, and pile up on the sheltered side of the fence. This is one of the most common causes of blockades. High, close, board fences have been built by railway companies at certain points to prevent drifting, and have proven very effective, but the cost is prohibitive for ordinary highways. The drifting snow is close to the ground, and such fences obstruct the velocity and change the current of the wind, causing the snow to fall beside the fence, before passing over or through it into the roadway.

Open wire fences along the roads have been encouraged by many municipalities, the council generally providing the wire. Wire fences prevent the snow from accumulating so as to form banks and pitch holes, and is a decided improvement, but by no means a complete preventative. The uninterrupted current of the wind constantly moves the loose snow into and fills up the beaten sleigh path, making traffic heavy and gradually raising the centre of the road to such a height as to render turning out into the loose snow dangerous, especially with loaded teams.

A necessary adjunct to open wire fences along the highways is tree planting to break the current of wind, and to prevent the sweep across wide clearings in which it gathers a large amount of snow. By this I do not mean planting trees along the roadway only, but along the division fences on the farms to interrupt the velocity of the wind before it reaches the roadway. Observation will prove, wherever this plan has been adopted, or where orchards, groves or hedges exist near the road on the side from which the wind prevails, that banking occurs to a much less extent.

Tree planting is not required for this special purpose only, but it is of great importance otherwise to the country. The rapidity with which the older townships are being stripped of the forest is alarming, and no time should be lost in restoring that protection from exposure which is proving so injurious. The severe gales of winter, uninterrupted in their course, are trying to man and beast. They are destructive to fences, orchards and buildings. Trees interrupt the air currents and moderate the severity of winter gales, protecting inhabitants, stock and property. They arrest the moisture gathered up by free currents of wind in summer. They tend to modify the extreme rays of the sun which frequently, during long, dry seasons in June and July, cause so much injury to vegetation. They relieve the parched and dusty appearance of the country in summer, and break the dreary monotony of the winter landscape.

CONVIOT LABOR.

The employment of prison labor in the work of highway construction is a matter frequently suggested in connection with the road reform movement. Convicts have been employed in this way in a number of the American States, notably North Carolina, where the system is said to operate with a considerable measure of success. In New Jersey, Florida, Delaware and California, prisoners are employed to some extent.

Were provision made for such a measure in this province there is little doubt that much might be gained in the advancement of road improvement. There are two classes of convicts—the short term prisoners confined in the county jails, and those confined in the provincial penitentiary at Kingston. If the latter of these, the Kingston convicts, were employed in quarrying and crushing stone, the cost would amount to about 35 cents per cord for use of machinery, explosives, fuel, etc., but not including the maintenance of the prisoners. The railways, which would be greatly benefited by roads such as will permit a more regular traffic, will doubtless be willing in Ontario, as in the United States, to transport the road metal at actual cost. The price of crushed stone could in this way be reduced very materially for many sections of the province, where road improvement is handicapped largely by the difficulty of obtaining suitable road metal. With railway transportation at a cost of \$1.00 or \$1.50 per cord, this, with the cost of quarry-

ing and crushing and handling, would bring the cost of the product at any railroad station, siding, or perhaps road crossing, to \$2.00 or \$2.50 per cord. This does not include any provision for the purchase of the quarry or the maintenance of the convicts, which are items generally borne by the state.

The prisoners confined in the county jails might, in some cases, be employed in the preparation of gravel or broken stone. Or, as in North Carolina, they could be given the work of grading, draining and spreading the road metal. With the latter course, there is an expense incurred in guarding the prisoners. Some counties have too few prisoners to warrant this expenditure, but in some cases arrangements might be made to collect the prisoners of several adjoining counties for the work. A county jail, fortunately, is usually in the most important town of the county, and the roads radiating from it are those which must support heavy traffic and, consequently, are those most in need of such work as prison labor could perform.

Prison labor in highway construction is advocated on the ground that it would thereby be withdrawn from competition with regular employments and legitimate factory labor. It is a healthful out-of-door work, and, with the majority of convicts, would be of decided physical and moral benefit. With tramps and vagrants, the measure is one which exerts a very deterrent effect.

The state's relation to prisoners is one in which revenge has no part. The object of legal penalty is the prevention of crime, and the object of punishment should be the reformation of the prisoner. The employment of convicts, in prison garb on the highway, is opposed by some on humanitarian grounds, with the belief that the effect on the public and prisoners alike would be injurious to morals. This, however, is a consideration on which the opinion of penalogists will be of value. From the standpoint of one interested in highway improvement, there can be little doubt that with the Kingston labor and the labor available from the jails of large cities, much might be accomplished.

THE ARTISTIC TREATMENT OF COUNTRY ROADS.

The artistic treatment of roads is a matter in which we have been entirely deficient; more than this, the beautiful has been neglected and sacrificed even when it might have been retained without additional labor and with no loss of the useful. Whatever beauty the country highways of Ontario possess has been bestowed upon them by nature in such a manner, seemingly, as to defy the ever militant hand of the despoiler. For an explanation it is only necessary to remember that the construction of roads in the province has scarcely yet passed out of the hands of those who hewed the first wagon tracks through the wilderness, and who are constantly engaged in a stern struggle for the bare necessities of life. It is no cause for surprise that, choosing between the useful and the beautiful, the former has invariably gained the ascendancy. There is a very prevalent but mistaken idea that anything which savors in the least degree of the ornamental in road-making belongs only to the city boulevard.

To what extent the treatment of a roadside should be conventional must depend on circumstances. The most of us can recall with pleasure some roadway leading through a thinly settled, swampy lowland, and closely bordered with woods. There are very few who would wish to so vandalize the works of nature as to go with a scythe among the golden rod and asters, the flags and the grasses that fill the angles of the moss-grown rail fence. Nor would we hew away the ivy-grown stumps, nor replace the picturesque snake fence with one that is "neater." Passing, however, from the region of log houses, with their forest-encircling clearings, to the location where handsome stone and brick country villas predominate, where the woods have been almost obliterated and the fields have been brought to a condition of perfect cultivation, we must give the Queen's highway a corresponding degree of attention to bring it into harmony with its surroundings.

A century ago the first highways of any importance were laid out in the province. The forest was then the enemy alike to agriculture and roads, and the pioneer settler quickly learned, too, that it was the foe to his own means of sustenance. To-day we may, in many localities, pass farm after farm without seeing any of the original trees remaining or any new ones planted for ornamental purposes.

Trees are a necessary adjunct to a beautiful highway. To make this unqualified statement causes a civil engineer to feel some tremors of conscience, accustomed as he is in this utilitarian age to study only the economic side of construction. Trees are, as a rule, anything but a benefit to a roadway. Masses of foliage and shade, so grateful to the traveller, keep the driveway constantly damp—the bane of good roads. If, however, beauty is desired at the expense of utility, highways can scarcely be too much shaded by over-arching boughs. The happy medium will suffice in the majority of cases, and the evil effect of an avenue of trees will be more than made up by the additional pleasure obtained. Trees, however, need not be planted very close to the carriageway, but may be within the private property, or, if on the road allowance, as close to the fence as practicable. The branches should be trimmed so as not to materially interfere with the paved carriageway.

The varieties of trees suitable for the ornamentation of highways in this climate are almost infinite. Maples are most commonly used in Canada, and so universal have they become that many trees having equal or greater claims for beauty are overlooked. The elm, with its graceful, arching branches, and delicate lace-like foliage, is unsurpassed. The oak, renowned in England, is rarely used here. And so we might enumerate walnut, butternut, hickory, beech, chestnut, poplar, pine, ranging from the most delicate to the most sombre and rugged, each more or less adapted to particular requirements and circumstances. Trees such as the willow, the roots of which spread extensively or seek water, should not be permitted to grow near tile drains, as the small fibres, entering the tile, are a frequent source of obstruction.

The matter of fences is a very puzzling one. We have not yet found a shrub that will enable us to copy the hedge rows of England; and to stretch a few strands of wire is easier than to construct a stone wall. Masonry is very common in England and in the New England States. Only occasionally may it be seen here; and when overgrown with Virginia creeper or other vines the effect is all that can be desired. The wire fence appears to suit the present stage of road improvement; and in sections of the province where snow is apt to drift during a part of the year there is sometimes no alternative. Where wire is used, however, a very trim appearance may be maintained; and if the fences are made so as to be as inconspicuous as possible, and a generous use is made of trees and shrubs, the result will not be at all disastrous.

Our system of surveys, which lays off concessions in parallel blocks, with the roadway a rigid line of demarcation, and places cross roads at regular intervals, is responsible for the many cuts and embankments which mar the appearance of the highways. Could roads be laid out as are railways, in a manner that would permit the most convenient route to be taken, much more frequently would we follow pleasant, winding lines and graceful curves. Not only would roads laid out in this manner be more cheaply made, and more artistic in every way, but they would in many instances be shorter. The curve which we make in going down into a hollow and up again, or up a hill and down, is sometimes greater than the horizontal curve that would enable us to avoid the hill altogether.

Very common is the belief that in order to have a beautiful highway an expensive form of paving material must be used. The ideal roadway is of crushed stone or gravel, such as is found in the parks of large cities. A popular feeling is that asphalt has a better appearance than these materials, but this is largely due to the fact that the possibilities of gravel and crushed stone are seldom understood. Horsemen and wheelmen are unanimously in favor of macadam roads, and but very little study of the question is

needed to convince one that only when traffic becomes so great as to render the maintenance of gravel or crushed stone roadways impossible, or excessively expensive, asphalt need be used. So that its domain is the busiest thoroughfares of cities.

It is imperative, however, that the driveway shall be as perfect as possible. We are forced to form our ideas of the beautiful largely from the associations with which we clothe an object. It is difficult to harmonize with our conception of beauty a swampy roadway, into which the wheels of our carriage may have on some dismal autumn day, settled immovably, or over which, if walking, we were compelled to carefully pick our way, in danger with each step of sinking to our shoetops in mud. Poets, it is true, find their inspirations in such country lanes; and a driveway which is largely a pool of water doubtless gathers and reflects the shadows which the painter uses so effectively in his pictures. Roads, however, are not for the exclusive benefit of poets and artists; so that the work of increasing their utility will continue.

The roadside, further, must be shaped and, above all, covered with rich sod. No simple or definite rules can be laid down for this portion of the treatment of highways, and the best law-giver in this respect is nature. Nature does nothing stiffly, with rigid and abrupt lines, but has an infinity of gradations and shadings. An attempt to obtain a perfect level will be futile, and we would only secure awkward grades and stiff transitions. Long, easy, swelling lines should be sought.

The science of roads is principally a matter of drainage. Not that the shaping of the roadway, the gravel covering, and other details are unimportant, but that these are a part of the system of drainage. The deep, dangerous and unsightly open ditches that are so frequently to be seen on either side of the roadway, however, are a great impediment to beauty, and must, in the artistic treatment of our highway, be replaced by under-drains of common field tile, which expedient affords an inexpensive remedy. Shallow gutters must of course be provided to carry away the surface water, but they need not disfigure the highway.

A great mistake is made in grading into a carriageway too wide a portion of the road allowance. For the great majority of the country highways in the province, twenty-four feet between gutters is ample, the central eight feet only being macadamized. We admire of course, broad and smoothly rounded driveways, but wide stretches of sod are equally handsome. The driveway must not be confounded with the road allowance, the statutory width of which is sixty-six feet; and this for artistic effects, as well as on sanitary grounds, should never be less, preferably greater.

In no particular are there better opportunities for the artistic treatment of highways than in the class of bridges and culverts employed. The introduction of steel for this purpose permits us to do away with the clumsy and awkward wooden structures which so seldom in their youth are pleasing, and which in old age become grotesque rather than picturesque, replacing them with bridges that are graceful and slender, but strong. Substantial arches of stone are, beyond question, the handsomest that can be employed, their great strength and durability appealing forcefully to one's aesthetic feelings.

To render our highways beautiful at a stroke is an herculean task not to be attempted. To impress upon a certain section of the community the value of beautiful highways is the first step in the much needed reform. Having thus gained the point at which this class of improvements will be systematically brought about, the beauty of the highways will in their turn teach the people their desirability. Men are instinctively better citizens for being surrounded with that which is pure and beautiful. The artistic treatment of highways would be a constant reproach to the shiftless; neglected lawns would become fewer; ramshackle houses and barns would be less common; the eye refreshed and educated at every point, a journey along our country roads would mean to us a journey into a vast park.

DEVELOPMENT OF ROADS IN ONTARIO.

The story of a nation's highways is the story of a nation's progress in civilization, wealth and commerce. A country which has no roads is the home of the savage; a country, the roads of which are typical of the highest stage of engineering science, has attained an equal standard in other branches of refinement. A review of the development of roads in Ontario, as with the roads of other civilized lands, takes us back to the period when the province was a portion of "darkest" America, covered with a primeval wilderness, peopled by that unfortunate race of which but a fast decaying remnant remains.

An Indian trail was merely a path winding through the wilderness. Obstacles were avoided in preference to removing them, and so we find the trails encircling swampy lowlands, or places subjected to inundation, or such ground as would be likely to retain a footprint—a matter of serious moment to the secretive nature of the American savage. The trails passed through dense thickets of underbrush and over fallen logs. Streams were forded, for there were no bridges. They very commonly followed the line of the lake shores and the valleys of rivers. The more important were portages between the head waters of streams. Along these the Indian carried his birch canoe, by means of which he ascended and descended rivers in making his journeys. The most frequently travelled trails became worn by the habitually stealthy step of moccasined feet until more or less distinct, but others were scarcely to be distinguished even by the observant eye of an Indian, to whom a turned leaf, a broken twig or a displaced stone was ample guide post.

The most important of these routes, frequently referred to by early writers, led from Burlington bay to the head waters of the Thames, down which the traveller might descend by canoe to Lake St. Clair. A branch of this trail, mentioned by Galinée, led from the ford at the Grand river to Long Point. An early map shows an Indian path following the north shore of Lake Erie. Trails led from Lake Erie to Chatham and London. From the northern branch of the Thames, a short portage and canoe route led to Lake Huron. A trail led from the Don to Burlington bay. Nottawasaga Bay was reached from Lake Ontario by the Trent valley; by the Don or Humber and portages to Lake Simcoe, from which several routes might be taken. One of these last led from Kempenfeldt bay to the Nottawasaga river; another passed by way of the River Severn.

The first explorers, traders and settlers who entered this portion of Canada, in the absence of roads, had to adopt the methods and means of travel used by the aborigines. Champlain, who in 1615 made the first tour of discovery, went by canoe up the Rivers Ottawa and Madawaska, across Lake Nipissing, and thence by the French river to Georgian bay, a journey which necessitated about forty portages. This was the route generally chosen by fur traders for many years. Champlain, from Georgian bay, reached Lake Ontario by the Trent valley trail. The pioneer settlements were located around the forts at Detroit and Niagara. The United Empire Loyalists, who were among the first to enter the province, about 1785 began to make clearings along the River St. Lawrence, the Bay of Quinte and in the vicinity of Niagara and Detroit. Those who went inland usually chose property so situated that they might reside near a stream, down which in summer they could journey by canoe, and which, when frozen in winter, afforded a convenient means of travel on the ice. Others frequently located in the vicinity of an Indian trail. As the number of settlers increased the trail became a bridle path, then a waggon track, and ultimately resulted in one of the many "trespass" roads of to-day.

The first serious attempt at road construction in Ontario was commenced immediately after the separation of Canada into the Upper and Lower Provinces in 1792, and was one of the chief considerations of Lord Simcoe upon his arrival as the first Lieutenant-Governor. At the first Parliament in 1793, a statute labor law was passed, in spirit very similar to the present law. Guided by his military ideas, which prompted him to see in the present site of London the future capital of the Province, in Chatham a naval dock yard, in Turkey Point and Penetanguishene, naval stations, he accordingly planned a system of military routes. When on one of his tours of exploration he stood on the present site of London, then known merely as the "Forks of the Thames," in the midst of

an unbroken wilderness, and proclaimed his intention of opening a road "straight as the crow could fly" to the Head of the Lake (Burlington Bay). Work was commenced on this road on September 10th, 1793, by a detachment of the Queen's Rangers. This was first called Dundas Street, but is now known as the Governor's Road. From the Provincial seat of Government, York, now Toronto, Yonge Street was opened to Lake Simcoe by troops, a work which was completed in February of 1796. It was the intention of Simcoe that the road through the Province from York to Kingston should be opened by the settlers, and travel from the latter place to Montreal was for a time to be by water. Had Simcoe remained as Governor, it is altogether probable that the work of opening roads would have been carried on more energetically, but with his recall in 1796 ended very largely the era of military construction.

The *Canada Constitution* of September 13th, 1799, contains a reference to the settlements of that period between Burlington Bay and the Thames, in which it says:—

"The settlement from the head of the lake began about four years ago, excepting the Mohawk village; in this period townships have so far increased in population and cultivation that there are several neighborhoods at convenient distances until we come to the mouth of the La Tranche or Thames. . . . These settlers being aware of the importance of roads in raising the value of their property, early set about to open and extend them; notwithstanding the numerous discouragements, and the immediate necessities of their families, they, in one year, at the expense of Mr. Ingersoll, cut and bridged a road from Burford to La Trenche, through a wilderness of twenty-five or thirty miles. This was done previous to the escheating of the townships. Mr. Elijah Putman, of that town, (Oxford), by subscription, has since continued the road thence thirty miles to Allan's (Delaware) township. Here is a villa, and a church raised and now finishing by Mr. E. Allan after the model of that at the Mohawk village. The subscription being very inadequate to completely finishing the work, it was left unfinished but passable for sleighs. He has been by no means discouraged, but issues a subscription to cut a road from Allan's to the Moravian Grant, a further distance, as it must run of fifty miles, to be ten feet wide, and the logs laying crossways to be cut twelve feet long. Without waiting the issue of the subscription, and relying on the patriotism of his neighbors and gentlemen in other parts of the Province who hold lands upon the river, he began and has already opened half the distance, and promises, if the liberality of his friends be equal to it, that he shall immediately complete the whole. The Moravians will extend it seven miles when it shall form a junction with the old road, from whence there will be a good wagon road forty miles to the mouth of the river. Thus we shall have by the ensuing winter a land communication with Detroit and not a day's ride without settlements. Such is the enterprise of our western inhabitants, that one hundred and fifty miles is made without the least allowance from the Government."

The foregoing chronicles the origin of the present London and Hamilton Gravel Road and a portion of the Longwoods Road. It is altogether probable that the former followed in part, the famous Indian trail leading from Burlington to the Thames. The forest along the Thames, from which the Longwoods Road derived its name, extended thirty-five miles westerly from Delaware. This road was improved by troops during the war of 1812-15 so as to permit the transportation of military stores and artillery.

The location of most of the early roads has been very much altered from time to time for the purpose of straightening them or to avoid hills or fords. Thus in 1831 the Longwoods Road was altered in Ekfrid so as to avoid six fords on the Ten-mile creek and two on the Eighteen-mile creek. A map contained in the "History of Scarborough," by David Boyle, illustrates the many changes which have almost imperceptibly taken place in the route of these old trespass roads.

The Danforth road was one of early prominence. Of it, the *Gazette* of April 13th, 1799, says: "The contracts for opening the road between this place (Toronto) and the Bay of Quinte are entered into; the road will be thirty-three feet wide, creeks will be bridged, swamps, etc., will be causewayed, the whole to be completed by July, 1890. A circumstance highly beneficial to the town, and what ought to draw from the inhabitants

the warmest expression of thanks to the Government, as the expense is paid by it." Again the *Gazette* of December 14th, 1799, contains the following: "The road from this town to the Midland District is completed as far as the township of Hope, about sixty miles, so that sleighs, wagons, etc., may travel it with safety. The report which has been made to the Government by the gentlemen appointed to inspect the work is highly favorable to Mr. Danforth, the undertaker, and less imperfection could not be pointed out in so extensive a work. The remaining part will be accomplished by the first of July next." Tradition says of the very crooked line followed by the Danforth road, that Mr. Danforth, being paid by the mile, wished to make the road as long as possible. It is more probable, however, that it followed the irregular windings of an Indian trail.

The road from Toronto to Burlington Bay through the Mississauga Reserve, was for many years merely a trail and difficult to travel. With the opening of Dundas street this trail was lost, but the location is roughly indicated on a map attached to an Indian treaty. This, with the Governor's Road and the Longwoods Road to the west, and the Kingston Road to the east was esteemed an important trunk line through the Province; and with Yonge street to the north, and the road around the head of Lake Ontario to Niagara, were the most important of the early roads.

The roads between Burlington and Niagara was interrupted by the "Black Swamp." A subscription list was opened with considerable success as the following announcement from the *Gazette* of March 15th, 1798, indicates:

"The trustees respectfully inform the subscribers and the public at large that there are 387 rods causewayed and 240 only remain undone, which at one and one half dollars will make 360 dollars. Thus that tremendous Black Swamp bridge, that has been estimated at so many thousand dollars and calculated to take up so much time, has been nearly two-thirds done in about one month, that too in a severe winter. Upper Canada cannot boast a work of the kind so far advanced in so short a time. Three hundred and fifty dollars more, gentlemen, and the work is completed. The money that has been subscribed being all expended, the trustees hope the subscribers will be good enough to pay into the hands of either of them, their respective donations, and further donations will be thankfully received and carefully expended."

By 1796 there were roads from Montreal to Lake Francis, and from Cornwall to Prescott. The intermediate stage of about fifty miles was travelled by boat owing to the wet water-front in Glengarry, and the consequent difficulty of constructing a road. As late as 1807 the mails were carried from Montreal to Toronto, Niagara and Amherstburg by pedestrians who carried an axe to assist them in the journey. In 1816 the first stage was run between Montreal and Kingston, and in 1817 a stage ran between Kingston and York. This, however, was only in the winter, and during summer navigation the stages were discontinued. In 1826 the first stage passed between Niagara and Toronto, and in 1828 a stage route was established between Ancaster and the Detroit River.

In the early history of Ontario, certain provincial roads, such as the Talbot Road, the Sunnidale Road and others, were laid out and administered by the Surveyor General. A great number of the roads first surveyed are supposed to follow the lines of Indian paths. The first Roads Act was passed in 1803. Under this, commissioners were appointed by proclamation of Lieutenant-Governor Peter Hunter, to have charge of their construction and maintenance. By an Act of 1810, that of the 1803 was repealed and the quarter sessions given control of the roads. Colonization roads as now understood were first authorized by an order-in-council of 1848, to be constructed under the Commissioner of Crown Lands.

The first statute labor law of 1793 provided that every landholder or free-holder might be required to work on the roads from two to twelve days, as directed. This was soon found to be unjust, as landowners possessing an entire township were in this way taxed no more than a person owning a few acres. After a few years, a scale was arranged according to assessment, which was afterwards varied from time to time. The work of opening roads was retarded for a time by the clergy reserves, also by the absentee landowners who were not at first required to perform road work.

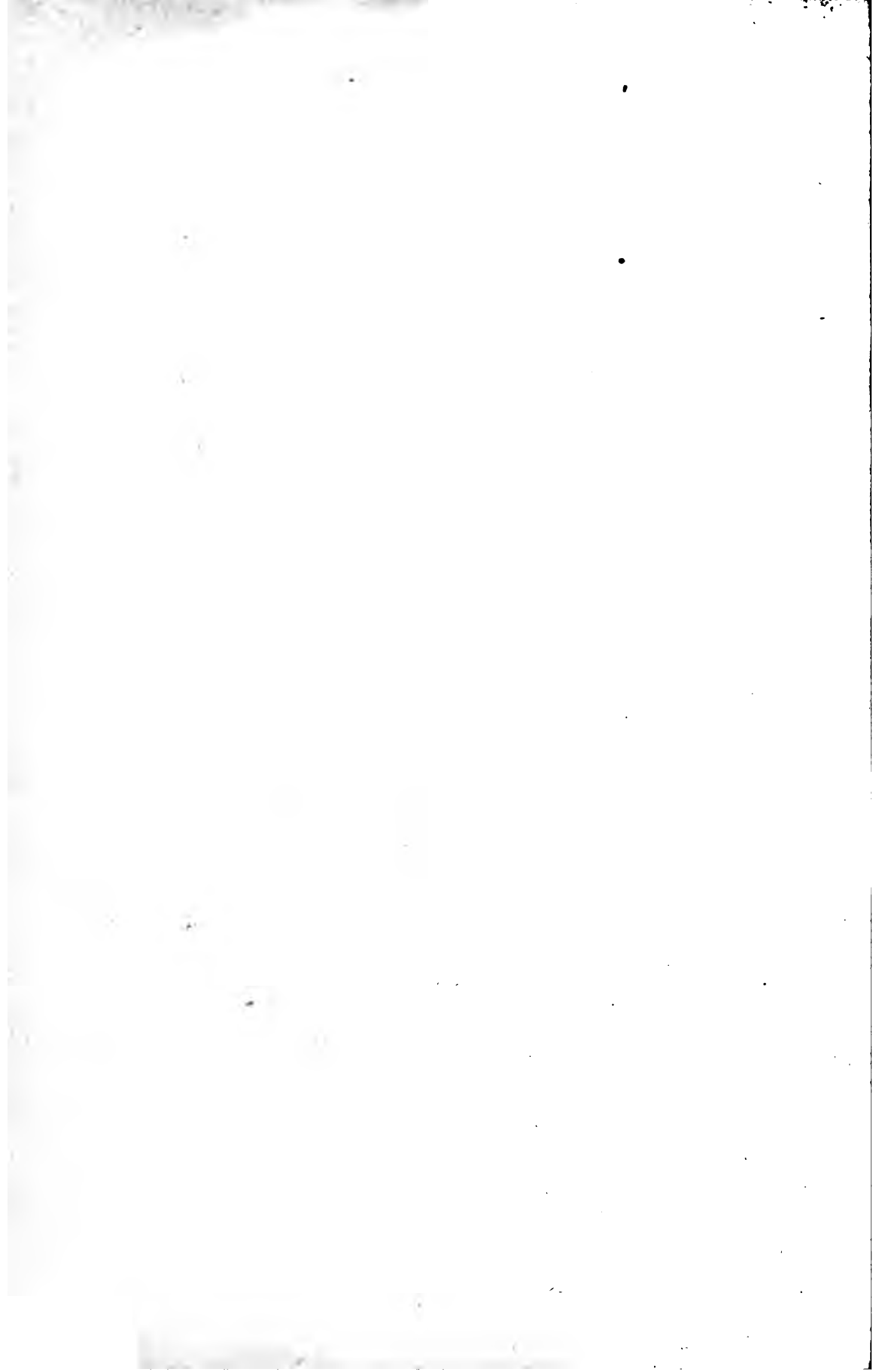
An Act of 1829 authorizing the formation of the Dundas and Waterloo Turnpike Co., was the first Toll Road Act. The Joint Stock Companies' Act of 1849 greatly fostered this principle and permitted many of the main roads to pass into the hands of private companies to be operated as toll roads. Before the union of the Provinces in 1841, a considerable part of the revenue was spent on roads and bridges. It was estimated in 1840, by Chief Justice Robinson, that £200,000 had been thus appropriated. The system of making provincial grants was abused, however, dissatisfaction was created, and it was found necessary to place the roads entirely under municipal control.

Attention began to be diverted from roadmaking by the construction of canals, the first canal, the Welland, being opened in 1824. The railway legislation of 1849-52 whereby over 1,300 miles of railway were, before the close of 1860, built and equipped at a cost of about one hundred million dollars, also withdrew a great amount of energy from highway construction.

The first roads opened were usually so obstructed by stumps and logs that they were merely winter roads, passable only for sleighs, when deep with snow. The troops, in opening the Governor's Road, chopped off the stumps close to the ground with an adze. Col. G. T. Denison, sen., when opening Dundas street between Toronto and Burlington, took the more radical course of tearing the stumps out by the roots. The Goderich Road opened for a distance of about one hundred miles through the Canada Company's lands was regarded by Sir John Galt as the one undertaking of which he might be proud. The cost of this work was \$5,000, no inconsiderable expenditure in Ontario at that time. In 1825, William Lyon Mackenzie described the road between Toronto and Kingston as the worst human foot ever trod. The Talbot Roads were commenced shortly after the coming of the Hon. Thomas Talbot in 1803, and, extending ultimately along the north shore of Lake Erie from Niagara to Detroit, became noted for their excellence.

Excellence at that time, however, was not what would be now regarded as such. The corderoy road on low and swampy land was the common means of overcoming the absence of drainage. The spaces between the logs were not always even filled with earth, so that the legs of horses were occasionally broken. Roads between dense walls of forest which shaded the roadway and kept it moist, the roadway itself composed largely of vegetable mould—such conditions could not but produce the discomfort and hardship experienced by the early travellers. It was only when the corderoy roads, the bogs, stumps and logs were covered with snow, that journeys could be made with comfort. When the frozen rivers could be travelled with sleighs, the height of luxury in Canadian travel was obtained.

A review of the development of roads in Ontario draws attention to the fact that they have by no means been built by statute labor alone. The pioneers, as a part of their settlement duties and quite distinct from the statute labor, were required to clear the road allowance of trees and remove sufficient stumps to permit a wagon to pass. It was as easy to build the corderoy roads where needed as to adopt the remaining alternative of burning the logs. The stumps not taken out as part of the settlement duties in providing a wagon track, were left largely for time and nature to destroy. The necessity for roads was so urgently felt, that labor and money were voluntarily subscribed, as instanced by the London and Hamilton, and Black Swamp roads, to which reference has been made. The regular troops and militia was employed in opening and constructing roads. The Canada Company opened a number of important roads through the Huron tract. Extensive provincial grants were made from time to time for opening new roads, and for the improvement of roads already opened. Later, many of the main highways became toll roads, maintained by private companies, permitting the municipalities to withdraw the statute labor from them and devote it to the less important roads. Since the roads have been under local control, they have been the object of large and constantly increasing annual money appropriations from the township and county funds. The road improvement which can reasonably be credited to the statute labor system consists mainly of a little grading and a little gravelling, neither of a very high order. So that, for the construction and maintenance of highways in this Province, we have been very much less dependent upon statute labor than is now popularly supposed.



PART II.

TOWN STREETS.

STREET IMPROVEMENT.

The improving of town streets is a question of considerable importance. The majority of the citizens of every town, the best and most progressive men of the community, understand the value of improving the appearance of private residences and lawns, their places of business, the churches, halls and public buildings. Work of this description is unfortunately handicapped in most Ontario towns by the neglected condition of the public thoroughfares. Instead of furnishing, as they should, an incentive to the improvement of private property, they are a constant object lesson in shiftlessness. Improved streets quickly pay for themselves in the encouragement they give to the making of better lawns, and the erection and better maintenance of houses. Well made roadways with boulevards nicely sodded and levelled, sidewalks in good repair, fences removed and trees abundant, give an air of prosperity to a town which is a constant source of pleasure to the citizens, and most favorably impresses a visitor.

A street properly constructed, and paid for in say ten years, is by no means a worn out property. The boulevarding, the grading, the draining and the foundation should be as good as when first constructed. It will remain merely to keep the surface in repair. A street designed on proper principles is, in its ultimate cost, a measure of economy, and will in the end decrease taxation. There are those who ask what revenue is to be derived from street improvement. If the increased property value is not sufficient, the same argument will largely condemn waterworks, sewers, sidewalks and almost any other public work. So long as streets are necessary it will pay to build them well. The majority of people still form their opinions from external appearances. A man's character and influence are largely judged from the house he lives in. On a larger scale a town is subject to the same criticism.

The majority of towns in Ontario pay for work on streets by making annual appropriations from the general funds of the municipality. The amount varies greatly with the different towns, according to the size, wealth, requirements and liberality in this respect, ranging usually from one to ten thousand dollars. This money is distributed over the town in several ways. In some cases it is sub-divided among the wards; the basis of this may be the assessed value of the property in each ward, or it may depend on the skill of the ward representative in wire-pulling. In other cases the ward boundaries are overlooked and the money is divided as the council may deem advisable; in this instance, influence is usually brought to bear directly upon the council by individual citizens or bodies of citizens.

In whatever form, the final tendency of this method of making expenditure invariably is to scatter the appropriation over the whole street area, not in accordance with the actual improvements needed, but according to a councillor's idea of equity, modified by his desire to retain or gain the support of his constituents. The manner in which the appropriation is spent makes it a sort of legitimate election fund; the people expect it and the council has no other course to pursue. The character of the work done is merely temporary patchwork, an effort to keep in repair mud roads and gravel roads which have never been suitably built. No durable improvements are attempted, and the plans are

usually those belonging to townships, as they are commonly developed by a bad phase of statute labor. While such roads are the cheapest in first cost, they are the most expensive to maintain.

It is impossible to condemn in a manner too pointed, the inefficiency of this system of providing for the improvement of streets, a fact made apparent by their wretched state. It is difficult to find a township, with a scattered and scant population, which has worse roads than the average town, notwithstanding the greater ability of the latter to pay for the proper construction of a comparatively short street mileage. A good system of street management must embody two ideas :

- (1) Capable oversight, so that work will be performed on proper principles of construction and in accordance with carefully prepared plans.
- (2) The concentration of funds so that work of a permanent nature can be undertaken.

In accordance with the foregoing, the first requirement, in the most of towns, is a capable street overseer. He should possess a good knowledge of the principles of road and street construction. Whenever work is to be undertaken, he should first report on it to the street committee, or Board of Works, with plans and specifications. The committee having rejected, accepted or amended this report, the council should sanction or discard it. Work should be carried out under the direct supervision of the street overseer, preferably by contract labor, if the extent of the work warrants it. The overseer should be in constant touch with the chairman of the Board of Works, and should be available at all times for consultation, attending all meetings of the board for this purpose. The necessity of such an officer has been pointed out in connection with township work. His position should be as secure as that of the average clerk or treasurer. Work left to the direction of councils and councillors, with their conflicting interests and ideas, cannot be efficiently performed.

If the system of making annual appropriations is adhered to in carrying on street improvement, the disposition of the money should be such that a certain amount of permanent roadway will be built every year, devoting the smallest sum possible to repairing unimproved streets. This, however, is not the business-like policy, the policy on which great enterprises and improvements have been most successfully and economically brought about.

One man never undertakes to build and pay for a railroad, but a company is formed, bonds are issued, the money obtained to be paid back in the course of a number of years. The work as soon as finished is an asset, to be balanced against the bonded indebtedness. The company which manages and directs the railroad expects it to earn a sufficient revenue to keep itself in repair, and to ultimately pay for itself. This is the method whereby all our greatest and best business concerns have been built up. It is the only method whereby permanent and durable results can be economically obtained. This is the method adopted in establishing municipal waterworks systems, electric light and sewer systems, and is quite as necessary in street improvement. The annual taxes should be used to pay for the interest, for maintenance and to create a sinking fund to meet the principal when due. Improvements made and maintained in this way are not a direct financial loss. But these improvements are an asset to be placed to the credit of the town, to be balanced against the bonded indebtedness just as is the case in the construction of railroads. A town which owns a public hall worth \$20,000, as a corporation is none the poorer because it is in debt to this amount for its construction. The hall is an asset, placed to the town's credit, and balances the indebtedness.

In assessing the cost of these improvements the annual payments may be met by the general funds of the town, in which case, in order to render justice to all, it would be necessary to raise sufficient money to suitably improve at least the most important streets of the town ; or a local improvement by-law may be adopted, this being usually framed on the frontage assessment system. Further reference to this system will be found in the appended report to Woodstock. The Municipal Act provides for this system in sections 612-628.

The principle of paying for a public work immediately on its completion is not a just one. The benefit derived from improved streets, a town hall, a sewer system, has only commenced when the work is completed. This benefit extends over a term of years and the only fair system is to ask payment from the citizens as the benefit of the work is received by them.

Nor is the full benefit of street improvement derived when only short sections of street are constructed annually. It is not until the whole town is provided with well designed thoroughfares that the full benefit commences. Short sections are merely scattered links of a chain; the usefulness of the chain is not realized until all the sections are joined, to the full length required. The "pay as you go" system is not the most profitable, if under it the work can be performed in limited sections only.

CHOICE OF PAVEMENTS.

The principle of economic fitness between the paving material and the wear to which it will be subjected, is an important consideration in designing streets as well as country highways. We have to consider the character of the natural soil on which the metal is to be laid—whether loose and porous, or retentive and difficult to drain; whether a light sand or a dense clay; whether stony or close to bed rock. We have to consider the qualities which the traffic over the road will require of the pavement; we must consider the availability of material for surfacing the road, whether economy advises the use of the metal at hand, or the bringing of a more suitable material from a distance.

All kinds of pavement are good when laid in the right way and in the right place. It is useless to condemn without proviso any one variety of pavement, for however defective it may be found under one set of circumstances, under other circumstances it is altogether probable it would be the most economical and serviceable that could be used. Cedar block is commonly looked upon as an entire failure for paving purposes. Cedar block has failed in a great many instances on account of the weak and undrained foundation on which it was placed. It has failed in other instances on account of the poor material used. In other cases it has failed by reason of a damp and unfavorable climate. It has failed frequently because of the excessive traffic to which it was subjected. While the use of cedar block is not advisable to any great extent in Ontario, yet it is easy to conceive of a dry climate, a loose, porous sub-soil and other facilities for securing a good foundation, an abundance of choice material close at hand, all of which would go far to render cedar block the most desirable material in that locality.

Macadam is pre-eminently the favorite pavement among horsemen, and if properly built is considered by many to be equal in appearance to asphalt. In hot weather it is scarcely possible to subdue the dust on an asphalt pavement by sprinkling, but Macadam can easily be kept moist. Macadam should in Ontario supplant asphalt on residential streets where the traffic is not too great. Brick is not, in appearance, equal to well kept macadam, and should be used on residential streets only when, for economy, a stronger pavement is required, and when the handsome effect of asphalt is not required.

One is sometimes asked for an expression of opinion regarding the relative merits of brick and asphalt, irrespective of the traffic, the climate and other matters affecting their durability. The only reply which one could give to such a question is, that both are good; or to look at it from the other standpoint, both are bad. Brick pavement appears to be the ideal pavement for the business streets of towns and smaller cities. The cost is from one-third to one-half less than asphalt, is not objectionably noisy, does not decay readily in a moist climate, and will support a considerable weight of traffic.

Asphalt is suitable for a business thoroughfare, where a slightly stronger pavement than brick is needed to support a somewhat heavier and more constant traffic. It is suitable too, for a residential street on which there is too much travel to permit the laying of a macadam pavement and where a more costly effect is desired.

The streets of towns in the immediate business portions are subjected to wear not imposed upon residential streets. If broken stone roadways are retained on business streets the gutters should be cobblestoned to protect them from the feet of horses when tied or standing close to the curb, and a covering of one foot in thickness should be used over the whole surface. Macadam is not the most desirable material, however, for this class of street. A pavement is needed which will not absorb street filth, which can be readily cleaned, which will not be dusty, nor muddy.

The choice of pavements for the business streets of towns and cities of Ontario appears to rest at present between asphalt and brick, the former so far obtaining the preference. Cedar block should be used so rarely that it need not be considered in a general discussion of the subject.

ASPHALT.

Asphalt is more noiseless than brick, is smooth, sanitary, and presents a good appearance, but it is not the ideal pavement, in all respects, that it was at one time supposed to be. In summer its smooth, stony surface radiates an intense heat; any amount of sprinkling cannot keep the surface moist so as to allay the dust, while the glare which it reflects on a bright day becomes painful to one driving or walking over it. The wet weather of fall and spring in this climate keeps an asphalt pavement in a very slippery condition, unsafe for horses, while at any time the foothold is insufficient.

The most common causes of the failure of asphalt are found to be weak foundations, which, settling, permit water to stand in hollows on the surface until the pavement is rotted; tearing up the pavement for sewer connections; surface cracks, caused by contraction in cold weather; insufficient traffic and excessive traffic. The destruction of asphalt arises more from atmospheric effects than from wear, a certain amount of traffic being beneficial rather than detrimental.

Asphalt is seldom worn out by too much traffic, the decay generally arising from other causes, some of which have been enumerated, and which the specifications should rigidly guard against as far as possible. A good asphalt pavement should be maintained by the company building it for seven years, and at the expiration of that time delivered to the municipality in perfect order.

VITRIFIED BRICK.

The only competitor of asphalt is vitrified brick. This pavement has come in a most timely way to take the place of the decayed cedar block pavements which are disgracing the streets of so many towns and cities. Vitrified brick is becoming popular, and presents features which tend to cause it to become more so. It offers a better foothold for horses than does asphalt. The surface is not so smooth, and in consequence radiates less heat and light, is quite as sanitary, with less liability to become dusty. Among bicyclists it is much more popular than asphalt. If the joints are filled with suitable cement, brick pavement is but little more noisy than asphalt.

The majority of failures which have occurred with brick, have been traced to defects which the material or better construction could have obviated. Its ease of construction and repair offers a great advantage over asphalt, ordinary laborers being easily taught to do the work. Few repairs are needed if good brick is used, and in the first cost as well as in maintenance, brick should be, and generally is, cheaper than asphalt.

Although brick is one of the oldest paving materials, it has been used on this continent for only about a quarter of a century; and only within the last ten years has it attracted widespread attention. In the United States, it has been used very extensively, but in Ontario, experience with the modern vitrified brick is very limited.

Of its success as a paving material little remains in question. Brick pavements have been in existence in the United States for eighteen years, remaining in good condition. It was feared that the climate of northern countries with severe frosts and rapidly alter-

ating conditions of moisture and temperature, would be unfavorable to its use, but the experience of various northern cities shows that vitrified brick of a good quality is a most valuable addition to our list of paving materials.

The best vitrified brick is made of shale or clay, or a mixture of the two. It is not "vitrified" as the name indicates, but is raised by intense heat just to the point of fusion. More than this fuses or melts the clay, permits it to run together and the product is then glassy or vitrified, and brittle in consequence. The process of cooling must be very gradual. A brick if too rapidly cooled or "annealed," will be brittle, but with a thoroughly pulverized and well-mixed shale, brought to the proper temperature and then slowly annealed, the resultant brick should be sufficiently hard and tough to scratch steel.

The chemical composition is a matter of importance. The chief defect is likely to arise from too much lime. Brick which contains an objectionable amount of lime, or other soluble substance, when immersed in water for three consecutive days, and then kept in a dry atmosphere for a corresponding length of time will show signs of spalling and pitting, and indicate a faulty composition of the clays of which they are composed.

When broken, brick should show a smooth and straight fracture, and the texture and color should be uniform throughout. A granular appearance is very objectionable. Bricks externally should be smooth and straight, not warped, and should be free from fire cracks and checks.

Brick should absorb the least possible amount of moisture. A large amount of water in brick, freezing, has a bursting effect which in this climate cannot be guarded against too carefully. A good brick will not absorb more than two per cent. of its weight in water. To absorb three per cent. may be permissible if in other respects it reaches a high standard.

The quality of brick is fairly well shown by the rattler or tumbling barrel test, as this most nearly approximates to the wear from the blows and chipping of horses feet. A tumbling barrel or rattler consists of a cylinder about three feet long, and two feet in diameter, placed on a shaft. Samples of the different makes of brick are first weighed and are then put in the barrel, together with from 150 to 300 pounds of scrap iron in from two to eight pound pieces. The rattler is then revolved for two or three hours at the rate of about thirty revolutions per minute. The comparative loss in weight will indicate the relative efficiency of the brick in resisting this class of wear.

A valuable test is that of transverse strength. In this the brick is laid upon two knife edges, placed about six inches apart. An upper knife edge centrally between the two lower blades is slowly brought to bear on the brick, and the weight required to break it is carefully noted.

Other tests are sometimes used, to determine crushing strength, the depth to which oil will penetrate the surface, etc. The specific gravity of a brick is valuable as indicating density, and therefore the extent of probable absorption. The chief tests, however, are those above described; that of absorption, representing the probable effect of atmospheric action; the rattler test, showing the effect of impact and abrasion as found in the chipping of horses hoofs and the grinding of wheels; the transverse strength, showing the power to resist the breaking strain of heavy loads. Where, in towns, the facilities for performing these experiments are not available, and only short sections of pavement are to be constructed, the experience of cities will generally afford a safe guide in choosing between different makes of brick. A further safeguard may be had by requiring the contractor or manufacturer to maintain the pavement up to a certain standard for at least five years.

There is a tendency to endeavor to reduce the cost of brick pavements by the use of a weak foundation or no foundation at all. Brick pavements laid on gravel and sand have been successful, but this has been the case only when the subsoil has been of such a kind as to be very porous, easily drained, and naturally firm. The experiment in Ontario, in view of fall and spring conditions with alternate freezing and thawing, is a very dangerous one, a lesson which has been strongly impressed by the experience with cedar blocks.

While a brick pavement may give satisfaction for a few years on sand and gravel foundations, there is every probability that the brick will settle irregularly, and, rendered thereby more susceptible to wear and strain, the bricks will be broken and the life of the pavement very much shortened.

A concrete foundation should almost invariably be employed. A four inch layer will, where a brick surface is suitable, be sufficiently durable. This forms a stiff monolith base which distributes the weight of traffic. There cannot be irregular settlements of brick, as is the tendency with yielding materials, such as sand and gravel. It also prevents water percolating beneath the road, not a very important feature in the south, but in freezing climates a matter of considerable importance.

Between the bed of cement and the surface covering of brick a thin cushion of sand is necessary. By this means the brick can be laid evenly, a certain amount of spring is obtained which lessens the effect of blows on the brick, and it overcomes the rumbling noise otherwise created. It is a common practice to merely fill the joints of the brick with sand. While this is not at all objectionable, by the use of a cement composed of pitch and sand the pavement becomes less noisy, absorbs less street filth, and corners and edges of the brick are strengthened.

MACADAMIZED TOWN STREETS.

There is a common impression that macadamized roadways are all very well for country highways, but that, for town streets, a higher grade of paving material, brick, or wood, or asphalt is necessary. This opinion, so adverse to macadam, has no doubt arisen from the common practice of terming a "macadam" road, any allowance between concessions that has had an irregular layer of gravel dropped along the centre. This was somewhat the idea the people had of roads before the time of Macadam, the main feature of whose road-making, however, was a firm, unyielding sub-soil, maintained in that condition by a thorough system of deep drainage: and a covering of stone, impenetrable to moisture and so shaped so as to shed water quickly to the gutters. All kinds and classes of paving have their proper place, and, in its proper place, there is none more serviceable or capable of giving greater satisfaction than Macadam.

RESIDENTIAL STREETS.

One place in which macadam can be rightly employed is in paving certain town and city streets—those of a residential character, not subjected to traffic so great as to render the maintenance of macadam excessively expensive. There are, of course other limitations which individual cases will suggest. Business thoroughfares, or a street in its business section, requires a less absorbative material, and one which presents a smoother surface, more easily cleaned. With horses standing, moving slowly, tied and pawing as so frequently is the case on a business street, and generally with an excess of heavy traffic, a macadam pavement retains a great amount of street filth, is difficult to maintain, and should be, if possible, replaced with vitrified brick or asphalt. A well kept macadam driveway is in keeping with well-kept boulevards, lawns and shade trees, the characteristics of a residential street; it has a cool appearance, the dust can readily be kept down by sprinkling, and for light driving is the favorite among horsemen. Bicyclists, now an important section of the community, usually favor macadam, in preference to the more costly classes of pavement. A comparison of macadam with asphalt or vitrified brick, in point of utility and beauty, will not result unfavorably to the former.

GRADES AND EARTHWORK.

Preliminary to undertaking the improvement of town streets, levels should be taken on all the streets to establish a system of grades which will provide a proper union of street intersections. Care should, of course, be taken to equalize cuts and fills as far as practi-

able, and to utilize surplus earth in filling up the boulevards or low lots adjoining the street. In this, the handling of considerable earth is often necessary to obtain the best results.

In the finished street, it is ordinarily advisable to have the crown of the roadway at about the same elevation as the surface of the sidewalk ; and, in any case, the sidewalk should not be lower than the crown of the roadway. It will be necessary to excavate below this elevation in the centre of the allowance to provide for the reception of the road metal. The depth of this excavation must provide for the thickness of metal to be used, and for surface drainage.

WIDTH OF ROADWAY.

The present tendency is to narrow the width of the driveway. It is found that to occupy a 66 foot allowance with a four foot walk on each side, and to devote the remainder to the driveway, is a needless expense, both in cost of first construction and in maintenance. From 22 to 26 feet is, on the great majority of the residential streets of towns, ample to accommodate traffic. A broad driveway is very handsome, but so also are broad stretches of nicely sodded boulevard, ornamented with shade trees.

SIDEWALKS AND BOULEVARDS.

At present, on improved streets, we ordinarily find a row of shade trees outside the walk. In commencing the reconstruction of a street, it is generally advisable, if not necessary, to take up the sidewalk to permit a proper grading of the road allowance. When the earthwork is finished, the sidewalks may be placed immediately outside the row of trees, and the space originally occupied by it, sodded ; and if the fences are removed the strip is, in appearance, added to the depth of the lawn. This arrangement will usually leave a space for a strip of sod between the sidewalk and the carriageway. If this part of the boulevard can have a width of about two feet or more, it takes away a certain dusty, business appearance which the street would otherwise retain. The sidewalk should have a distinct elevation above the sod, sloping slightly toward the roadway, to provide for a proper drainage. The strip of sod between the walk and curb should have a fall toward the curb of six or more inches, and on a 24 foot roadway having a one foot crown at the same elevation as the walk, this will leave from six to eight inches of curb exposed.

THE CROWN.

The crown referred to above, about one inch to the foot, will seem to many, perhaps, excessive. This applies to the newly-made roadway and provides for settlement. Two-thirds of an inch is the convexity ordinarily adopted on newly-constructed English or French roads, but with the material available in most localities of this province it is not sufficient. The limestone, gneiss, field stone and gravel ordinarily obtainable in Ontario are not the most durable, hollows are apt to appear, dust accumulates, and unless there is ample fall for the water, it will be found lying upon the road surface. The tendency invariably is for traffic to use the centre of the roadway, and this is always increased with the narrower driveways. But in choosing between two evils it is better practice to provide ample crown. It is better that travel should be less distributed than that water should lie in pools on the road.

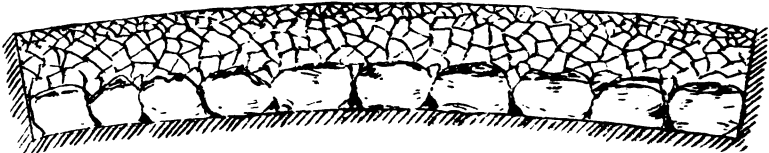
The shape of the crown has been a subject of some discussion. A circular rise is believed to be the best principle to follow in practice. Two planes, joined at the top by a short curve do not provide for settlements and wear. A flat ellipse, sometimes advocated, does not provide for settlement and gives an unduly steep fall at the edges increasing the wear at the side, and practically narrowing the roadway.

CURBING.

For curbing, flagstone, which is easily obtained in many districts, is the more handsome and more durable material. A good substitute is 3x10 cedar, which should be spiked to cedar posts 2½ feet long and 6 inches in diameter. By bevelling the posts the curbing may be inclined at an angle of about 30 degrees. It presents a better appearance than when perpendicular, the tops of the posts are protected and there is less liability to decay.

MATERIAL.

In nearly every locality throughout the province good material will be found within easy distance. Crushed stone is usually regarded as the only material for macadam roads, but Macadam really used and advocated any material which would provide a good wearing surface not readily penetrated by moisture. Thus we have in Ontario a choice of gneisses, limestones, field boulders, pit and creek gravels; or if



CROSS SECTION.

Showing a broken stone driveway, with a foundation layer of flake-stone.

we wish the best material, trap rock is available. Of these materials we have all qualities, from that which is exceedingly good to that which is little better than clay. The gneisses are usually a harder and tougher rock than limestone, but the latter offsets this defect largely by better cementing qualities. Field stone makes a very good metal if care is taken in its selection. Pit gravel usually needs screening and crushing to remove sand and earthy matter and to reduce the large stones to suitable dimensions. Creek gravel is often sufficiently clean to be applied directly to the road but some attention should be given to breaking large stones. In choosing the metal, a judicious selection must usually be made between a cheaper and poorer material in the immediate vicinity and an expensive but more durable metal from a distance. The selection will be based on the expenditure permissible and the nature of the traffic and the quality of stone used. Twelve inches is sufficient for the heaviest travel, and a thickness of seven inches is the least advisable. The coating should be heavier at the centre than at the curb. These measurements are after consolidation with a roller. A fuller discussion of this portion of the subject will be found in connection with country roads.

BINDING MATERIAL *vs.* ROLLING.

The use of binders is another question of considerable importance. The best that can be had is usually the clean chips and dust, the screenings of the metal used. The amount of vacuum in a surface covering consolidated without a binder is very considerable, but the mechanical grasp which one stone takes upon another under pressure is infinitely preferable to the consolidation obtained by a mixture of sand and stone. Sand attracts and retains moisture, while a vacuum will permit the water to pass away, at the same time providing space for the expansion of the frozen water retained. The action of frost on a clean metal is much less serious than upon a roadway having a temporary bond of sand. Rolling is an absolute necessity in obtaining a proper surface covering of crushed stone. Broken stone dropped loosely upon a roadway permits water to pass into the sub-soil as through a sieve and the process of consolidation under traffic not only results in a very great wear and tear of metal, but mixes it with the earth which of itself is very injuri-

ous and in so doing destroys the crowning and therefore the proper surface drainage of the roadway. The sub-soil should be crowned and, like the covering, thoroughly consolidated by rolling. The metal should be applied in layers three or four inches thick, each layer being thoroughly rolled before the next is applied.

THE ROLLER.

The weight of the roller should be from ten to fifteen tons, the former generally producing the better results. An excessive weight tends to crush the metal, especially if of the softer local varieties, instead of working it into position. The heavy rollers produce consolidation more quickly, but the lesser weights have more permanent results. Less than ten tons is not advisable except in the instance of a horse roller, a cheaper but not very satisfactory substitute for a steam roller.

UNDERDRAINAGE.

The necessity of underdrainage has been referred to fully in connection with country roads. The means must usually be common field tile. The location and extent of drainage must depend on various circumstances, the nature of the soil, the opportunity for outlets, and whether or not there is a sewer beneath the roadbed. Ordinarily a three or four inch tile placed beneath the edge of the roadway on each side, and below frost line, is the best rule to follow, but this, if the soil is loose and porous, and has a natural outlet for sub-soil water, may be more than is necessary. A line of tile placed in the centre of the road allowance disturbs the earth foundation, and settlements are likely to occur, a condition which is difficult to repair and is very injurious to the stability of the road.

GUTTERS.

If gravel or broken stone is used on a business thoroughfare, the gutters should be cobble-stoned to protect them from the stamping of horses; but on residential streets this is not necessary, the angle between the road surface and curb forming a sufficient waterway. Outlets for surface and sub-drainage must be procured as frequently as possible so as to dispose of the water before it gains force and headway. If a system of sewers exists, with provision for storm water, the matter is very much simplified. When necessary the water may be admitted to the sewer or tile drain through settling basins which should be very carefully guarded, to prevent obstruction.

The principles of road-making touched upon in this discussion of macadam streets for towns, will be found more fully developed in connection with the articles on country roads.

FORM OF SPECIFICATION FOR MACADAMIZING AND BOULEVARDING RESIDENTIAL STREETS.

In giving the following specification for macadam streets, it is not intended that it shall be rigidly adopted by any municipality. It indicates, however, the form which a specification should take, and may be modified to suit local circumstances.

EARTHWORK.

The excavating and grading is to be made to the depth and width shown on the drawing having preference thereto and to the levels given on the ground by the engineer or the person in charge of the work; and also to such greater depth and width as may be directed, to enable the curbing to be properly set; the levels and cross sections may be varied at the discretion of the engineer or other person in charge of the work, to conform to the elevation of lots and grades of intersecting streets.

The earth taken from the excavation for the roadway is to be used in properly grading up the boulevards and filling in any portion of the roadbed which is beneath the grade line on the proposed improvement; and the surplus earth is to be teamed from one point of the street to another as may be required in making the said boulevards, where there is not sufficient earth, and at any particular point in this excavation to complete the work; and the remainder of the excavated earth is to be deposited wherever directed by the engineer at a distance of not more than 1,500 feet on either side of the said street.

THE SUB-GRADE.

The sub-grade when complete must conform truly to the required cross section as shown on the drawing.

Should the contractor, in grading, excavate below the proper level on the sub-grade, the deficiency shall be made up with gravel or other approved material well sprinkled and rolled at the contractor's expense.

SIDEWALKS.

The contractor is to carefully take up the present old sidewalk and crossings on the street, running in an east and west (or north and south) direction, and deposit the material; and the contractor, where required, shall take up all crossings over the streets running in a north and south (or east and west) direction and pile the material by the side of the intersecting street where it will be out of the way of travel.

The contractor is to excavate, or form in the excavated earth, a trench or opening for the sidewalk feet wide and inches deep. The bottom of the said trench is to be level with the top of the line of the curb except where otherwise directed by the engineer.

All the sidewalks on the said street will be laid down by the contractor for that purpose; but the contractor for grading, etc., is to make the proper excavation as shown on the plan for said sidewalks, and after the sidewalks are laid the contractor for street grading must do any necessary filling or lowering of the boulevards in order to properly meet the grade of the sidewalk.

BOULEVARDS.

The grade of the boulevard at the side of the sidewalk nearest the line of curb is to be six inches above the top of the line of curb. The boulevard between the curb line and the place made to receive the sidewalk is to be regularly bevelled off from the grade line at the top of the sidewalk to the top of the curb. The boulevard between the sidewalk and the street limit is to be regularly and evenly graded by cutting down or filling in as may be required, so as to conform to the grade of the sidewalk, except where otherwise directed by the engineer, in order to conform to the elevation of the lawns along the said street. The boulevards are to be left smooth by raking or otherwise levelling to the satisfaction of the engineer or other person in charge of the work.

TREES.

The contractor, in doing the work, must excavate or fill in around trees on the said street in a careful manner so as not to bark or injure the said trees.

CURBING.

Each side of the roadway throughout the whole length of the street is to be curbed with sound, live, white cedar, mill-made planks three inches in thickness by ten inches in

depth, and not less than twelve feet long, The posts are to be of round cedar, barked, good and sound, having a finished length of three feet, and to be not less than six inches in diameter. The posts are to be firmly set in the earth to a depth of two feet, centre to centre and cut off flush to the top of the curb. The planks are to be secured to the posts with seven inch wrought spikes, three spikes to each plank at each bearing. The planks must be neatly and closely joined, the joints to be made truly to the centre of the posts. The whole of the posts are to be set in an accurate manner, and true to the line and level given, and at such elevation as to show a face of ten inches above the subgrade, and with such incline from the roadway as may be directed.

At each street, lane, alley, private way, etc., the curbing shall, unless otherwise directed, be returned to the sidewalk, the returns to be placed at an angle of forty five degrees with the line of the curbing, each return to have a post where the curb meets the sidewalk. The earth at the back of the curbing is to be properly rammed so as to ensure stability of the curbing. The material and workmanship used must be in conformity with the specifications and plans, and to the satisfaction of the engineer, or other person in charge of the work.

SEWER AND WATERWORKS FIXTURES ON STREET.

Returns and off-sets, if necessary, must be made in the line of the curb around any of the water gullies on the street. The levelling of the top of the sewer gullies, man-holes, etc, and the building up or lowering of all waterworks standpipes in such manner as the engineer may direct to suit the grade and crown of the roadbed, will be done by the contractor.

PRIVATE ENTRANCES AND STREET INTERSECTIONS.

All intersections of private lanes are to be properly made and graded in the boulevard by the contractor at a gradual slope from the top of the sidewalk to the bottom of the gutter, and all street intersections are to be graded to conform to the finished grade of the street.

BREAKING UP THE STREET.

The contractor is to break up the street in such lengths and at such points as may be directed by the engineer in charge of the work, and no part of the street shall be broken or torn up, or no portion of the old sidewalk removed, without the order of the engineer.

WORK TO BE CONTINUOUS.

The contractor will be required to carry out the work continuously from the time of the commencement until the completion of the contract.

RIGHT OF WAY.

The contractor will keep, if so ordered by the engineer, ten feet of the roadway nearest the curb open for use of the public from 7 a.m. to 7 p.m., and will keep the work properly barricaded and protected at night with lanterns.

ROAD COVERING.

The surface of the roadway over the said road is to be covered with crushed stone to the depth of inches in the centre, and inches at the curb, to be regularly and perfectly spread over the whole of the roadbed to a depth to conform to the cross section shown on the drawings, and proportionate to that specified for the centre and curb. The crushed stone is to be furnished by at

PLACING THE METAL.

The whole of the sub-grade, when levelled, shall be thoroughly rolled with a roller weighing ... tons, to be passed over the surface at least times. The roller shall be provided by, the work to be performed by the contractor. Crushed stone of a size to pass through a two and a half inch ring is to be placed over the whole of the surface of the sub-grade to a depth of inches in the centre and inches at the curb, and rolled with the said roller times; crushed stone to pass through an inch and a half ring is then to be placed over the whole of the surface of the road to a depth of inches in the centre and inches at the curb, and this rolled with the said roller times. A sufficient quantity of crushed stone of one inch in diameter, is then to be placed over the whole of the roadbed to bring it to the line of the finished grade as specified on drawings, this to be rolled by passing the said roller over it times. A coating of fine screenings is to be placed over the surface of the road and worked into the interstices of the mass of stone by thoroughly saturating and flooding with water, either from street hose or properly constructed sprinkling carts, and by passing a short harrow over the surface of the whole mass until the engineer is satisfied that the interstices are properly filled. The roller must be again passed over the whole of the road times, or to the satisfaction of the engineer.

WATER.

During the whole of the rolling herein specified, a sprinkling cart is to pass immediately in front of the roller so that at all times the surface of the road will be saturated with water. The water is to be obtained from the street hydrants for which a charge of one cent per lineal foot must be paid by the contractor to the water commissioners.

SOD.

All sod on the street between the curb line and the line of the new sidewalk is to be carefully taken up by the contractor and neatly piled at one side of the work. After the boulevards have been built to the required grade, the sod is to be relaid by the contractor in a perfect manner.

TILE DRAINS.

The contractor is to furnish the tile and construct a four-inch field drain along the inside or roadside of the curb line on each side of the street. The tile is to be placed inches under the subgrade of the roadway, and after it is laid the trench made to receive them is to be filled in with the excavated earth to the surface of the subgrade, and thoroughly consolidated.

EXTRA EXCAVATIONS.

Any extra excavations that may be ordered by the engineer will be paid for at the rate of fifteen cents per cubic yard.

TENDERS.

Tenders are to be made in the form of a lump sum which will be taken to cover the cost of furnishing all necessary material (except stone), implements (except roller), labor, etc., of every kind to complete the work of grading, boulevarding and macadamizing the said street.

PAYMENTS.

Payments to the extent of per cent. of the work done each week shall be made by the certificate of the engineer, such certificate to be binding only as regards progress; and weeks after the completion of the work to the satisfaction of the engineer, the engineer will issue certificate for the balance.

A P P E N D I X .

TOWN REPORT.

To His Worship the Mayor and Members of the Council of the Town of Woodstock :

GENTLEMEN,—Having visited your town on the 19th of October last, in accordance with the invitation extended me, I drove over a number of the streets in company with the Board of Works, examined the methods used in their construction and maintenance, the system of management and expenditure, and beg to submit the following report :

ROAD EXPENDITURE.

Money for this purpose is appropriated from the general funds of the town to the amount of \$2,000 annually. It is distributed among the various wards, and the sum given to each is under the control of the councillor representing that ward on the Board of Works. Under this system the effect invariably is to again subdivide the ward appropriation to such an extent that no work of a durable nature can be attempted. The people expect the amount to be scattered and the council has no other course to pursue. The system is fundamentally bad. Money is not spent in accordance with the requirements of street improvement, and the result is waste, extravagance and bad roads. An annual appropriation from the general funds for the repair of roads is found to work satisfactorily and equitably. To construct roads, however, under this method is practically an impossibility, as the expenditure cannot be sufficiently concentrated.

FRONTAGE TAX.

The most suitable remedy for existing conditions in Woodstock is the frontage tax system. When work is undertaken under it, money is raised by the issue of debentures extending over a term of years. The amount is assessed against the property abutting on the work, or benefited by it, according to the frontage of the lot, or according to its superficial area, or according to the assessed value of the property. Which ever one of these plans is chosen by the council should be stated in a general by-law submitted to the people for the adoption of the system. By means of the frontage tax system sufficient money can be raised to do durable, serviceable and economical work, and it is the most satisfactory method of consolidating road expenditure. Money thus raised is obtained at a very low rate of interest, and payment being extended over a term of years, the annual rate of taxation is small. The ultimate cost is no greater than under the old system of patchwork, the difference being that less money is wasted. The durable improvements obtained are at once a benefit to the individual property owner, the value of property is increased, and the town as a whole becomes a more desirable place of residence.

Under this system work may be undertaken in three ways : (1) On the petition of at least two-thirds of the property owners affected, representing at least one-half of the value of the real property benefited. (2) On the initiation of the council, unless petitioned against by a majority of the property holders affected, representing at least one-half of the property to be benefited. (3) On the recommendation of the board of health for sanitary reasons. In assessing the cost, ratepayers are notified of the amount, etc., and courts of revision are held giving an opportunity to appeal, and to adjust any errors in the assessment.

As has been intimated, a general by-law for the introduction of the system should be submitted to the citizens, and should be sanctioned by a majority vote. In framing this general by-law very great care must be taken to adjust it to local circumstances as far as possible. When this is done, and the by-law provides for a just and equitable assessment, street improvement is invariably stimulated. To this end peculiar cases, such as corner lots, triangular or irregularly shaped lots, side-hills and similar property should be taken into consideration; also the matter of street intersections, which are sometimes paid for by the property owners on the street, sometimes by the municipality. The frontage tax has in some cases unquestionably worked injustice, but when the by-law is judiciously framed there is no fairer way of paying for street improvement.

DIRECTION AND SUPERVISION OF STREET IMPROVEMENT.

In the direction and supervision of street improvement in Woodstock there are very serious defects. This branch of public work is in the hands of the ward representative on the board of works. There are five wards in Woodstock, and each member of the board of works is subjected to the possibility of a change annually. The result of such a system is that there is an entire lack of uniformity in street improvement. The work of one year is generally done without reference to the work of succeeding years, in view of which it may be rendered useless. Almost every street presents instances of this defect, in its sidewalks, its drainage, grade and road metalling. For economical and satisfactory results, there should be someone with a knowledge of the principles of road-making and street improvement over this branch of public work, who would examine, stake out and report to the board of works with plans and specifications; and works of importance should be let by contract. All work should be completed under the supervision of this official. He should be in constant touch with the chairman and board of works, and should attend each of their meetings for consultation. Even if all who have been members of your board of works had been expert road-makers, the constant change and interchange of plans would necessarily result in patchwork.

WASTED LABOR.

A considerable quantity of gravel was being placed on Reeve street, High street, and a number of other streets at the time of my visit, but owing to the poor quality of the material and the method of placing it on the road, the expenditure is practically wasted. A quantity of broken stone has been placed on McQueen street. The sub-soil was not prepared so as to form a suitable foundation and the metal was scattered loosely over the road. Traffic will soon force the stone down into the mud and the result will be very unsatisfactory. A broken stone road built in this way is less serviceable than gravel.

CONDITION OF THE STREETS.

The inefficiency of the system of expenditure and supervision employed in your town requires no other demonstration than the present condition of the streets. Gravel has been heaped on the centre of the road without reference to the objects to be attained in road construction. Very much of the gravel has been of an exceedingly poor quality. The drainage, where there is any, is crude and imperfect. The people have shown a disposition to boulevard the streets, but owing to the absence of a suitable curb line or established grade, these boulevards are undulating and irregular in width and elevation.

It is really a source of surprise that money, drawn from the pockets of the people, should be so long lavished in such a haphazard manner. To all other branches of construction, waterworks, sewerage, and public and private buildings, you have given serious consideration, but on streets the object appears to be to spend money. As with other enterprises, street construction, in order to obtain serviceable and economical results, must be treated as a matter of business.

DRAINAGE.

Good pavements are largely a matter of good drainage. Not that the shape of the roadway, the material of which the surface is composed, or the way in which it is laid, are unimportant—but that these are very largely a part of the system of drainage. With the River Thames in the westerly portion of the town, and Cedar Creek passing through the southern portion, with natural depressions or watercourses leading to these streams throughout the town, admirable facilities for street drainage are afforded.

Underdrainage is one of the first points to consider. In making roads it is the native soil which must really support the weight of the traffic, no matter what paving material is used to surface it. Gravel, stone, brick or asphalt are not sufficiently strong to bridge over a wet and yielding subsoil. But if this natural soil is kept in a dry state it can support any weight, and to this end underdrainage is necessary. Underdrainage of common field tile, four to six inches in diameter, should be placed on each side of the carriageway underneath the gutters, and below frost. This "lowers the water line" and secures a good foundation.

There must be surface drainage and for this the roadway should be crowned or rounded up, covered with suitable surface material, and open gutters provided to carry away this surface water. The surface metal (gravel or other material) of course resists wear, but on streets which are lightly travelled the main object is to provide a cover which will prevent the water penetrating the natural soil underneath, making it unfit to support traffic. By crowning the surface of the road, water is shed at once to the side, where provision should be made to carry it away immediately in open gutters.

Gutters or under-drains are useless unless outlets are provided, and care should be taken to see that these do not become obstructed. Surface drains may have outlets into the tile drains through catch basins, or into the sewers, where capacity for storm water is provided.

Springs underneath roadways should be tapped with blind drains and the water carried diagonally to the underdrains at the side of the road.

Cedar street, a leading entrance to the town, receives a great amount of soakage water from the uplying land. A very large quantity of gravel has been placed on this street, but without a complete system of underdrainage the roadway can never be economically maintained.

CROWNING THE ROADWAY.

To secure perfect drainage the roads should be uniformly crowned or rounded. On the streets of Woodstock very little attention appears to have been given to this important matter, as on the majority of streets no crown whatever is made, while in one or two instances, notably Vansittart avenue, it is considerably above the adjoining property, driving the traffic to the gutters and making it almost dangerous to turn from the centre. On gravel or macadam streets this crown should be one inch of rise to each foot horizontal from the gutter to the centre; that is, the centre of a roadway twenty-four feet wide should be twelve inches higher than the outer edge. It is well to first establish the grade of the sidewalk and then make the centre of the roadway conform to this as nearly as possible, and with the same elevation. On hills the crown should be sharper, say one and one-half inches rise to one foot horizontal, so as to draw the water quickly to the gutters and prevent it following the wheel tracks, deepening them to ruts. Obtain the crown chiefly by rounding up the sub-grade or earth foundation, the remainder to be made up by the additional quantity of road metal placed in the centre of the roadway. Thus, for a roadway twenty-four feet wide, the crown of the sub grade should be nine inches, the remaining three inches to be secured by the difference in thickness of gravel between the centre and the sides. Care should be taken to make the crown perfectly circular.

FORMING THE ROADBED.

The practice in your town has been to pile gravel in the centre of the roadway on top of the natural soil, or to bring the earth from the gutters to the centre of the roadway, placing the gravel on top of this. This is the plan usually followed in grading township roads, but is totally unsuited to street construction.

The roadbed should be excavated to the required width to receive the gravel, broken stone or other material, and the excavated earth used in making boulevards or filling in the low lots to bring them to the grade of the street. The side of the street should be levelled to conform to the surface of the roadway. As far as practicable the crown of the finished street should not be higher than the adjoining property, but should conform to certain established grades.

ROLLING.

I do not understand how the town of Woodstock can expect to build roads without proper tools. In examining the outfit of roadmaking machinery, I found it to consist of wagons to haul gravel and shovels to spread it, and two small stone hammers. The wagons and shovels belong to the workmen. Roads cannot be built without tools.

For economical, durable and serviceable road-making, a heavy roller is indispensable. A road should be sufficiently smooth and compact to shed the water readily to the side gutters. If the gravel or other road metal is dropped from the wagon loosely on a soft earth foundation, as is your practice, water passes into the subsoil as through a sieve. Wheels passing over the road when in such a condition at once sink into and rut not only the gravel, but the earth beneath. Water is held in the ruts and each succeeding vehicle renders their condition worse. The road is less durable, since, the gravel being mixed with the dirt beneath, it obtains, when finally consolidated, a dusty, easily worn surface.

The weight of roller used must depend upon various circumstances—the amount of work it will be required to do, the quantity of road metal used, the strength of the bridges and culverts over which it must pass.

A steam roller costs much more than a horse roller, but does so much better and faster work that it is more economical. Municipalities surrounding your town would find it to their advantage to rent such a machine, thus keeping it employed, as it is too costly to remain idle. A weight of twelve tons does satisfactory work, and it is not too heavy for the majority of bridges.

Rolling should commence at the side of the road, approaching the centre gradually. If the roller is first passed over the centre the loose metal is crowded out and the shape of the road destroyed. It is best to roll the earth foundation in dry weather, and each succeeding layer up to the top dressing. When the latter is put on, the rolling should be continued in wet weather (or the metal thoroughly soaked from a hydrant or with an ordinary watering cart) until the metal is thoroughly compact and solid, able to resist, without displacement, the heaviest load passing over it.

ESTABLISHED GRADES.

Without a complete plan of fixed grades it is impossible to do perfect work, and as far as practicable when any work is undertaken it should embody the whole street, both roadway and sidewalk. The grade of the street should be so fixed as to utilize all excavated earth in making boulevards, bringing the sides of the streets up to the grade and filling in low-lying property so as to conform to such grade. Without this it is impossible to fix a grade other than that which will conform to the elevations of the lots. One piece of street about half a mile in length was pointed out to me on which the sidewalks were constructed in six different sections under six different by-laws within the last four years, and without any road improvement being attempted. The effect is that a

different grade has been given for each piece of walk, the best that could possibly be given under the circumstances, but by no means as good as could have been given if the work had been carried out as a whole. Also streets are improved and graded without reference to each other. The result is that the intersections are very uneven, and are not only inconvenient, but have an awkward and unfinished appearance.

Slopes should in all cases be reduced as far as practicable, having due regard to the proper draining of water in the gutters, and having regard to the elevation of the adjoining property, for in fixing this grade it is bad practice to injure such property by excessive cuts or fills in front of it to satisfy a uniform grade of the finished road.

CURBING.

On each side of the roadbed it is advisable to place a line of flagstone curbing four inches thick and two and one-half feet in depth. A good substitute on residence streets is three inch by ten inch cedar plank. It should be spiked to six inch in diameter cedar posts, two and one-half feet in length, which are placed at intervals of eight feet and bevelled at the top so that the curb will slope slightly towards the boulevard. This curbing defines the roadway, protects the boulevard and gutters, and keeps the road metal in place.

WIDTH OF PAVEMENT.

The usual width of your street allowance is sixty-six feet. This width is always advisable for sanitary reasons, besides adding greatly to the appearance of the street. It is, however, an unnecessary expense both in first cost and maintenance to have a wider driveway on residential streets than is necessary to accommodate the traffic. A broad carriage way is very beautiful, but so also are wide stretches of nicely sodded boulevard ornamented with handsome shade trees which are a notable feature of your town. Carriageways of from twenty to twenty-six feet are now being adopted on the residential streets of cities and are meeting with general approval.

HILLS.

The Dundas Street hill and the Mill Street hill are in very bad condition. A large quantity of gravel has been placed on them, but no attempt whatever has been made to prepare the foundation, or to provide gutters. The course of the water is now down the centre of the roadway, the finer particles of gravel have been washed out and, in consequence, the large stones are exposed, forming a rough surface, checked and lined with crevices. Hills need very careful treatment. No water should be allowed to pass over the hill by the roadside other than that which falls on or exudes from the hill itself. Water from above the hill should be carried in underdrains, through private property, if possible, to an outlet. Neglect of this results in dangerous and expensive washouts. Gutters on hills should be protected with cobble stones from rushes of water. Where water oozes from the side hills or from land lying higher than the surface of the road, this soakage water should be intercepted and conveyed to the tile drains. It is good practice to fill the trench over tile drains with broken stone, gravel or other porous material.

GRAVEL.

My attention was directed to the gravel in three pits, Rapson's, one on Joint Street, and another on the north side of the town. The material in the two former is of a very inferior quality in its natural state, being very largely mixed with sand, earthy matter and large boulders. In order to be of service in road construction, material of this kind should be screened to remove large stones, sand and earth. A means sometimes adopted is to place in the pit a stone crusher with screen attachment, whereby the refuse is removed; at the same time the large stones are broken and form a very valuable part of the road metal resulting.

The gravel in the pit on the north side of the town is excellent. Much of it is fit for application to the road without any treatment other than to remove occasional large stones.

Cleanliness of material is absolutely necessary. Sand and earth are very injurious to the roadway when mixed with gravel, as they attract and retain moisture and permit it to pass through to the sub-soil. A covering of this kind is not only less serviceable while it lasts, but is less durable than one composed of clean metal. Gravel should be placed on the road in layers of not more than four inches in depth, and each layer consolidated with a roller. The depth of gravel required on the street varies in proportion to the nature and extent of the traffic. On residential streets, little travelled, eight inches at the centre and five at the sides will be sufficient. Residential streets considerably travelled, with occasional heavy loads, will require nine inches in the centre and six inches at the sides. On other streets, largely travelled, ten inches in the centre and seven at the sides will be needed.

BROKEN STONE.

Broken stone is a much more durable material than gravel, providing, of course, that a suitable quality of stone is used. The best stone is that which is hard, tough and which will not readily decay on exposure to the atmosphere. The only stone to be had in the vicinity of Woodstock is field stone and river boulders. If these are crushed care should be taken to exclude rocks of a poor quality, the most common of which are "weathered" sandstones and granites. Some limestones also weather very rapidly. Slatey rocks are too brittle to be used on the road. There are heavily travelled streets in your town approaching the business centre which it would be well to pave with broken stone in place of gravel; and for preparing gravel and for breaking field stone, a stone crusher would be a very valuable machine for your town to possess.

The stone should be crushed and screened into sizes varying from one inch in diameter to two and one-half inches. The largest stone should be placed in the bottom of the roadbed and the smallest at the top. A road surface of a mixture of large and small stones, in time becomes very rough owing to the smaller wearing more rapidly than the larger, while large stones at the surface have a tendency to become loose.

The stone should be placed on the roadbed in layers and each layer thoroughly consolidated with a roller before the next is applied. The depth of the stone may vary, according to the traffic, from nine inches at the centre with six inches at the side, to twelve inches at the centre and nine inches at the side. As with gravel, it is very important that the material should be clean. No "binder" is needed with limestones. If an exceedingly hard metal such as trap is used, the fine "screenings" of the stone will be the best aid to consolidation. In rolling, the lower courses of stone should be perfectly dry, but in finishing the road, water may be used to flush a dressing of stone screenings into interstices.

GUTTERS.

When gravel or broken stone is used on business streets, the gutters should be cobblestoned similarly to those on Dundas street, which were evidently well laid; but on account of tearing up to make sewer and gas connections they have become somewhat injured. In opening the streets for this purpose, great care should be taken in replacing the earth and pavement. Pipe for sewer, gas and water connections should as far as possible be laid by drifting with an auger large enough to accommodate the pipe, excavations only large enough to perform this work being made. On residential streets, the angle between the curb and the road-surface usually forms a sufficient waterway for surface drainage.

VANSITTART AVE.

This avenue could be made one of the most magnificent residential streets in the province. It is 132 feet wide, the residences on each side are handsome, and the lawns surrounding them spacious, well-kept and tastefully designed. A very large amount of

money has been spent on the roadway, enough to have provided an asphalt pavement. Artificial stone sidewalks have been laid on each side of the street, boulevards made and neatly kept. But the dirty, imperfectly constructed and shapeless roadway is entirely out of keeping with its surroundings. While this street, on account of being a leading avenue, and on account of its exceptional width and beauty, attracted my attention specially, there are numerous other streets equally deserving of improvement.

SIDEWALKS.

Considerable attention has been given to the matter of sidewalks and they are now, generally speaking, in good condition. A large number of flag and artificial stone walks have been laid and add very much to the appearance of the town. The plan of laying the walks outside of the trees on residential streets has been largely followed, and is commendable. There should, however, be a narrow boulevard between the curb and the sidewalk, but, where there is not sufficient width of street allowance to permit of this, an exception should be made. If this boulevard does not exist between the sidewalk and the road, the beauty of the improvement is largely lost by giving the street a dry, dusty and business appearance, and frequently places the sidewalk so far from the trees as to be away from the shade. No sidewalk should be laid without a plan of the whole street being made, to show its location with respect to the roadway and other walks on the same street, in order to secure regular and uniform results.

SEWERS AND SEWER CONNECTIONS.

Where a sewer is necessary on a street it should be constructed before the improvement of the road is undertaken; and as far as practicable all sewer, water and gas connections should be laid to the street line in order to prevent the breaking up of the roadway for this purpose. A macadam road cannot be broken up with injury, and the damage to a brick or asphalt pavement is correspondingly great. I would advise that special attention be given to this matter before Dundas street is repaved.

DUNDAS STREET.

The life of the cedar block pavement on Dundas street is about expired, the material is decayed, has become uneven and rough, and must in a very short time be renewed; but although in this condition, it will, I presume, be used for another year, during which time your council should carefully consider the merits of the different pavements. Neither gravel nor macadam make a suitable pavement for the business thoroughfares of a town of the size of Woodstock. The pavement on Dundas street, in the business portions, should be surfaced with a much more durable material than either of these, a material which will be easily kept clean, so that it may be crossed at any point, will be sanitary, and will in every way give good service. A dirty, muddy, rough main street is certainly a poor advertisement for a town.

The qualities essential to a first-class pavement are:

- (1) A secure and pleasant footing for horses.
- (2) Smooth, so as to render travelling and traction agreeable, easy and noiseless.
- (3) It should be sanitary; the form and material such that it will be impervious, liquids will not have permanent lodgement, and dust will not be easily produced.
- (4) The durability and service rendered will be commensurate with the cost of construction and maintenance; that is, it must be economical.
- (5) It must be easy of removal, replacement and repair and at reasonable cost, and with the appliances and materials within the control of the corporation.

In view of the above it will be apparent, I believe, that brick and asphalt are the two competing materials for use on Dundas street. With regard to the first quality, the

foothold afforded to horses, bricks must have the preference. It is one of the objectionable features of asphalt that it is exceedingly slippery when wet, and even when dry it is not always safe.

As to the second quality, asphalt must take first place since it is in a slight degree smoother and less noiseless than brick. It is doubtful, however, if traction is any easier, owing to the insecure footing afforded to horses.

There is little if any difference between the sanitary status of the two pavements. Both are, of course, impermeable and offer little resistance to the flow of liquids; the joints of the brick are just sufficient to retain moisture and subdue dust. With the smooth asphalt surface no amount of sprinkling will keep the surface moist in hot weather.

As to cost, asphalt is from one-half to one-third more than brick, and experience has not proven its life to be any greater. As to durability, there must always be the proviso, that good material and proper plans and methods of construction are used in any case.

The laying or repairing of brick does not require skilled labor as does asphalt, and this difficulty in connection with the latter is felt more particularly in places where only a short section of asphalt is used. In large cities the inconvenience is not so great.

While asphalt is extensively used in the United States and Canada, and is unquestionably a good paving material for certain streets of large cities, I believe brick more suited to the requirements of your town.

The quality of a brick pavement is not to be gauged by the best brick used in its construction, but by the poorest. For this reason it is necessary, before deciding on the kind to use, to see that it comes up to the standard of scientific tests. More than this, while the building of the pavement is in progress there should be careful inspection to see that no brick of an inferior quality is used.

There is a tendency also, to endeavor to reduce the cost of a pavement by having a cheap foundation. Foundations of gravel, sand, and macadam have proven successful in a number of cases, but only where the natural sub-soil is of a loose and porous nature. In this climate, where we are subjected to alternatives of frost and slush, the experiment is a dangerous one. A foundation of four inches of concrete should be used. On this, place a one-inch cushion of sand and fill the joints of the brick with a cement of tar or pitch and sand. The earth sub-soil should, of course, have been previously graded, and consolidated with a heavy roller.

TIRES.

To consider the cost of the construction of pavements is important, but the cost of their maintenance is not to be overlooked; and this latter will necessarily be very great so long as the use of narrow tires is permitted. The cost of maintenance falls on the same individuals as does the cost of construction, and as the users of vehicles are members of the community, it is hard to understand why they should wish to tax themselves for the additional cost of maintenance, and the additional cost of transportation caused by bad roads, the result of narrow tires. The use of narrow tires delays the era of good roads to a greater extent than does any other cause. The roads of European countries owe their perfection very largely to the tires which are sometimes as broad as ten inches, the tire of the ordinary market wagon being six inches in width. No town should be without a by-law regulating the width of tires, and no drays used for delivering heavy loads from the depots should have tires of less width than eight inches. Various towns in the province are now taking steps to regulate this matter.

PUBLIC WORKS.

Very rarely does the average citizen consider the meaning of a municipal corporation. It is merely a society organized to carry on certain institutions in a more serviceable and economical manner that would be possible for the individual member. The

character therefore of a town's public works are an index to the good judgment and taste of the citizens as a body. The handsome lawns and well kept private properties so numerous in Woodstock are indicative of the aspirations of the citizens toward rendering the town as perfect a place of residence as it can be made, and point to the conclusion that they are willing and prepared to bear their respective portions of the cost, as soon as they find that progress at a reasonable cost can be made. Every improvement made in a town helps to build up the town, and is in this respect a benefit to every ratepayer. While sewers and waterworks are necessary to a town's welfare on sanitary grounds, the proper construction and maintenance, at the same time beautifying the public thoroughfares, is a matter which you cannot in the future, as in the past, afford to overlook.

I was very much pleased with the active interest taken in the question by your council, and their evident desire to place this important part of municipal affairs on a business-like basis. The interest exhibited also at the public meetings of the citizens is an evidence that they have been considering the matter and are awake to the inefficiency of present methods.

TOWNSHIP REPORT.

To the Reeves and Councils of the Townships of East Nissouri, West Zorra and East Zorra :

GENTLEMEN: In accordance with your invitation, I visited your townships, drove over the roads, inspected their present condition, examined the material available for road construction, the system of expenditure, the methods and plans used in doing road work, the administration of the statute labor law, and beg to submit the following report and recommendations :

LABOR AND ROAD EXPENDITURE.

The roads of your townships are built and maintained under the statute labor system, supplemented by grants from the municipal funds. The statute labor system has some commendable features. Under it good work can be done, and has been done, at less cost to the taxpayer than could perhaps have been obtained otherwise. It is, however, peculiarly subject to neglect and evasion, and this has been taken advantage of to such an extent that the roads of your district have suffered in consequence. The manner in which road work is slighted is too well known to need description. The wagons, horses and implements often brought out for use on the road, make up an assortment that would disgust any intelligent farmer, if he had to pay for them by the day at one-quarter rates for work on his farm. The work is too frequently done in a spiritless way, and a great part of the time is spent in talk. Some townships in various parts of the province commute the statute labor at such a reduced rate as to be practically an acknowledgement that statute labor has been shamefully neglected and thus brought into disrepute. The experience of one municipality is that, commuting at thirty-five cents per day, better results are obtained. That such could be the case is certainly a reflection on any municipality, and rather than commute at so low or even at a reduced rate, it is worth a struggle on the part of a township to make each day of statute labor worth one dollar in actual results on the roads.

Were the statute labor system economically administered, it would take only a short time to work a complete transformation in the condition of your roads. The true spirit in which statute labor should be performed was exhibited in the work of the pioneers, in road-making. In their day, population was sparse and the road allowance had to be cleared of forest. Stumps had to be grubbed, bogs burned, the roadway graded, swamps corduroyed, and streams bridged. Drainage was exceedingly difficult and there was no municipal fund from which to draw, for mending a culvert or filling in ruts. Labor was

then concentrated, and good, honest labor was given. Now, however, with the increase of population and wealth, roadwork is scattered in patch-work, and is done in a half-hearted way.

Statute labor is performed at a period of the year when the time of every farmer is of great value to him on his own farm. If the work of improving the roads is neglected, and a portion of the time spent in statute labor is wasted, it must be apparent to every citizen of the Province that there is a tremendous amount of energy lost to the country. We are obtaining our roads at an enormous expense, and if they are still to be bad roads, the expense is many times increased indirectly. Sufficient money is now being expended in your townships for material, if it were carefully prepared and skilfully applied.

NUMBER OF PATHMASTERS.

In order to bring the roads to proper condition, the operation of the statute labor system in your municipalities must be placed on a more systematic basis. One of the first steps would be to reduce the number of pathmasters. In each of your townships there are about 100 pathmasters, and, in consequence, as many different ideas of road building. The result, as shown on the ground, is that one pathmaster has considered sixteen feet a sufficient width of roadway, another eighteen, another twenty, and so on up to forty feet; one thinks a roadway should be flat; another that it should be rounded up a few inches; another, a few feet. Instances of these differences of opinion might be multiplied to almost any extent. Certainly all cannot be right, and this points to the fact that the control of the roads should be concentrated in fewer pathmasters under one central head. One man should have general oversight of the pathmasters and all work and expenditure on roads and bridges; and he should be required to report to the council at each meeting. He should have charge of all township machinery and materials. He should be a man specially adapted to the work, and his tenure of office should be as secure as that of the clerk or treasurer.

Besides being encumbered by the number of pathmasters, none of whom have sufficient means at their disposal to do permanent work, each pathmaster is surrounded by a few neighbors, and rather than incur disfavor by insisting on a fair day's work, frequently invites popularity by allowing or even encouraging neglect of the work. Were the beats enlarged, every pathmaster would become more independent and would have a sufficient number of days of labor under his control to undertake durable and complete work. But with their knowledge and experience in road-making, it cannot be in the interest of the townships for each pathmaster to be given the power to carry out his own ideas and experience on the road. To believe that every man is a born road-maker is a great mistake, and in order to get over the evil it is necessary that every citizen should be willing to join hands with his neighbor, and to work under a man specially instructed. The work cannot be planned nor carried out to suit everybody. An army composed entirely of commanding officers, or a party of workmen made up of foremen, cannot make much headway.

PATHMASTERS.

Pathmasters should be chosen from among the most respected, intelligent men in the community, and they should be kept in office. Even then, their individual opinions and plans should be subject to revision by the council, which body should exercise a direct oversight of the roads, and their improvement by statute labor.

That there are men who could fill the office of pathmaster in a creditable manner, is apparent from the fact that, in driving over your roads, there were several sections which have been well done, and good pieces of road constructed by statute labor alone. That many are poor pathmasters is indicated by other sections which show no evidence of results except those produced by the expenditure of municipal taxes, contributed very largely by those who had worked with a will in performing their own statute labor on other beats.

When men of suitable ability, knowledge and experience have been once obtained, they should be retained in office. By the present system of annual rotation, as soon as the pathmaster has learned something in his year of apprenticeship, his experience is lost and another is appointed, who begins in his turn to take lessons in road-making at the expense of their condition. In other occupations an apprenticeship of some years is necessary before a person is qualified to practice, but a pathmaster, the moment he is chosen, is thought fit to direct a work requiring much science, at the expense of the township's capital of time, labor and money. Men are taken from the occupation in which they are skilled, and are transferred to one of which they know nothing. An honest man knows this, and immediately strives, from his personal experience, from the advice of others trained by practice and instruction, from experts, and from every available source, to properly qualify himself; with this and with continued experience he can become a most valuable officer. Such a man realizes the responsibility of his position and the duty he owes his fellow citizens whose property is placed in his charge, the expenditure of whose money he directs, and whose interests he is expected to advance. He commands the respect of those who have to serve under him, his order is obeyed, public interest is created and maintained, and the question of making good roads is rendered simple, requiring but a small outlay of money.

There is another man who strives for the office of pathmaster, and looks upon his appointment merely as a victory over a neighbor. He exercises his authority in performing some act to aggravate a citizen with whom he does not see eye to eye. The improvement of the road is by no means the prime object; injury rather than betterment is the probable result, and those who are obliged to perform the work become disgusted.

Some men look upon their appointment as an acknowledgement by the council and community of their special skill as road-makers. The solicitation or acceptance of any advice from experienced men would be looked upon as a confession of unfitness, and certainly any suggestion to accept instruction from an expert would be regarded with contempt. The competent pathmaster, as well as the competent business man, is he who eagerly seeks, from every source, the fullest possible information pertaining to his special line, believing that his knowledge never can be too perfect.

LOYALTY TO PATHMASTERS.

It is one of the weaknesses of the statute labor system, that it is very difficult for a pathmaster to enforce prompt obedience to his orders. If a neighbor persists in idleness, or if, instead of doing as he is directed, he stops and argues for an hour on the wisdom of so doing, the pathmaster cannot obtain redress except at the expense of his neighbor's good-will, a step very few pathmasters are willing to take. A farmer could and would discharge an employee for pursuing such a course, but in the case of a pathmaster the roads must suffer. Under a proper administration of the statute labor law, it is apparent that it is the duty of every man, when performing statute labor, to set aside his personal feelings and opinions, and promptly do the work allotted him. This is the relation of the tax-payer to the pathmaster. On the other hand, the pathmaster (and going further back, the council), owes it to the tax-payer that all plans shall be carefully prepared and considered before the work is commenced, so that improvement will be made along the right line, and with the greatest economy of labor.

GRAVEL.

There appears to be a desire on the part of the people of your townships to put in their time by drawing gravel; and gravel is therefore hauled and dumped on the roads before the foundations has been prepared to receive it. I learned that, in West Zorra, \$600 was paid this year for gravel (that is, just for the material), but most of the stuff placed on the roads is almost worthless, although first-class gravel is to be found in various pits in each township. Judging from the immense excavations, many thousands

of loads have been drawn, and farms have been paid for with the amount received for the gravel taken from one small corner. Contrasting the amount of material placed on the roads, with their condition as a whole, it is evident that there has been great waste.

Buying gravel by the load is like buying water by the bucket. Borings and test-pits should be sunk where gravel is known to exist, and the amount estimated from these borings. The material should be purchased in a business-like way for a fair amount; and in case of opposition from the owner, the land may be expropriated if necessary.

One source of the use of poor and dirty gravel arises from insufficient care in taking it from the pit. The social feature of statute labor is carried to the extent that those drawing gravel want to load up together, drive together to the place of dumping and return in company to the pit. There may be a dozen teams, and only room for one or two where good material is to be had in the pit. The remaining ten, however, drive in and fill up the wagon boxes with anything in sight, whether sand, clay or boulders, and this is the kind of "gravel" with which a good many roads are surfaced.

Not only should gravel be chosen from the best part of the pit, but care should be taken to sort out any bad stuff, sand, boulders, etc., that may be mixed with this. Gravel pits are covered usually with a deep layer of clay, loam or earthy material, useless on a road. While this is not shovelled directly into the wagons, it is undermined, falls down on the good gravel, and then goes along with it to the road.

CLAY AND SAND IN GRAVEL.

Dirt mixed with gravel is not only useless, but is hurtful to the road. Evidence of this is found on your roads where a few loads of clean, hard gravel, have been placed on the road, followed by a load or two of dirty stuff, then clean metal, then dirty, and so on. The surface of such a road is a series of ups and downs, some smooth spots, then ruts.

It is contended that a certain amount of clay and sand is needed to act as a binder, and sections of roadway are pointed out which have consolidated readily, and make a nice smooth drive. This is undoubtedly the case during summer, but it will be found that a pleasant summer road is frequently a very poor fall and spring road. The bond made by sand and clay is weak, and there is not the mechanical hold which one stone takes upon another by pressure, as is the case when clean material is used and well rolled. The result is that an extended rain turns the road into slush, mud and ruts.

When clay or sand is mixed with gravel, it absorbs and retains moisture, the whole yielding readily under traffic. It has not the power to support wheels that good clean material has. In a number of townships throughout the Province, the practice of screening the gravel is successfully followed.

STONE CRUSHERS.

Where such a large amount of money is spent annually for gravel, which, when laid on the road in its natural condition is largely useless, it appears to me that one of the first considerations of your council should be the purchase of a crusher whereby the gravel might be properly prepared, and nothing hauled but first-class material. By placing a crusher in the pit, all material could be passed through, and with a rotary screen attached, all sand, clay and fine stuff would be removed, leaving only the clean road metal. In some sections of your townships, field stone is plentiful, and gravel has to be hauled several miles. Where such is the case, the field stone might be crushed and placed on the roadway, making a more durable road at less cost than gravel. The work of crushing could be done at any suitable season of the year, in the pit or field, and the material left in piles ready to be placed on the road whenever needed.

DRAINAGE.

The drainage of a road is quite as important as the gravelling, but the best way to obtain this must be one adapted to the nature and elevation of the soil, and the direction and extent of natural watercourses, etc. It is imperative, however, that provision shall be made wherever the soil is in a low or wet location, or is of a retentive nature, that it shall be drained both above the surface and below it. Sandy or gravelly soils very frequently do not need sub-drainage, while clay always requires it. Unless there is natural sub-drainage, artificial underdrainage must be resorted to.

Considerable has been done in making open drains by the roadsides in your townships, but a great many of these need deepening and cleaning to provide better sub-drainage. A much better practice than having deep open drains, is to use tile placed below frost line. These should, wherever practicable, be placed beneath the present open drains on each side of the road. No matter how good the material placed on the surface of the roadway, unless the natural soil beneath is kept dry the metal is forced down and the mud comes to the surface. When roads in this country are gravelled, they are frequently termed "Macadam." The important feature of the roads built by Macadam, however, was the drainage, not the surface covering.

Tile and open drains must be kept free, with a good fall to an unobstructed outlet. It is useless to drain water from the roadway and then keep it standing in drains until it soaks into and softens the natural soil under the gravel. At all seasons of the year, particularly spring and fall, when obstructions of snow and ice are likely to occur, the outlets should receive special attention.

CROWNING THE ROAD.

Too much crowning is as bad as too little; but the road must be crowned or rounded to shed the water to side gutters. A great many of your gravelled roads are perfectly flat, and some are lower at the centre than at the side. To provide for wear and settling, the roads, when in a proper state of repair, have a rise of about one inch to the foot from the side to the centre. To crown the existing gravel roads which are too flat, the square shoulders at the side should be cut off. If these shoulders are composed of clean gravel, forced from the centre of the road by the wheels, the material may be drawn in towards the centre. As a rule, however, it is of such a quality as will turn to mud immediately the fall rains commence, and should be used in levelling the sides of the road, or in any other way that circumstances may make advisable. This stuff from the shoulders of the road has, in some instances, been brought indiscriminately to the crown of the road, with the result that a firm gravel foundation has received a coating of mud. During the summer it may have made a very pleasant surface, but in wet weather it becomes slushy, holds water and destroys the foundation beneath it. The shoulders having been disposed of, clean material from the pit should be placed in the centre. It will frequently be found that a sufficient depth of gravel is already on the road, requiring only the cutting off of the sides, and shaping up with a light coating of clean material properly applied. To do this work perfectly, quickly and economically, road graders should be used.

APPLYING GRAVEL TO THE ROADWAY.

The gravel placed on the roads of your townships has been merely dumped in the centre without any further treatment. The first effect of this is to drive vehicles to the side of the road. These side tracks are used until late in the fall when they have become cut up and flattened to such an extent as to destroy the shape of the road, preventing surface drainage. Water is held at the side, soaks into the roadway, and the gravel is forced down into the mud and is largely wasted.

Gravel should be nicely spread on the roadway and thoroughly rolled so that travel will at once use it. The objection is made to the use of clean gravel that it will not con-

solidate, a difficulty entirely removed by the use of a roller. If a roller is not used, the gravel should be raked into the wheel tracks and into the hollow worn by single horses, until the whole roadway is hard and smooth under traffic.

WIDTH OF ROADWAY.

The width of driveway on the majority of the roads in your townships has already been fixed by sharp, open ditches, and where the road has been coated with gravel, it will usually be found impracticable to alter it now. A gravelled strip, however, of eight feet in the centre is sufficient to accommodate the traffic on the greater number of your roads, and a greater width than twenty-four feet between ditches will generally be found needlessly expensive to maintain. If the roads of your townships were properly underdrained so that the open drains could be very shallow, a less width than twenty-four feet would generally be not only cheaper but better. A less width than eighteen feet is rarely advisable.

ROAD GRADERS.

Road graders are as necessary on the road as are self-binders in the harvest field, and if true economy is to be considered, are absolutely indispensable. When properly managed, a grader will do the work of from fifty to seventy-five men in shaping, crowning and scraping a road. One capable man should be employed to operate it whenever it is in use, as it requires considerable experience to handle it to the best advantage. The operator should know something about roads, and should be capable of following minutely instructions received from the Council. A grading machine of itself has no knowledge of road-making, and unless the operator has, satisfactory results cannot be obtained. By using such an implement for a few days on each statute labor beat, it is apparent that very much could be accomplished in shaping up the edges of existing gravel roads, and in grading dirt roads which have had but little attention.

LOCATION OF ROADS.

As the sustaining power of a chain is no greater than the strength of its weakest link, so the utility of a road is measured largely by the steepest grade or poorest section. For this reason, hills, cuts and steep grades should be avoided, and the best ground for a road foundation should be chosen. Such pieces of road as are on hills should be judiciously altered to obtain easy grades, provision being made in the Municipal Act for such changes of location. A mistake sometimes arises from the idea that the distance over a hill is necessarily shorter than the distance round it; and that the distance down the valley and up, is shorter than the detour necessary to avoid it; whereas the reverse may be the case; and in any event, the steep grades, necessitating smaller loads, loss of time and expense for maintenance, form a very great consideration.

This is a very important matter which should be considered as far as possible at once. The longer the delay, the more difficult it will become to rectify errors. Numerous instances will be found throughout your townships in which a slight change in the course of a road would result not only in a great saving in building and maintaining it, but would add to the convenience of traffic.

HILLS.

Hills need very careful treatment. Water is very commonly carried long distances by the side of the road, past natural water courses and through deep cuts in order to dispose of it over a hill. The secret of successful drainage is to dispose of water in small quantities, without allowing it to gain force and headway. Neglect of this principle is the cause of the bad condition of so many hills, making deep washouts and necessitating expensive repairs.

Hills are frequently springy, or are kept soft by water oozing from them. In such cases a line of common farm tile should be placed on each side of the roadway, underneath the gutter, and below frost. Wet points should always be tapped by blind drains leading diagonally across the roadway to side under drains. The crown of the roadway on hills should be sharper than on level ground, otherwise water will follow the wheel tracks, deepening them to ruts, instead of being shed to the side gutters.

BRIDGES AND CULVERTS.

The repairing and renewing of wooden bridges, sluiceways and culverts is a constant drain on the funds of a municipality. Timber is becoming expensive and iron is each year decreasing in price. It is encouraging to find your councils studying true economy in the construction of bridges by the use of stone and iron. The first cost of such permanent structures is much greater than that of wood, but with the greater term of duration and the fewer repairs needed, they will in the end be found less costly than wood, at the same time giving better service.

Cement culverts are used largely, particularly in East Nissouri, and will be found much more economical and more serviceable than wood. When properly constructed, with a free outlet, they are permanent, and in a few years your councils will be largely relieved of expenditure from this source.

TIRES.

People tax themselves in time, labor and money for constructing roads, and then purchase narrow tired wagons to destroy them. If tires of from four to six inches were used on lumber wagons in your townships, a very great change would be quickly noticed in the condition of the roads. The difference between a narrow tire and a wide tire on a road, is about the same as the difference between a pick and a pounder. The one tears up, the other consolidates. When wide tires come into general use, a great part of the question of good roads will be solved, as the wear of the road is nearly as important an item to consider as its construction. With wide tires, the cost of keeping roads gravelled would be reduced one-half.

It is objected to broad tires that they are heavy to draw over a rutted or stoney road. It is, however, the narrow tires which are so largely responsible for the ruts and for working the stones loose; instead of which, wide tires would keep the roadbed smooth and would compress stones into it.

MAINTENANCE OF ROADS.

The roads of your townships receive very little attention except at the time of performing statute labor. Roads are very much more cheaply and easily kept in good condition if repairs are made as soon as wear appears, for ruts and wheel tracks interfere with surface drainage, hold the water, and quickly cause the road to "break up" in wet weather. Repairs of this kind are of such a simple nature, requiring a little raking, filling a rut, the freeing of an obstructed drain, at the most a load of gravel, that every farmer could, with no loss of time, see that the half of the road allowance passing his farm was not neglected in such trifling matters. Attention of this kind would result in an immense benefit to every resident of the townships.

Instead, however, of men having sufficient public spirit to volunteer work of this description, we find them adopting all sorts of means to avoid doing even their just amount of statute labor. These men, after idling away their time, will wait in deputations on the councils, complaining of the bad and dangerous condition of their roads, asking for municipal grants to do the work which they should, in justice to other taxpayers, have done by their statute labor. No encouragement should be given in such cases by making money grants where the ratepayers neglect to make good use of their statute labor.

An excellent system has been adopted in some townships which has been productive of good results. Under a by-law of the municipality, where any section subscribes an amount of money for road improvement, the municipal council assists with an equal amount from the general funds. In other townships, no money is paid out for gravel unless the road division has first drained, graded and formed the roadway according to the set specifications of the municipal council.

If, however, a section of the people persist in looking on their statute labor as a joke, if the present waste and injustice continues, and the roads are permitted to remain in an unsatisfactory condition, a hindrance to individual and national progress; if, after a fair trial, it is found that the statute labor law cannot be operated in your townships on business principles, it will be the inevitable result that your ratepayers will demand its abolition and the substitution of a law that will be less easy to evade. This has been the result in numerous other townships, and if the old system is to exist still in your townships, its friends must make a radical change in the present methods of administration and prove its worth.

TOLL ROADS.

In a long settled and wealthy community such as Oxford, it is usual to find the main roads in fairly good condition. I was quite surprised to find, however, that this is rarely the case in your townships, and more so when I was informed that a number of these roads, especially near the business centres, are in the hands of chartered companies. The people have been taxed directly and heavily for these roads for nearly half a century. They were nevertheless in a deplorable state at the time of my visit.

The user of the toll-road is supposed to be taxed for travelling on a good road, not a bad one, and if there is any reason for these roads being in the hands of private corporations, it is that the public shall be benefitted. These roads on which heavy toll has been collected for so many years, in a populous and wealthy community, should be models for the construction of roads under municipal control. The only features which seemed to distinguish them from some of your concession roads were the toll gates at frequent intervals.

It may be urged by the owners of these toll roads, that a fair proportion of the earnings is being spent on them to place them in good condition, but if so, those in charge of this expenditure should also take lessons in road-making. The paying of toll is a system of taxation both antiquated and annoying, and becomes especially so when the roads are badly maintained.

GOOD ROADS.

There is a tendency on the part of some to look upon the good roads agitation as a stupendous enterprise, and to wonder how "they" can ever make all the roads of Ontario. In order to bring the roads of the entire Province to a good condition, no pathmaster and his men would have to go beyond their own beat.

In some divisions under incompetent pathmasters, men rake and scrape the mud, and idle away their time, expecting to receive grants from the council to do their work. When good roads are mentioned, they sit down on the bank of the ditch and ask what they will cost, and where the money will come from, wasting additional time in arguing against the movement. If those who built up the pioneer settlements, who chopped trees, grubbed the stumps, logged, burned and cleared the allowance, graded the roads, corduroyed the swamps, and bridged the streams by their united, zealous toil, without the aid of municipal grants, if these had sat down and estimated the cost, and cried for money, we would to-day have no roads, and your townships would still be a wilderness. Surely, with allowances cleared, the larger amount of grading done, material plentiful, increased population, knowledge and ability, and large amounts appropriated annually for road improvement, it cannot be a question of money.

A reorganization of the statute labor system bringing it to a business-like basis, gravel taken from the pit and applied more carefully, a better acquaintance with the principles of road-making, better drainage, proper implements, wider tires, a personal interest on the part of every citizen in keeping roads in a proper state of repair—these are the reforms necessary to improve your roads economically, durably and efficiently. These constitute the true objects of the good roads movement.

It would scarcely seem necessary to urge upon the residents of the rich and long settled townships of Oxford (a county which for many years has claimed the distinction of being a most progressive district, the garden of the Dominion), the benefits to be derived from, and the great need of better roads. While the county is a model one in many respects, in its handsome farm residences, its splendid barns, its reputation for high-grade stock, the roads nevertheless, are far from being such as I could recommend to other municipalities as models. That a community which thoroughly comprehends the advantages to be derived from modern improvements in agriculture, should be so far behind in matters pertaining to road improvement, is certainly to be regretted. But with a live interest now being taken by the citizens generally, in this branch of public improvement, I believe your councils will find the task of reform a comparatively easy one.

CEMENT-CONCRETE SIDEWALKS.

In connection with the improvement of streets, there is a widespread tendency to replace wooden sidewalks with others of more permanent material. In fixing grades, it will generally be necessary to undertake the construction of the entire street—roadway, sidewalks, boulevards—if the best results are to be attained, and for the purpose of economically and easily handling the excavated earth. Wooden sidewalks are lacking in durability, need constant attention to keep them in a safe condition, are not agreeable to walk on when a little worn, and will, in their ultimate cost, be found more costly than those of cement-concrete.

In constructing this latter class of walk, the space over which it is to be laid should be excavated to a depth of twelve inches below the top of the proposed walk, perishable or objectionable material being removed to a greater depth if necessary. The bottom of the excavation is then compacted by pounding or rolling until as firm as possible.

On the bed so prepared a layer of clean gravel or broken stone is spread to such a depth as will give, on thorough consolidation, a thickness of seven inches. Temporary curbs of pine should be set on each side of the excavation, to be removed after the walk has hardened.

On this foundation, and in the mold formed by the curbs, is placed four inches of concrete which should be composed of small broken stone, of a size not greater than one and a quarter inches in any direction, and entirely free from dirt or dust; clean screened gravel, sharp sand and Portland cement. This should be thoroughly rammed, and may be of the following proportions:

Cement	1 part
Broken stone	5 parts
Clean gravel	2½ parts

Slab or flag divisions are then marked off, from sixteen to twenty feet in length. This provides for settlement, expansion and contraction from frost, and lessens the tendency of the walk to crack irregularly from these causes. The joints should be filled with sand or other approved separating material.

Before the concrete has set, and while it is still adhesive, there should be laid on it a wearing surface one inch in thickness. This surface coating may be composed, if a sand finish is desired, of:

Portland cement	2 parts
Sharp sand	3 "

Over this should be sifted a layer of the best Portland cement, the whole to be then levelled, neatly trowelled, and rolled with a tooth roller to make a surface which will not be slippery. A more durable surface coating, known as granolithic, may be had by using broken granite in place of the sand. This broken stone should have a square or cubical fracture, be clean and fresh, and of sizes three-eighths of an inch downward. The composition will then be :

Portland cement	2 parts
Broken granite.....	3 "

The work should be kept moist, and protected from the direct rays of the sun, with canvas, until thoroughly set. Traffic should be kept off for a period of about fifteen days.

The various parts, cement, sand and broken stone, should be measured and mixed when dry, water then added as required, and the whole again thoroughly manipulated and mixed. There is a great necessity for careful and skilled workmanship, and an experienced man should have charge of the work.

If the sub-soil is of a nature to retain water, is low-lying and not sufficiently under-drained, a line of common field tile should first be placed below the centre of the walk. The surface of the finished walk should have a slope towards the gutter of one-quarter of an inch to each foot of width.

When built by a contractor, he should be required to guarantee that the walk shall last and remain in all respects in perfect order and condition, and free from all cracks and defects, and of perfect hardness consistency, smoothness and finish, for at least the term of five years from the date of its completion, unless subjected to other influence than the ordinary effects of wear and weather.

STATUTE LABOR LAW IN TOWNSHIPS.

The main provisions of the statute labor laws are the following ;

(1) Every male inhabitant of the age of twenty-one years and upwards, and under sixty years, who is not otherwise assessed, and who is not in her Majesty's Military or Naval Service, on full pay or actual service, is liable to one day of statute labor on the roads and highways.

(2) Every person assessed upon the assessment roll of a township shall, if his property is assessed at not more than \$300, be liable to two days statute labor ; at more than \$300, but not more than \$500, three days ; at more than \$500, but not more than \$700, four days ; at more than \$700, and not more than \$900, five days ; and for every \$300 over \$900, or any fractional part thereof over \$150, one additional day.

(3) In townships where farm lots have been subdivided into park or village lots, and the owners are not resident, and have not required their names to be entered on the assessment roll, the statute labor shall be commuted, where such lots are under the value of \$200, to a rate not exceeding one-half per centum on the valuation, but the council may direct a less rate to be imposed by a general by-law affecting such lots.

(4) A farmer's son rated and entered as such on the assessment roll is liable to perform statute labor as if he were not so assessed ; but a tenant farmer's son, bona fide resident on the farm of his father or mother, is exempt from statute labor in the same manner as if he were the son of an owner and jointly assessed.

(5) A non-resident, wishing to perform his statute labor, must require his name to be entered on the roll, otherwise a commutation tax is charged against his property according to the assessed value.

(6) Where a resident owner, tenant, or occupant, who has caused his name to be entered upon the assessment roll, after notice or demand, makes default in performing his statute labor or in payment of commutation for the same, the overseer of the highways in whose division he is placed shall return him as a defaulter to the clerk of the

municipality before the fifteenth day of August, and the clerk shall, in that case, enter the commutation for statute labor against his name in the collector's roll, and the same shall be collected by the collector.

(7) In every case the clerk shall notify the overseer of highways, that may be appointed for such division, in the following year, of the amount of such commutation, and the overseer shall expend the amount of such commutation upon the roads in the statute labor division where the property is situate, and shall give an order upon the treasurer of the municipality to the person performing the work.

BY-LAWS.

The council of every township may pass by-laws.

(1) For empowering any person, resident or non-resident, liable to statute labor within the municipality to compound for such labor, for any term not exceeding five years, at any sum not exceeding \$1 for each day's labor ;

(2) For providing that a sum of money, not exceeding \$1 for each day's labor, may or shall be paid in commutation of such statute labor ;

(3) For increasing or reducing the number of days labor, to which the persons rated on the assessment roll shall be liable, in proportion to the statute labor to which such persons are liable, in respect of the amounts at which they are assessed, or otherwise, respectively ;

(4) For enforcing the performance of statute labor, or payment of a commutation in money in lieu thereof, when not otherwise provided by law ;

(5) For regulating the manner and the divisions in which statute labor or commutation money shall be performed or expended ;

(6) For reducing the amount of statute labor to be performed by the ratepayers or others within the municipality, or for entirely abolishing such statute labor ;

(7) For reducing or varying the amount of statute labor to be performed by the ratepayers or others within certain defined areas in the municipality when, in the opinion of the council, exceptional circumstances exist, rendering such reduction or variation equitable, and upon such conditions as may be imposed by the by-law ;

(8) For providing for the making and keeping open of township roads during the season of sleighing in each year, and for appointing overseers of highways, or pathmasters to perform that duty, and such overseers and pathmasters shall have full power to call out persons liable to perform statute labor within their respective municipalities to assist in keeping open such roads, and may give to such persons as may be employed in so doing certificates of having performed statute labor to the amount of the day's work done, and such work shall be allowed for to such persons in their next season's statute labor.

(9) For providing for the application of so much of the commutation of the statute labor fund, as may be necessary for keeping open such roads as last aforesaid, within such respective municipalities.

(10) For providing that the pathmasters in their several road divisions, or other officers appointed for the purpose, shall enforce the provisions of by-laws respecting obstructions placed on the roads.

(11) For appointing overseers of highways, road surveyors, road commissioners, or other officers required by the corporation ; and for regulating the remuneration, and duties of such officers.

The foregoing provisions will be found in the Consolidated Assessment Act of 1892, sections 87 to 101 ; the Consolidated Municipal Act of 1892, sections 479 and 521 ; the Municipal Amendment Acts of 1894 and 1895.

BOULEVARDS.

The Municipal Amendment Act 1894 sec. 9, enacts that the councils of cities, towns and incorporated villages may pass by-laws :—

“For setting apart, upon such terms as to resumption by the municipality and otherwise as the by-law may state, portions of the public streets at or near the sides thereof for the purpose of boulevarding, and regulating the construction and maintenance of such boulevards, or for permitting the owners of property abutting on any street to construct at their own expense such boulevards, but so as not to unreasonably confine, impede or incommode public traffic, and for regulating and confirming boulevards already constructed on any street.

“For making regulations for the protection of all boulevards constructed in the public streets.”

PROCURING MATERIAL FOR ROADS.

The council of any municipality may, in accordance with 55 Vic. chap. 42 sec. 550 pass a by-law as follows :—

“For searching for and taking such timber, gravel, stone or other material or materials (within the municipality) as may be necessary for keeping in repair any road or highway within the municipality ; and, for the purpose aforesaid, with the consent of the council of an adjoining municipality (by resolution expressed), for searching for and taking gravel within the limits of such adjoining municipality, and the right of entry upon such lands, as well as the price or damage to be paid to any person for such timber or materials, shall, if not agreed upon by the parties concerned, be settled by arbitration under the provisions of this Act ;

But no such gravel shall be taken or removed from the premises of any person in an adjoining municipality until the price or damage has been agreed upon between the parties, or settled by arbitration.”

OPENING AND ALTERING ROADS..

County councils have power to pass by-laws under Sec. 566 S.S. 2 of the Consolidated Municipal Act as follows :—

For opening, making, preserving, improving, repairing widening, altering, diverting and stopping up roads, streets, squares, alleys, lanes, bridges or other public communications, running or being within one or more townships, or between two or more townships of the county ; or any bridge required to be built or made across any river over 100 feet in width within any incorporated village in the county connecting any public highway leading through the county, and which is in continuation of a country road, or between the county and any adjoining county or city or separated town, on the bounds of any town or incorporated village, within the boundries of the county, as the interests of the inhabitants of the county, in the opinion of the council, require to be so opened, made, preserved and improved ; and for entering upon, breaking upon, taking or using any land in any way necessary or convenient for the said purpose, subject to the restrictions herein contained.

AN ACT TO PREVENT THE SPREAD OF NOXIOUS WEEDS.

Chapter 202, Revised Statutes of Ontario, 1887 (Section 9).—It shall be the duty of the Overseer of Highways in any Municipality to see that the provisions of this Act relating to Noxious Weeds, are carried out within their respective Highway Divisions, by cutting down or destroying or causing to be cut down or destroyed at the proper times,

to prevent the ripening of their seed, all Noxious Weeds* as part of the ordinary statute labor, or to be paid for at a reasonable rate by the treasurer of the municipality, as the council of the municipality may direct.

Sec. 10, (Sub-Sec. 4)—Every overseer of highways who refuses to discharge the duties imposed upon him by this Act, shall, upon conviction, be liable to a fine of not less than \$10 or more than \$20.

ROADS AS LOCAL IMPROVEMENTS IN TOWNSHIPS.

Consolidated Municipal Act of 1892, Section 611a.—In case all the owners of the property or lots abutting according to the original survey by the Crown, on the road, street or public way hereinafter mentioned to be benefitted thereby, in any part of any township, petition the council for the macadamizing, gravelling, planking or otherwise improving by approved material, and draining any road, street or public way (describing it), or building a bridge in connection therewith, the council may procure an engineer or provincial land surveyor to make an examination of the said road, street or public way so proposed to be improved, and may procure plans and estimates to be made of the said work by such engineer or surveyor of the real property, municipalities and corporations to be benefited by such work, or the owners or occupants of which real property may or can use the same, stating as nearly as may be, in the opinion of such engineer or surveyor, the proportion of benefit to be derived therefrom by every road and lot or portion of lot, and of any railway or street railway, or municipality or corporation; and if the council is of the opinion that the proposed work or a portion thereof would be desirable, the council may pass by-laws:

1. For providing for the proposed work or a portion thereof being done, as the case may be.
2. For borrowing on the credit of the municipality the funds necessary for the work, although the same extends beyond the limits of the municipality (subject in that case to be reimbursed as hereinafter mentioned), and for issuing the debentures of the municipality to the requisite amount, including the costs of reference, if any, in sums of not less than \$100 each, and payable within twenty years or less from date, with interest at the rate of not less than four per cent. per annum.

611b.—The several provisions of this Act, from section 569 to section 611a, both inclusive, and the amendments thereto not inconsistent with the last preceding section, and in aid thereof, shall, *mutatis mutandis*, be applicable, as far as possible, to the making and improvement of the said road, street or public way, and the drainage and other work connected therewith, in manner hereinbefore provided, as if the said several sections related to roads and the improvement thereof, so as to make the said clauses efficient for the construction of roads in substantially the same way as drains are now constructed.

COUNTY ROADS AND BRIDGES.

Consolidated Municipal Act of 1892, section 532.—The county council shall have exclusive jurisdiction over all roads and bridges lying within any township, town or village in the county, and which the council by by-law assumes with the assent of such township, town or village municipality as a county road or bridge, until the by-law has been repealed by the council, and over all bridges across streams or ponds or lakes separating two townships in the county, and over all bridges crossing streams or rivers over 100 feet in width, within the limits of any incorporated village in the county, and connecting any main highway leading through the county, and over all bridges over rivers or ponds or lakes forming or crossing boundary lines between two municipalities.

* Noxious Weeds include Canada Thistles, Oxeye Daisy, Wild Oats, Ragweed and Burdock.

534.—When a county council assumes by by-law, any road or bridge within a township as a county road or bridge, the council shall, with as little delay as reasonably may be, and at the expense of the county, cause the road to be planked, gravelled or macadamized, or the bridge to be built in a good and substantial manner; and further, the county council shall cause to be built and maintained in like manner, all bridges on any river or stream over 100 feet in width, within the limits of any incorporated village in the county, necessary to connect any main public highway leading through the county.

SIDEWALK CONSTRUCTION IN TOWNS.

57 V. chap. 50, sec. 16.—It shall be lawful for the council of the corporation of any city or town which has not adopted the local improvement system in respect of sidewalks, by by-law to provide from the general funds of the municipality, or to raise by way of loan on the credit of the debentures of the municipality, forty per cent. of that part of the construction of granolithic, stone, asphalt, or brick sidewalks upon leading or principal business streets of such city or town as local improvements falling on the property benefited, in addition to the part of such cost to be provided by the municipality, and to add said forty per cent. to the said part of the cost of the construction of the said sidewalks to be provided by the municipality, and to issue debentures for the amount thus obtained, and it shall not be necessary to submit the by-laws of the said corporation authorizing the issuing of the said debentures to, or to have the assent of, the electors of the municipality before the final passing thereof, but any by-law for carrying the provisions of this section into effect, both as to selecting streets and providing the necessary funds, shall require an affirmative vote of three-fourths of the members of the council.

(1)—The remainder of that part of the cost of constructing the said sidewalks, falling on the property benefited, after the said forty per cent. has been deducted, shall be assessed for and dealt with in the manner provided as to assessments for local improvements in other cases.



THE RESULT OF RIGHT PRINCIPLES.

Increased property values, and, in ultimate cost, a measure of economy and decreased taxation.



MONEY SCATTERED AND MISAPPLIED.

Few back townships have worse roads than the average town.



A ROCK-CRUSHER NEEDED.
Stone for fences but none for roads.



THE COMMON TYPE OF HILL.
Constructed without reference to drainage.



A NICELY CROWNED COUNTRY ROAD.



AFTER A CENTURY OF ROAD-MAKING.

The effect of narrow tires on a flat and badly drained road.



SAMPLE STREET IN ORANGEVILLE.



A STREET WHICH WAS NOT ROLLED



STREET IMPROVED IN THE OLD WAY.



A STREET WELL ROLLED, WELL DRAINED.



STEAM ROLLER.
The property of the Town of Galt.



STONE CRUSHER WITH ROTARY SCREEN ATTACHED.

REPORT

OF THE

PROVINCIAL INSTRUCTOR IN ROAD-MAKING

ONTARIO

1897.


(PUBLISHED BY THE ONTARIO DEPARTMENT OF AGRICULTURE.)

PRINTED BY ORDER OF
THE LEGISLATIVE ASSEMBLY OF ONTARIO.



TORONTO:
WARWICK BRO'S & RUTTER, PRINTERS, ETC., 68 AND 70 FRONT STREET WEST.
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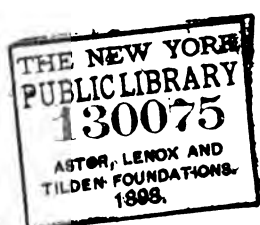


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REPORT
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PROVINCIAL INSTRUCTOR IN ROAD-MAKING,

1897.

To the Honorable JOHN DRYDEN,
Minister of Agriculture.

SIR,—I herewith beg to submit to you my report for the year 1897, on Road and Street Improvement in Ontario. It is gratifying to record the increasing interest which is being taken in this branch of public work, and to observe the efforts in many districts and in many ways, to bring about reform. A large number of municipalities were visited by me during the year, addresses delivered and reports made (as outlined in my report of 1896), the requests for my services in this respect being very numerous, so numerous, I regret, that it was impossible to comply with all. The fullest information appears to be everywhere desired, and any means suggested for bettering the condition of the common highways, is given the most earnest consideration.

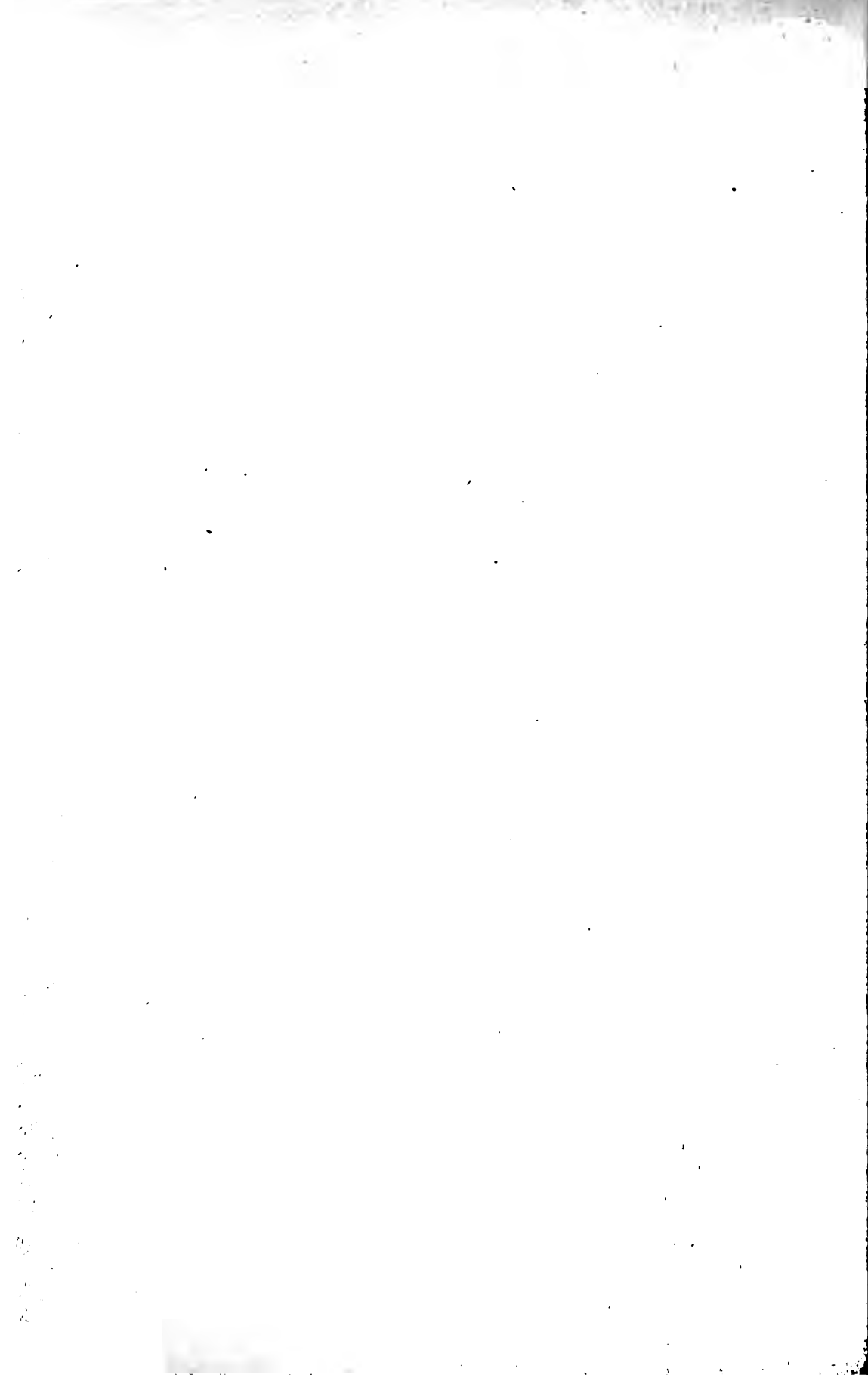
I have the honor to be,

Sir,

Your obedient servant,

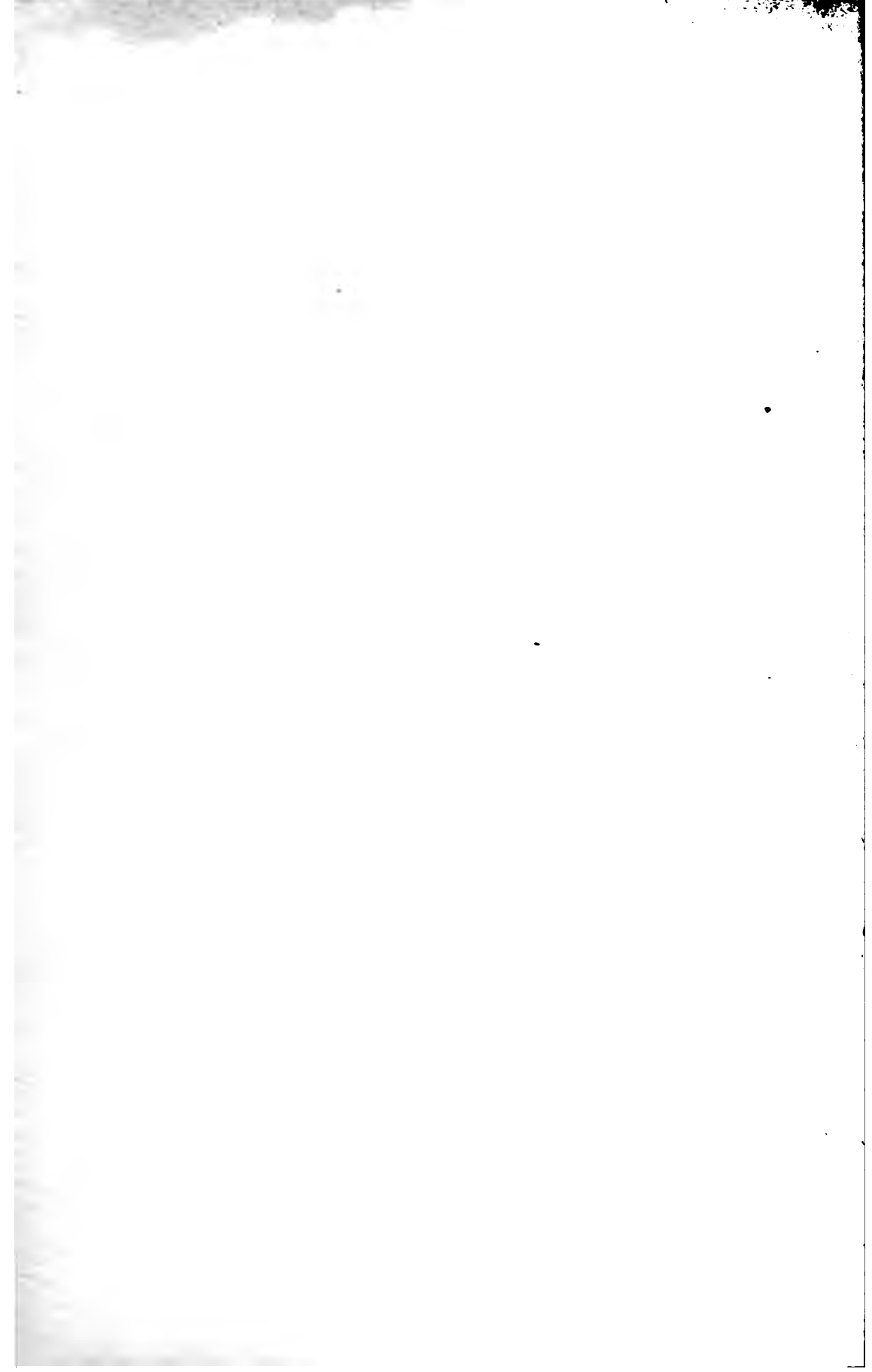
A. W. CAMPBELL,
Provincial Instructor in Road-making.

PARLIAMENT BUILDINGS,
TORONTO, Ontario, 10th January, 1898.



CONTENTS.

	PAGE
LETTER OF TRANSMISSION	3
GOOD ROADS—A NATIONAL PROBLEM	7
COUNTRY ROADS :	
Statute Labor—Waste of Money—Utilizing Statute Labor—Duties of Supervisor—Path- masters—Gravel Roads—Purchase of Gravel—Broken Stone—Placing the Road Metal— Crowning the Road—Consolidating the Material—A Smooth Surface—Keep the Road Dry— The Open Drain—Foundation—Underdraining—The Effect of Frost—Maintenance—Road Graders—A Road Roller—A Stone Crusher—Wagon Tires—Dimensions of Roads—Hills— Location of Roads—Culverts—Bridges—In Brief	12
THE GOOD ROADS MOVEMENT :	
Canada, Massachusetts, Michigan, New York, New Jersey, Connecticut, Rhode Island, Wisconsin, Quebec	28
MUNICIPAL REPORTS	31
REPORTS FROM CITIES, TOWNS AND VILLAGES	32
TOWNSHIP REPORTS	41
COUNTY REPORTS	53
BRANTFORD BOARD OF TRADE	55
REPORT ON THE STREETS OF THE CITY OF GUELPH, ONTARIO :	
The Expenditure—Oversight—Classifying the Streets—Choice of Pavement—Earthwork— Drainage of Roadways—Foundation—A Road Roller—A Crusher—Sewers—Some Illus- trations—In Conclusion	56
EXTRACTS FROM A REPORT TO THE CITY OF ST. JOHN, N.B.:	
Stone Block Pavements—Asphalt—Vitrified Brick—Wood Pavements—Cost of Pavements...	62
THE FRONTAGE TAX SYSTEM—FORM OF BY-LAW	66



REPORT

OF THE

Provincial Instructor in Road-Making.

GOOD ROADS—A NATIONAL PROBLEM.

The demand for increased transportation facilities, for more rapid transit, for better and shorter routes is arising from all quarters, is far reaching and urgent. In response to this, railways are improving the character of their roadbeds and rolling stock. Canals are being deepened and widened, and harbors improved. Vessels of greater capacity and speed are being constructed. Cold storage is being provided. In the cities electric street railway systems have displaced the slower horse cars, and are stretching their tendrils along the country highways; the bicycle has obtained a prominent place in the transaction of business; motor carriages may be seen on the streets of the larger cities, and present indications are that they will soon become an important means of travel and transportation. Nations are growing greater and stronger, and trade is forcing its way more rapidly and with greater volume along the veins and arteries of commerce.

Addressing the Board of Trade of Toronto, the chief commercial city of Ontario, the Premier of the Dominion recently said: "Let us cheapen the cost of transportation, and from that moment we have in the markets of Great Britain a preference for the products of Canada." Referring more directly to the importance of the common highway and its relation to trade, the Premier of New Brunswick, the Hon. H. R. Emerson, in an address at a banquet recently tendered him in the city of St. John said: "I have already given you assurance on behalf of my Department that the good roads movement will be furthered and encouraged as far as in our power lies. The betterment of our public highways which are avenues of transportation and form an essential part of the great system which is necessary to transfer the products of our farms from the farm granaries to the busy markets of the old land, is a most desirable end to be attained, from the standpoint of economics, as well as from social or other standpoints."

The question of road improvement has been too much regarded as a matter of individual loss or gain. It is necessary to convince farmer Jones that he is losing yearly by bad roads \$50, Brown that he is losing \$75, Smith that he is losing \$100. This is the phase which the good roads advocate most frequently has to meet. Seldom is the problem regarded in the broad light of its influence upon national prosperity. The total production of Ontario farms has a value annually, it is estimated, of \$200,000,000. All this must first pass over the common highways before reaching the markets. It is the basis of Ontario's wealth. The amount is far in excess of that needed for home consumption and the only resource is to obtain a market in foreign countries. This market is available only so far as we can sell more cheaply, and produce a better quality than other

competing countries. It is not the effect on present conditions which constitutes the evil of bad roads; but it is in the conditions which they prevent that the loss lies. It is not on the principle that it is so expensive to draw one load of produce to market, but in the principle that, with every means provided for easy, quick and good transportation, a market would be created for two loads. Many links in the system of transportation are being perfected, but the chain will not be complete without cheap transportation over the first part of the journey, the common country highways.

GOOD ROADS AND AGRICULTURE.

Good roads will be of vast benefit to agriculture. They increase the profits of the farm by decreasing the cost and difficulty of transportation. It is cheaper to draw produce to market in one load than in two; on the same principle that it is cheaper that one team and a driver to do the work of two teams and two drivers. Good roads will increase the opportunities of farming and extend the range of marketable produce. Fruit and other perishable articles can be marketed in better condition, and a purchaser will be obtained for produce which now cannot be taken to market because of the difficulty of transporting it from the farm to the city or railway station.

A market in England is awaiting all kinds of Canadian fruit as soon as the difficulty of carriage is overcome and the fruit offered for sale in sound condition. Rapid transit on steamship and railway, with cold storage, is being provided, but until the fruit can be taken from the farm to the railway in the best possible condition; that is, with the greatest speed and least amount of jolting, we must labor at a serious disadvantage. With cheese, the constant complaint is that the quality of the Canadian product is not what it should be. Cheesemakers are all aware that the quality of the cheese is injured by churning the milk over rough roads; and that a more uniform quality can be obtained by making the cheese in large factories, a condition which can only be brought about by making the roads such that the milk can be carried greater distances.

The dulness and isolation of farm life will be overcome by good roads, and they will thereby tend to elevate and bring about a better citizenship. The school, the church, the public meeting, the neighbor's house will be more easily reached. The effect of all this, the increased land value, the greater profits and pleasures of the farm, is to render the country attractive rather than repellant. The effect is to draw the people of the city to the country rather than drive the people of the country to the city. "How can we relieve the congested condition of our cities?" "How can we keep our boy on the farm?" are questions demanding an early solution. By making the farm profitable, by giving to energy and ambition sufficient business opportunities on the farm, these problems will be solved, and one of the most important factors in the solution is "Good Roads."

GOOD ROADS AND COMMERCE.

Good roads are of great benefit commercially. Agriculture is the foundation of the wealth of this country. As agriculture prospers so will business prosper. Better home markets and better foreign markets for the farmer will bring more money into the country. This means increased sales for the merchant and a greater demand on the manufacturer. Bad roads, beside the retarding influence on the country by lessening its ability to compete in foreign markets, cause produce to be marketed, not as prices or the demand require, but as the roads permit. The result is that the market is alternately glutted or starved; there is either unnatural excitement or depression. The stores are either deserted or extra clerks have to be employed. Railways either cannot provide cars to move the contents of the granaries, or else their cars and their crews are in idleness. Such fluctuations cannot but be hurtful.

When farmers can market their produce easily and regularly they have money with which to enable the merchant to turn over his stock and meet his obligations at the bank. The banks are thereby enabled to do business to greater advantage, and the whole

machinery of commerce moves with less friction, less useless expenditure of energy. The stringency which so often takes place in the fall as a result of bad roads; the bad roads of Christmas time which so often bring disappointment to the merchant; the resulting failures which act and react on all classes of society are all of too frequent occurrence.

ONTARIO ROADS.

It is doubtful if there is a mile of true macadam road in Ontario, outside of a few towns and cities. There are miles of road which are covered with dirty gravel or rough, broken stone, and are popularly supposed to be macadamized, but they more resemble the roads Macadam found in England three-quarters of a century ago, the kind he removed, rather than those he constructed.

A century ago the roads of Ontario, such as existed, and they were very few, were mere trails. To-day, notwithstanding the amount of money and labor placed on them, the majority are little better than trails. This criticism may seem overdrawn to those who drive over some of the best gravel roads during the summer season; but if the journey had been made in the fall or spring, the rainy seasons, its fairness would become more apparent.

While the majority of highways in Canada are strikingly like the trails from which they have sprung, nevertheless we have highways which are of surpassing magnificence. The Canadian Pacific Railway spanning the continent is, in extent, daring and durability of construction, unequalled in the world. The populous districts of Canada are covered with networks of steel; while other lines are being projected wherever opportunity presents itself. But while capital has exhibited so much enterprise in providing the great lines of commerce, the lesser routes have been overlooked and neglected. Common highways, while less in extent and cost, are not one whit less in importance. Over them must first pass all the freight which feeds the greater railways, and it is this first cost of transportation which so enormously increases the cost of production, and lessens our ability to compete in the markets of the world. The improvement of country highways is an investment, not a tax, and will be a most profitable one, commercially and socially.

From the middle of October until the end of December, and from the first of March to the middle of May, a period of five months, by far the greatest part of the mileage of the Province is mud, ruts and pitch-holes. This may vary somewhat at the more northerly and more southerly parts of the Province, but it is the general rule of the average year. Of this period of five months there are at least two months of the year when the roads are practically impassable for loads. From the middle of November until the middle of December; from the middle of March until the middle of April, the agricultural trade of the country is practically cut off. For the remaining three months of the five the roads are barely passable. To this may be added January and February, for part of the Province, those southerly counties of Western Ontario where sleighing exists for two or three weeks at the most. If we include a winter month (and during the winter there are many days when travel is shut off), there are six months of the year when travel and teaming in Ontario is retarded by bad roads. Part of that time, at least two months, as pointed out, country traffic is practically impossible. If during those two months fortune favors us with a few days when teaming can be done, there are plenty of days during the remainder of the year to make two months of impassable roads over the Province generally.

AN ESTIMATE.

It is difficult to fully estimate what this means to each farmer individually, as it affects the value of his farm, the marketing of his produce, his business affairs, social intercourse of himself and family with his neighbors, attendance at church, at public meetings, the attendance of the children at school, and numerous other similar matters.

Horses are standing idle in the barn eating hay and oats when they would otherwise be at work earning their food as well as eating it. The idleness of horses means in the great majority of cases idleness of the teamster.

In order to forestall the bad roads the farmer goes about his business, does his teaming and marketing while roads are good in the early fall. His plowing is left until the roads are bad, but when the roads are bad the ground is soaked and unfit for cultivation. It is on clay land where the roads are generally the worst; in such districts it is generally found that marketing and cultivation come in the order described, and it is clay soil which is most injured by being worked when saturated with moisture. It cannot be argued that these matters are of no consequence. Many individual farmers may be willing to put up with all these drawbacks. But when aggregated, the loss is one which the country as a whole cannot afford.

The statistics compiled in the Bureau of Industries show that in 1896 there were in Ontario 434,384 working horses owned by the farmers of Ontario. This number does not include the unbroken horses nor breeding mares, but represents the number actually used for work. Let us assume, and the assumption is a very moderate one indeed, that one-half of these horses, which would otherwise work, are prevented from doing so for two months or sixty days of the year, and that their teamsters are also left in idleness. This means that for sixty days 108,596 teams and 108,596 teamsters are idle. The price paid generally is \$3 per day for a team with a driver, and is a proper estimate of the value to the country. The direct financial loss daily, therefore, is at least \$325,788, and for sixty days, \$19,547,280—no inconsiderable tax for one Province with a population of about two million. At three per cent. this is the interest on a capital of \$651,576,000.

There is nothing in this of the annual loss through the interference with trade. There is nothing of the loss of time, there is nothing of the smaller loads which have to be drawn at all seasons of the year because of merely rough roads. There is nothing of the increase in land values that would result through good roads. There are many thousand of acres in Ontario which are remote from the market towns and with only common dirt roads over which the market can be reached. These isolated farms are of very little value, worth not one-half nor one-quarter what they will be when first-class roads are provided. Instances are not lacking to show that property in the immediate vicinity of even a city market has increased fifty per cent. by the construction of good roads. When we consider that very few roads in even the oldest and best settled districts are what they should be, and that many are decidedly what they should not be, it is apparent that the increase in land values throughout the Province by a complete system of roads would be very great.

THE OBJECTS OF REFORM

The means whereby the roads of Ontario have been brought to their present state have been wasteful in the extreme. There has undoubtedly been earnest effort and honest motive on the part of many. But in the absence of systematic management little could be expected. Statute labor has been assessed, the nominal value of which is one dollar per day. It is safe to say that the average day of statute labor as actually performed is not worth fifty cents. Even if honest work had been given, it has been applied in a manner so disconnected and with so little knowledge of the true principles of road-making, that economical results could not be anticipated. To bring order out of chaos is the work of the road reformer. If we are to have good roads in the country, it is only by building them in a business-like way, on business principles.

The general system of roads required by the Province is not of an expensive kind. Gravel roads, or in the absence of gravel, broken stone roads, built cheaply but well, are needed. They should be built in such a way as to suit the traffic over them. Roads which are either insufficiently strong, or which are stronger than necessary, are each of them wasteful. There is no immediate likelihood of roads of the latter kind being built. The great defect which now characterizes them is that they are not strong enough, are of

the kind which "break up." Roads should be constructed and kept in such a state of repair that they will not "break up." Roads which are rutted, cut up and converted into a canal of mud for any part of the year, in that time are injured to such an extent that the labor placed on them is largely thrown away. The "breaking up" process forces the stone down into the mud, mixes the mud with the stone, and takes away the durability of the road. The gravel and broken stone should be placed on the roadway and so maintained that at all times it will form a smooth, hard crust. To provide roads for the light and moderate traffic which passes over the generality of our country highways, and to do so in the most economical manner, there is need of the most skilful application of the principles of road-making as to choice and preparation of materials; drainage; the use of proper implements; re-locating the road in some instances so as to best avoid steep grades, cuts, fills and bad ground; grading, crowning, etc. All this, and much more, is now left unreservedly to the pathmaster. Until good business ability, system, and a knowledge of road-making are combined there cannot be good roads in this Province.

How to place before the people of this Province roads which will educate, roads which will by their merit compel their extension widely and within a reasonable period, is a problem deserving of an early solution. Only those who have been active in promoting the cause of good roads can appreciate the difficulty. While there is a demand for good roads by the general public, there is still a prejudice existing in favor of old systems and old methods which it is difficult to counteract by ordinary measures. The good roads movement in Ontario has reached the stage where a united effort of the foremost citizens should be made. What is needed is that there should be measures taken to improve in a large way the main roads of the Province. Whether this should be done by each township municipality, or whether a county, or even a provincial road system should be adopted to attain this end, are questions which are to be considered.

Township systems are in force throughout the greater portion of the Province, and the result we have with us. Hastings is the only county which has a well developed county system, and the means employed there have a decidedly beneficial effect on the roads. Wherever roads, however, have been brought to the highest excellence, as in France, England or Germany, and wherever on this continent the good roads movement has reached its most successful stage, as in Vermont, New Jersey, Connecticut or Massachusetts, state aid has been given to greater or less extent.

One mile of good road will secure more converts to the cause of improved roads than will ten miles of bad roads. The people of this country have for years been driving over bad roads. They know what bad roads are, but do not know what good roads are. A few years ago town streets which could boast of coal oil lamps possessed a luxury, and we were contented for we knew nothing of gas. When gas replaced the kerosene it was pronounced a marvel. To-day so far are we from returning to the dim light of coal oil that electricity is not brilliant enough. When the people see them, experience them, feel and know the benefits of good roads, there will be no possibility of a return to the mud and ruts that now prevail.

The object of the good roads movement must be attained by a process of education. Good roads are their own best advocates. That is wherein the strength of the road reform movement lies. One step in advance is never lost, but lends impetus for the next. A single example of a good road will argue more for the cause of road improvement than will a volume of words. A good road must be seen to be fully appreciated. The benefits cannot be stated in dollars and cents, but have to be experienced in order to be realized.

COUNTRY ROADS.

Do we actually want good roads? Or are bad roads preferable? Is the cry that has been raised throughout the length and breadth of Canada and of this continent: "We want good roads," the demand of men in their sober senses? Or has labor and money been placed on our roads for a century past merely to fill in time, and keep our surplus capital in circulation. If we do not want good roads, if bad roads are preferable, why should we want roads at all?

We must have roads. That necessity having been placed upon us, the experience which has taught us the wisdom of building other structures substantially, teaches us the economy of having roads that are good. We want roads which will withstand wear. We want the labor and money spent on them to be a paying investment. We want roads which will be good no matter what the state of the weather. We want roads which will not become rutted immediately the fall rains come on or when the frost leaves the ground in the spring, remaining in rough ridges for a considerable part of the summer. A road which does this is a bad road. The money and labor spent on it is largely forced down into the mud, is plowed under within a year and wasted. A good road is an economical road.

In building an economical road, improvements must be made in such a way that they will last. Roads in Ontario have been built on the same principle as is a wagon which breaks down under the first load, and is used for firewood after a year of service. Most of the leading roads of Ontario have been made and remade a score of times and are still bad roads. They are of the kind that "break up." A road that "breaks up," like anything else that breaks up, is a poor investment. When road building is rightly understood in this country, township councillors will no more think of building roads that will break up in the spring than they will think of constructing houses that break up in the spring, barns that break up in the spring, or fences that break up in the spring.

The road builders of this country have not given sufficient consideration to the effect of building bad roads. Year after year work of a flimsy, shiftless character is placed on the roads. The results are only temporary and are destroyed by a very little wear and traffic. In a very short time the work has to be done over again. But the evil does not end with this. This annual demand for repairs is so great that no township can respond to it. The roads instead of being repaired when they need it are neglected, grow worse and worse, and all the evils of bad roads follow.

What bad roads are doing for this country is only one side of the evil. The other side is what they are not doing. The loss does not arise so much from the money and labor wasted every year as it does from the absence of the benefits which good roads would bring. Our loss must be measured not so much by the money and labor we are throwing away on bad roads, as by the opportunities which would come to us if the roads were good.

One of the greatest obstacles in the way of road improvement is the narrow view taken of the question by so many citizens of this country. They have been accustomed to think of roads merely as incidental to statute labor; and statute labor they consider as a means whereby each pathmaster can get a little work done in front of his own farm which will be of direct personal advantage.

They do not see nor appreciate the benefits which would accrue to the township, county and province. They overlook the public good. But public good is merely individual benefit conferred on every citizen. Money and labor spent on the roads of the township will enhance the value of every farm by increasing the demand for farm land; it will increase the profits of the farm by reducing the expenses of the farm. The dairy-industry would be immensely benefited by good roads, fruit growing would become more remunerative; sale would be obtained for products which now are not grown because the market cannot be reached easily and quickly.

STATUTE LABOR.

The great majority of roads in Canada are under the control of township councils, and are built by statute labor supplemented by money grants. The statute labor system is suited to a pioneer age. It suits the abilities of the people making a home in a new country, and it suits the spirit of their circumstances. They feel keenly the need of improving their roads, and work with a will earnestly and faithfully. That was the history of statute labor in the pioneer days of what are now the populous districts of Canada. But where the townships have grown wealthy and well populated, a different condition exists. To such an extent has statute labor degenerated that some townships find that they can do more work by commuting all the labor at thirty-five cents per day.

If statute labor has not outgrown its usefulness, there is certainly need for reform when a man's labor is worth less than thirty-five cents a day. A feeling in favor of statute labor still holds in some localities but is growing weaker. There is not a country in the world, characterized by good roads, where a system of statute labor is in use. To a slight, but very slight extent, it may be said to be used in France, but with very different methods of applying it. If it is to be retained in Ontario, the present feeling of the people strongly indicates that it will have to be placed on a basis whereby satisfactory results can be obtained.

There is a prevalent opinion that a resident on a concession road knows better than anyone else the requirements of that particular road. A remarkable feature, however, is that no two citizens agree in their ideas of what work should be done, or how it should be done. No doubt there are cases in which one man is right, but when one man is right all the rest must be wrong. Pathmasters are appointed in turn from among these to plan and superintend the roadwork. The only result which could reasonably follow from such a system is that which we find—waste and bad roads.

The result of this lack of supervision is mismanagement in actual road construction. A description of the actual work done sounds like ridicule, however earnest it may be. A statute labor day is short, of eight hours, but the men are rarely on the ground for more than six hours. A good part of the six hours is spent in gossip, in arguing as to what should be done, and in finding fault with what is being done. Another part is spent in looking up scrapers and plows that should have been provided before the work was commenced. When we consider the class of teams sent out by the rate-payers, the wagons which can carry only half a load, the boys who come to do a man's work, these and many other details, the friends of statute labor must necessarily frame numerous apologies.

Statute labor is performed at a time of year when every farmer can use his time to good advantage on his own land, but instead of remaining where he can do the most profitable work he uses his time in paying a road tax. A large part of the time spent in road work is unquestionably wasted. In addition a considerable part of our annual tax is also expended on the roads. A careful consideration of this expenditure will show that many of the leading roads, far from being good, cost much more than first-class roads should cost. Nor is this all; the most regrettable feature is that through improper plans and imperfect construction the most expensive form of maintenance has been imposed.

WASTE OF MONEY.

Councils commonly appropriate an amount of money each year to be spent on the roads. If this money were spent in making complete and durable work it would be of the greatest assistance in improving the roads. We find, however, that the money is distributed in small sums among the pathmasters, is spent in small sums for repairing temporary culverts: in doing a little draining in front of the farm of some discontented ratepayer to appease him; in doing no particular work at all, but merely to give some voter an opportunity to earn a few dollars. In such ways as these is the money which should be spent on durable improvements, scattered, wasted and misapplied.

IMPROVING PRESENT CONDITION.

If a supervisor were appointed, if pathmasters, (and fewer of them) were appointed for a term of years to carry out the directions of the supervisor, and if to these positions the right men were appointed, a considerable step would be taken towards the better management of roads in the townships. Under systematic management, money and labor could be made to work together to the best possible advantage. At present each pathmaster is given a few dollars and a few days of labor. He is surrounded by neighbors. Nothing but patchwork can be expected. By consolidating this money and labor; by using the statute labor in the ways in which it can be turned to the best advantage; by using the money in purchasing material and doing work to which statute labor is not adapted; by seeing that all this work is properly done, and at the right time, much would be done to improve present conditions.

UTILIZING STATUTE LABOR.

One of the most profitable methods of using statute labor is to utilize it in hauling gravel. In this work there is less opportunity for wasting time; it is work which is more agreeable than much of the other work of grading and ditching; it is work which the average farmer understands to be of direct benefit and is entered into heartily.

It follows that it is better for other means to be used in performing other work. For the grading of the roads there is machinery which every township should possess. For the operation of these machines one man should be hired, as skill and experience are absolutely necessary. It cannot be passed around from farmer to farmer like a scraper or a plow. These machines do not know how a road should be shaped, but in the hands of a man who does, they are exceedingly valuable. For draining and ditching it is usually best to hire the work done by men who are accustomed to this class of work. One of the most necessary steps to be taken so as to receive the greatest benefit from statute labor is to see that the material, whether gravel or broken stone, is prepared in the pit or quarry, ready to be drawn to the road. If screening, crushing or stripping a pit is necessary, it should be done before statute labor commences. Gravel may be plentiful, but of a very inferior quality. Usually no effort is made by the council to screen the gravel, remove the surface soil from the pit or in any way prepare the gravel for being placed on the road. Teamsters go to the pit, one or two at a time; it is of inferior quality, but they cannot undertake the task of removing the earth and clay, in order to satisfy their few days of labor. In other sections there is no gravel, but field and quarry stone is to be found in abundance, frequently along the road to be improved. But with this material unprepared for road work, nothing can be done by the farmers except to plow and scrape the mud, and in other ways put in their time in useless effort. If the council would purchase or otherwise employ a rock crusher to prepare this stone and leave it ready for hauling, if they were to purchase screens, strip pits, etc., and see that nothing but first-class gravel was provided, farmers would be encouraged and would willingly spend their time to the fullest extent in hauling this material.

A SUPERVISOR.

The first and greatest need of every and any system of road construction, is to have one man at the head of the work to act as a general supervisor. He must be a practical man, and if possible, experienced in road construction, and acquainted with the principles underlying it. The advantage of having such an officer, appointed by by-law, with term of office as secure as that of a township clerk, will be best explained by stating his duties:

- 1st. The supervisor should prepare a plan of his district, or township, showing all roads, and the location on them of all culverts, bridges and water courses, classifying roads according to their requirements.

2nd. He should prepare a list of all culverts and bridges, showing dimensions, material used in construction, their condition and the direction of the watercourses passing through them, with memoranda as to the course of the water and location of the outlet.

3rd. The plans and records should show, as to roads, whether they are of gravel, stone or earth, graded or ungraded, the system of drainage, and nature and extent of traffic upon them. He should carefully study the present and future requirements of traffic on all roads, the class of roads best suited to such traffic, the width and depth of metal, width of graded portion, amount of crown and other details of construction.

4th. He should possess full information as to location, extent and quality of material suitable for road construction, and amount of plank and timber obtainable from ratepayers in the township or district, suitable for repairs.

5th. He should report to the council as early as possible in each year, showing the number and location of culverts and small bridges to be re-built or repaired, with a detailed statement of all material required for this work, and an estimate of the cost. It would also be advantageous to have a probable estimate of material required during the following year, presented each fall so that, if thought advisable, it may be purchased and delivered on the ground during winter months or other most convenient season, and in this way utilize as much as possible the labor of ratepayers during the slack season.

6th. His report should specify the condition of all bridges, indicating those which require repairs or reconstruction, together with an estimate of cost, and a statement dealing with such special protection work on streams as he may deem worth the council's consideration. There should also be reference to any needed re-location or deviation of existing roads with a view to doing away with bridges, culverts, expensive grades, cuts or other features which tend to prevent permanent work and economic maintenance.

7th. He should consult with all pathmasters and report to the council, showing the number of days labor in each division, the work to be undertaken, and the amount of money which should be appropriated by the council to properly utilize the statute labor.

8th. He should arrange with divisions desiring to compound statute labor for a term of years, with a view to the construction of permanent and finished work.

9th. He should take stock annually and report to the council on all machinery and implements, showing their condition and where kept.

10th. He should carefully examine all parts of the township where gravel and stone exist, and should by borings and tests determine the quality and extent, and report thereon to the council. All material, stone, plank, gravel, etc., should be purchased by the supervisor in large quantities and under instructions from the council, the required amount to be determined by his estimates referred to above. As far as possible the material should be purchased by tender, and due consideration should be given to any ratepayer having such material for sale. When purchased it should be delivered and stored at convenient points and placed in charge of and used by the pathmaster, subject to the order of the supervisor, and in emergency work.

11th. He should prepare specifications of all work for which the council makes appropriations. Contracts should be awarded to the lowest bidder if proper security is given, but the work should be subject to the approval of the supervisor, and all accounts should be certified by him before payment.

PATHMASTERS.

There are many men in this province who are capable of taking the oversight of road construction, but the system of changing the pathmasters every one or two years is not likely to produce men who are well qualified in this respect. Appointed in the spring, the pathmaster has no time to make a study of the subject such as it demands. Nor is the fact that he will be succeeded by someone else as pathmaster the following year an encouragement to effort in this direction.

Road divisions or "beats" should be from three to five miles in length. A path-master should be a permanent officer, and his division should preferably be such that the most of his travel will lead him over the greatest portion of it. He need not receive a salary but may as a slight recompense be preferred in doing small jobs under the supervisor, where the work is not considered of sufficient importance to be let by contract. He should in addition give special attention to all emergency work, such as washouts, broken culverts and bridges. If the time required to oversee the statute labor in his division is more than would be needed for his own statute labor, he should be paid for such excess under certificate of the supervisor or council, the object being to secure proper supervision of all work performed; the council or supervisor to determine whether the excess time was actually necessary to oversee the work of the division, in accordance with the local by-law of rules and regulations, which should be framed so as to include this matter.

GRAVEL ROADS.

Gravel is very plentiful in many parts of Canada, and where it can be obtained of a good quality within reasonable hauling distance, makes a cheap but good road surface. As previously pointed out, it should be clean, containing little sand and clay, since it is the stone, not the earthy materials, which are needed on the road. Nor should large stones and boulders be mixed with it, as they will work up and roll loosely under the feet of the horses and the wheels of vehicles.

In the preparation of gravel it is frequently advisable to place a stone crusher with screen attachment in the pit. By passing all the gravel through, the sand and clay are removed and the large stones broken by the one operation. If the gravel is fit to be placed on the road without such treatment, in nearly every case it will be necessary to send a man over the road to rake off large stones and break them by hand.

Much carelessness is exercised in taking gravel out of the pit. In the pit we may find, for example, the surface layer of from two to four feet composed of earthy matter, then a layer of four or five feet of good gravel, then a stratum of coarse sand one or two feet thick, and underlying this another stratum of fairly good gravel. The common practice followed by teamsters is to scrape down the face of the pit, causing the soil, clean gravel and sand to mix together at the bottom. This is put in the wagons and taken to the road.

Very few gravel pits provide material fit in its natural state for use on the road. Screening and crushing are often necessary, particularly the former, to remove sand and clay from among the stone. It is the stone which is wanted on the road—not the sand and clay. There is enough sand and clay already on the road without drawing it several miles from the gravel pit. Dirty gravel, while it unites readily and forms a good roadway in dry weather, dissolves, turns slushy and ruts with equal readiness in wet weather, whereas with clean material the stones assume a mechanical clasp the one of the other that will not yield to the same extent in wet weather.

Earth and sand attract moisture and after a few hours rain the road becomes softened. In this slight ruts are formed which hold water, the whole structure becomes saturated, weakening the bond and permitting each successive vehicle to churn the ruts deeper until the gravel coating is cut through.

PURCHASE OF GRAVEL.

A great many townships buy gravel by the load. This is very much like buying water by the pailful instead of digging a well. Gravel should be bought by the pit or by the acre, and should be available at all times for any farmer who wants to increase the value of his land by improving the road past it. Especial care should be taken by councils to see that, prior to the performance of statute labor, the pit is stripped and the gravel otherwise treated if necessary.

This material, purchased at from five to ten cents a load, when mixed with an excessive amount of sand or clay, is a most expensive road material for much travelled highways. When we consider the number of pits in some townships and their immense size, representing that thousands of loads have been taken out, and then consider the short mileage gravelled, we must readily see that something in the quality of material and the mode of construction is radically wrong. In some instances much more than the value of a whole farm has been paid for gravel purchased by the load and taken from a small corner. If first class material were used, under ordinary traffic the annual repairs rendered necessary would be very slight.

In searching for gravel, the clearest indications are usually to be found along the banks of streams, where any extensive strata is apt to be exposed. A post-hole auger affords a convenient means of making tests over the surface of the soil for gravel, but the best implement is generally a simple form of drill. There are cases in which gravel beds may be entered at the level of a stream bed, and water is thereby obtained for washing the metal by natural drainage, affording a cheaper means of freeing it from sand and earthy matter than by screening. Gravel is still being deposited in drifts and bars by the agency of streams; this will be found to partake of the character of the pit gravel of the locality but generally will contain less clay, although sand may easily be in excess. This is usually one of the best sources, as the gravel can be washed by natural drainage. Lake gravel is often a good metal but varies greatly. It is apt to be slaty, an undesirable quality. It will be free from dirt and clay, but contains sufficient sharp sand to secure consolidation, especially if a roller is used. Gravel which retains a perpendicular face in the spring, and shows no trace of slipping when thawing out may generally be assumed to be sufficiently clean and free from clay for use on the road without any treatment other than is necessary to remove stones greater than one inch and a quarter in diameter.

BROKEN STONE.

There are localities in Canada where good gravel is not obtainable, but where stone can be had, either as bed rock or as field boulders. Some townships have used stone broken by hand, but a stone crusher, with a screen attachment affords a much cheaper method.

The stone should be separated into grades according to size, the coarser stone to be placed in bottom of the road, and the finer at the top. This grading of the stone is done by means of the screen attachment. If the stones are placed in the road without being graded in this manner, the smaller stones wear more rapidly than the larger, and a rough surface results. Large stones at the surface, moreover, are more apt to become loose, to roll under the horses' feet or the wheels. For a country road there should be placed in the roadbed (1) a layer of stones such as will pass through a 2½ inch ring; (2) on this a layer of stones such as will pass through a one inch ring; (3) on this a coating of screenings—that is, the dust and chips created in crushing.

Care must be taken in choosing the stone to be used. Some limestones make good metal; but limestone shales of a slaty nature, or limestones which decay rapidly on exposure to the air should be rejected. Sandstones are brittle and do not unite well in the road. Granites, which are found in the northern parts of Ontario, make good road metal. Trap rock is the best obtainable. Gneiss is very frequently a good rock. The latter with other hardstones, are frequently found as boulders scattered over the southern parts of Canada. In using field boulders, care must be taken to reject such stones as are evidently softened by exposure. Rocks which crumble readily under successive blows of a hammer, or which show iron stains when broken, should be discarded. A little experience will quickly teach a judicious roadman to detect boulderstone which is unfit for road purposes.

There must be a sufficient body of broken stone to consolidate into a compact layer. A sprinkling of stones over the surface is useless. It merely impedes travel on what might otherwise be a good dirt road. Six inches of broken stone is the least which should be used in making a durable roadway for any purpose, and it should be the aim of councils to thicken this covering as circumstances will permit.

PLACING THE ROAD METAL.

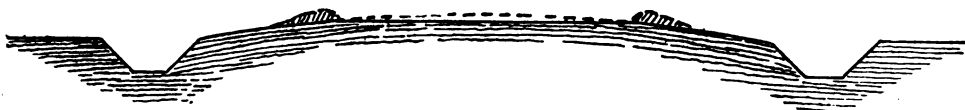
To know how gravel or stone should be placed on the road, it is necessary to have a knowledge of why it is placed on the road. This is a matter to which very few of our road-makers have given the slightest attention, and very few could give an intelligent answer to the question. The popular idea is that the stone makes a sort of carpet for a while, in a short time it will be forced down into the soil to form a bottom, and on this more gravel or stone will have to be placed, and that this process will have to be continued indefinitely until a good road is made. There is even a very general belief that it is not necessary to drain a road, but that the only means of accomplishing the desired end is to pile on gravel year after year, and that water, unless it actually floods over the top of the road, has little to do with the matter, and that so long as the actual surface of the road does not get wet it does not matter how boggy it may be underneath.

In the intelligent construction of a road, the intention of the gravel or stone coating is to form a waterproof covering for the soil underneath as well as to form a hard wearing surface. Of course, gravel and broken stone can not, as a matter of fact, be entirely impervious, but so far as the coating of these materials does prevent the water passing through to the sub-soil, it fulfills the greatest portion of its mission.

To accomplish this to the greatest possible extent there are several points which it is necessary to pay attention to: (1) The road must be crowned or rounded up in the centre. (2) The material must be as compact and solid as possible. (3) The surface of the road must be smooth.

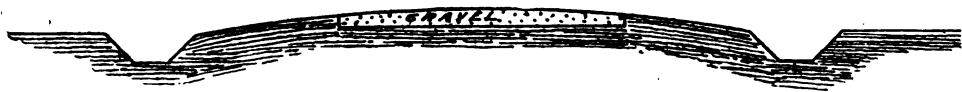
CROWNING THE ROAD.

By having the road crowned or rounded up in the centre, water is at once thrown to the sides where it can be carried away in the drains. If the road is flat on top, or if hollow, as many of the roads of Canada are, water stands on the road, soaks down through the road covering and softens the soil beneath. Then the trouble begins. There is nothing to support the gravel, so that when a loaded vehicle passes over it the wheels are forced down through the gravel and into the soil. The soil is plowed up, mixed with the gravel, and the serviceability of the road is largely destroyed.



"TURNING THE LOOSENED DIRT TO THE SIDE."

The means of providing a proper crown must depend on circumstances. For an average country road on which a grading machine is used, the best method will be to first round up the natural soil giving it a slightly less crown than it is intended the finished road shall have. This completed, pass the grader over one side of the centre cutting off the top and turning the loosened dirt to the side, then pass the grader back along the other side turning the loosened dirt to the side. This will leave a flat surface



THE GRAVEL IN PLACE.

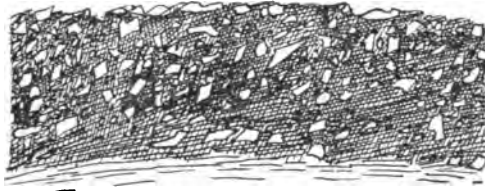
in the centre of the roadway, along each side of which is a shoulder of loose earth, forming a shallow trench. In this the gravel should be placed, spread with a rounded surface and the loose dirt at the sides levelled off to conform to the shape of the roadway.

Old gravel roads are commonly flat, in ridges, with square shoulders at the edge of the ditches. In this case the better plan is to cut off these shoulders throwing the loosened earth outward. The ditches are usually very wide and flat, the road having been graded by drawing the earth out of the ditches with a scraper, so that the shoulders thus turned outward merely widen the graded roadway without interfering with the drain. If, however, these ditches are sharp and deep, the loosened earth may drop down so as to obstruct the water, in which case it will have to be thrown across the drain to the roadside by hand, a proceeding seldom necessary.

Usually a sufficient depth of gravel will be found upon these roads, requiring only that the centre should be raised by cutting off the sides. After this is done as above described, a light coating of clean gravel to fill the ruts and depressions and restore the crown will make an excellent road.

CONSOLIDATING THE MATERIAL.

The road covering should be solid and compact in order to shed the water. Under present methods, the gravel or stone is dumped in the centre of the road and is left as it falls, a mound of loose material, avoided by the users of the road until late in the fall when the muddy and rutted state of the side of the road compels them to drive along

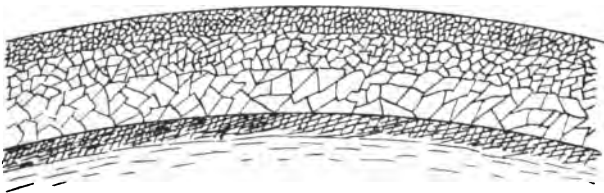


"MIXED WITH THE SOIL BENEATH."

this mound. Gradually it is flattened down and after a year or so, during which time it has been mixed largely with the soil beneath, it assumes the shape of a road. The utility of roads made in this way is largely wasted. Roads must be made for traffic not by it.

This loose stuff absorbs the rain as it falls even before it is cut into ridges by wheels and the feet of horses. When it has been cut into this condition it acts as a receptacle to hold all the moisture its surface will receive. In this way the whole surface and foundation of the road is softened, is readily cut up and destroyed.

The best remedy for this waste in road-making, is to spread the road metal to conform to the required surface of the finished road, and then thoroughly consolidate it by the use of a heavy roller. It can be largely remedied also by taking proper care of the



THE BROKEN STONE IS CONSOLIDATED INTO A DISTINCT COATING BY THE USE OF A ROLLER.

road, if a roller cannot be had. By raking the loose material into the ruts and wheel tracks as fast as they appear, nearly the same end will be accomplished but less perfectly and requiring a longer time. The first vehicle passing over the road does comparatively little injury; it is when ruts have been formed which hold water and other wheels follow in these tracks that the greatest damage is done.

A SMOOTH SURFACE.

It is evident that a smooth surface is essential to a good road. A rough surface is necessarily such as will impede the flow of water from the centre to the drains. To such roads rain is always an injury. With roads properly built, on the contrary, a good dash of rain will flush away the dust which has accumulated, and which, if it remains on the road in time of steady rain and alush, acts as a sponge to absorb moisture and soften the surface of the road.

KEEP THE ROAD DRY.

Keep the roads dry, and with very little attention they will remain good roads. A little moisture does comparatively little injury. Against some moisture it is impossible to guard. But with proper precautions the excess moisture can be removed before great damage is done. The means to be taken with regard to the actual surface of the road have already been commented upon.

THE OPEN DRAIN.

The open drains at the sides of the road should be sufficiently deep to hold water in times of freshets, and should have a sufficient fall to carry it quickly away. The fall should be uniform, not a series of rises and falls, knolls and holes.

Most important of all, in view of the neglect which it receives, the outlet must be ample and always free from obstruction. It is useless to dig a drain without providing an outlet for the water, a thing very commonly done. The water merely stands in the trench, soaks into and softens the roadbed. Unless a drain carries the water away it is useless.

THE FOUNDATION.

The importance of keeping the roadbed dry cannot be too thoroughly impressed. Clay in thick beds, when dry, will support from four to six tons per square foot of surface, according to the quality of the clay. If only moderately dry it will support only from two to four tons per square foot of surface. If the clay is wet and soft it will yield to almost any load. Gravel, if well compacted, forms a much stronger roadbed, is less yielding to the action of moisture, and for this reason, even for a thin surface coating, strengthens the road somewhat. But the real strength of the road must lie in the subsoil. Vegetable moulds and alluvial soils are weak, having a sustaining power of only one-half to one ton per square foot, and for this reason it is well to remove such soils, securing, if possible, a gravel, clay or sand foundation.

UNDERDRAINING.

In order to keep the roadbed dry and secure a strong foundation, it is frequently advisable to use tile underdrains. Owing to the weakening effect of water on clay, also to the retentive nature of clay, that soil is usually most in need of underdrainage. In gravel and sand water is not ordinarily so destructive unless when acted upon by frost; at the same time these latter soils usually provide better natural underdrainage, as they are more porous, and artificial means of underdrainage are usually less necessary.

Under-drainage may be had by means of common field tile. It is usually best to place two such drains, one on each side of the roadway underneath the open drains. One tile drain placed underneath the centre of the roadbed is sometimes used. The extra cost of two drains is largely reduced, however, by the lesser cost of excavation, since, on the majority of road allowances, deep open drains now exist, and the soil is softer and more cheaply handled than that in the hardened centre of the road. It is very unwise to excavate and soften the hardened centre of the road when it can be avoided, as settlement is

very slow. At the same time, two drains are more effective, carrying away water more rapidly and thoroughly. They intercept the soakage water from the adjoining land before it passes under the roadbed.

Any thoughtful farmer who knows the effect of underdraining in his field will at once recognize its usefulness in road-making. In the fall, water will be rapidly removed from the bed of the road and the destructive action of frost lessened. In the spring the frost will come out of the ground more quickly, and each of those periods in fall and spring are shortened, when, with the foundation and surface thoroughly saturated, the roads are not only impassable on account of the mud, but are injured by traffic more than in all the remaining nine or ten months of the year. One year's statute labor with the annual appropriation, spent in providing proper drains for the roads of the Province would do more than can be done in five years with the present method of merely filling the holes which appear, with gravel or crushed stone.

Underdraining and grading should be carried on simultaneously. Unless drained, the traffic during the ensuing autumn and spring will usually leave a graded road as shapeless and rough as a pioneer wagon track can be. If drained, there will be a foundation for the gravel or crushed stone to rest on when applied.

A porous soil, like a sponge, retains in its texture, by attraction, a certain amount of water. When water in excess of this is added it sinks to the first impenetrable strata, and from there it rises higher and higher until it finds a lateral outlet; just as water poured into a pail will rise higher and higher until it finds an outlet in the side of the pail, or until it flows over the top. Underdraining supplies the necessary outlet for this excess moisture at a proper depth from the surface; "it lowers the water line."

With plastic clays the process is slightly different. Clay will absorb nearly one-half its bulk and weight of water. In drying it shrinks and is torn in different directions. The fissures thus commenced by a tile drain become new drains to lead water to the tile; and so the process of contracting and cracking continues until a net work of fissures is produced, and the stiffest clay is thereby drained.

THE EFFECT OF FROST.

The injury done to roads by frost is caused entirely by the presence of water. Water expands on freezing, and the more there is under a road, and above frost line, the greater is the injury. The particles of soil in immediate contact with the water are first compacted. When room for expansion ceases within the body of the soil itself, the surface is upheaved. When thawing takes place the sub-soil will be found honey-combed, ready to settle and sink beneath traffic. It is therefore of the utmost importance that the soil should be relieved of the water of saturation as quickly as possible by underdrainage. The impassable condition of most roads in Canada during the spring, often axle deep in mud, is to be attributed very largely to a wet sub-soil which has been honey-combed by frost.

MAINTENANCE.

It is not merely necessary to make the roads good; they must be kept good. It is not sufficient to provide drains; care must be exercised to keep the drains open and free from obstructions. It is not enough to merely place the gravel or broken stone on the road as it ought to be placed; care must be taken to see that the covering is kept in place. It is not sufficient to build culverts; it is further necessary that the culverts be kept in order. It is folly to build roads properly if they are afterwards neglected and allowed to remain out of repair. Every farmer knows that if the repair of his barns, his outhouses, and his fences were neglected as is the repair of the roads, a great personal loss would result.

Probably one of the greatest defects of the present system of road construction is that the roads are not repaired when the need of repair first arises. Unless in a very dangerous state, work is done on the roads only once a year at the time of performing

statute labor ; ruts, hollows and other defects are permitted to remain without attention, and when these defects commence they increase with great rapidity. Culverts are permitted to fall to pieces for want of repair at the proper time. Drains become obstructed, and the roadway is allowed to be flooded and saturated for want of a little timely attention.

It is absolutely essential to the maintenance of a good and economical system of roads that provision be made by the township for their repair as soon as signs of wear appear. However extravagant it may appear at first sight to recommend that a man be constantly employed to go over such a mileage of roads as he can attend to, devoting his whole time to the work, there can be no doubt that it would in the end be the most economical plan. It is the same system pursued by railways in the care of their roadbed, and railway corporations are noted for their economical methods.

A man constantly employed in this way could fill up ruts and wheel tracks as soon as they appear, before water has been permitted to stand in them to assist in deepening them. A decayed plank would be removed from the culvert or bridge before an accident was caused thereby, and before the weakening of the bridge at this point had caused other portions to be destroyed. An obstructed drain would be opened before injury had resulted to the roadway. Loose stones would be removed from the road where they are rolling under the wheels and feet of horses. Loose gravel and stone would be kept raked into place until it had become consolidated. By these and many other simple means the roads would be at all times kept in a more serviceable condition, and of greater importance, repairs would be made in time to save the road from injury which could not be remedied ; and in time to save much labor and expense in making possible repairs.

Some municipalities have adopted the plan of employing a foreman and a couple of laborers to devote their whole time to the roads of a district or township, and in such cases a grader and other road machinery is employed. Whatever the details of the system, the principle should be the same throughout, that wear must be repaired as soon as signs of it appear, if economy and good service are to be had.

ROAD GRADERS.

A road grader is one of the most necessary implements for a township to possess. To depend upon manual labor for the first grading of roads, and the repair of others that require re-shaping, is a useless waste of labor and money. Improved road machinery is as great a saving in road-making as is the self-binder or the steam thresher in farming operations. To neglect the use of a road grader is as unwise as it would be to return to the old time cradle and flail. If every ratepayer took the same personal interest in the roads that he does in his farm, no municipalities would be without modern machinery for roadwork. The grader in the hands of a skilful operator will do the work of fifty to seventy-five men in grading and levelling the roadway.

A ROAD ROLLER.

A most valuable implement in road construction is the road roller. On town streets it is indispensable. To thoroughly consolidate the gravel or stone into a smooth, hard layer, before it can be mixed up with the sub-soil, renders the surface coating more durable and serviceable in every way. A steam roller of ten tons is too expensive for the majority of rural municipalities, but in some instances, townships could rent from a town in the district which owns one. A horse roller of six or eight tons is less expensive and some municipalities may see fit to purchase. In the construction of broken stone roads the loose stone consolidates under traffic less readily than does gravel, and without rolling remains for some time very loose and rough.

A STONE CRUSHER.

Wherever good gravel cannot be had, and where stone for crushing is obtainable, a stone crusher is most useful. Stone can be broken by this means at a price within the range of every well populated township, although a very expensive work when performed by hand. A crusher can be operated by the steam engine used for a threshing machine, which can generally be rented.

A rotary screen attached to the crusher will separate the stone into grades according to size, ready to be placed on the road in layers, the coarser in the bottom of the road.

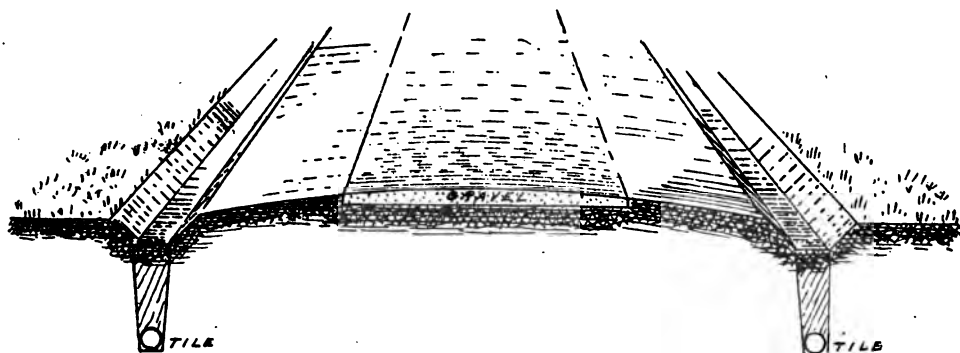
WAGON TIRES.

It would seem as though in everything the present methods in regard to roads in Ontario are contrary to good judgment. Gravel or broken stone is dumped loosely, without even spreading, on a badly graded, badly drained sub-soil. In the use of these roads the same recklessness is glaringly apparent. When wide tires have universally replaced the narrow tires which are now found on farm wagons, a great part of the road question will be solved. Narrow tires of two and one-half inches in width have only one-half of the bearing on the road which would be provided by tires of proper width. By referring to the supporting power of soils discussed in the paragraph on "Foundations," the effect of this is more apparent. By the use of a six inch tire, the roadway will support, without yielding, twice the load which it could support with a three inch tire.

Narrow tires cannot be too strongly condemned. They cut and grind the road, plow and upheave it. Wide tires, on the contrary, are a benefit rather than an injury to the road, inasmuch as they act as rollers to preserve a smooth, hard surface. In some localities wide tires are objected to under the argument that they increase the draft required to move the load. This may occur under certain occasional conditions of very wet and soft roads. But when wide tires are universally used this objection will disappear, as the increased draft is due to the ruts and mud caused by narrow tires.

DIMENSIONS OF ROADS.

For the average country road, a graded roadway twenty-four feet in width between the inside edges of the open drains, will be ample to accommodate travel. For the average road, if the central eight feet is metalled with gravel or broken stone, it will be



"FOR THE AVERAGE COUNTRY ROAD."

sufficient. The depth and width of the open drains will have to be governed by circumstances. Sufficient capacity must be provided to carry away all surface water. The depth must be dependent also on the fall obtainable. With tile under drains, deep open

ditches are not needed to drain the road foundation. The use of tile does away with the deep and dangerous open ditches which may otherwise be necessary. The crown of the road should be such as to give a fall of one inch to the foot from the centre to the edge of the ditch.

HILLS.

Hills are among the difficult portions of the road to construct, and are a constant source of expense for repairs. The reason of this usually is, that the drainage is imperfect. Water is brought long distances in open drains by the roadside, and poured over the hills, frequently to flood over the whole surface. It is not uncommon to find the centre of the road over the hill lower than open drains at its side—if there are drains at all. The natural result is that washouts are constantly occurring. For conditions of this kind the simple remedy is to dispose of the water before it reaches the hill, by conveying it through the adjoining fields if necessary. The probability is that a great amount of water has been carried in deep ditches past watercourse after watercourse in order to dispose of it over the hill, thereby avoiding the necessity of constructing drains through farm land in the natural watercourses. The secret of successful drainage with respect to roads, is to dispose of water in small quantities before it can gain force and headway.

Another common occurrence is to find water oozing from the surface of the road on hills. This is especially noticeable after the frost leaves these spots. The surface is soft and spongy and is cut readily by wheels. Such hills should be drained by placing a line of tile down each side of the roadway between the gutter and the gravel, carrying these underdrains to proper outlets. Cross drains should be laid in the wet spot leading to the side underdrains in a diagonal course. Well concaved gutters should be made on each side of the roadway, and at regular intervals catch basins should be placed to arrest the flow of water in these gutters, leading it into the tile underdrains.

The roadway on a hill should be well crowned. This will draw the water quickly to the drains at the side of the road, instead of permitting it to follow the wheel tracks, deepening them to ruts.

LOCATION OF ROADS.

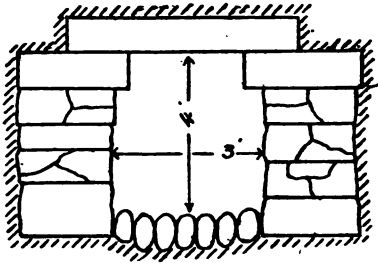
There are very many instances where, by changing the course of a road slightly municipalities would save a large sum in construction, and at the same time produce a better road. A slight deviation would frequently avoid swampy or wet ground, or would do away with the necessity of expensive cuts and fills. A hill can sometimes be avoided or the grade very much reduced by altering the location of the road. There is a prejudice against taking the roads from the lines laid down in the original survey, and property owners prefer to have their farms bounded by straight lines. At the same time the value of good roads to the farm should not be overlooked, and whenever a change in the road allowance means the change from a bad to a good road, or a change from a steep to a gentle grade, the slight inconvenience created by the alteration of boundary lines will be many times repaid.

CULVERTS.

~~It is~~ In no branch of municipal work is so much money wasted as in the construction and maintenance of sluices and culverts. In most townships these are built of timber.

Timber is perishable, culverts are subjected to repeated changes of wet and dry weather, the severest test to which timber could be subjected. Each year a large number of these culverts are renewed at a cost of from \$5 to \$50 each, in some townships aggregating from \$500 to \$1,500, and this is an annual outlay. The life of these structures is so short that it is not more than five years before repairs are required, and these repairs

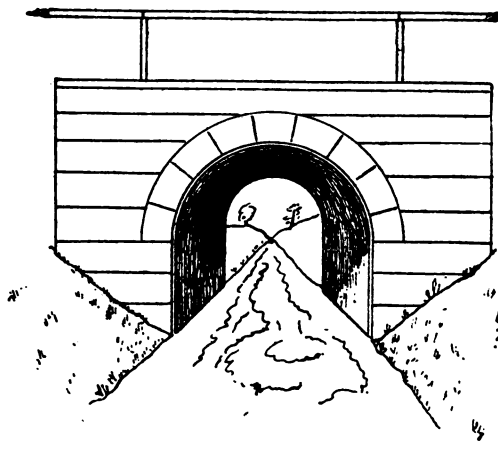
in a short time amount almost to renewal, and that of the most expensive kind. A broken plank, or stringer, a rotten log or any timber replaced with new at different periods, makes the maintenance very costly, and this class of structure the most temporary and expensive. No sooner have they all been rebuilt than we must again com-



FOR SOME LOCALITIES A CHEAP, STONE CULVERT OF THIS DESIGN WILL ANSWER.

mence the reconstruction of the first; in this way the expenditure becomes perpetual, and fixes a large percentage of our annual tax. If these culverts are in their proper locations, natural watercourses and other fixed places, they will always be required, and their construction in the most durable manner is the best and most economical plan.

For small culverts there is very little difference in the cost of timber and vitrified pipe. If properly laid the latter will withstand the frost and is durable. These pipes may be used up to eighteen inches in diameter, and the capacity may be increased by laying two or more rows, but the pipes should have at least one foot of earth or other filling between them. Culverts of five or ten feet span should be cement-concrete arches, which is permanent if the concrete is properly made. The concrete should be composed of first-class cement, clean, sharp, silicious sand, free from earthy particles and coarse enough to pass through a twenty mesh, and be retained on a thirty mesh sieve; clean gravel screened through an inch and a half screen, the largest stones to be not more than two and one-half inches in diameter; or, in place of gravel, broken stones that will pass



A CONCRETE CULVERT WITH WING-WALLS OF CONCRETE.

through a two and one-half inch ring. These materials should be mixed in the proportion of one cubic foot of cement, two cubic feet of sand and three cubic feet of gravel or broken stone, with just enough water to make the whole into a plastic mass. The sand and cement must be first mixed dry, then a sufficient quantity of water added to make it

into a thick paste. It should then be thoroughly mixed again, spread out, the stone or gravel added, and the whole thoroughly mixed until every stone is coated with the mortar, then put it in place. The walls should extend well below the frost line and have a wing at the ends to protect the embankment from wash.

BRIDGES.

Wooden bridges, except where timber is very plentiful in the immediate locality, are not a good investment in view of the reduced cost of iron and steel, and the increasing cost of timber. Timber decays quickly, and while cheaper than steel in first cost, is more expensive after a term of years, since the cost of repairs is very great.

Generally speaking, the cost of an iron superstructure is more than that of wood. The substructure of stone or concrete is more expensive than pile or crib work, but as in other structures, a firm foundation is most serviceable and economical. Wooden foundations, from decay and other causes, settle, and the least settlement in the foundation twists the timber, causing a disarrangement of the strains, frequently transferring the greatest load to the weakest point. Wherever timbers have a seat or bearing exposed, decay soon commences, and when least expected the bridge collapses under a heavy load.

Wherever timber is used in bridges it should be in members of from four to six inches in thickness, the strength of the beam or cord being obtained by building several members together, breaking joints, and coating each bearing with lead. A further protection is to cover these built timbers with galvanized iron to protect the numerous joints and bearings from moisture. All caps, corbels, chords, braces and floor beams should be made in this way so that the thickness of no timber will be more than six inches. A wooden bridge should be painted one year after erection; an iron bridge at the time of erection, and care should be taken to see that they are kept painted and that all nuts are kept tightened so that each member may carry its fair share of the load.

The cost of renewing a wooden bridge, in which a man has to be sent to put in a new timber from time to time, will amount to twice the initial cost of the bridge. In this way the ultimate cost of a timber structure becomes very great.

The course pursued by some, indeed most municipalities, in erecting iron bridges is likely, however, to result disastrously, and throw iron and steel into disrepute. A council advertises for tenders. The companies responding supply their own plans and specifications. Thus far the procedure is entirely satisfactory. The difficulty arises when councils accept the lowest tender without obtaining the advice of an experienced builder of iron bridges as to the plans and specifications submitted. This is a matter in which few township engineers and surveyors are qualified to decide, and certainly the wisdom of councillors entirely without professional training in such matters is not to be trusted. Cases have occurred in which a difference of five dollars has influenced a council to accept a tender for a bridge which was manifestly, to a man of experience, worth less than the other by several hundred dollars; and which was indeed unsafe, offering every likelihood of failure with attendant loss of life and great expense for reconstruction. It is difficult to understand the action of councillors, shrewd in other matters, in the construction of bridges and other public works proceeding with such apparent disregard for the true interests of those whom they represent. A small sum spent in securing reliable advice is as much a matter of economy in public as in private affairs.

IN BRIEF.

Roads that "break up" are bad roads.

Make road improvements in such a way that they will be permanent.

Whether by statute labor or other means undertake road work systematically.

Appoint a supervisor who will have charge of all the road work.

Make road beats five miles in length, choose the best men as pathmasters, and keep them in office.

Classify the roads according to the nature and extent of traffic over them.

Specify the width of grade, amount of crown, plan of drainage, kind, width and depth of material to be used, and see that these specifications are carried out.

Purchase gravel by the pit, not by the load.

Use clean road material.

Strip the clay and earth from over the gravel pit before the time of performing statute labor.

If screening or crushing is necessary let this be done before the time of statute labor.

Do not scatter money in making trifling repairs to temporary structures.

If statute labor is to be made successful the work must be systematically planned and some definite end kept in view.

Have the work properly laid out before the day appointed to commence work. Call out only a sufficient number of men and teams to properly carry out the work in hand and notify them of the implements each will be required to bring.

Let no pathmaster return a ratepayer's statute labor as performed unless it has been done to his satisfaction.

In justice to others, make the returns clearly, showing what work has not been done.

See that the council collects the amount from the delinquent parties and have it expended the next year.

The pathmaster should inspect the roads under his charge after every heavy rain-storm. A few minutes' work in freeing drains from obstructions, filling holes, diverting a current of water may amount to several days' work if neglected.

With the money which can be spent, build permanent culverts, permanent bridges, buy machinery, buy gravel pits, prepare gravel for hauling, construct drains, operate the machinery.

Use the statute labor as far as possible in drawing gravel or broken stone.

Do not leave the gravel or broken stone just as it drops from the wagon. Spread the metal.

Crown the road with a rise of one inch to the foot from side to centre so as to shed water from the roadway to the drains.

Give the open drains a good fall to a free outlet. Lay tile underdrains where needed.

Drain thoroughly. Keep the road surface dry. Keep the earth underneath the surface dry.

Use road machinery.

Use graders, stone crushers and road rollers.

Improved machinery is as necessary for good and economical work as are self-binders and steam-threshers.

Employ a man to take charge of the machinery. He will become experienced and do better and cheaper work.

The same teams should be employed to operate the graders. They become accustomed to the work and give better service.

Do not cover an old gravel road with sod and earth from the sides of the road. Turn this earth and sod outward and raise the centre with new gravel.

- Adopt every means to secure a hard, smooth, water-proof surface.
- Do not let stones roll loosely on the road.
- Do not let ruts remain. They make travelling difficult, and spoil the road by holding water.
- Make repairs as soon as the defects appear.
- Use wide tires.
- Improve the drainage of the hills. Make the crown of the roadway higher than on level ground.
- Change the location of the road if a steep hill can be avoided.
- Do not use wood for culverts. Use concrete, vitrified pipe or stone.
- Do not build wooden bridges. Use iron, stone or concrete.
- Build good roads.

THE GOOD ROADS MOVEMENT.

The demand for better roads is one which is far reaching and urgent. From Nova Scotia to British Columbia the question is being discussed and active measures are being taken to place road improvement on a more satisfactory basis. Throughout Canada statute labor still prevails but in all the provinces evidence is not lacking to show that a change in this regard will take place within a short term of years. It is not a matter in which any of the legislatures are likely to interfere until the people themselves make such changes as the power with which they are now vested permits. The legislatures are, however, using educational means to influence the people to better their condition in this respect. Ontario and Quebec have appointed commissioners for this purpose. In New Brunswick and Quebec the provincial governments are granting aid to the Good Roads Associations and in the actual construction of roads, chiefly, as yet, for educational purposes. In Nova Scotia measures are being advanced leading to the expenditure of Provincial grants, and to regulating the width of waggon tires. Manitoba has as yet taken no active steps, but in the North West Territories organization has been given to a statute labor law and in British Columbia the agitation is brisk.

The movement has grown out of actual conditions. The roads are bad. They are the result of a certain system. That system is therefore insufficient, and better methods are being sought. There is no desire on the part of any to increase taxation. The reverse is entirely the case. But where taxation may be increased to a slight extent, it is claimed, and fully substantiated, that the amount of taxation is many times made up by profit derived from the improved roads. The movement is on a firm basis, is one which is supported by the best economic principles borne out by the experience of other countries.

The possibilities of agriculture in Canada are unsurpassed by those of any country of the world. A comparatively small proportion of our land available for farming purposes, is under cultivation; and such lands as are under cultivation are rarely productive to the full extent of their ability. To enable us to receive the greatest advantage from the resources surrounding us, there is every need that the portion of the community engaged in farming shall be provided with the best machinery of agriculture. This does not consist merely in the seeders that assist in sowing; the threshers that separate the wheat from the chaff. To obtain sale for his produce at the highest price is equally important for the farmer with raising his produce cheaply and of the best quality. The former, however, is a matter which the farmers have too long neglected, too long left to the good will of others. The real market of the Canadian farmer is not the railway station or country town where he delivers his produce. The price of the article is fixed at the great distributing centres in accordance with law of supply and demand. The

amount received by the farmer is the price at the distributing centres, less the cost of transportation. To realize increased profits, the farmer must produce more cheaply and transport more cheaply. The first part of the carriage is entirely in the farmer's hands.

The first link in the chain of transportation, the country road, is abnormally expensive and inefficient. To remedy this is to assist very materially in improving agricultural conditions, and in a great measure increase the prosperity of the country as a whole. There is not an industry nor profession which would not be benefited by good roads, for the basis of Canada's wealth is agriculture. The State of Massachusetts is one of those which has taken most advanced steps in road improvement. On petition of a county, the State road commission may, with the assent of the Legislature, adopt any road within the county as a State highway. Except that the grading and bridging is done by the county, the work thereafter, both construction and maintenance, is under the authority of the state commission. Also on petition of two or more cities or towns, a road between them may be made a state highway. The "State Commission" is composed of three commissioners who compile statistics, make investigations, advise regarding road construction and maintenance, and hold public meetings for the discussion of road matters. One-fourth the cost of construction is paid by the county, the remaining three-fourths being paid by the State. In 1894, the State spent \$300,000 in this way; in 1895, \$400,000; and in 1896, \$600,000. It is intended that ultimately about one-tenth of the entire road mileage shall be built in this way.

In Michigan, upon a majority vote of the ratepayers in any county, a county road system may be adopted. A board of commissioners, five in number, are elected by the people to lay out, and construct certain of the leading roads, to be paid for and thereafter maintained by a county rate.

A bill has just passed the New York State Legislature providing that, on petition of a county council, certain roads may be adopted as state roads. The petition is first presented to the State Engineer. If he approves of the section of road thus sought to be improved, he prepares plans, specifications and estimates. These are presented to the legislature and if approved by that body, fifty per cent. of the cost of construction is paid by the state.

The New Jersey Highway Law provides that, on the petition of the owners of two-thirds of the land bordering on a road, the state Commissioner of Public Roads will cause the road to be improved in accordance with plans and specifications prepared by him subject to the approval of the Legislature. The owners of the land affected by the improvement pay one-tenth of the cost; the county pays six-tenths; and the state three-tenths.

Connecticut has introduced a plan of highway improvement providing for the appointment of three state commissioners. When a township votes in favor of constructing a road under the provisions of the State Highway Act, specifications are prepared and submitted to the state commissioners. If the commission approves, the township council lets contracts for the work to be performed under the supervision of the state commissioners. One-third of the cost is paid by the state; one-third by the county; and one-third by the township. The expenditure by the state in this way is limited to \$75,000 annually.

The State of Rhode Island has appointed a Commissioner of Highways. When a council represents to the Commissioner the need for improving a certain road, an examination is made by him. If he considers the work necessary he prepares plans, specifications and estimates, and reports to the municipalities affected, also to the State Legislature as to the proportion in which the expense should be met by the state, and the municipalities benefited. If the State Legislature approves, the work is performed by contract.

Vermont and California also contribute largely in the form of state aid, while Indiana, Kentucky and others contribute in a less degree. Only the bare outlines of the systems have been stated with the object of showing the prominence the question of road improvement has attained of recent years. All the above legislation has been adopted

within the past five years. In all these systems safeguards are placed to prevent the expenditure exceeding, for any state or any locality, certain reasonable limits, according to requirements and ability to meet the payments. In most of these states, the tax is so levied that the towns and cities pay the greater proportion of the cost of state road construction; for example, in the State of New York, it is estimated that the people outside of the towns and cities will pay only 10% of the cost.

The last report of the Wisconsin League for good roads concludes with the following summary:—

"Interest in the crusade for better roads has again made great progress during the past year. Road machines are multiplying in number, better methods are gradually being adopted, the sentiment for substantial macadam roads is growing rapidly, and it may truly be said that the work of the League for Good Roads during the past two years has borne fruit. Still, only a beginning has been made during those two years. We must continue in our efforts to bring about state aid to road building, which is the best practical method of securing the construction of short pieces of model road in each country, to serve as object lessons. Such pieces of roads would serve to teach the practical value of substantial highways, and would give a wonderful impetus to the good roads sentiments. State aid to roads is employed in New Jersey, in Connecticut, Washington, Rhode Island, Vermont and Massachusetts. Public sentiment in Wisconsin seems to be drifting to state aid as the most practical reform in connection with road building. Our league has the heartiest co-operation of the farmer's institutes, the agricultural societies, the public press, and of many organizations whose character permits them to aid in our work. With the continued assistance of our members who have manifested their interest upon numerous occasions, we believe confidently that our organization will continue to succeed in its labors as heretofore."

The inducements offered to municipalities by the Provincial Government of Quebec, are stated in a circular issued from the Department of Agriculture. While the means adopted may in a measure be termed "state aid," they in effect are similar to those of Ontario, in as much as they encourage the people themselves to undertake the work; are educative in results, rather than affording substantial assistance such as is generally understood by "state aid." The circular is as follows:—

"In 1897 the Department of Agriculture inaugurated a new policy for encouraging improvements to roads by granting aid to rural municipalities to enable them to purchase special machines for repairing roads.

Notwithstanding the bad weather that prevailed during the summer and last autumn, and was unfavorable for effecting repairs to roads, the results obtained have been most satisfactory. Seventy-seven municipalities have had their names entered so as to benefit by the advantage offered them, and according to the official reports received up to date at the Department of Agriculture, over 150 miles of road have been repaired since last June.

These results justify the Department in carrying out the system inaugurated last summer with the few following modifications:—

A special grant of \$300 was placed last summer at the disposal of each county, to be divided into premiums of \$125, \$100 and \$75 between the first three municipalities that availed themselves of the same. The same grant will be given this year to the counties which shall have claimed it in whole or part between this and the month of July next.

In counties where the total amount of \$300 shall have been distributed between now and July, the Department will again grant from and after that date three premiums of \$75 each to the first three municipalities applying for the same. If a single municipality in a county has then availed itself of the offer made to each county, the department will place at the disposal of the same county a bonus of \$100 and two of \$75.

Moreover, in order to encourage the stoning of roads, the Department has decided to grant each county municipality, aid towards the purchase of stone breakers with engine, roller, and sorter, provided that the total cost of such machines do not exceed \$2,400 per county. An amendment to the Municipal Code, passed during the last session of the Legislature, authorizes county councils to make arrangements with each parish, village or town municipality of the same electoral division, for repairing the roads in those various municipalities. The general act respecting town corporations has likewise been amended in the same sense.

Municipal councils of parishes or of counties desiring to avail themselves of the above advantages must draw up their application in the form of a resolution, a copy whereof must be sent to the Department of Agriculture. The premium to which they shall be entitled shall be

paid to them after at least two miles of earth road shall have been repaired, and after a certificate to that effect shall have been forwarded to the Department with a copy of the invoice from the company that sold them the machines.

In the case of the purchase of a stone crusher, half a mile of macadamized road must have been completed and accepted by an officer of the Department of Agriculture, before the Government contribution shall be paid.

Machines for repairing earth roads that are bought by the municipalities, with the help of the Government, shall remain the property of such municipalities for at least three years, and the county councils that purchase stone breakers cannot sell such machines before they have been at least five years in use in the same county."

MUNICIPAL REPORTS.

During the year an effort was made to secure reliable statistics from municipal clerks respecting roads and streets throughout the Province. The importance attached to the question of road and street improvement was evinced by the interest taken in the matter by municipal clerks. The replies to questions asked, in the great majority of cases, were painstaking, with an evident desire to give the fullest and most correct information. In spite of this, however, the lack of system which characterizes this branch of public work is so deplorable that nothing of a definite character, respecting the Province as a whole, can be tabulated. Very few municipalities have any records showing the number of miles of roadway to be maintained; some clerks could not even venture an approximate estimate. Scarcely a township could separate the expenditure on roads from that on bridges. Rarely could a reliable answer be given as to the number of miles of road improved in any year, or the amount of work done. In regard to the general condition of roads in any municipality, the majority of clerks affirmed that they were bad. Others believe that their roads compare very favorably with those of adjoining townships—not always indicative, it is to be observed, of good roads.

In towns, very much the same condition prevails. Only an approximate estimate of the street mileage could be given. Many could not state the number of miles gravelled or otherwise improved. While they could give the total amount spent on roads, sidewalks and bridges, very few could separate the items under these heads; nor had they any record of the work done in any year.

If the roads had received any kind of systematic treatment whatever, these are all matters which could be known without difficulty. The result of the enquiry is a most emphatic avowal of the neglect which road improvement, under present methods, has received. Councils are spending money in amounts which are very often large, but no proper record of the expenditure is kept. The money is merely scattered in patchwork and temporary repairs, and at the end of the year, for one thousand dollars spent in this way, very rarely can satisfactory results be shown. A proper record of the money and labor placed on the highways, together with the work performed, if presented to the people would be an important factor in correcting many of the slipshod methods which now prevail. The following is the circular sent to township clerks. Those sent to towns, cities and counties were of a similar nature:

DEPARTMENT OF AGRICULTURE, GOOD ROADS BRANCH, TORONTO, OCTOBER 11TH, 1897.

To the Clerk of.....

The information possessed by the Department respecting the roads of Ontario is of a very indefinite character; and in view of the importance of the subject, and the interest now being taken in road improvement, it is earnestly requested that you will favor the Department with replies to the questions in this circular. Kindly fill in the blanks as far as possible, and return in the enclosed envelope without sealing it. Any further particulars respecting the roads or road system of your municipality, not touched upon by the questions, will be gladly received.

A. W. CAMPBELL,
Provincial Instructor in Roadmaking.

1. What is the total road mileage maintained by the township ?
2. How many miles are metalled with gravel or broken stone ?
3. Is the road metal (gravel, stone, etc.) plentiful ? What is the average length of haul ? What is the quality ?
4. Is statute labor employed ?
5. What number of days' statute labor was on the rolls in 1896 ?
6. What is the number of pathmasters appointed ?
7. What is the commutation rate ?
8. What number of days were commuted in 1896 ?
9. Is commutation of statute labor compulsory ?
10. What amount of money was appropriated by the township for roads in 1896 ? For bridges ? Total.
11. Is any aid received from the county ? If so, what amount was received in 1896 ?
12. Were any actions for damages brought against the township in 1896 caused by defective highways ? What was paid by the township in verdict, fees or settlement ? Are any such actions now pending ?
13. What roadmaking implements or machines are owned by the township, and when were they purchased ?
14. What extent of road mileage was graded or otherwise improved in 1896 ? In 1897 ?
15. What is the general condition of the roads ? Has any special effort been made recently to improve them ?
16. Further particulars :

CITIES, TOWNS AND VILLAGES.

The following extracts from reports received from the clerks and engineers of cities, towns and villages are of interest, and contain very suggestive statements. Special reports were asked from certain municipalities, which are also given :

BARRIE.

In 1896 the Council of the Town of Barrie voted an appropriation for the purchase of machinery and the construction of macadam pavements.

In that year and the spring of 1897 there was constructed 7,050 yards of Telford and 20,000 yards of simple macadam.

Under the Telford, the sub-grade was a light loam and sand, and under the macadam pure sand for the greater part of the way.

Some years previously there had been a specification drawn up for a Telford road, estimated to cost \$1.20 per square yard.

The authorities desired to put down a cheaper pavement, although there was no discussion as to discarding the Telford foundation.

The roadbed actually laid was a very substantial one, quite as heavy as the original design, but having cheaper features in the foundation and costing about half the estimated cost of the original.

The Foundation consisted of six to eight inches of field stone. The interstices were filled by the broken stone working into them. A large amount of broken stone was used in this way, but except where a swampy sub-grade or other reasons called for this procedure, I would not advise it, on the score of the attendant expense. It left a very open foundation no doubt and solid, but in general the gravel from the old roadbed might be hauled back to serve the purpose or else if that were not fairly clear, fresh pit gravel. This forms a good cushion also for the metal proper.

If excavation has to be made for the foundation, as would be the case in general on a built up business street, there is a certain amount of expense in that, but in cases the excavated earth serves as a fill, and where stone is as cheap as we had it here, the expense of the foundation,

for stone and setting the same, is but a small proportion of the total cost. Here, this item amounted to about ten or eleven cents per square yard, but the excavation and the broken stone to fill the interstices added considerably to the outlay chargeable to this portion of the work.

After a small piece of the pavement had been finished a tremendous rain storm swept it as clean and smooth as a floor and caused a citizen "*agin the Government*," to remark "I do not mind paying for a thing when it is well done."

The Road Metal was then spread in three grades, coarse, medium and screenings. The medium grade is practically the wearing surface, as the screenings are intended for binding only, but were only used over the top layer. Water was used liberally with the fire hose where possible, otherwise with the water cart used for sprinkling the town's streets.

The traffic was never stopped. The best piece of work was done, however, on a section of road where the traffic could be diverted and plenty of water laid on from a fire hydrant.

The intention was, on our work, to make the metal on the crown of the road six or seven inches thick after compacting, and the edges four or five inches. I think that this was approximately obtained. It is false economy not to grade the sub-grade carefully, as there is a waste of stone otherwise.

We estimated that the stone as measured in the wagons and as paid for ought to cover the roadway, if placed unbroken, to a depth of about ten inches, so that an allowance—say 66 per cent,—ought to be made in calculating the stone required for a given area.

The circumstances were such as to prevent any very exhaustive tests being made, but it would appear that the stone when broken and measured separately, increased very little in bulk. As has been mentioned, the greater part of the simple macadam was laid on pure sand.

It was difficult to maintain a decent cross-section, as the wagons and carts bringing the stone and the ordinary traffic cut into the sub-grade, the soft sand spreading out almost like water. Generally, however, a thin crust was formed by spreading clay, gravel or sharp sand, as it could be obtained, and often the loam underneath the sods at the roadside, where cuts were made was spread over the sand, but when the expense of this is likely to be at all large, I would not advise it. Even if the lower stratum of metal is to some extent buried in sand, there is the weight and solidity of the stone to the good.

The stone placed as this soft sand was rather coarse and had a good many spalls amongst it. We considered that, owing to the soft bed, these spalls were not very objectionable. When the jaws, crushing plates or dies are worn a good deal, there is apt to be a large proportion of flat pieces. These can be broken by napping hammers, and should be broken if they do not rest directly on a soft bed such as I have described.

A new set of jaws will help the difficulty of spalls, of which, however, there will always be some, and the napping hammers will always be in requisition for the upper course at least.

Engineering. To provide ample time for preparing careful plans is common sense. Due notification to the corporation of changes of grade affecting properties will give them time to act expediently. A good deal more of the construction engineer should be seen on the road than was arranged for in our work. Engineering cost money to the client, but the expenditure is conservative and legitimate and always tends to saving in the end, often of large sums.

The Cost of pavements of this kind must differ widely according to the locality. The cost of the stone and of breaking it are factors, and these vary considerably—the first to the greatest degree.

We found that the 4,220 square yards of simple macadam constructed in the spring of 1897 cost thirty-three cents per square yard, including all repairs to machinery, oil, waste, etc., etc., and crushing plates, but not including depreciation, interest on capital, nor engineering. The excavation was not responsible for more than two cents per square yard. The amount to add for depreciation and interest might be estimated roughly at two cents per square yard on the supposition that say 15,000 square yards were laid per annum. See also under "*Machinery*" for cost.

Cleaning. The pavement on the main street has been swept by hand, a large stable broom having been used, with shovel and hand scraper, no labor-saving machinery having been purchased.

Expensive as this method is, the road lasts well under it, as it is dealt gently with. The low spots are kept filled and prevented from being worn deeper, while the whole roadway is left covered with a very light crust protecting the stone from abrasion.

The crust would be dust except for the use of the water cart, which is thus an important factor in the preservation of the pavement.

Gutter. A cobble stone gutter about four feet wide is perhaps the best and most economical to edge the roadway with in small towns. The cost will be from twenty-five to thirty cents per square yard if not grouted with cement. One advantage of cement is that grass cannot grow in the interstices; the removal of which from the gutter on streets of little traffic being an item of expense; otherwise, however, the gutter seems amply stable under the light vehicular traffic of small towns.

After the macadam has been laid it is a source of great annoyance to have holes pawed in it by standing horses, and the cost of the gutter is little if any more; the above price not including excavation and grading, which would have to be done to prepare for macadam. When the macadam is thoroughly compacted by travel, this source of annoyance is lessened very much, but not altogether obviated unless by a gutter of cobble stones, flag stones or concrete.

Machinery. The roller used was a five and one-half ton horse roller. We purchased two and one-half tons of pig iron to weight it with. The old jaws of the crusher may be used for that, but they are too heavy to place or replace at convenience.

In setting up the crusher it is a great matter to view the operation from a labor-saving point of view. Let gravity help the men to bring the stone to the crusher and the gang may be reduced to a minimum.

The regular screening attachment to turn out three grades was purchased with the crusher, consisting of an endless chain of buckets raising the stone to the revolving perforated drum, which is set up in a frame to which are attached chutes. We constructed larger chutes to hold a wagon load or more each, and the labor of shovelling up the stone was saved. Great care should be exercised in making everything convenient at first. Our crusher, although built to be convenient for frequent moving, was set up twice; the distance macadamized was a mile and a half.

It was set up solidly each time and a shed built for the engine. When wagons with stone to be broken came in while the crusher was running, they were unloaded directly on to the platform beside the man or men feeding. Otherwise the stone was wheeled on to the platform from the pile. The amount of money to be spent on labor-saving conveniences must be determined by the circumstances solely.

Since the product of the crusher as to amount varies with the size and hardness of the stone to be crushed, and also with the set of the crushing plates, etc., etc., the results obtained in one locality and under one set of circumstances cannot be offered as a guide for all localities and circumstances.

Turning to one test in 1896—Four loads averaging 2.67 cubic yards each were broken in an average of twelve and one-half minutes each, or at the rate of 12.78 cubic yards per hour. The greater part of the stone brought to the machine was field stone just as it was picked up, from say five inches in diameter to the limit imposed by the opening, viz., 6 x 14 inches. It is true that larger stones were often brought (although we endeavored to get the teamsters to sledge them to size before bringing), and a certain amount of sledging was always in order. One of the loads in the above test was all sledged stone (blue hard-heads, etc.) measuring 2.85 cubic yards, weighing 4.06 tons, broken in fifteen minutes. The weight of a cubic yard as measured in wagon of this stone would be 2,850 lbs., but we might assume in general the weight of the ordinary field stone as picked up to average not far from 2,500 lbs. to the cubic yard.

In 1897 with another engine and not so much of the largest size turned out, a test shows that three loads averaging 2.41 cubic yards and 6,020 lbs., or three tons each, took an average of sixteen minutes to grind. The weight per cubic yard was 2,492 lbs., and cubic contents of a ton 0.80 cubic yard; rate per hour, 9.19 cubic yards or 11.46 tons. The product of the 7.24 cubic yards was 7.61 cubic yards of three grades measured separately. This latter consisted of 0.59 cubic yards of the first grade or largest size which passed out of the end of the drum; 4.63 yards of the second grade which passed through the two-inch circular holes in the screen; and 2.40 cubic yards of "screenings" which passed through the one-inch circular hole.

The product contained very little of the coarsest grade it will be noticed; while in the first case perhaps half the product passed over the screen.

However, the average per hour including delays of all kinds is much less. I find in 1897 that only six yards were broken per hour paid; and the cost, including breakages, (of which we had one of some magnitude) and accounts, and delivery on the roadbed, amounted to 43 cents per cubic yard unbroken, or 11½ cents per square yard. Stone cost us 37½ cents per cubic yard delivered at crusher. Deducting the expense of repairing, rebabbiting and all accounts, 34 cents would cover the cost of breaking and delivering per cubic yard on the road.

Concrete Walks. While the crusher was in operation 11,300 square feet of concrete pavements were laid. The screenings from the crusher were used for the concrete. It would

have been profitable to use some of the medium grade on a heavier walk. It is much easier to level the bed for the topping when the material composing the concrete is not very rough and so the workmen object to the latter quality.

Extra screens may be attached under the revolving drum referred to under "Machinery," so that the screenings may be passed over a five-eighth inch mesh. That which passes the mesh falls on a fine screen which removes the flour-like dust, and the clean stone and sharp sand passes to the cart to be taken to the mixing board to be used for the "fine" concrete or topping. The rough screenings with only the dust eliminated may be used for the rough concrete.

One of the pavements of a total depth of twelve inches, built on a sub-grade of porous soil, on a business street and turning out to be a thoroughly satisfactory job, cost 9½ cents per square foot.

The foundation consisted of five or six inches of cobble stones topped with two or three inches of gravel to make eight inches in all; and the concrete consisted of three inches of "rough" and one inch of "fine."

The curb was of concrete six inches by fourteen inches, placed on a bed of small cobble stones and gravel, well tamped, six inches deep by twelve wide. This was considered better than making the concrete curb itself deeper to avoid frost as it made a drain and of course was much cheaper and quite solid. This sidewalk was designed for a business street and was ten feet wide. I am an advocate of three or four foot sidewalks on residential streets in small towns in order to procure the spread of permanent pavements by lessening the cost per lineal foot. The concrete may be lighter the smaller the "boy" or square. In place of laying three foot board sidewalk in out-lying districts I have been advocating the use of broken stone, nicely crowned, say four or five feet in width, but we have laid none as yet. There are a few hundred yards of gravel walk.

(Signed) ARTHUR G. ARDAGH,
Civil Engineer.

CORNWALL.

I do not know of any town in Canada, in which it is so difficult to construct a permanent street, as Cornwall. We have no gravel available, for one thing, and the sub-soil that we have here, is a sort of mixture of soapy clay and quicksand in which any material or metal sinks quickly out of sight.

The Town Council has for years past put tons of costly metal upon our streets, and each spring a small army of men was employed to keep our roads in some sort of shape, and many thousands of dollars have been spent in this way.

Upwards of thirty thousand dollars have been spent on one street alone in the past twenty-five years, and still our roads have always been very bad; some blocks being almost impassable, for even a moderate load, during the spring months.

This year, however, our Council decided to adopt the views set out in your various Bulletins, and set apart one block for experimental purposes.

The block selected was one having a great deal of heavy traffic over it, and one that we have always had great difficulty in keeping in shape.

The first thing we did, was to put down four inch tile drains on each side of the proposed street, and as the soil was very wet, we put in a number of diagonal drains. These tile drains emptied into a sewer at the end of the block.

After allowing about four weeks to dry the roadbed, we had our engineer lay out a roadway twenty-four feet in width.

The roadbed was then excavated to a grade line, the excavated earth being thrown on each side to form a boulevard. Instead of curb stones, we used what might be described as an ordinary pine plank.

This plank served three purposes. It retained the broken stone in the roadbed proper; prevented the loose earth from being mixed with the stone, and formed one side of the gutter. These gutters emptied into catch basins placed at the end of the block.

When our roadbed got dry we covered it with a layer of stone, broken to cubes of about two inches in size, to a depth of eight inches in the centre, running to six inches at the sides.

This as it was being put down we rolled thoroughly with a fifteen ton steam roller which the town of Brockville kindly allowed us the use of.

When this foundation was thoroughly compacted, we placed another layer of slightly smaller stone, four inches in the centre running to three inches at the sides, and also rolled this thoroughly, until the roller would make no impression.

Over this we sprinkled common field sand, and then turned our street sprinkler on, following up with the roller again. The water washed the sand into the interstice of the stones, and this packed by the roller, formed an exceptionally good bond. The street was kept closed two days longer to permit of its drying.

It now resists the wear and tear of traffic admirably and sheds water like the roof of a house. All our people are satisfied with the experiment, and are demanding that we continue this improvement.

I might add, that we get our broken stone from our County Council ; it is broken by the prisoners in our county jail ; we can get it in any size we wish, and at a very reasonable price.

There is a very strong feeling in our town that our Council for next year should buy a steam roller, and I am inclined to think the purchase will be made.

General regret is expressed that this system of making roads was not adopted years ago.

JOHN A. CHISHOLM,
Member of Board of Works.

STRATFORD.

The question of road-making has received considerable attention in the city of Stratford during the last year, and if the policy adopted by the present council is carried out in the future, it will result in decreased expenditure, through a saving in the cost of maintenance, and a vast improvement in the roads.

The expenditure in the past has been for the most part for maintenance, and very little permanent work has been done. The soil is clay and the gravel obtainable contains more or less clay, the roads are therefore uniformly bad in wet weather and require a coating of gravel each year to keep them from breaking up.

The money appropriated for street work, about \$12,000 per year, has been used up in patch-work and no improvement was possible so long as the ordinary system of expenditure by wards was followed.

It was recognized by the council of 1897 that no improvement or saving in expense could be made without a radical change both in policy and methods of doing work. Road-making machinery was therefore purchased, the money being taken from the appropriation for the year, and an engineer was appointed to advise the council and to take charge of the work and expenditure of the money.

The road-making machinery now in possession of the city consists of a fifteen ton road roller, a rock crusher with elevator and screens capable of crushing about ninety yards of field stones per day, and a reversible road grader.

The streets were graded, drained and macadamized during the latter half of the year 1897, and several heavily gravelled roads were broken up, regraded and rolled at a small cost.

It is found that much good work can be done on heavily gravelled roads by this method provided that the metal on the road is not too fine and that the grading is thoroughly done. Roads with a small depth of gravel or with worn out gravel cannot be much improved without the addition of a considerable quantity of new metal.

The council has decided that no permanent roads will be built until such time as all drains, sewers, water and gas pipes are put in, and all sewers that are now being put in have house drains put in at the same time. This work makes a grid of trenches on the streets ; but when once completed the roads will not require to be broken up again and the cost of maintenance of roads will be very small.

W. F. VAN BUSKIRK,
City Engineer.

INGERSOLL.

In response to a request in December last for information relating to the streets of Ingersoll, W. H. Jones, at that time mayor, writes :

You will remember we invited you up a couple of years ago to exchange views with you regarding certain improvements we were contemplating in our roadways, and our council was so impressed with the importance of the ideas you suggested that they determined to at once pur-

chase the necessary machinery and make a start toward the permanent work you had advocated. We bought a street roller, plow and stone crusher and elevator, and in order to educate our citizens as to the advantages of having good roads, we decided to take a block about fifty feet long by thirty feet wide on one main street, and build it according to your plans as submitted to us. We excavated the road bed for a depth of twelve inches and thoroughly rolled the surface until it was perfectly solid; we then put on five inches of broken stone and thoroughly rolled that course; then we put on another course of broken stone slightly smaller than the first course; and after thoroughly wetting and soaking it, rolled as before until the whole became compact. For our top course we put on a course of about a couple of inches of stone chips and dust, thoroughly wetting and rolling until the whole mass became as solid as cement, and making, as you are aware, one of the finest pieces of road that could be built. I may mention that before this was done this piece of road had given us very great trouble in the spring and fall, on account of it being so wet, and at times it was almost impassable for heavy loads, although for years we had been continually putting on gravel through the summer, and scraping it off as mud in the spring. I also wish to mention that after making one excavation we put down twelve-inch tile drains on each side of the road with catch-basins to take off the surface water, and the consequence has been that our citizens were so well pleased with the smoothness and durability of the road after a winter's traffic had passed over it, that when the spring came and they saw that there was not the slightest rut or damage done to it, they decided to at once go on and build the balance of our two main streets in like manner. A by-law was submitted to the people for \$10,000; it carried, and we have now nearly completed the work and I have not the slightest doubt that when finished we shall have two as good roads as there are in the country. This has been done under the general rates, on account of the two roads being leading thoroughfares, but in the future we may alter this and have our side streets done on the frontage tax plan. One thing however is certain, we are so well satisfied with our experiment that we shall never go back to the old plan of hauling dirt and gravel on our streets.

You will also remember complimenting us on the appearance of our sidewalks in comparison to the state of our roads as they then were. For the last seven or eight years we have entirely discarded the use of laying down wooden sidewalks. A by-law was passed that all sidewalks should be laid on the frontage tax plan, and giving the citizens the power to petition the town council to lay down artificial stone walks instead of the plank walks. This has been availed of to such an extent that we have now over twelve miles of silica-barytic stone sidewalks laid down, and we are adding to it year by year. We have done it on the plan of the town assuming the cost of all crossings and intersections, and in the case of corner lots, the town pays the cost of one-half of the depth of the lot, the property owner paying the rest. This has been done on the principle of twenty year payments divided equally, so that the burden should not come too heavy. So that as you are aware, we are not only adding to the appearance of the town, but I am sure that the durability and the permanent character of the work will result in a large saving in the years to come.

BROCKVILLE.

Brockville has twenty-five miles of streets, 10.65 of which have received a coating of broken stone, hard limestone and granite being plentiful. The work on the streets is performed by day labour under the immediate supervision of an engineer and a street inspector, and directed by a Board of Works. In 1896, \$5,974 was spent on roadwork; \$3,410 on sidewalks; and \$50 on bridges. Respecting the street work, the Mayor, Mr. D. W. Downey, says:—

A portion of our Main Street extending over 2,000 feet was a good deal cut up by the making of numerous water, gas and sewer connections previous to the laying of permanent sidewalks and during next summer we contemplate putting in a better class of roadway on this street, than any we have had hitherto, and hope to profit from the admirable address you delivered here on the subject of "Good Roads".

A very great improvement has taken place in our streets since we obtained the use of a grader, steam road roller and other implements, and considering the great saving of labor thereby, I am often greatly surprised that our adjoining townships have not availed themselves of their use.

As you are aware we only commenced putting down permanent sidewalks during the past summer. These extend to 3,150 lineal feet, covering in all 30,750 superficial feet, at a cost of \$6,200.

NIAGARA FALLS.

The Council did not set apart any but a very small sum for actual street improvement during the year, on account of the construction of the sewers which have been under way; however, they purchased valuable road machinery, including a fifteen ton steam roller and a new grader, and have now in contemplation the purchase of a rock-crusher.

There have been, however, several streets, on which sewers were built last year, which have been macadamized and improved during the past year. The length of street thus improved would not exceed half a mile, but the work which has been done on these has been entirely satisfactory.

We expect next season to go extensively into macadam pavements, and probably some brick on business streets. There are arrangements being made at the present time to provide for the construction of cement or brick walks on an extensive scale, but nothing definite can yet be said in this regard. We have laid none during the past year.

We have obtained a large amount of rock from the sewer construction which is fairly suitable for road metal, though a little soft. This we will have broken during the winter by hand, to give our town people employment.

Whatever work we have had done by our road machinery, has given entire satisfaction and the only thing which now remains to obtain good roads here, is attention to the work by an energetic committee, and the employment of a first-class foreman who understands road construction in its details.

(Signed) CHARLES H. MITCHELL,
Civil & Hydraulic Engineer.

LONDON.

The following extracts from the annual report of the Street Commissioner of London, Ontario, Robt. Ironside, to A. O. Graydon, City Engineer, contain information of an instructive nature :—

Sidewalks. I would recommend that the specifications for walks be amended so as not to allow any planks of a less width than eight inches to be used as it has been found that the lesser widths contain sap and are therefore not fit for sidewalk purposes.

I would also recommend the appointment of an inspector for this work as it is impossible at present to make a close inspection of the material as it is brought on the work, and should a contractor so desire, it is an easy matter for him to substitute inferior scantling between the visits of the Street Commissioner and the Engineer. The presence of an inspector would be a safeguard against this and other devices contractors are prone to practice in the carrying out of City contracts. The contractors for this work for this year have given general satisfaction and there is little or no complaint about their work or material used, but considering everything, I think an inspector necessary.

As in the past the repair of walks is one of the principal expenditures of this department and while careful supervision has been exercised and repairs made when necessary, yet there are several sidewalks throughout the city that are in a positively dangerous condition and almost beyond repair. In the repairs to sidewalks this year we have used 115,000 feet B. M. of lumber and 8,000 pounds of nails. This includes the material used in relaying old plank saved from old walks that were taken up and replaced with a new walk and relaid in stretches where there were no sidewalks. About 1,200 feet of this class of walk was laid. From the foregoing it will be seen that the "*Plank Sidewalks*" are by no means of a permanent nature and the constant adding each year to our already large mileage of plank sidewalk, together with the annexation of London West, means added increase of expenditure for this work each year; and with a view of keeping down this expenditure, and at the same time putting our sidewalks in an up-to-date condition I would recommend that permanent walks only shall be laid on our principal streets and avenues.

Pavements. The superiority of broken stone roadways over the ordinary gravel roadways is apparent and more of this work should be done each year. There would then be no necessity for scraping off in the spring, the material that was placed there the year previous. I would therefore recommend that less gravel and more broken stone be used in the future, throughout the city; then good roads, not mud roads, would be the rule and not the exception.

The cedar block pavements are, as a rule, in poor condition, some are even beyond repair, and nothing can be done except replacing them by another and better class of pavement.

The asphalt pavement is in fairly good condition, and with the exception of that portion along the street railway tracks, seems to be wearing well. The great objection to this pavement is its slippery condition at times. The cause of this is the too frequent use of the watering cart. Washing down should be substituted as is done in other cities. Washing down thoroughly cleanses the pavement, while on the other hand the moisture of water from the watering cart and the dirt on the pavement renders it dangerous to both drivers and wheelmen, and at the same time destroys the life of the pavement.

A constant source of repairs has been caused by the delivery of material to new buildings and the removal of debris from their site. The teamsters engaged in this work drive over our sidewalks regardless of consequences. The ward foremen have received instructions to make crossings for them but as a general rule the damage is done before their arrival on the scene. I would therefore recommend that a deposit be collected from all who obtain building permits and the amount of damage deducted therefrom.

Culverts and Sewer Repairs. The culverts throughout the city have received the necessary care and attention called for, and every effort has been made to keep them in repair and in a state of cleanliness. The use of the tile pipe in replacing the old wooden culverts is gradually doing away with one of the greatest sources of annoyance and worry to this department. There are several of the present wood culverts and bridges that demand an almost constant supervision, and these should be replaced with structures of a permanent nature without further delay.

Stone Crusher. This machine has, with but few delays, been in operation since May 26th up to December 1st. During that time it has crushed nearly 1,350 cor's of stone. The actual crushing time was 110 days, nine hours each, about twelve cords per day.

The cost of the stone, the crushing expenses and the delivery upon the streets, including repairs to crusher, rent of engine, fuel, etc., was about \$5.60 per cord.

This is about \$1.15 per cord cheaper than we can purchase it already broken from the quarries. The crusher has been removed to the storage yard and is now under cover. I would recommend that this machine be overhauled as soon as possible and some of the old parts that are worn out, replaced with new ones so that it will be ready to start operations as soon as possible.

Gas and Water Trenches. There has been little or no improvement in the manner in which gas and water trenches are left by the employees of those departments or companies. No care whatever is taken to preserve the road metal so that it can be replaced again on the surface, but with the utmost carelessness and indifference it was thrown into the trench in any manner, and at times they have been left in a most dangerous condition. Effective means should be taken to stop this practice. There should be no difficulty whatever in refilling in a proper manner such shallow trenches as these, leaving the roadways in as good a condition as formerly.

Sweeping Block Pavements. These pavements have been swept as regularly as the funds would permit, but owing to the unevenness of the surface they are never in a state of cleanliness. The crevices, even after the most thorough sweeping, retain the horse droppings and other filth, and on a warm day, especially after watering carts go over them, the stench arising from them is, to say the least, simply awful. The health of the city requires that steps be taken to replace these pavements, especially in the heart of the city, with asphalt or brick. They are beyond repair, and it is practically impossible to keep them clean.

Street Watering. In concluding this report, I would again call your attention to the manner in which the street watering through the city has been done. Good roads have been built, well rolled with the steam-roller and left hard and smooth, and in a first-class condition. In a short time, owing to the excessive quantity of water put upon them by the contractors for street watering, they have become soft and are easily cut up into ruts.

Unless a change is made in the mode of street watering, money is only thrown away in improving the roads. The street watering inspectors either cannot or will not make the contractors do the work as it should be done, and personally I have repeatedly stopped the drivers of the water-carts from watering mud puddles. If this work was placed under the control of this department and a good foreman and teams employed, I have no hesitation in saying that little or no complaints would be made and the street watering would then be done in such a manner that no injury would be done the roadways, and at the same time satisfy the ratepayers who pay for the work.

ST. CATHARINES.

About \$6,000 is appropriated yearly for repairs, street cleaning, repairs to sidewalks and some small pieces of permanent improvement, under the control of Board of Works and street superintendent.

A street commissioner is employed during the whole year at a salary of \$650. We have this year purchased a grading machine and also a steam road roller and hope to have a stone crusher early next year so that we will be able to follow out Mr. Campbell's instructions.

We have laid this fall about three miles and a half of concrete walk under the local improvement plan which we hope to continue another year.

KINGSTON.

The frontage tax system has been adopted on sewers, and on granolithic and asphalt sidewalks; for sewers in 1835, for walks in 1893.

The City Council Committee, called the Board of Works, has control of all streets and the City Engineer has full oversight. The City Engineer is employed at a salary of \$1500.

The streets are in fairly good order but for the last two or three years the appropriation has been too small and the streets are beginning to run down. We used to get \$25,000 but only get \$10,000 or \$12,000 now and the most of this goes in street cleaning, etc. We were getting up fine macadam streets till the reduction. We have about two and one half miles of granolithic walks, one mile of asphalt and the balance plank. Our chief trouble with our streets is the frequent digging of trenches for sewers, water and gas pipes, and we never can keep a street in proper shape.

NORWICH.

During spring and fall the condition of the streets is bad. Only two main streets are gravelled, and as they lead to railway stations as well as being main roads from the country, the heavy teaming over them cuts them badly. Other streets are only earth roads, graded. About five years ago nearly \$1,000 were spent for graveling these main streets, and patching has been done since. Most of expenditure for streets has been for sidewalks recently.

Although our rules of council require committees to report in writing stating work completed, location and cost of same before expense is incurred, they do not adhere to that rule. So that because of this lack of system the figures given are approximate, but are probably sufficiently correct to answer your purpose.

Our gravel costs fifty cents per cord, and \$2.50 for drawing and laying—and not very good at that. For the past three or four years we have used a good deal of brick-bats, which we get from the brick yards within two miles of the village. They make a good road and appear to wear well. The past summer about a quarter of a mile of Main street was coated with these and covered with gravel. The bats are, of course, broken with a hammer to about two inches in size.

STAYNER.

The way the work is done by the Commissioner and the manner of paying accounts is such that I cannot give definite answers.

CLINTON.

The general condition of the streets is not good, but since the purchase of the road machine a number of the streets have been graded and made narrower.

COBourg.

Streets are in better condition than they have been in for some years, some of the gravel roads have been broken up and regraded and are very fine.

Arlsa Craig.

The streets were put in excellent permanent repair fifteen to twenty years ago and have only needed patching since that time, with an occasional coat of gravel on the heavily travelled streets.

WINGHAM.

A greater effort has been made this year to improve the streets than formerly. Three miles of Portland cement or granolithic sidewalks are now built. We expect to build a large quantity next year. Had we a stone crusher along with our road machine we would have good streets.

LONDON.

We are using stone and cedar curb and narrowing up streets, and putting down macadam roads as much as possible, using broken stone, underdraining them and rolling with steam roller 12½ (long) tons empty. Plank, asphalt, granolithic and stone are used in the construction of sidewalks.

PICTON.

Money is taken from the general funds of the corporation and expended on roads, walks, bridges and culverts where required "or where the indicator denotes the biggest pull."

We require the appointment of a qualified practical man independent of the bias of the wire pulling always in vogue in councils, to superintend the work on streets, walks, bridges. Such a system would be far better for this, and, I believe, for every municipality; and would be a benefit from a monetary standpoint.

LINDSAY.

Some streets in fair condition, others bad. It is the intention of the council to improve streets in accordance with your ideas.

WATFORD.

Fairly good improvement is being made in the streets. The only special effort was two years ago when we bought forty car loads of gravel from the G. T. R. Co. and had it taken from Komoka gravel pit.

We need more uniformity in our streets, some of the councillors grade too high and narrow and we need a good instructor.

SAULT STE. MARIE.

Streets are not in good condition. None of them that do not cut up in wet weather, though of course their condition is improving every year. It is not the result of any good system though. The worst portions are improved each year, but no systematic attempt.

LAKEFIELD.

About three miles of grading was done in 1896-97, and broken stone was used, the average depth of stone being fifteen to eighteen inches. About 800 yards of stone was also used to fill up a wharf owned by corporation and to grade it uniformly with adjacent street; part was done in December, 1896, the balance in January, 1897.

The roads are first-class; nearly five miles have been covered with stone from the Trent Valley canal during the past two years. Stone or tile culverts has replaced wooden ones, streets graded to uniform level and water tables cleaned and improved. About \$1,000 has been spent in 1897 on roads. Stone was delivered to us free by canal contractor and was broken by contract about six inches deep at 1½ cents per square yard.

Our Road Master is employed by the year and has other duties to attend to for which he receives \$350 per annum.

TILSONBURG.

Steps are being taken to improve the principal street called Broadway street, and the Provincial Instructor's assistance and advice will be asked by the council.

FORT ERIE.

Plank has always been used for sidewalks until this season when granolithic was adopted and about 400 yards laid down. If satisfactory it will be adopted entirely for the future.

EXTRACTS FROM TOWNSHIP REPORTS.

MACHINERY MAY BE PURCHASED.

Adjala, Simcoe Co.—Special effort has been made recently in that gravelling has been started and the purchasing of road machines is under consideration. I believe that if the money could be judiciously expended, and thirty or forty cents a day charged, we would have better roads than under the present system of statute labor.

TILE CULVERTS SATISFACTORY.

Adelaide, Middlesex Co.—Our leading roads are gravelled and are as good as you will find. Culverts are mostly of tile, which give good satisfaction. The tile are manufactured in the municipality.

WOULD LIKE TO COMMUTE.

Ancaster, Wentworth Co.—The council would like to be in a position to commute a part or the whole of the statute labor as deemed advisable.

DOING BETTER WORK.

Arthur, Wellington Co.—The general condition of the roads is fairly good. The county road is good. The most of the roads in the township are fairly well graded and a great deal of gravel is being put on every year by statute labor. The people are getting interested in the matter and are doing better work. It costs the township close on \$300 each year for gravel. They pay six cents per load for statute labor work and ten cents per yard for contract work. This is for gravel alone. Hauling is extra.

BECOMING INTERESTED.

Ameliastburgh, Prince Edward Co.—The roads are in a very good condition ; gravel beds are plentiful in many parts of the township and the inhabitants are becoming interested in the roads and bridges of the municipality.

The doing away with the present system of statute labor and replacing it with a tax on the ratable property of the township, the money to be applied under the supervision of the best available overseer, would be more in the right direction, but public opinion moves slowly, sometimes.

NO SYSTEM.

Albemarle, Bruce Co.—Our roads are very bad ; we have not two miles in the whole township that could be classed as really good. Our roads are constructed without any regard to plan or system, and in consequence the improvement will be slow.

IT WOULD PAY.

Amabel, Bruce Co.—Statute labor as it is worked in our municipality is a complete farce. If the municipal council would only levy twenty-five cents per day (even 25 cents) and appoint a Road Commissioner, it would pay every time.

A STONE CRUSHER.

Bertie, Welland Co.—The road machine is constantly in use during the road-making season. The general condition of the roads is not good. Efforts are now being made to secure a stone crusher.

IN DEBT FOR RAILWAYS \$7,000

Bexley, Victoria Co.—The general condition is not to say bad. Pathmasters draw out every summer gravel where it can be got handy, or do some grading with plows. However, this time of year (fall) the roads appear as bad as though there were no work put on them in the summer. If stone could be utilized, there is any quantity of lime stones close along the roads. I have never seen any of them drawn on the travelled roads and broken up. It is always gravel, or turnpike with the plow. The hammers we got for breaking the tops off the stones striking up too high. The township is \$7,000 in debt to railways and cannot spare much money to lay out on roads. This year (1897) hardly anything has been laid out.

DO NOT GRADE MORE THAN CAN BE GRAVELLED.

Blenheim, Oxford Co.—The grader has been pretty steadily at work in 1896 and 1897. We do not care to grade, however, more than we can gravel, as graded roads become very bad roads in time of rain. The roads are fairly good. The outlay on bridges each year is very heavy. We have two large streams winding through our township causing us to keep up over fifty bridges of large size, besides many small bridges.

SHUT OUT FROM MARKET.

Binbrook, Wentworth Co.—We feel very much the need of a good road metal. The road leading through the township is so bad that we are practically shut out from market with heavy loads. The difficulty of procuring stone makes it almost impossible to better ourselves.

MORE SUBSTANTIAL WORK.

Brudenell, Renfrew Co.—Roads are rough and in a poor condition generally all over the municipality. We have no means to improve them except the statute labor and some small grants by council to repair places that sometimes become dangerous to public travel, and sometimes a colonization grant, but these grants are becoming fewer every year. The main reliance is statute labor, and this, in a great many cases, does not amount to much, and sometimes is done with a great lack of judgment. In my opinion it would be far better to levy a commutation tax at about fifty cents per day and do away with statute labor. There would be more substantial work done.

A ROCK CRUSHER NEEDED.

Burleigh and Anstruther, Peterborough Co.—The statute labor system is bad, but the settlers are too poor to remedy the evil at present. If more broken stone was used it would be better. This country here requires a rock crusher which could be utilized along the main roads on the spot.

THE ROADS WOULD BE BETTER.

Barton, Wentworth Co.—It is the opinion here that roads would be better kept if statute labor by commutation or otherwise were abolished and roads kept in order out of the general funds of the municipality. All stone roads belong to road companies. There is plenty of stone in this township and of good quality.

AN EFFORT TO IMPROVE.

Beckwith, Lanark Co.—Some of the roads are very good and some are very bad, but an effort is being made to improve them by grading and stoning with stone from hammers and stone crushers.

CONSIDERABLE AGITATION.

Brant, Bruce Co.—There has been no special effort to improve the roads other than by county grants which had to be supplemented by an equal amount by the township this year (1897.) Considerable discussion and agitation pro. and con. has taken place regarding better roads. We considered it advisable to place literature on the subject of road-making in the hands of the pathmasters soon after their appointment. This should be done early, say April.

THE TOWNSHIP MAY BUY.

Brighton, Northumberland Co.—The township has not ventured as yet to buy road grading machinery, but it is possible they may do so before long, if from machine work in neighboring township they are thought to be a saving in labor and a better mode of making and repairing roads. Then the township may buy. It is hard to apply statute labor on machines, and the time to force commutation and hire all road work has hardly yet come in this community, although some think this would be a gain.

TALK THE MATTER UP.

Chapman, Parry Sound District.—The roads are in a very bad condition, especially during the months of April, October and November. There are millions of tons of rock (granite and feldspar) in this and adjacent townships. If we had some good man to talk the matter up, the ratepayers who are afraid of higher taxation might demand the purchase of a rock crusher by the Council.

THE RATEPAYERS ADMIT IT.

Charlottetown, Norfolk Co.—The roads are passably good and improving slowly all the time. Most of the roads in this township are sandy roads and very heavy in summer, some parts have been clayed and keep in pretty fair condition. In the vicinity of the gravel beds the roads are improving, but gravel being so scarce very few pathmasters can make use of it. The old way of performing statute labor I consider a fraud, the system is *passe*, much of the labor is wasted under the present system—the ratepayers admit it.

FOR FORTY CENTS A DAY.

Chinguacousy, Peel Co.—The general condition of the roads is first class in dry weather, but bad in spring. Statute labor is a fraud, more work and better roads could be made for forty cents a day.

THIS LAST TWO YEARS.

Carnarvon, Manitoulin District.—The roads are good in dry weather being dirt roads. They cut up in the fall. Our roads are getting better since we have been using the machine this last two years.

SNOW DRIFTS.

Charlottenburg, Glengarry Co.—For some years a good deal has been done to substitute wire fences for timber fences on north and south roads and other roads subject to snow drifts. The effect has been very good not only for winter travel, but also in making a dryer and firmer road in spring and early summer.

IT IS THE NARROW TIRE.

Culross, Bruce Co.—We have pretty fair roads in dry weather. In spring and fall they sometimes get very bad, especially when freezing and thawing. There has been a little more than usual spent on the gravel roads by the county this year. I think that if we could get the wide tire and the unequal axle introduced it would do more to improve our roads than anything we have had yet, or rather save them from destruction, for it is the narrow tire that cuts them up in soft weather.

ONE DRAWBACK.

Colchester North, Essex Co.—There is one great drawback we labor under—that is the failure to get the statute labor performed early in the season so that grading and gravelling has time to settle before our autumn rains.

BY NOT GETTING THE WATER OFF.

Calvin, Nipissing District.—The roads are rutty with steep grades. The pathmasters put in most of their time trying to ditch and crown the roads. Some succeed, more make them worse, if anything, by not getting the water off, still all do the best they know how with what time they have. Twenty miles of these roads have been made by grants from the Government, the Mattawa and Caladar and Shield Pit roads, but there is hardly sufficient statute labor to keep them in repair. Where the ditches carry off the water the roads are passable, (a piece of good and a piece of bad road sandwiched.)

SUPPLEMENTARY LABOR.

Caradoc, Middlesex Co.—The roads are very good. The Council, for the improvement of roads, has adopted the system of supplementary labor, viz., where a road division puts on a certain amount of gravel over and above statute labor, the council supplements the amount. Some are in favor of doing away with statute labor and having the money raised by commutation, say at fifty cents, expended by a road commissioner, while others oppose anything like increased taxation.

A PUBLIC HOLIDAY.

Christie, Parry Sound District.—There is so much rock and swamp in this district that it is and always will be impossible to make good roads, but they can be vastly improved. One of the first steps would be to abolish statute labor which is simply a public holiday.

WATER FOLLOWS THE WHEEL TRACKS.

Cardiff, Haliburton Co.—The roads are in a fearfully bad state all over the township. Water follows the wheel tracks washes out the dirt leaving the stones and boulders for wheels to climb.

IT IS THE PREVAILING OPINION.

Clarke, Durham Co.—The roads are generally good. There is a general improvement made during June and that has to do, unless some part is washed away by floods, and pathmasters have to see that they are repaired. It is the prevailing opinion that the roads are not in as good a condition as they should be considering the amount of statute labor expended or employed thereon and money spent.

EFFORTS HAVE BEEN MADE.

Clarence, Russell Co.—Roads are in a pretty fair state and efforts have been made to improve them still more. We had our road machine rather late last summer and it is expected that more improvements to roads will be made next year.

THE OLD SYSTEM.

Cornwall, Tp., Stormont Co.—The roads are not good, nothing outside of the old system of statute labor is employed.

DISCUSSING ABOLITION.

Derby, Grey Co.—The Council are contemplating the purchasing of improved, up-to-date machinery and are also discussing the abolition of statute labor.

A GREAT IMPROVEMENT.

Dunn, Haldimand Co.—The roads are pretty good. The road grader makes a great improvement on our roads.

SOME LEADING ROADS.

Dummer, Peterboro' Co.—The general condition of the roads is fair, some of the leading roads extra good. Five road divisions are kept up by \$500 road appropriation besides about nearly 700 days statute labor yearly.

THEY SHOULD BE ELECTED.

Dorchester, North, Middlesex Co.—The roads are not very good. There has been \$1,000 expended for gravel on roads that had none on before. The Council of 1897 if elected for 1898, I think, would favor buying a grader.

MORE REAL WORK.

Dumfries, North, Waterloo Co.—It is likely a road machine will be purchased this season and then more real work will be done on the roads.

ABOUT THREE THOUSAND DOLLARS.

Dereham, Oxford Co.—No special effort has been made to improve the roads this year. About \$3,000 per annum is usually expended for the improvement of roads.

BY CONTRACTS.

Downie, Perth Co.—The roads, especially the leading ones, are fairly good, being kept up by township grants and gravel put on by contracts. The other roads are repaired and gravelled by statute labor, while some road beats, which have far to haul, get grants from the Council. We have a large quantity of gravel, some road beats being close to pits, thus the roads can be kept and even made by statute labor; other beats have as far, in some cases, as five miles to draw gravel. What we want in this township, in my opinion, is a stone crusher, but as yet the Council does not see their way clear to buy one.

THE RAILWAY BONUS.

Eldon, Victoria Co.—This township is burdened with a debenture debt incurred in bonusing a railway and until that debt is paid off our Council, fairly representing the feeling of the rate-payers, are averse to expending much money on roads, more than is absolutely necessary to keep them in safe condition.

SHOULD BE COUNCILLORS ONLY.

Eramosa, Wellington Co.—No special effort and with yearly statute labor, and usual annual grants, the roads seem to be running down. There seems a general willingness to start out afresh, purchase road machinery, appoint competent road commissioners, etc.

I feel satisfied it would facilitate improvement if Councillors were either debarred from acting as road commissioners, or else only legally acting, when appointed by by-law, the same as any other officer. I think the law is defective; the Council as Councillors only should be elected by popular vote.

PROPER VALUE FOR THE EXPENDITURE.

Easthope North, Perth Co.—The general condition of the roads is "fair to middling," a great deal better than they were fifteen years ago, but not so good as they should be. There is no system of road-making, and consequently there are many different "styles and shapes" of roads. Every pathmaster will have his own way. No special effort made but there is gradual improvement. The whole system of statute labor should be abolished and then, and not till then, will roads assume their proper shape and condition and proper value be gotten for the expenditure made thereon.

GRADING AND GRAVELLING.

Easthope South, Perth Co.—This year (1897) the road grader was used over all the township roads, rounding off the road surface, grading lower parts higher, and the newly graded parts were gravelled.

A QUALIFIED PERSON.

Egremont, Grey Co.—In some localities the roads are fair, according to the skill of the pathmaster ; in others they are simply horrible. No special effort has been made to improve them, and I can see that legislation on this subject is imperatively necessary. I am fully satisfied that if we paid 40 cents a day and had this money expended by a qualified person we would soon have superior roads. I think the general feeling is that way and our present system is now considered a relic of bygone days.

HEAVY LOADS, NARROW TIRES.

Fredericksburgh South, Lennox Co.—The cheese factories open early before the roads become dry ; with heavy loads and narrow tires the roads are badly damaged.

IT SHOULD BE SUFFICIENT.

Glenelg, Grey Co.—The statute labor should be quite sufficient to put the roads in good shape, but many of the pathmasters are very careless in the performance of their duties, and think if they put in their time that is all that is necessary.

A GREATER INTEREST.

Gwillimbury East, York Co.—A greater interest is being taken which will doubtless lead to more systematic action.

THE MACHINE KEPT AT WORK.

Gwillimbury West, Simcoe Co.—The purchase of a road machine at a cost of \$250 was a special effort in 1897. The machine was kept constantly at work for two months.

OUT-OF-DATE.

Goulbourne, Carleton Co.—Every corporation should have two or more road graders, and if the out-of-date statute labor system was done away with and the road graders substituted the change for the better would soon be realized.

AWFUL IN SPRING.

Gosfield North, Essex Co.—There are about ten miles of natural gravel ridge, one of the best roads in the Province. About twenty more are gravelled and in good condition ; the balance are clay roads, fair in continued dry weather, but awful in spring and during a wet fall.

NEED A ROAD TO MARKET.

Gower North, Carleton Co.—Our leading roads some years ago were gravelled, but they have sunk down and become too flat. Since we got our grader we are covering up the old gravel and starting in places to gravel again the next year after grading them. If I might be permitted I would just like to suggest the propriety of the Government amending the Municipal Act in such a way as to make townships lying next to a city give something of a decent road through their territory to the city or market. We are living about twenty-three miles from the City of Ottawa, and I might safely say that about six or seven miles of that road through the Township of Nepean is simply impassable in spring and fall. A very large part of it is in a natural state.

OTHERS DO NOT CARE.

Glanford, Wentworth Co.—Some pathmasters try to keep the road in good condition ; others do not seem to care. I believe the proportion of careless ones is on the increase or has been in recent years.

VERY LITTLE EXPENDED.

Georgina, York Co.—This township has for twenty years (1875 to 1895) been paying a very heavy tax, the result of which was that very little was expended on the highways.

FOR COMMUTATION.

Garafraxa East, Dufferin Co.—The council is going to submit a by-law to the vote of the electors for the commutation of statute labor.

VERY PRIMITIVE.

Glamorgan, Haliburton Co.—Statute labor is nearly all we have to depend on to keep the roads even passable, and the majority of pathmasters' ideas as to roads and the repairing of them are very primitive.

VERY OFTEN DANGEROUS.

Grattan, Renfrew Co.—The greater part of the roads are rough and in the spring and fall they get cut up so that public travel is much impeded and very often dangerous. All that can be done is barely to keep the roads in a sort of repair.

ONE ACRE OF GRAVEL.

Gosfield South, Essex Co.—We have a gravel ridge about one mile long and about sixty rods wide and fifty feet deep of excellent quality. We purchased one acre some fifteen or twenty years ago at a cost of \$300, and we are using it yet.

IN GOOD ROUNDED SHAPE.

Grantham, Lincoln Co.—The roads are nearly all gone over every season with the road machine or large scraper; and I consider the clay roads in good rounded shape.

THE BULLETINS HAD A GOOD EFFECT.

Harvey, Peterboro' Co.—Condition much improved the last two years. The colonization grants from the Government have helped these roads very much for several years past. The little Bulletins which were sent me for distribution in 1896, and which I enclosed in pathmasters' lists, seem to have had a good effect wherever they have been intelligently used by the pathmaster. I would suggest that you cause about fifty copies of the same to be sent for the pathmasters for 1898. It is noticed that the grading, pitch in centre and the width of turnpikes or roadbed is much improved.

STONE CRUSHERS WANTED.

Howland, Manitoulin District.—The council think the Government should give us stone crushers and withhold Government grants to the roads, as stones are plentiful and gravel scarce, and it is the only way to make permanent roads (crushed stone).

WHEN THEY LEARN HOW.

Hallowell, Prince Edward Co.—Our roads in this township are going back and are not so good as they were years ago, but I believe as we learn to use the road machine the roads will improve. Last year we almost spoiled every piece of road we fixed with the machine but this year we have very much improved them. I think we are going to like the machine when we learn how and when to use it, but last fall the people cursed it.

THE PLAN WORKED ADMIRABLY.

Hilton, Algoma District.—The general condition of the roads in this township is good. The municipal council this year handled all commutation moneys and gave out jobs on the different road divisions and inspected the work when completed and found this plan to work admirably. The work was done according to specification and a very much larger amount was done for the same amount of money. Heretofore the pathmasters superintended the laying out of the commutation moneys with very poor results.

RENT THEIR MACHINES.

Hibbert, Perth Co.—The council are employing road scrapers and machines from outside parties to do road work. A considerable amount of road grading was done this year.

ONE MILL ON THE DOLLAR.

Hullett, Huron Co.—One mill on the dollar as a commutation tax payable when other taxes are paid would give us in Hullett about \$2,000 which sum could be expended on the roads in a much more profitable manner than under the present system.

SHOULD BE SCREENED.

Hay, Huron Co.—Under fair conditions the roads are usually good, but during extremely wet weather they get soft and muddy owing to the sandy nature of our gravel.

THEY IMPROVE LEADING ROADS.

Huron, Bruce Co.—A road machine was used this year for about ten days in grading gravel roads. On an average \$1,000 per annum is expended by the township council in improving the two leading gravel roads in the township, viz.: The Saugeen and Sideline Fifteen.

SHOULD BE ABOLISHED.

Hamilton, Northumberland Co.—The roads are not what they should be, but they are as good as can be expected under statute labor which should be abolished.

LABOR WASTED.

Johnson and Tarbutt, Algoma District.—Mud roads, labor wasted for want of road machinery, the council afraid of a little extra tax.

SHOULD BUY A CRUSHER.

Keppel, Grey Co.—We have limestone suitable for crushing all over the township and scarcely any road is over a mile from some. There are also granite boulders scattered over a large part. There is talk of buying a stone crusher.

THE RIGHT WAY.

Kincardine, Bruce Co.—The roads are in a fairly good condition. The grader has been used in rounding up the gravel roads by gathering the gravel in the centre and cutting off the sides and putting it in the opposite direction. The roads are made a good deal narrower than formerly, especially on hills, in order to prevent water from running down the centre of the road.

FOR SOMEONE ELSE.

Lochiel, Glengarry Co.—I would say if our council would assess the ratepayers fifty cents per day or less for statute labor and expend that on the roads, the improvement in the roads would be seen in a very short time, as most of those who perform statute labor think they are working for someone else and not for themselves.

STARTED IN THE RIGHT DIRECTION.

Front of Leeds and Lansdowne, Leeds Co.—The roads are bad. Since the road grader was purchased a very decided interest in making good roads has been awakened throughout the township. Your literature this spring was very helpful in getting road-making started in the right direction. The small pamphlets issued by you in time to be sent out with the road lists are much appreciated. The council are contemplating the purchasing of a stone crusher and road roller but are not fully decided as to the crusher thinking they may be broken by hand at a less rate.

LOANED BY THE COUNTY.

Luther East, Dufferin Co.—The township got the loan of a road machine from the county and have been doing a considerable amount of grading with it.

THEY WERE TOO HARD.

Loughboro', Frontenac Co.—In the name of common sense how does the department expect a clerk to be in a position to answer all these questions? It's absurd.

LEVY ON THE ACREAGE.

McGillivray, Middlesex Co.—Would suggest that the Legislature enact that any municipality may levy statute labor on acreage in place of assessed value, as the vicinity where land is lowest assessed generally has the worst roads and least statute labor to perform.

STATUTE LABOR ANSWERABLE.

McKellar, Parry Sound District.—Statute labor system is answerable for all the bad roads in Ontario. Most of the work done is an injury to the roads.

TOO MANY PATHMASTERS.

McKillop, Huron Co.—The general condition of the roads is good. The last two summers a grader has been hired to round some of the old graded roads. In repairing the roads I think the gravel is put on too deep and it is not packed until late in the fall and the sides are badly cut up before the centre is packed. I think a statute to only allow not over twenty pathmasters and pay them would be a good law.

WOODEN CULVERTS AND BRIDGES.

Macaulay, Muskoka District.—The culverts and bridges have been mostly constructed of wood and are all crumbling to pieces and will cost a large amount to re-build.

AS MUCH HARM AS GOOD.

Mountain, Dundas Co.—Statute labor is in some parts faithfully done, in some places it does about as much harm as good. Roads are made wide and flat and soon are full of holes. Many roads are insufficiently drained.

COMMUTATION AT ONCE.

Mara, Ontario Co.—I believe forty cents for statute labor commutation would do more work than working the labor in the usual way and I would make commutation of statute labor compulsory at once.

STATUTE LABOR HAS BEEN ABOLISHED.

Malden, Essex Co.—Statute labor has been abolished in this township. Every year the township spends about \$1,400 on the general improvement of the roads. We have been getting some crushed stone by going about three miles for it, but now the crusher is in the courts and we fear will not be operated again. Gravel is scarce and of a poor quality. The land in our township is very flat and it requires an enormous amount of ditching to get the water off quickly. Stock is prohibited from running at large. There are about sixty-five miles of road maintained by the township, of which twenty-five are metalled with either gravel or broken stone.

IF CASH COULD BE HAD.

Machar, Parry Sound District.—If cash could be had in place of statute labor I am persuaded that much better roads could be made for the expenditure.

A BIG IMPROVEMENT.

Mariposa, Victoria Co.—We have fairly good roads in most places, as the bulk of them have had some gravel, and the main roads several coats of gravel. I think there might be some improvements in the manner of grading roads as they are too flat on top in most places. The new road built in Lindsay, under, I presume, your instruction, will be a good lesson for this township, as one half of the people passing from this township into Lindsay will pass over that road, and I can testify that it is a big improvement over the old road. The Council spent \$675 in purchasing gravel pits last year but that was exceptional.

THE MAIN ROADS NEED GREATER CARE.

Melancthon, Dufferin Co.—The travelled roads in this township are fairly good, but we would require about thirty miles of road which lead to Dundalk, Corbetton, Melancthon, and Shelburne railway stations, to be given extra care in the way of building, so that the road would be able to bear the heavy traffic.

IMPROVING.

Mono, Dufferin Co.—Between \$1,500 and \$2,000 is expended on the roads each year besides the statute labor. The Township of Mono is rather rough and any amount of hills, and it is hard to keep the roads in good order, but they are improving every year.

IS IT THE SOIL?

Maidstone, Essex Co.—The general condition of the roads is fair in a favorable season. In spring and fall, however, owing to the nature of the soil, we seldom have good roads. In those seasons even our gravel roads get considerably worked up, and this makes travel over them disagreeable.

THOSE WHO WORK HAVE TO PAY.

Monck, Muskoka District.—In some places statute labor is performed faithfully, and in other places it is not. The council has generally to expend an amount of money equal to fifty per cent. of the statute labor. Statute labor is every year getting more of a farce, and those who work faithfully have to put their hand in their pocket to pay for those who do not work faithfully, causing a great deal of discussion at nominations. Some here think that statute labor is only the relic of the old feudal times as they think they are not working for themselves but for the Queen.

BEGINNING TO SEE IT.

Minto, Wellington Co.—Fairly good, but too much gravel has been put on before the roads were properly graded, the consequence is the roads are too flat on top and retain the water in the spring and fall. I firmly believe a great deal more work would be done, and of a more uniform quality, were the statute labor commuted even as low as sixty cents per day, and many are beginning to see it in the same light.

MORE IMPROVEMENT THIS YEAR.

Medonte, Simcoe Co.—The road machine was kept in use for some time this year and gives satisfaction. More improvement was made in the roads this year than in any previous year, owing to the use made of the machine; and after grading, gravel was applied where needed. Some good substantial pieces of road were made this year.

THIS IS THE SYSTEM.

Monaghan North, Peterborough Co.—The roads are in a fair condition. A very decided effort is on foot to give good roads. We have commuted statute labor at fifty cents per day, and divided the township into three districts, each having a commissioner who supervises all work performed outside of grading which is done by a permanent gang upon the grader, under the foreman who is the operator and directed by the Council. This is working well and in another year, when the people see the results, it will be a great improvement upon the old system.

OWNED BY A PRIVATE PARTY.

Mornington, Perth Co.—There is a road machine owned by a private party in the township and the council employs it for making the roads.

A COMPETENT ENGINEER.

Norwich South, Oxford Co.—Have had a good deal of drainage done under a competent engineer in the past few years, which has a tendency to keep the roads dry, as the water gets away more quickly. In one village, which is incorporated, the road work has been commuted for three years, and the ratepayers find that they get better roads, better jobs, and more work done at a less amount than formerly under the old statute labor.

MORE ATTENTION WILL BE GIVEN.

Nottawasaga, Simcoe Co.—The general condition of the roads is fairly good. Our township is very rough and hilly, and consequently the roads are very hard to maintain, culverts being carried away by freshets, and huge washouts formed on the face of hills. Our roads, notwithstanding, are gradually improving. This is more apparent since the purchase of the grader. There has been no special effort at improvement recently as the township has heretofore been burdened by a railway debt; this being now paid, more attention will be given to roads.

THE ROADS HAVE IMPROVED.

Niagara, Lincoln Co.—The road machine is worked every summer and is very good for grading. The roads have improved since we commenced using it.

ONE-THIRD FASTER.

Oxford East, Oxford Co.—I think statute labor should be done away with, and collect fifty cents per day and let jobs. The road would then improve one-third faster than by statute labor.

POOR, AND GOOD REASON FOR IT.

Pelham, Welland Co.—The roads are poor. Many of them too wide and too flat. When gravel is hauled that which loads the easiest and hence the poorest is taken. The road machine is useful, but there is no uniform system of operating it, each overseer managing it on his own beat. Roads are not, in many cases, put in proper shape before being gravelled. Often they are flat or even hollow in the middle when gravel is dumped in to fill the centre.

CREEK GRAVEL IS BEST.

Plympton, Lambton Co.—Special effort has been made by the purchase and more extensive use of road graders; and more creek gravel used than ever before, as it is found to be very much the best.

LIKE ALL OTHERS.

Peles Island, Essex Co.—Stoney ridges, with thin stones, one to six inches thick, and any quantity of stone below that, are conveniently situated all over the Island. The west shore has good clean gravel. Pathmasters are like all others—statute labor is a farce. Some pathmasters allow the work to be done late in the fall.

BETTER TO COMMUTE.

Portland, Frontenac Co.—We would like you to give us a talk on road-making. We think that it would be of great benefit to our roads and road-makers. We believe it would be better if it was compulsory to commute the statute labor if at no more than twenty-five cents per day.

THEY WANT GOOD ROADS.

Rawdon, Hastings Co.—The Municipal Council pays for the gravel and thereby encourages the making of permanent roads. Some of our concession lines equal the county gravel roads. Generally speaking nearly every ratepayer wants better roads.

MONEY THROWN AWAY.

Roxborough, Stormont Co.—We need a change in first preparing the roadbed properly and by having good ditches on both sides of the road. Much money has been spent for years, in fact thrown away, owing to this neglect.

SINCE THE AGITATION COMMENCED.

Reach, Ontario Co.—The roads are not first-class, although since the good roads agitation commenced an effort is being made to get our pathmasters to adopt a uniform system, but so far without much success. I am glad the department has sent out this circular, as I believe it will influence councillors to have fuller reports from pathmasters and others engaged in road-making to give statistics which heretofore have not been given in rural municipalities.

ONLY HALF A DOZEN.

Storrington, Frontenac Co.—About sixty-seven pathmasters are appointed, and out of that number perhaps half a dozen know how to make or repair a road. If the statute labor was commuted at forty cents a day and that sum expended by a competent man, I think that the roads would be much better than they now are.

MAY DO IN A NEW COUNTRY.

Sullivan, Grey Co.—The statute labor system does very well in a new country, but it is unsatisfactory and detrimental to good roads in the older townships. There is no definite system in making roads by statute labor, every pathmaster has his own plan—the roads do not show the improvement they should for the amount of labor expended. I am confident if statute labor was commuted at fifty cents for each day's labor and competent men and road machinery employed, more and better work would be performed.

TILE DRAINAGE.

Southwold, Elgin Co.—Grading and gravelling is the order in this township according to the number of miles to be kept in repair. The roads are, in general, good. There is considerable tile drainage along sides of road.

SHOULD HAVE A CRUSHER.

Sydenham, Grey Co.—Our roads are in a poor state, not properly graded. We have very little gravel and find stone broken with the hammer very expensive, and when done not a success, owing to not being able to break it fine enough on top for travel.

MORE WORK IS DONE.

Sarawak, Grey Co.—On two roads, 823 days were commuted at sixty cents per day, and more work is done for sixty cents than for one day's statute labor. All statute labor in older sections should be commuted at fifty cents to sixty cents per day, and road butchering done away with by fifteen butchers for 569 days.

IN THE WAY OF DRAINAGE.

Sombra, Lambton Co.—Considerable improvement has been made in the way of drainage under by-law, a large number of drains being located on the highways throughout the municipality.

THE REPAIR SHOULD BE THOROUGH.

Stephen, Huron Co.—We have as good roads as can be found. I advocated better roads twenty-seven years ago, the result was about fifty miles of gravel road. I had charge of most of the leading roads and although they are fairly good, I could now make improvements on my system. We lay out large sums annually to repair our gravel roads but it is not done to my satisfaction; instead of spreading a certain amount of expenditure over the whole, not adequate to properly repair, I would repair as far as the money would go to make a thorough repair.

GRATIS WORK GIVEN.

Sarnia, Lambton Co.—There has been some extra gravelling done this year and considerable gratis work given.

A SAVING OF GRAVEL.

Turnberry, Huron Co.—Good average of country roads. No special effort has been made except buying a new road machine at a cost of \$235.00 and using it on the roads at a considerable extra cost, but at a considerable saving of gravel. These machines should be highly recommended wherever they could be used. Our council believe they will save their cost in one year in improving the roads.

AN EFFORT MADE.

Tecumseth, Simcoe Co.—Roads in bad condition, but an effort has been made to improve them in purchasing a road machine.

GETTING TOO WISE.

Toronto, Peel Co.—I think the statute labor should be done away with. The people are getting too wise to work faithfully on the roads. I think a commutation of forty cents per day expended by commutation would get a great deal more work done and in a great deal more satisfactory manner.

A GOOD TOWNSHIP OVERSEER.

Thurlow, Hastings Co.—The county has purchased a grading machine which is kindly loaned to aid when a large amount of grading is to be done on a township job. We have a good practical man as road overseer for the township and a very energetic man as county inspector, and I think I can say advisedly that our roads are fairly good.

WELL PLEASED WITH PIPE CULVERTS.

Tuckersmith, Huron Co.—Our roads are said by travellers to be among the best in western Ontario. Some of them are too flat and could be improved by draining. Our main roads are in good condition. Our councils have been very careful of them for a long time. We have commenced this year to use sewer pipe extensively. Have been using some for years and are well pleased with them.

THEY HIRE A MAN.

Thessalon, Algoma District.—All our roads are done by statute labor. We have a road grader and hire a man to run it. It is taken from one road division to another and worked by those using teams to do their statute labor. We could do one-third more work if all the statute labor were commuted.

A RADICAL CHANGE NEEDED.

Tehkummah, Manitoulin District.—I have advocated the doing away of the statute labor system, cut down the commutation tax and purchase a road machine. We need a radical change in our road system. The municipal machinery moves very slowly in the matter. Trust a change will soon take place.

NO AMOUNT ON TOP.

Verulam, Victoria Co.—In those sections where gravel is plentiful and of a good quality the general condition of the roads is very good. In a number of places they lack proper drainage and no amount on the top will make a good road in such cases. In these places the roads are in a very poor condition, being almost impassable. No special effort has been made to improve such roads.

WILL GROW WORSE.

Wallace, Perth Co.—The roads in Wallace are not bad. The grading and gravelling of a central sideroad leading from Listowel to Palmerston since the purchase of the road machine is of great benefit, as the heaviest teaming and greatest traffic is between those towns. The system of pathmasters and statute labor is wretched and will continue to grow worse. A few say twenty of the 105 discharge their duties conscientiously and the others as may be expected, by men who are not paid for their work, and who can not be even reprimanded by the members of a council, who might thereafter suffer from their enmity. A compulsory commutation of even twenty-five cents per day, if judiciously expended, would make such an improvement on our highways as would astonish the natives, *i. e.*, supposing statute labor abolished, as it ought to be.

THEY WANT GOOD ROADS.

Williams, East, Middlesex Co.—Our roads in general are excellent. Concession roads, I may say, are all gravelled, and also many of the side roads. Our people are alive to the advantage of good roads, and will not rest contented with poor roads.

SYSTEM NEEDED.

Whitby, Ontario Co.—Better results should accrue from a recognized, continuous system of road repairing, with larger road divisions; or commuting for statute labor whereby longer continuous stretches could be made or repaired in one season, obviating the ordinary patch system and giving continuous good roads on leading lines.

RECEIVING MORE ATTENTION.

Wilmot, Waterloo Co.—A more uniform system of grading is desirable, and since we have the road machine this is receiving more attention. I believe better results would be had if statute labor were abolished and a rate of say fifty cents per diem were charged instead, when road commissioners could be appointed who would carry out a uniform system of grading, draining and gravelling.

EXTRACTS FROM COUNTY REPORTS.

HASTINGS Between 400 and 500 miles of road are maintained exclusively by the county. The County Council appoints a man, during its pleasure, called "the superintendent of gravel roads and county surveyor." He has charge of all gravel roads and county bridges. He devotes all his time to it. Under him are gangs of men each with a foreman. They are employed most of the year and are paid monthly. Over thirty-five years ago, county gravel roads were built and all company roads bought by the county and all roads made free of all tolls.

FRONTENAC. About seven miles of road are maintained exclusively by the county. The County Council appoints an engineer who attends to repairs and keeps the road in an excellent shape.

WELLINGTON. One hundred and forty-eight (148) miles of road are maintained by the county. The roads are managed by commissioners composed of members of the County Council; the roads within each county council division being under control of members of the division. At the January session each year an appropriation of \$50 per mile is granted for maintenance (exclusive of bridges and culverts). Accounts with vouchers are produced at end of year and audited by county auditors. All the foregoing 148 miles have been assumed by the county and purchased from road companies and municipalities. These roads have been owned and maintained by the county for the past thirty-three or thirty-four years. Roads under county management are all macadamized and in first-class state of repair. With the aforesaid annual appropriation they are kept in very good condition.

LENNOX AND ADDINGTON. A number of roads were from time to time assumed by the county, but in January, 1893, the County Council repealed all by-laws assuming county roads. The roads are in a bad condition and rapidly growing worse. No effort has been made by County Council to improve their condition. Good opportunity here for instructor in road-making.

HURON. Some leading roads and all boundary lines are partially maintained by the county.

WENTWORTH. About thirty miles of road are maintained by the county. They are managed by commissioners appointed by the County Council, one road being generally assigned to three commissioners. In 1880, '81, '82 and '83, a net expenditure of about \$16,500 was incurred on two roads of a length of about five miles each, assumed by the county and maintained for some years as toll roads. One of the roads were abandoned in 1891 and the other is now maintained as a free road. The roads maintained by the county are, I believe, in fairly good condition.

VICTORIA. There is much need for improvement. The County Council purchased a stone crusher for the use of such local municipalities as will avail themselves of its use to provide better and well prepared metal. It has already been used in the Town of Lindsay. Mr. Campbell, Provincial Road Commissioner, visited Lindsay in June last and addressed the County Council and the Council of Lindsay. Great interest was aroused and a condition of apparent hopelessness has given place to great hopefulness and determination that hereafter expenditure shall be made under proper supervision according to an approved method.

PETERBOROUGH. In January, 1896, a special committee was appointed by the county council to investigate the system of building and maintaining roads in the County of Hastings, and report with suggestions. This report concludes with the following:

"Having viewed the roads, we are thoroughly convinced that the system of construction and maintenance of roads as adopted by the County of Hastings is good and meets with our approval; and we would recommend, if thought advisable, that this system be adopted by the County of Peterborough.

OXFORD. It has been the custom to appropriate \$3,000 each year to be spent by the reeve or councillor in accordance with the equalized assessment or dividing lines between municipalities. At the June (1897) session of the council a commission was appointed to gain information respecting toll roads with a view to purchase.

ONTARIO. The plan for road improvements has been the construction of three roads through the length of the county. The county is seventy miles in length and eighteen in breadth. One road commences at Lake Ontario and runs through the Townships of Pickering, Uxbridge and Scott, with some slight deviations; another road commences on the Whitby and Port Perry gravel road at Manchester, and runs north through Reach, Brock and Thorah to Beaverton; a third road runs from Oshawa north through Port Perry to Beaverton, following in some places the county line between Ontario and Victoria. There were expenditures on a road connecting the towns of Uxbridge and Epsom, which are on the secondly above mentioned road. There was also a road extending from Beaverton north through the townships of Thorah, Mara and Rama.

Scugog Island, lying to the east of the Township of Reach, was connected to the mainland and very considerable expenditures were made from this roadway down to the foot of Scugog Island. It was only last year that any sum could be obtained for the improvement of the Kingston road passing through Pickering, Whitby Town and Township, East Whitby and Oshawa. This expenditure, as were many of the other expenditures, was obtained upon condition that the local municipalities would make grants of double the amount granted by the county. There are a large number of isolated grants, not connected with any plan of road improvement. The expenditure for bridges on the boundary line between this county and

York, Simcoe and Victoria have been very large indeed, as has also the expenditure upon bridges over the Black, Talbot and branches of the Severn River.

The road across Scugog Lake, three-quarters of a mile, is maintained exclusively by the county, and is managed by a commissioner who makes reports to the county council as to estimated costs of needed repairs, and also makes repairs which are urgently needed under the advice of the reeves of the municipalities between which it lies. This is a roadway across the head of Lake Scugog, with one small bridge, which cost several thousand dollars.

BRANTFORD BOARD OF TRADE.

A report to the City Council on the streets of Brantford was taken up by the Board of Trade, the following memorandum being prepared by a special committee:

Your Committee on City Improvements, to whom was referred the report of Mr. Archibald Campbell, C.E., the Government Roads Commissioner of Ontario, begs to report:

That it views with satisfaction the interest that is now being displayed in this Province, and elsewhere, in the improvement of the public highways, and is convinced that the action of the Ontario Government in creating the office of Roads Commissioner will be productive of very great benefit. In Mr. Campbell, a man eminently well qualified for the important duties with which he is entrusted has evidently been found.

The suggestions that the Roads Commissioner is making from time to time for the betterment of the condition of the thoroughfares of the various municipalities in the Province cannot fail to be productive of valuable results. So far as Brantford is concerned, your committee believes that his recommendations are deserving of most careful consideration by those who are interested in seeing the expenditure upon our public streets made in such a manner as to secure the most permanent results, and at the same time add to the attractiveness of a city of which we are all so proud. Acting upon the authority given us, we have ordered that copies of the report be printed for distribution among the members of the Board of Trade.

It is a source of satisfaction that Mr. Campbell finds so much in Brantford of which he can approve. For our sidewalks, waterworks and sewer systems he has only words of praise, while even in the matter of road-making the concession has been made that more attention has been given to the subject than in the majority of places in Ontario. At the same time, he finds that the system that has been pursued in the past has been faulty, indeed that there has been a want of system, and that there has been a large expenditure of money without adequate results. In this connection, it may be observed that Mr. Campbell probably does the city an unintentional injustice in saying that since its incorporation the sum of \$185,035 has been expended upon its streets, because, prior to 1890, when the local improvement system came into operation, much of the expenditure under this heading was made on sidewalks, bridges, culverts, etc.; this, in addition to the usual expenditure for cleaning, removing snow, etc., from which permanent results are not looked for.

It would be presumptuous on the part of your committee to express an opinion on many of the technical recommendations made by the Roads Commissioner. It is justified, however, in coming to the conclusion, from a purely business standpoint, that it would be the path of wisdom to have our streets permanently made, rather than that considerable sums of money should be annually spent in repair and patchwork without enduring results being obtained. It is certainly deserving of consideration if, in the not distant future, a by-law should not be submitted to the ratepayers, which would provide for the proper construction of some of our principal streets. Whether such permanent work can be done to advantage at the present time is open to question, because so much of the work on our sewers, waterworks, gas service, etc., has still to be done. Had the system been adopted of completing all street connections for such services, as the main work progressed, the proper care of our streets would have been a much easier matter than it is at present. Meantime, as rapidly as circumstances will permit, as much permanent street-making as it is wise to do should be undertaken.

Your committee heartily endorses the recommendation that the city should acquire a steam roller, believing that the experience of other cities has been that proper macadam roads cannot otherwise be made. This is a very important matter, because, in the very nature of things, the great majority of our streets must be made of macadam, and it is desirable that we should have for the purpose the very best appliances. With the roller it would be easy to adopt Mr. Campbell's suggestion that the use of sand and cinders to aid consolidation of the road material be done away with and crushed stone substituted. The results would certainly be more satisfactory, having regard both to the condition of the roadway and the comfort of the citizens. In this connection, it is understood, that the purchase of a rotary screen for the stone-crusher has already been decided upon.

The importance of having established grades for the streets is readily apparent, and uniform grades should as rapidly as possible be adopted.

Without any large additional cost, the appearance of the streets of the city, even as they are at present, would be greatly improved if they were more frequently cleaned than is now done. The proper cleaning of the streets is always an incentive to householders to improve the condition of the lawns and boulevards abutting. Besides, clean streets would give our local wheelmen much less excuse for riding on the sidewalks than they have at the present time.

A matter that is deserving of more serious attention than has yet been given it in Brantford is the proper curbing of all the streets and the laying out of boulevards upon uniform lines. At present, nearly all work that has been done under these headings has been at private expense, which is not only unjust in itself, but there is in consequence a great deal of raggedness and irregularity even on our best residential streets. With the boulevarding once properly made, the public spiritedness of the citizens might fairly be appealed to to maintain it in order. The up-keep of lawns and boulevards would also be facilitated if the hours for lawn watering were made more liberal.

As to the recommendation that the driveway on our streets should be narrowed so as to lessen the cost of construction and maintenance, while at the same time giving greater room for boulevarding there are many things to recommend this course, and so far as our present system of street drainage makes it possible, this plan should be adopted.

Before leaving this subject, your committee would call attention to the statement made by Mr. Campbell, that the funds placed at the disposal of the civic officers entrusted with the maintenance of the city's streets are altogether inadequate for the end in view. For the future it will undoubtedly be necessary to deal with this important department more liberally than has been done in the past.

As to bicycle side-paths, your committee would endorse Mr. Campbell's position, that the best way to please the wheelmen is to improve the general condition of all the city streets, rather than to make narrow paths here and there. There should not be a street in the city upon which a wheel might not be ridden with comfort and with safety.

All of which is respectfully submitted.

REPORT ON THE STREETS OF THE CITY OF GUELPH, ONTARIO.

To His Worship the Mayor and Members of the Council of the City of Guelph:

The importance of well paved streets has not been sufficiently appreciated by the cities and towns of this Province, and it is gratifying to find that Guelph, among other progressive cities and towns, is becoming interested in the matter. Much money has been wasted in many municipalities through ill-advised expenditure arising out of an indifference founded on the idea that the streets are as good as they can be made; that the methods which have held ever since the time of the first settlers are the only methods, and that streets are of little consequence in any case.

Good streets are of prime necessity to the welfare of a city. They are the object of an annual outlay which, if wasted, reacts in a two-fold manner by increasing taxes, and at the same time permitting the evils of bad streets to remain; and it should be understood at the outset that the object of the road and street reform movement is not to urge increased expenditure but to obtain a better use of the money now expended.

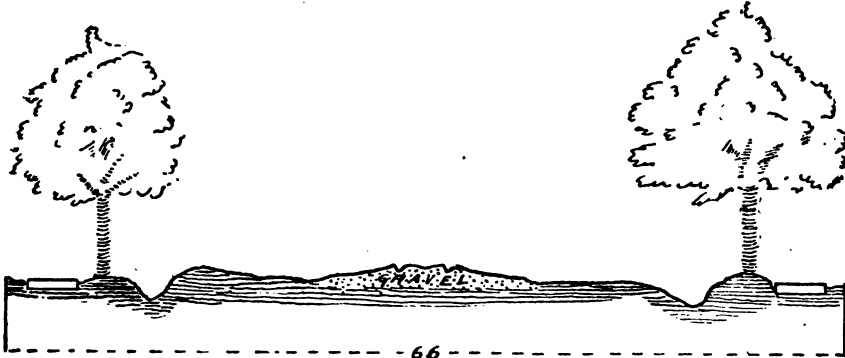
The defects most observable in the street system of Guelph arise from the fact that durable and permanent work is not undertaken; in order to correct which there is need of reforming the present system of street management in two particulars—the method of expenditure, and the method of oversight.

THE EXPENDITURE.

The expenditure should not be distributed over the whole street area in patchwork and repairs, but a reasonable amount should be provided for permanent work. Small sums of five, ten or one hundred dollars quickly eat up an entire yearly appropriation, whereas one-half this appropriation spent in properly macadamizing a few blocks would in a few years revolutionize the condition of the streets in Guelph.

An expenditure to the amount of about \$5,000 annually is divided among the different streets and wards of the city, and this is again subdivided by the ward representative in doing odd jobs here and there. It is not spent in accordance with the needs of the work, but as certain electors think it should be spent. The logical outcome of the system is that this money becomes a legitimate campaign fund; the people expect it and the Council has no other course to pursue. It is the inevitable result of such a system that too much money is provided for one piece of work, and not enough is devoted to another—usually the latter. It is productive of shoddy roadways, and is always wasteful in the end.

Guelph has been making an effort to keep in repair a class of roadways not suited to the traffic over them. They might be considered fairly good township roads. Cheap in first construction, they are expensive to maintain, and after a term of years are very costly. The repairs made to these streets are supposed to be such as will eventually provide a solid roadway; but this method of sinking stone in the mud year by year, and in



Cross-section of a Street under the Old System.

the spring carting off the mud which has been forced to the surface, is an extravagant and useless process, which will not make streets. The waste that arises is of a two-fold nature, combining high taxation and bad streets. It is not to be supposed that streets can be built without money, but when the expenditure is made, as now, it should be to provide good streets.

In order to get the best results in street construction, the work has to be undertaken on a proper scale. A roadway, like a house or any other structure, should be built from the foundation upwards, and should be completed, if only in short sections, before it is used. The roadbed should first be graded, under-drained, and otherwise prepared to receive the gravel, broken stone or other road metal; which last should be placed on the roadway with proper machinery and in accordance with the best principles of road-making. To do this, the expenditure, now extended over a term of years on a badly formed roadway, should be concentrated so as to secure permanent and durable work.

To this end there are three courses open: 1. To set apart a portion of the present annual appropriation. 2. To issue debentures for the amount necessary to do finished work. 3. To adopt the frontage tax system.

OVERSIGHT.

The oversight should be delegated to a competent supervisor, preferably an engineer, instead of being left to the council or a committee of the council.

On business principles there is every reason for placing this work in the hands of one man. Street construction is a matter requiring experience and special training. The plan of leaving it to the councillors is the pathmaster system of the towns. It is even more objectionable in the towns, since the streets demand more skill than do country roads. The supervision of street construction should rest with one who has a knowledge of the subject, together with good business ability.

The engineer would prepare plans and specifications for all work. These having been submitted to, and passed upon by the board of works and council, he would further relieve the council by taking full direction of the work. An engineer is now in the employ of your city for certain purposes, and his duties should be extended to include the management of street improvement and maintenance.

CLASSIFYING THE STREETS.

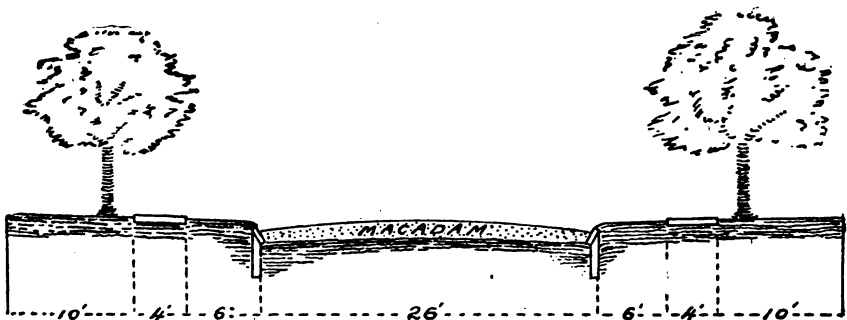
One of the first duties of the engineer in arriving at a plan whereby street improvement may be undertaken systematically would be to classify the streets according to the traffic over them, the character of the street, whether a residential or business thoroughfare, the nature of the soil, grades, etc., with the object of discovering the strength, etc., of roadway required.

It will be seen that such streets as Wyndham, Macdonnell and Quebec, the main business streets, have a large amount of heavy traffic over them to the depot, and that a strong form of pavement is needed.

Another class would evidently include such thoroughfares as Eramosa road, Woolwich street, Paisley street, Gordon and Norfolk streets, over which traffic from the country reaches the centre of the city. That these are residence streets, and that the travel is less severe than on the main streets, should evidently influence the character of the pavement.

A third class would include such streets as are residential, but are not called upon to support either heavy or frequent travel, and the roadway should be built accordingly, at a correspondingly less cost.

By placing before the council such a report upon the streets, showing the present requirements, the probable future requirements, the special improvements in the way of culverts, grading, etc., your council would be in a position to undertake the improvement



A WELL-DESIGNED STREET.

Cross-section showing sidewalks outside the trees and roadway curbed with cedar.

of the city streets on an intelligent basis. At present there is no definite object in view. When permanent improvements are undertaken, much of the work now being done will be found premature or unnecessary, and will be torn up. This is specially true of the grading and the coatings of broken stone and gravel.

There should be definite plans, which successive councils can follow, and toward which all work and expenditure will tend.

CHOICE OF PAVEMENT.

With an abundance of stone easily obtained, broken stone pavements are the most suitable which can be had for heavily travelled streets in Guelph. The excessive width of the street allowances is adverse to the use of vitrified brick or asphalt in the business

sections, as the cost would thereby be much greater than in the majority of towns. For less travelled streets, gravel would be suitable. Macadam roadways are growing in favor in towns. They are pleasant for driving, wheeling, and, if properly constructed and maintained, are handsome in appearance.

EARTHWORK.

The present practice in Guelph is to merely spread the crushed stone over the travelled roadway, without having prepared the subsoil to receive it.

The first step in properly constructing a macadam roadway is for the engineer to take a series of levels on the streets, so that a suitable grade may be decided upon. The width of roadway to be metalled can then be staked out, to be excavated in accordance with the required grade, and to form a receptacle for the stone.

To properly grade and excavate will frequently necessitate the handling of a good deal of earth. This earth should be used as far as possible in levelling the sides of the streets and filling in low lots adjacent to the roadway, disposing of the remainder as circumstances may dictate.

DRAINAGE OF ROADWAYS.

Good drainage is essential for all classes of pavement, and none more than macadam. It must be understood that it is the natural soil which really sustains the weight of traffic, and that broken stone piled on a wet soil cannot be made sufficiently strong to support heavy vehicles, and water permitted to accumulate and stand on the surface of the roadway will cause the pavement to be rapidly churned into ruts and mud.

Underdrainage by means of common field tile is very often necessary. Water falling in the form of rain passes at once through the soil until it reaches a strata of rock, compact clay or other impenetrable layer. If there is an outlet along the strata or rock or clay it will issue in the form of springs. If not, it will rise higher and higher in the soil above the strata through which it cannot pass, just as water poured into a vessel will rise higher and higher until it reaches the top, when it will overflow. Tile drains provide an outlet for this water before it reaches the surface of the soil, and thereby maintain a firmer foundation of earth on which the broken stone may rest. These tile drains should be placed at a depth of about three feet.

Surface draining is provided by rounding up (crowning) the surface of the roadway, thus draining the water to the gutters at the sides of the road, which latter are so graded as to permit the water to flow along them to suitable outlets. Deep ditches are not needed, but the road should be curbed and the angle between the surface of the road and the curb will serve as a sufficient waterway. The surface drainage is further aided by a coating of broken stone or gravel. And it is of prime importance that this coating should be made smooth, hard and compact by rolling so that water will not pass through, thereby softening the earth supporting it.

FOUNDATION.

A firm, unyielding foundation is an absolute necessity for any kind of pavement. This is one of the chief defects in the roadways of Guelph. All road coverings, whether broken stone, asphalt, stone blocks, vitrified bricks, wooden block, form merely a wearing surface. The weight of the load must be borne by the foundation beneath. The natural soil, if kept in a dry state, will support any load. The practice in Guelph appears to have been an effort to cover and fill up wet places by dumping into them loads of broken stone, rather than by removing the water in drains. In this way, the roadways on many of the streets are raised away above the elevation of the sidewalks and adjoining property.

A foundation for a macadam road is obtained by drainage of the roadbed, and thoroughly consolidating it with a roller. In certain cases, especially on clay soils, and

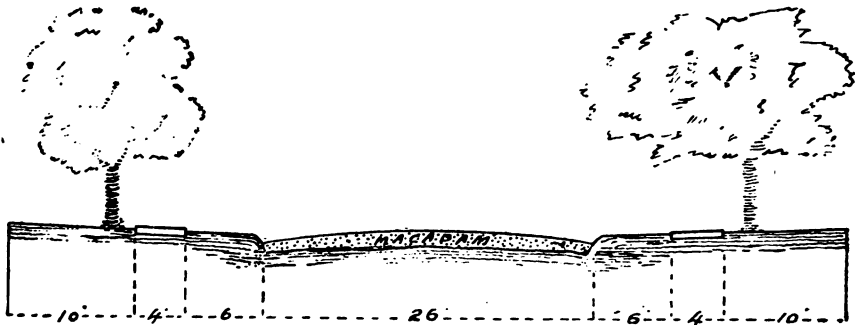
for the most durable class of broken stone roads, a foundation layer of large stones may be placed on edge, the projecting points being chipped off and wedged into the interstices. But for ordinary cases, the layers of crushed stone may be placed directly on the natural earth.

A ROAD ROLLER.

A roller is of first necessity in the construction of macadam roadways. It is needed first to compact the earth on which the broken stone rests. The stone or gravel is then placed in the roadbed in successive layers of about four inches thickness, each layer being consolidated before the next is applied.

The present method is to spread the stone over the roadbed, leaving it for traffic to consolidate. When done in this way the stone is forced down into the soil, and the soil comes to the surface in the form of dust, which has to be carted away every spring. There is no real bond between the stones thus mixed with earth.

But when, by rolling, the subsoil has been thoroughly consolidated and the metal has been placed on the road in a compact coating, each stone of which is wedged firmly, there is a mechanical clasp which will not yield to at all the same degree. The stone



Cross-section of an Improved Street Without Curbing.

covering becomes, under a roller, a roof as well as a floor, and will not permit the water to pass through and soften the earth foundation below ; but, being smooth, and properly crowned, the water is at once shed to the gutters.

A CRUSHER.

Stone could be broken to much better advantage for the streets of Guelph by the use of a crusher.

The stone crusher should have a screen attachment whereby the stone could be separated into various grades according to size, the larger stones to be placed in the lower layers of the roadway and the finer on top. The grades frequently are :

- 1st. Such as will pass through a $2\frac{1}{2}$ inch ring.
- 2nd. Such as will pass through a $1\frac{1}{2}$ inch ring.
- 3rd. Such as will pass through a 1 inch ring.
- 4th. Chips and dust screenings.

When stones are laid on the roadway, large and small indiscriminately, the smaller wear more quickly than the larger and a rough surface will result. The larger stones have a tendency to loosen, as their bearing is not so perfect, and they will be found rolling under the feet of horses and under the wheels of vehicles.

A crusher with screen attachment is needed also to properly utilize the gravel of the vicinity of Guelph. It is of very irregular size, ranging from large boulders to sand, and

contains a large amount of clay. One pit on the western side of the town is a complete mass of boulders. By placing a crusher and screen in the pits, and passing the material through, the material would be broken into a suitable size, and the clay and sand would be removed. In dry weather clay and sand assist in making a smooth road, but in wet weather they retain moisture, soften, diminish the strength of the roadway, ruts rapidly appear and the roads "break up." Clean material of proper size is necessary, and, to this end, a crusher with screen attachment should be obtained.

SEWERS.

Sewers have not yet been laid in your city. This is a matter which is closely connected with sanitary conditions and deserves the very serious consideration of your council. That sewers are needed, especially in the central portions of the city, there can be little doubt, and further delay can be but very small economy, if it can in any sense be considered economy. The streets most in need of sewers are those most in need of pavements. It therefore becomes of greatest importance to give these matters early attention.

Sewers should be laid before the streets are paved, since to open a macadamized street to lay sewer pipe causes an injury to the pavement which cannot be easily repaired. Sewers and paving are matters which should be considered together for various reasons, one worthy of mention being that drainage of the roadbed can be very much aided by a sewerage system.

If a sewerage system were designed for the city, and portions laid in the main streets, these could at once be paved. The sewers could then be extended and the paving done concurrently on the lateral or less important streets.

The fact that sewers are needed and that permanent pavements cannot be laid immediately on the main streets does not lay aside the urgent need of a road roller and a rock crusher. There are some streets which will not require sewers for a number of years to come, and these could be properly macadamized at any time. In any case these machines can be used on streets of the kind at present used in Guelph to very great advantage, a plan which has given much satisfaction in Brockville for a number of years, which city possesses a fifteen ton steam roller.

SOME ILLUSTRATIONS.

It may emphasize the foregoing remarks to point out that a large sum of money was spent last year in resurfacing Carden street, but for the want of a crusher to properly break and grade the material, and a roller with which it might be consolidated, the roadway is a practical illustration of the wastefulness of trying to do work without proper implements. Fine gravel and sand were used in covering the stone. This became very smooth during the dry weather, but is being rapidly destroyed in the wet seasons, turning already into mud and ruts.

A grading machine is used in Guelph, but it is almost useless to go to the expense of grading a street without rolling it. By the use of a grader alone a good shape can be given to the road, but when rain comes this loose material is at once saturated, wheels churn it, and the condition of the street is often injured rather than improved. If, after the street is properly shaped by the grader, a heavy roller were used to compact the material a smooth hard surface would be formed. Such a surface would throw the water to the gutters and would make a very serviceable roadway for light travelled streets.

With an immense amount of stone in every part of the city, the large sums of money which have been expended on the streets should, all must agree, have placed them in very much better condition. They are without shape or form and indicate thoughtless, careless management.

IN CONCLUSION.

The City of Guelph bears a royal name, the location is not an accidental one, but was chosen because of its many advantages and the excellence of the surrounding coun-

try. This has evidently been appreciated by your citizens who, in their public buildings, private residences, and the general features of a city, have studied closely the civic welfare. In the matter of streets, which should, under proper management, be converted into park-like stretches, something has been done in the way of narrowing the roadways, laying artificial stone walks, and providing handsome shade trees.

Much, however, remains to be done. While your citizens may feel that the streets are as good as those of many other towns and cities of the Province, the streets of other towns and cities are generally bad. Under the favorable conditions which exist in Guelph the streets should be superior to those of other towns. The people of other towns are becoming aroused to the necessity of improved streets, and there is every probability that this satisfactory feeling of equality will be endangered. The towns of Galt, Stratford, Woodstock, St. Catharines, Niagara Falls and others have purchased steam rollers and crushers, while a number have decided upon the same or equally important steps in the near future.

In the preparation of this report the object has been to outline briefly the essential features of a good system of management. Many important points have received mere mention, while details have been pretty generally avoided. The document will have served its purpose if successful in directing your attention to a very important matter and creating a more lively interest which, if aroused among the citizens as well as in the council, will result in great benefit to your city.

I have the honor to be, gentlemen,

Your obedient servant,

A. W. CAMPBELL,
Provincial Road Commissioner.

DEPARTMENT OF AGRICULTURE, ONTARIO,
22nd day of December, 1897.

EXTRACTS FROM A REPORT TO THE CITY OF ST. JOHN, N.B.

STONE BLOCK PAVEMENTS.

Stone block is the oldest of paving materials; is extensively used in cities, and is the strongest and most durable that can be had. It is well adapted to grades up to ten per cent, yields little dust, requires little repair and suits all classes of traffic. It is, however, very noisy and is rather rough. It is therefore not suited to residence streets nor business streets on which there are retail stores and offices. It is best adapted to streets such as Water street, on which there is a large amount of slow, heavy traffic. It should be used also on steep grades in place of asphalt.

The stone generally used is granite or trap. Excessively hard stone wears to a smooth surface and becomes slippery. No examination or test which can be made of stone is perfectly satisfactory in distinguishing the best variety. Different kinds of the same stone and even stone taken from different parts of the same quarry, have different wearing qualities. The trap rocks of Carleton Heights, or the quartz diorites at the end of Sandy Point, will prove satisfactory. Quartzites, of which varieties are found at the west of Lily lake, on the slope of the ridge north of Douglas avenue and on the road to the lake after passing Seely street, are apt to wear away too rapidly. White marble, of which some is found back of Lily lake and elsewhere in the vicinity, is too soft.

The stone blocks should be cut into rectangular blocks about seven inches deep, three inches wide and nine inches long. The price paid for quarrying and making these blocks will average about thirty dollars per thousand. Slabs of a kind which can be handled by one man are split out in the usual manner. These are sub-divided into sections corresponding to the size of the paving blocks, which are then trimmed and finished.

In constructing a stone block pavement the natural earth is first prepared by draining, grading and rolling with a steam roller. On this a layer of concrete is laid, say six or eight inches in thickness, according to the traffic to be supported. On this is spread a layer of sand about one inch in thickness, and in this the stone blocks are embedded.

The blocks are laid stone to stone in courses at right angles to the street line and so that the joints will be broken. A slight variation in the size of the blocks is permissible as regards depth and length, but the width (if three inches as previously specified) should be exact.

On hills and grades a better foothold for horses may be obtained by using rough-finished blocks, or the blocks may be so embedded in the layer of sand on a slight incline in such a way as to present a series of steps. At street intersections the blocks are laid obliquely in what is termed the "herring-bone" fashion so as to give a secure foothold to horses turning the corners. The joints between the blocks are filled with sand and tar cement.

The State of Maine, adjoining New Brunswick, is one of the largest producers of stone paving blocks.

ASPHALT.

The materials of which asphalt pavements are composed may be either natural or artificial. Natural asphalt is obtained by grinding to powder bituminous limestones found in Texas, Utah and elsewhere, or the bituminous sandstones found in California, Kentucky, Texas, etc. This powder is then heated until soft and it spread while hot on the roadway. The chief source of artificial asphalt is the Island of Trinidad, W. I., where crude asphaltum is obtained, is then refined and mixed with sand and stone dust; is heated and applied to the roadway.

Owing to the skilled labor and machinery needed in laying this pavement it is, in the great majority of cases, laid and kept in repair by contract. When properly laid its durability cannot be questioned, but there is some difficulty in surrounding a contract with such safeguards as will ensure first-class material and workmanship. A reliable company should be employed and the maintenance of the pavement guaranteed for ten or fifteen years. A common guarantee is for a term of five years, but this is not sufficient. Breaks in asphalt pavement must be immediately repaired, otherwise they quickly shear off under wheels, and the size of the hole increases with great rapidity.

On the business streets of St. John, where the traffic is severe, where noise is objectionable, and where smoothness, cleanness, ease of travelling, are desirable, stone blocks are not suitable. Asphalt is the most durable material filling these conditions. It is not, however, suitable for steep grades, and stone blocks would necessarily be retained for grades greater than three per cent. Nor should asphalt be used between and adjacent to street car tracks. The gutters, too, should be formed of concrete or flagstone, as asphalt decays rapidly from the effect of moisture.

VITRIFIED BRICK.

Vitrified bricks for street paving are different in composition and manufacture from the ordinary building brick. They are made from shale, or clay, or a mixture of the two, which is heated to the point of vitrification and then slowly and gradually cooled. The size of each brick is about $2\frac{1}{2} \times 4 \times 8\frac{1}{2}$ inches. They are laid in the same manner as stone blocks, viz., in courses at right angles to the direction of the street, with broken joints, etc. The durability is not equal to that of asphalt or stone blocks. But they are less noisy than stone blocks. The pavement is adapted to business or residential streets on which the qualities, but not the strength, of asphalt are required. They are manufactured in the Province of Ontario, in the States of Ohio, New York, and Pennsylvania and elsewhere. There is room for much variation in the quality of brick. The process of manufacture is one which requires an expensive plant and much skill in burning. The

composition of clay or shale used is of great importance. It may contain, for example, too much lime, which will destroy the brick on exposure to moisture. Care must therefore be taken in selecting the brick to be used.

WOOD PAVEMENTS.

Cedar blocks and pine pavements have been used in St. John to some extent. In Canada and the United States wooden pavements are very much in disrepute. They have been found to decay rapidly, settle unevenly, become rough and unsanitary, absorbing filth and giving off bad odors. Much of this is unquestionably due to the methods of constructing these wooden pavements in this country.

In England and France they are regarded with favor, but the timber used there is carefully selected, so as to exclude any blocks showing signs of decay. Oblong blocks are cut all of equal size. They are treated with creosote, tar, and other preservatives, and are laid on concrete foundations. Some soft woods are used, and the life of such pavements is about ten years. The best wooden pavements are made, however, from Australian hardwoods, particularly the jarrah, karri and other of the eucalyptus woods of South Australia.

In the absence of actual experience in this climate with wooden pavements constructed in the careful manner outlined in the foregoing paragraph, their use cannot be recommended. Certainly cedar block and pine pavements as commonly laid in this country are not a success and should not be tolerated.

COST OF PAVEMENTS.

In the matter of cost most pavements, including macadam, require pretty much the same expenditure up to a certain point—the completion of the grading, under-drainage, curbing and excavating, preparatory to receiving the paving materials. Up to the layer of concrete the cost, excluding macadam (that is for asphalt, stone block, vitrified brick and wood pavements), is very much the same, but may vary according to the depth of concrete used and the material for curbing, whether stone or wood. As I have no data for calculating the amount of grading, excavating and drainage needed in St. John, it becomes impossible for me to make an estimate of the cost of the work applicable solely to your city.

The following indicates the basis for a strong form of pavement, on which the cost of a macadam roadway twenty-seven feet wide, for a resident street, may be estimated per lineal foot :

Excavating and grading roadway	\$0 30
Grading boulevard	6
Water and sprinkling	4
Rolling	10
Crushed stone in the street at \$1 per cubic yard	1 00
Drainage	10
Total per lineal foot of street	\$1 60
Total per square yard	0 53½

With cedar curbing, 4 x 12, the cost would be \$1.72 per lineal foot or 57½ cents per square yard ; with stone curbing, \$2.80 per lineal foot or 86½ cents per square yard.

In estimating the cost of an asphalt pavement to replace the cedar block on such a street as Main street, with a roadway forty feet wide, the actual cost would be about as follows :

Removal of cedar blocks and excavation per lineal foot.....	\$0 44 4-9
Drainage per lineal foot	15
Concrete, six inch layer, per lineal foot.....	2 22 2-9
Asphalt surface, per lineal foot	4 44 4-9
Total per lineal foot.....	\$7 26 1-9
Total per square yard	1 63 $\frac{3}{4}$

The maintenance is usually estimated on a basis of eight cents per square yard per annum. The cost, therefore, for a five years' guarantee is 40 cents; ten years, 80 cents; fifteen years, \$1.20. The total cost then under a five years' guarantee is \$2.03 $\frac{3}{4}$; under a ten years' guarantee, \$2.43 $\frac{3}{4}$; under a fifteen years' guarantee, \$2.83 $\frac{3}{4}$. In the above estimate the profit of the contractor on initial construction is not included and is regulated by competition.

Asphalt pavements, as previously stated, are laid by contract. This material is one which can be greatly adulterated and it will usually be advisable to have the work done by a reliable company under a long term guarantee. Recent inquiry shows that Ottawa has had rock asphalt pavements laid, with maintenance guaranteed for fifteen years, for \$3.10 per square yard; Montreal has had rock asphalt pavement laid, guaranteed for a similar term, for \$3.09. Trinidad asphalt, maintenance guaranteed for ten years, has cost in Montreal \$3.43 per square yard; Toronto, \$2.60; in Hamilton, \$2.15. Montreal paid \$2.85 for Trinidad asphalt on a five years' guarantee. The apparent disparity in these prices may be accounted for in part by differences in specifications regarding drainage, concrete foundations, and as regards special work on gutters, curbing, etc.

For a roadway of the same description as that on which the cost for broken stone was computed, twenty-seven feet wide, an estimate for brick pavement would be as follows:

Excavation and grading roadway, per lineal foot.....	\$0 30
Grading boulevard.....	6
Concrete, six inch layer.....	1 50
Drainage.....	10
Brick surface laid	3 45
Total per lineal foot	\$5 41
Total per square yard	1 80 $\frac{1}{2}$

The following is an estimate of the actual cost of laying one square yard of stone block pavement, on a basis of the whole of the work being performed by the city. Labor on street railway allowances would be slightly more expensive:

Stone in quarry	\$0 10
Quarrying stone.....	25
Making blocks.	1 20
Hauling	10
Laying and grouting.....	75
Removal of old pavement, excavation, grading, etc.....	13
Drainage.....	5
Concrete, eight inch layer.....	67
Sand coating	5
Total per square yard	\$3 30

THE FRONTAGE TAX SYSTEM.

In connection with the frontage tax system, inquiry is frequently received respecting the form of by-laws necessary. The following by-law of Blenheim, Kent county, a town having about 2,000 population, appears to be very complete, and should serve as an excellent basis for other places adopting this system :

Whereas it is deemed desirable to provide for certain improvements, works and services being paid by special assessment on the property benefitted :

Be it therefore enacted by the municipal council of the corporation of the Town of Blenheim, in the County of Kent, as follows :

1. This by-law may be known and cited as "*A By-law respecting Local Improvements and special assessments therefor.*"

2. Unless the context otherwise requires, the following words and expressions in this by-law shall have the meaning hereby assigned to them respectively, that is to say :

The words "local improvement," "work," or "service" shall mean any improvement, work or service, the cost of which may be charged by the council under any of the provisions of "The Municipal Act of Ontario" against the real property immediately benefitted, by way of special assessment.

The word "owner" shall be construed and deemed to include a leaseholder, the unexpired term of whose lease (including any renewals therein provided for) extends over a period which is not less than the duration of the proposed assessment, if the lessee has covenanted in his lease to pay all municipal taxes on the demised property during the term of said lease, and would be liable for the taxes for the proposed improvement.

The word "Inspector" shall mean such person as may be appointed Inspector of Public Works by the Municipal Council of the Town of Blenheim.

The words "Engineer," "Clerk," "Treasurer" and "Board of Works" shall mean "Engineer," "Clerk," "Treasurer" and "Board of Works" of the Town of Blenheim.

The word "corporation" shall mean "The Municipal Corporation of the Town of Blenheim."

The word "council" shall mean "The Municipal Council of the Town of Blenheim."

3. From and after the first day of May, A.D. 1895, all future expenditure in the Town of Blenheim for the several classes of works, improvements and services herein-after mentioned, for which (amongst others) special provisions are made in section 612 of "The Consolidated Municipal Act of 1892," that is to say :

A. Making, enlarging or prolonging any common sewer.

B. Opening, widening, prolonging or altering, macadamizing, grading, levelling, paving or planking any street, lane, alley, public way or place, or constructing any sidewalk.

C. Curbing, sodding or planting any street, lane, alley, square or public place.

D. Re-constructing, as well as constructing, any of the said works or improvements shall be by special assessment on the property benefitted and not exempt by law from assessment.

4. In the opening, widening, straightening or extension of streets, where the whole cost of the improvement is assessable against the property fronting or abutting upon the street or portion of street to be opened, widened, straightened or extended, the owners of the land to be taken for such opening, widening, straightening or extension must dedicate the same to the town free of cost, and no such street shall be opened, widened, straightened or extended unless the land required therefor has been so dedicated to the town free of cost, notwithstanding the fact of a petition sufficiently signed having been presented for the said improvement; provided this prohibition shall not apply in any case as to which three-fourths of the members of the council present at any meeting thereof

shall vote that it is in the public interest that the street should be opened, widened, straightened or extended at the expense of the properties abutting thereon, notwithstanding the refusal of the owners of the property required therefor, or of some of them, to dedicate the property so required.

5. In making every assessment to defray the cost of the construction of pavements, sidewalks and sewers, the Engineer shall make an allowance on corner, triangular and other irregularly shaped pieces of land situate at the intersection or junction of streets, as follows :

On lots having an angle of ninety degrees, an allowance of one-third the number of feet on the side of the lot to be assessed ; on lots having an angle of more than ninety degrees, an allowance of less than one-third the number of feet on the side of the lot to be assessed ; on lots having an angle of less than ninety degrees, an allowance of more than one-third the number of feet on the side of the lot to be assessed, as may in each case, in his opinion, be deemed just and equitable, having due regard to the situation, value and superficial area of such lot, as compared with adjoining lots and pieces of land assessable for such improvements, works and services.

6. Any allowance made in pursuance of the next preceding section may be charged on the real property fronting on the improvements, or be assessed as a portion of the town's share of the costs of such improvements, in like manner as the intersection of streets, or partly on both, in such proportions as may be deemed just and equitable by the Engineer, and the cost of any portion assessed against the town shall be provided in like manner as in the case of street intersections.

6A. In every assessment to defray the cost of the construction of pavements, sidewalks, and sewers, where the said improvement fronts upon lands which are by law exempt from taxation for the said purposes, the amount necessary to construct that portion of the said improvement fronting upon the lands so exempt as aforesaid may be charged on the other real property fronting on the improvement, or be assessed as the town's share of the cost of such improvement in like manner as the intersection of streets or partly on both, in such proportions as may be deemed just and equitable by the Engineer, and the cost of any portion assessed against the town shall be provided in like manner as in the case of street intersections.

7. Any allowance or assessment made in pursuance of the last two preceding sections shall be subject to appeal to the Court of Revision, and from the Court of Revision to the County Judge, as hereinafter provided.

8. In the construction of any granolithic, stone, asphalt or brick sidewalk the municipality shall assume forty per cent. of the cost thereof, in addition to any other portion which may be assumed by or assessed against the municipality, and the said forty per cent. shall be added to that portion of the cost of the construction of said sidewalk to be provided by the municipality, as in the case of street intersections.

9. All matters of small pieces of sidewalks and other improvements, the cost of which can be estimated and reported upon by the Inspector, shall be referred to him, and shall be reported upon by him, in the same manner as if referred to the Engineer, and in such cases the word Inspector shall be taken as meant Engineer.

10. All costs incurred in carrying out the construction of any improvement, work or service, other than the construction itself, the overseeing of the same and providing the necessary funds, shall be borne by the municipality, and may be provided in the same manner as in the case of street intersections, or charged to the general fund, as the council may by resolution determine.

11. No work or improvement for which it is proposed to assess the real property immediately benefitted, as for a local improvement, shall be undertaken by the council unless and until the provisions of this by-law shall have first been complied with.

12. All petitions for local improvements, works or services to be made, done and performed under the provisions of this by-law, shall, as soon as received by the Clerk, be examined by him, and it shall be his duty to ascertain and finally determine whether the

same are signed by two-thirds in number of the owners, representing at least one-half in value, exclusive of the value of improvements, of the lands benefitted, according to the last revised assessment roll of the municipality, and liable to special assessment for the proposed improvements, works or services, and such petitions when found to be correct, as aforesaid, shall be numbered by him in the order they are received, and be entered at length in a book to be kept for that purpose, to be called the "Local Improvement Book," and the Clerk shall endorse upon such petition his certificate of the correctness thereof, and of the value of the whole of the real property (exclusive of improvements aforesaid) rateable for the proposed improvement, work or service, and shall forthwith so transmit the same to the Engineer. In case the petitions shall be found to be insufficiently signed, they shall be so certified by the said Clerk, and forwarded in like manner to the Engineer for his consideration.

13. Upon receipt of any such petition the Engineer shall forthwith examine into the subject matter of the petition, and report with as little delay as possible upon the necessity for, or the advisability of, undertaking the proposed improvement, work or service, and the reasons therefor, and particularly whether in the case of drains and sewers the same are desirable or necessary for sanitary or drainage purposes.

14. In the event of the Engineer reporting in favor of the undertaking of any such improvement, work or service, he shall also, after due and proper examination and inspection, report :—

A. What real property will be immediately benefitted by the proposed improvement, work or service, and the measurement of the frontages liable to the rates, and those exempt from taxation ;

B. The probable lifetime of the improvement or work ;

C. An estimate of the probable cost of the proposed improvement, work or service, and the amount thereof, which shall be assessed against the property to be immediately benefitted ;

D. The proportion in which the assessment is to be made on the various portions of real property so benefitted.

15. In the event of the adoption by the council of the report of the Engineer recommending the undertaking of any such improvements, work or service, it shall be the duty of the clerk to give all necessary notices to property owners as hereinafter provided and as provided by the statute in that respect, and make all necessary special assessments on such property immediately benefitted as aforesaid, pursuant to the statutes.

16. In any case when the engineer or board of works shall have recommended the undertaking of any improvement work, or service as a local improvement, and shall have recommended that the cost thereof shall be assessed against the real property immediately benefitted, although there be no petition therefor, or the petition therefor shall not have been sufficiently signed, and in any case when the said engineer or board of works shall have recommended the construction of any works, or the making of any improvement for sanitary or drainage purposes, the engineer shall, after due and proper examination and inspection, ascertain, determine and report, as hereinbefore set forth, and make the necessary assessment, and when such report and assessment has been approved of and accepted by the council, it shall be the duty of the clerk :

A. To cause a notice of the intention of the said council to undertake such proposed work, improvement or service, and to make such proposed special assessment, to be given in a public newspaper published in the Town of Blenheim, once in each week for two successive weeks, and the said notice shall state generally the nature of the proposed improvement, work or service, the estimated cost thereof, the property reported by the engineer as immediately benefitted, and estimated amount to be charged as a special assessment against the lands immediately benefitted, and that unless a petition against the proposed improvement, work or service and the proposed special assessment, signed by a majority of the property owners, representing at least one-half in value of the real property proposed to be assessed, according to the last revised assessment roll, be presented

to the council within one month from the last publication of the said notice, the proposed improvement, work or service will be undertaken, and the special assessment therefor will be made by the council.

17. In the event of a petition being presented to the council against any proposed improvement, work or service, and a special assessment therefor, within the time limited by the notice, the Clerk shall report to the council whether the same is sufficiently signed by a majority of the owners of the real property proposed to be assessed, representing at least one-half in value of said property, according to the last revised Assessment Roll.

18. In the event of no petition, or no petition sufficiently signed, being presented within the prescribed time, the clerk shall forthwith cause a notice at least fifteen days before the day appointed for the sitting of the Court of Revision, to be given or mailed to the owners and lessees or agents of such owners and lessees having a right to petition, of the real property mentioned in the report of the engineer as being immediately benefitted, in the form of Schedule "A" hereunto annexed, stating the time and place of meeting of the Court of Revision for the hearing of appeals, and ten days notice shall also be given in some newspaper published within the municipality, of the time and place of meeting of the said court, which notice shall specify generally what is the nature of the improvement, work or service, and what property is proposed to be specially assessed as immediately benefitted, and the time and manner in which the same is payable, and the said notice may be in the form "B" hereunto annexed.

19. The engineer, clerk and solicitor shall attend the meetings of the Court of Revision, and the said court shall sit at the time and place given in the notices given as aforesaid, and shall hear and determine all appeals which may be brought before it, pursuant to the provisions of the statute in that behalf.

20. In the event of any property owner appealing from the Court of Revision to the Judge of the County Court, the clerk shall proceed forthwith to procure an appointment from the Judge for the hearing and disposal of the appeal.

21. In the event of no appeals from the Court of Revision, or so soon as any such appeals shall have been disposed of by the judge, the clerk shall forward a certified copy of the report of the engineer, with any alterations or amendments which shall have been made by said Court of Revision or judge to the council.

22. Upon the receipt of such report as last before provided, it shall be the duty of the board of works to call for tenders for the construction, making or doing of the proposed work, improvement or service, and report the result to the council, recommending the awarding of the contract, or recommending the carrying on of the work by the inspector or engineer or under his supervision, and asking that funds may be provided for carrying on the work to completion.

23. In the event of the council adopting the report of the board of works awarding any such contract for any such local improvement, work or service, or the doing of any such work by the inspector or engineer, the mayor and treasurer shall make such arrangements with banks or other persons or bodies corporate, pursuant to the provisions of the statute in that behalf, as may be necessary to provide the amount of money required to carry on such local improvement, work or service, to completion, in anticipation of the special assessment therefor, and no such contract or agreement shall be executed, or work proceeded with, until such financial arrangements shall have been made.

24. The engineer, upon receipt of notice that a contract has been awarded, and the necessary financial arrangements therefor have been made, shall forward the specifications, plans, drawings and all other proper and necessary material, together with the accepted tender, to the town solicitor, who shall, with as little delay as possible, prepare the necessary contract and bond of security.

25. Upon the contract or agreement being duly executed, and not until then, the board of works may authorize the improvement, work or service to be proceeded with, and carried to completion.

26. After the completion of any such improvement, work or service, and after the entire cost thereof, including compensation for damages (if any) shall have been ascer-

tained, the engineer shall certify the total amount thereof to the treasurer, showing by such certificate what amount is chargeable to the property benefitted, and what amount is chargeable to the town at large.

27. The treasurer shall, as soon as possible after the receipt of the report of the engineer, under the last preceding section, ascertain the amount properly chargeable for interest on the advances made, and the estimated interest which will accrue thereon until the necessary assessment shall have been made, the necessary by-law passed, and the debentures to be issued thereunder shall have been disposed of, and moneys provided to retire the temporary loan, and certify the same, together with the amount shown by the engineer's report distributed as before provided, to the council.

28. Upon the adoption of the last mentioned report the council shall, with as little delay as possible, proceed to make the necessary special assessment upon the property immediately benefitted, and upon all other assessable persons, bodies corporate and property, pursuant to the provisions of the statute in that behalf.

29. The engineer, treasurer and clerk shall furnish the solicitor with all statements, calculations and other information, as may be required by him to enable him to prepare the necessary by-laws, providing for levying and collecting such special assessments, and for borrowing money by the issue and sale of debentures.

30. Any person whose property has been assessed for any improvement, work or service, under the provisions of this by-law, may pay the amount of such assessment, less the interest, at any time before the preparation of the debentures, in which case the amount of debentures shall be proportionately reduced.

31. When a by-law shall have been passed, making the construction and maintenance of works and improvements at the intersection of streets and opposite properties exempt from local rates, the subject of a general rate or charge, the exemption from general rates shall not extend to the item included in the by-law respecting yearly rates to meet the expense of such improvements and works opposite such exempt properties and at the intersection of streets.

32. In cases where the improvements or works are provided for by section 612 of "The Municipal Act," no real property or any owner thereof shall be entitled to the benefits of the provisions of the said Act and of this by-law in respect to exemption from any general rate during the year in which the by-law providing for any local improvement shall have been passed; and where any by-law for a local improvement provides for the issue of debentures, such debentures shall not be issued before the 30th day of December next after the passage of such by-law; and no special rate shall be collected in respect of any such by-law during the year in which the same shall have been passed.

33. This by-law shall come into force and take effect on, from and after the final passing thereof.

Finally passed, signed and sealed in open council this eleventh day of March, 1895.

JOHN W. GIBSON,
Clerk.

JOHN F. TITUS,
Mayor.

FORM A.

To

You are hereby notified that the Council of the Corporation of the Town of Blenheim proposes to pass a By-law providing for the
The total cost of the said improvement as estimated by the
is the sum of \$ and the said improvement is proposed to be carried on as a
local improvement, and be paid for by special assessment on the real property immediately
benefitted, as the same appears by the report of the
dated the day of 18 on file in the Clerk's office.
Your real property, which will be immediately benefitted by the said proposed improve-
ment, consists of feet frontage on the side of said
 upon which it is proposed to charge an equal proportionate part of the
cost of the said improvement, having regard for the whole assessable frontage on said
 , the same to form a special assessment upon your said lands,
according to the frontage thereof, and be payable in equal annual
payments, at the same time as your general municipal taxes, such payments to be suffi-
cient to cover annual interest and form a sinking fund for the principal sum or debt
incurred for said improvement. The Court of Revision will sit for the confirmation of
the Overseer's or Engineer's report upon the lands assessable and in which said special
assessments are to be made, for the said improvement, at the Town Hall, on day
the day of A.D. 18 at the hour of o'clock
in the noon, at which time and place all appeals against the same will be heard.

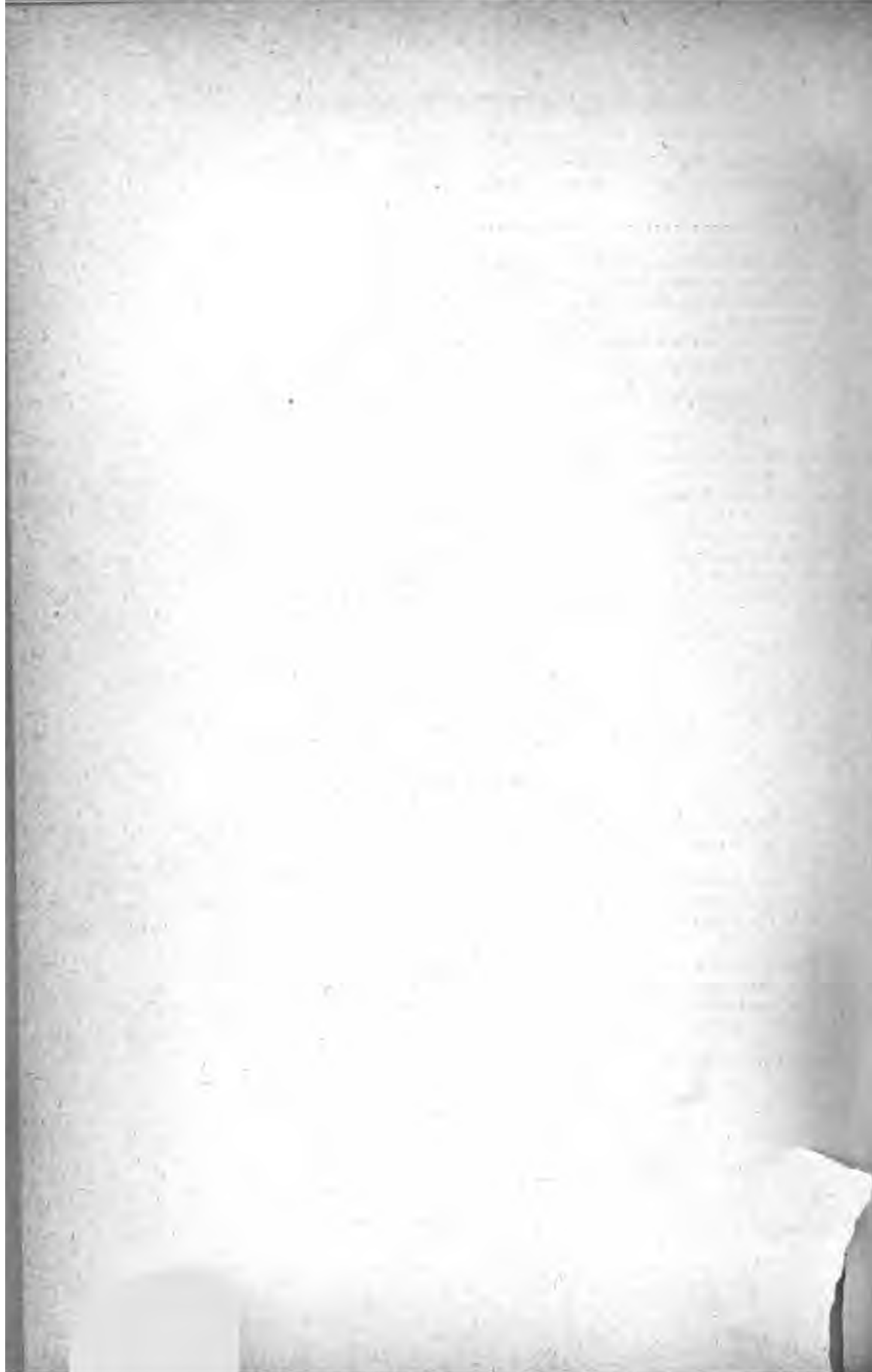
I am, your obedient servant,

FORM B.

Public notice is hereby given of the sitting of the Court of Revision at the Town
Hall, Blenheim, on day, the day of A.D. 18 , at the
hour of o'clock in the noon, for the hearing of appeals, pursuant to
the Statutes in that behalf, respecting the proposed
and the special assessment of the cost thereof upon the lands immediately benefitted, pur-
suant to the report of the Overseer or Engineer, dated the day of
 18 .

The estimated cost of the said improvement is \$ payable in
equal annual instalments, sufficient to cover interest at the rate of per
centum, per annum, and a sinking fund for the payment of the said principal sum, and
the lands proposed to be especially assessed therefor consist of

Dated at Blenheim, this day of 18 .





THAMES STREET, INGERSOLL : MACADAMIZED IN 1897.



A MACADAMIZED STREET IN INGERSOLL.



A STRATFORD STREET PAVED IN 1897.



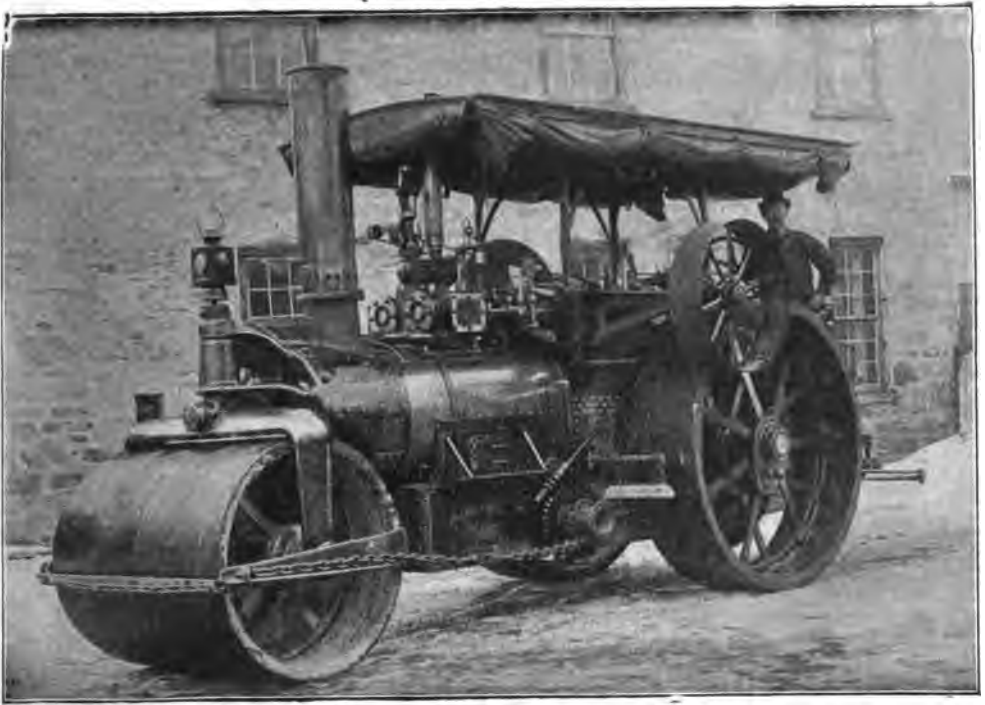
A STRATFORD STREET.

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STEAM ROLLER.

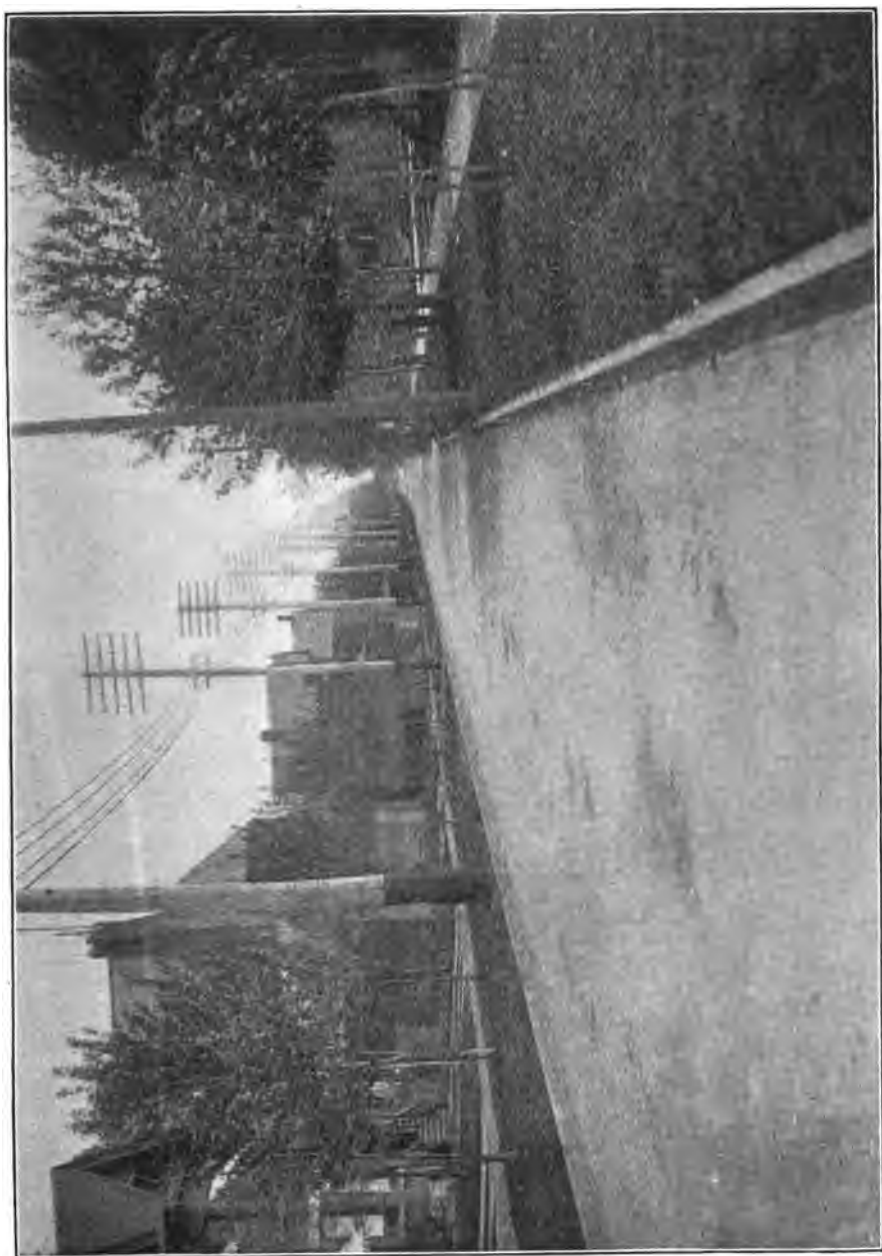


AS COUNTRY ROADS SHOULD NOT BE.



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THIRD ANNUAL REPORT

OF THE

PROVINCIAL INSTRUCTOR IN ROAD-MAKING

ONTARIO

1898.



(PUBLISHED BY THE ONTARIO DEPARTMENT OF AGRICULTURE, TORONTO.)



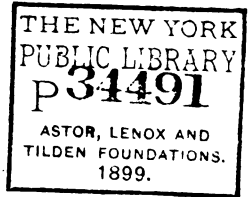
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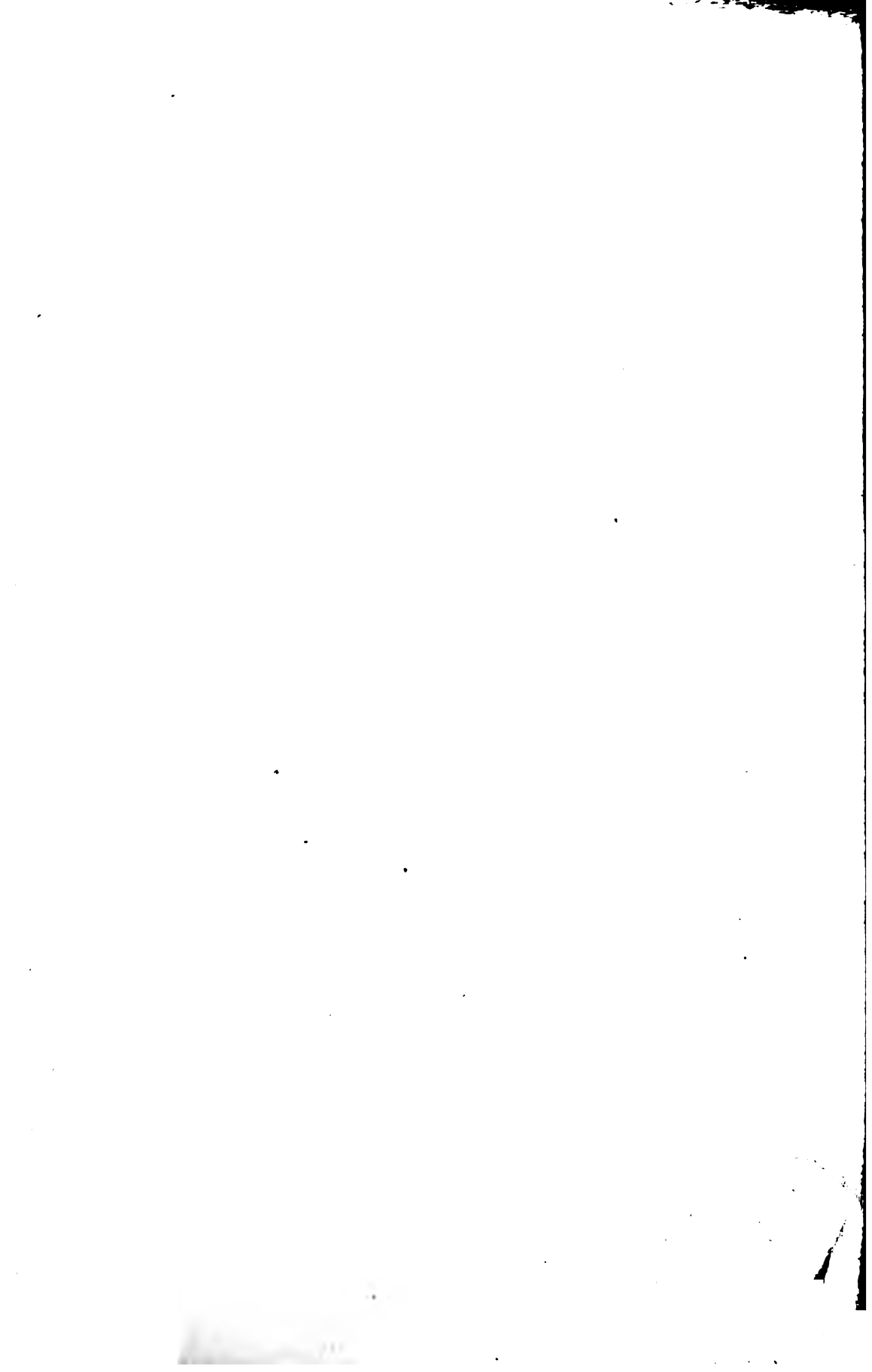


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1899.

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THIRD ANNUAL REPORT
OF THE
PROVINCIAL INSTRUCTOR IN ROAD-MAKING,
1898.

To the Honorable JOHN DRYDEN,
Minister of Agriculture.

SIR.—I have herewith the honor to submit to you my third annual report,
for the year 1898, on Road and Street Improvement in Ontario.

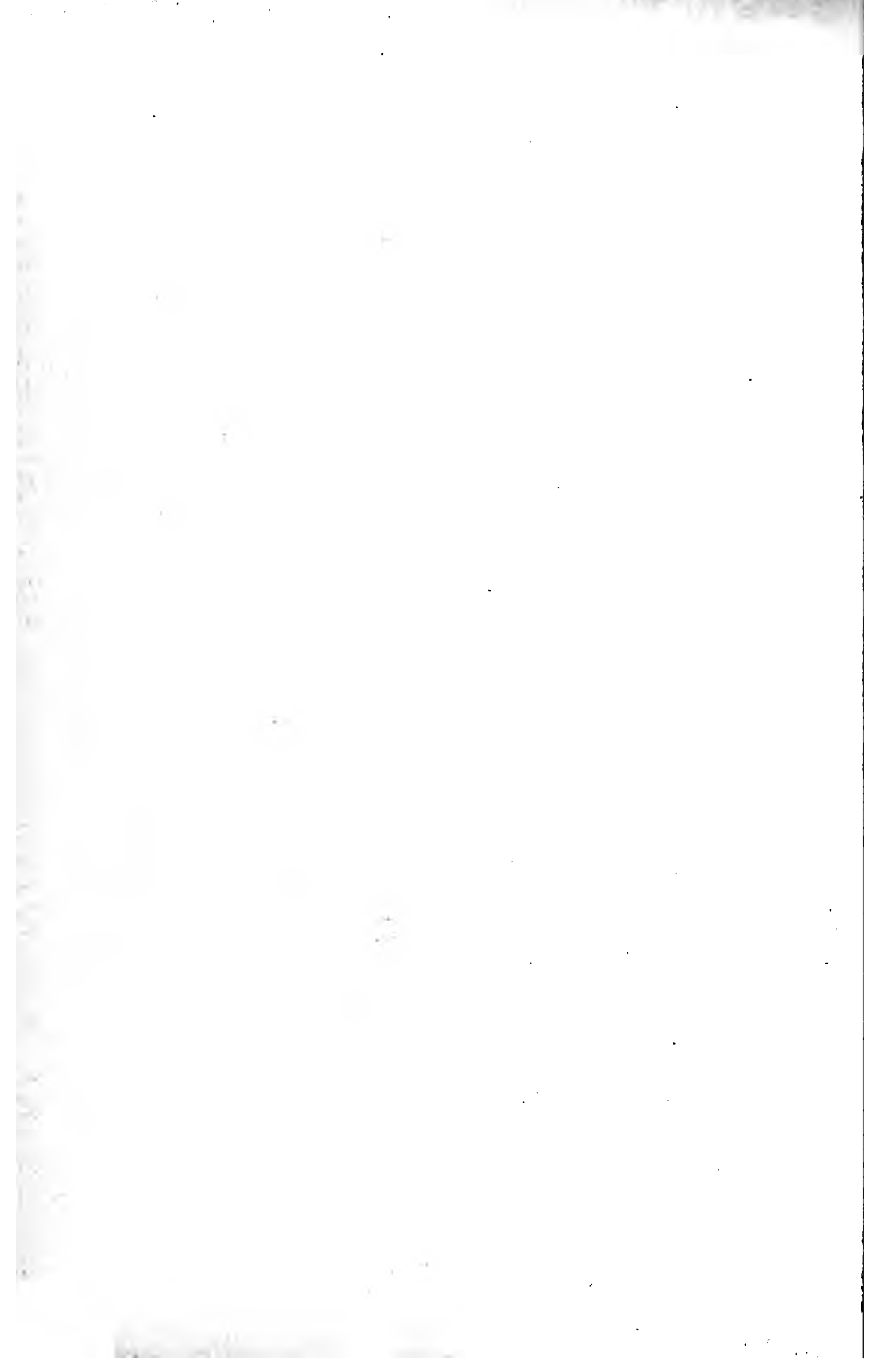
I have the honor to be,

Sir,

Your obedient servant,

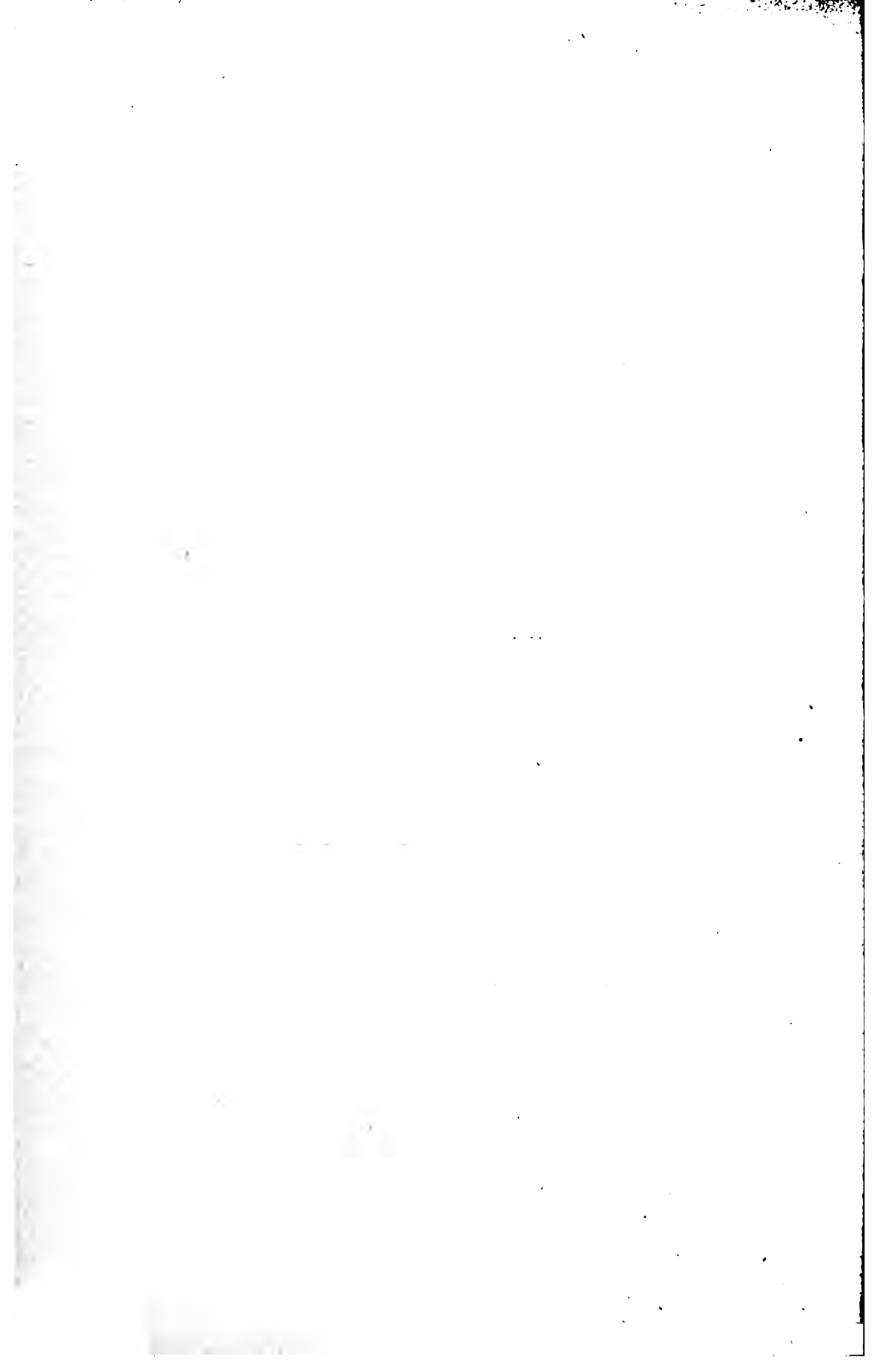
A. W. CAMPBELL,
Provincial Instructor in Road-making.

PARLIAMENT BUILDINGS, TORONTO,
ONTARIO, 20th March, 1899.



CONTENTS.

	PAGE.
Letter of Transmission	3
Transportation	7
Better Roads in Ontario	12
Road Reform	14
Should the Towns Assist ?	19
Petition of the Dufferin County Council	22
A Lesson from Scotland	23
A County System	24
Statute Labour	25
Road Reform in Outline	30
North Monaghan—Statute Labour abolished	31
Preparing for Roadwork	34
Directions for Pathmasters	35
The Best Use of Statute Labour	37
Gravel	37
The Essential Feature of Good Roads—Drainage	39
The Road Grader	42
Rock Crushers	44
"A Steam Roller an Absolute Necessity"	47
Weight of a Road Roller	51
Repairing with a Roller	52
Highway Culverts and Bridges :	
Location, Materials, Sewer Pipe, Cement-concrete Pipe, Arch Culverts of Stone and Concrete, Bridge Abutments, Design of Iron Bridges, Painting Iron Bridges, Painting Wooden Bridges, Economy of Iron Bridges	52
The Influence of Pavements on Public Health	61
Town Streets—Report to Windsor :	
Present Condition of Streets, Expenditure, Temporary Work, Frontage Tax, The Ward System, A Standard Pavement, Cedar Block, Asphalt and Brick, Cost of Pavements, Classification of Streets, Construction of Broken Stone Road- ways, Rolling, Curbing, Drainage, Main Entrances to the City, Sidewalks, Recommendations	66
Specifications for Cement-Concrete Walks	77





THIRD ANNUAL REPORT

OF THE

PROVINCIAL INSTRUCTOR IN ROAD-MAKING.

TRANSPORTATION.

The question of transportation is every year assuming a greater prominence, a deeper meaning to the people of Canada. Every year is turning greater activity renewed energy towards the improvement and extension of canals and railways, harbors, and all facilities for carriage and traffic associated with them. The inauguration of a domestic and foreign penny postage, which marked the beginning of the present year, was but another expression of the desire for free and uninterrupted communication, not only within the country but with other nations as well; so also the introduction of the telephone and telegraph of comparatively recent years. Lands excepted, the value to man of most material things is created, in part at least, by transferring them from their original position to another. Transportation is one of the most important problems occupying the minds of engineers to-day, and no field offers more reliable and remunerative opportunities for investment.

The national life of Canada is but in its infancy, and we have a country of magnificent distances and resources. The feeling that our means of communication cannot be too numerous is steadily on the increase, and the leading minds of the country have taken this view of the situation. New railways are needed, and new canals are needed to develop areas not yet reached, to provide less expensive, more capacious and more rapid transportation, to regulate rates, and to stimulate invention and enterprise towards securing cheaper methods and means of carriage.

Public opinion has moved rapidly and energetically along the policy of improving the larger means of transportation, ocean carriage, canals, railways; on the other hand the arteries and veins leading to and from these, and which are of such vital importance to the Ontario farmer and through him to the country as a whole, have been sadly overlooked, with the result that a bushel of wheat belonging to the Manitoba farmer can be carried by boat to Montreal or New York, a distance of over 1,200 miles, and handled at each terminus for about four cents; or only twice the amount it costs the Ontario farmer to deliver his wheat at the nearest market over the country road.

That there is an important relation between the common highway and the greater avenues of trade is plainly apparent; an importance which is greatly intensified by its bearing upon agriculture, and the value of agricultural prosperity to the Province. Fruit culture, the dairy, grain production, every department of farm enterprise is intimately associated with and dependent upon transportation, though in varied matters of detail.

Wheat, to which reference has been made, is but an example. The great waterways extending from the head of Lake Superior to the Atlantic Ocean the Canadian Pacific Railway extending across the continent, have brought into competition with the farmers of Ontario the wide grain-producing areas of the North-West. The wheat of Central Russia has easy access to the markets of the world by means of the Volga; Hungary possesses the Danube as a cheap water route; the Argentine Republic is served by the Rio de la Platte and the Parana, whereby ocean vessels may ascend a distance of 500 miles in the interior of the country. Africa with its great water routes from the interior of the continent to the seaboard, with its railways now under rapid construction, will no doubt soon enter into active production from its almost illimitable wheat areas. The surplus from these countries reaches the markets of the world by systems of transportation, the cost of which is being steadily lessened, and the result upon the agriculture of Ontario, the basis of our wealth, must be very marked indeed.

The farmer pays the cost of transportation. When the price of wheat at Fort William is one dollar a bushel, the farmer living at that place receives that amount for the grain he delivers at the elevator. On the other hand the farmer delivering his wheat at Edmonton receives one dollar, less the cost of carriage and handling between Edmonton and Fort William. In the same way an English farmer who can haul his wheat to the market in London when wheat there is at \$1.12 per bushel receives that amount for his grain; while on the other hand the Ontario farmer receives that amount less the cost of transportation and hauling between Ontario and London. The farmer of Argentina who raises his wheat as cheaply as the Ontario farmer and who delivers his wheat to the ocean vessel on the River Parana, has a cheaper means of transportation than has the Ontario farmer who has to pay for an expensive carriage by railway before reaching the seaboard.

The cost of labor and production is more nearly approximating the world over year by year; the value of the surplus supplied to the markets of the world is fixed by the lowest price accepted, not the highest or intermediate amount; the home market is governed largely by the foreign market. The profit, therefore, which the Ontario farmer can derive from his produce depends to a very great extent upon a decrease in the cost of transportation. There is frequent complaint that the rates of railway and water communication are high, that they absorb too great a proportion of the farmer's profits. Nevertheless these rates are to-day much less than they were a quarter of a century ago—much less probably than could have been anticipated. And while the rate of railway and steamboat transportation is regarded as high, we are met with the very striking comparison that one ton can be carried by water for a thousand miles or more, and two hundred and fifty miles by rail, for the cost of moving one ton only five miles over the common roads.

While this great change and improvement has been effected with regard to rail and water carriage, the case has remained practically the same with regard to travel over the common roads. Ontario has 65,000 miles of country roads, and slightly over 6,500 miles of railway. With commendable enterprise, an immense amount of capital, both public and private, has been invested in the latter, with profit to the country and to the stockholder. It will scarcely be pretended, invaluable as the railways are, that they are of more importance to the country than is the primary system of transportation, the common highway. Yet so important have the railways been considered that had an equal amount been spent on the common roads these latter would have been built at an average cost of \$3,000 per mile, obviously an expenditure which, judiciously applied to the roads in accordance with the requirements of traffic, would have made this Province one of the best paved countries in the world.

In minor ways the more local means of transportation are being facilitated. Bicycles are becoming more universally adopted as a convenient means of travel. Electric railways are radiating more and more from the centres of population. The automobile carriage has made progress during the year. They are to be counted by the hundreds in London, England; and by the thousands in Paris, France. Their manufacture has been vigorously started in Ontario, and a dozen or more are in use in Toronto, while their number is rapidly increasing. Although scarcely past the experimental stage, their great utility has been demonstrated. A brief period promises a more nearly perfected vehicle at a price much reduced from that for which they are now offered. When their use has more fully developed, they are likely to exercise a considerable influence upon the public roads, by creating a much greater demand for road improvement from the people of the towns. Present indications are that they will provide a cheap and convenient means of travel and transport, not merely within the towns, but over the country roads as well.

A most erroneous view has been taken by a few, of the influence of electric radial railways upon the use of the common highway, the belief being advanced that they would eventually lessen the importance of the common road. On the contrary, the electric road is more likely, by increasing the suburban and rural population, to increase the need of good roads. The roads are the feeders of the railroads and canals, and will retain the same relation to electric roads. Electric roads will no more do away with the necessity for the best common roads, than have railways done away with the necessity for deepening and widening our canals. All our means of communication and transportation are needed, all are more certain to increase rather than to decrease in importance. The presence of electric roads in a district will increase population, increase travel, increase the use of the common road; and, inversely, the improvement of the common road will increase the use of the electric road, acting as a feeder to it.

All our modes of travel, all means of conveyance have a decided tendency to remain, merely adjusting themselves to circumstances. Over the common road must first pass all the freight received by the railways, steam and electric; over the common road must pass all the freight delivered by the railways. The horse, in spite of other means of conveyance, must remain for use in farming operations, and as such must remain for use on the common roads to compete with and regulate the cost of transportation in a manner similar to the action of railways and canals upon one another.

That our roads are bad is due to various causes, some of which could have been avoided, while others could not. Chief among these latter, the unavoidable causes, are the sparseness of population whereby a long road mileage is necessary to serve a few farms; and the character of our climate, which, with great extremes of heat, frost and moisture, is not favorable to the easy maintenance of roads. Causes which could have been avoided are the unsatisfactory systems of road construction and control which, satisfactory at one time, have outlived their usefulness; the invariable tendency in a new country is towards work of a temporary kind to answer immediate wants without a view to permanency; and the fact that the great majority of the people of this country, not having travelled in European countries, have no conception of what they should achieve and demand in the way of good roads.

A most unfortunate lethargy has existed throughout Canada as to the need for better farm roads. The farmer has said: "We have horses which are often in idleness, we ourselves have plenty of time, and we can therefore afford to drive slowly through mud, carry small loads, be entirely shut in at times by impassable roads, cut off even from the post office, the market, the school, the church, the neighbor, rather than pay money, of which we have little, for better roads." The townsman has said: "If the farmer wishes to travel in the mud, live in isolation,

reach the town only once a month, he can do so. It is none of my concern. I he comes but seldom, when he does come he must buy more of my merchandise, that is all."

All this is radically wrong. It is mutually important to farmer and townsman that the country roads should be of the best, that traffic and intercommunication of all kinds should be brisk. Free and uninterrupted intercourse for the farmer in his commercial and social relations will advance the intelligence of the farming community, will create for the farmer more ways of money-making. The greater prosperity and refinement brings in their train greater needs, together with the means of obtaining them. Along the highways will flow wealth and culture, to be participated in by the villages, towns and cities which directly or indirectly, through the retail or wholesale merchants, are mainly dependent for support upon the agriculture of the Province.

The value of agricultural land is shown by experience to be in direct ratio to the condition of the roads by which it is reached. With no roads leading to it, land has no value. When the first wagon track is made, land first obtains a real worth. In proportion to the improvements made to the wagon track, the value of the land advances, and only when property possesses the best means of communication over first-class roads does it attain its greatest worth.

Distance is measured by time and ease of travel, not by miles. Land situated one mile from the market town, but with a boggy road intervening, is frequently of less value than an equally fertile property ten miles from the market, but having good roads communicating with it. A ten mile drive over a good road is easily less disagreeable than a drive of one mile over an excessively bad road. A wagon load of produce can frequently be drawn to market over ten miles of good road; while one mile of swampy, bad road, can render the moving of marketable produce an impossibility.



BETTER ROADS IN ONTARIO.

Road and street improvement has continued to be actively urged and discussed throughout Ontario during the year 1898 with most beneficial results. The procedure outlined in the first annual report of this department has not been materially altered. Public meetings have been held in all quarters of the Province, written reports made in a considerable number of instances, and literature distributed.

The most noticeable effect of educational influences has been, perhaps, a lessening to a remarkable degree of the prejudice and suspicion with which the movement was at first met, particularly in the agricultural sections. The desultory manner in which road improvement had been urged prior to the appointment of a Provincial Instructor in Road-making had done much to create a misapprehension with regard to the object of the reform. There was a feeling that the sole aim was the abolition of statute labor. There was an apprehension that in place of this, a burdensome system of taxation was proposed, and that the townships would be bonded for the purpose of constructing the roads in an expensive manner.

It is gratifying to record the very different feeling with which the problem is now met throughout the Province. There has been every success in drawing the attention of the people to the great benefit of good roads, and this has resulted in a desire to adopt the most approved methods for improving them, both as regards actual construction, and the system of control.

There has been some misconception with regard to the class of roads advocated for this Province. "Good Roads" so readily suggests the highest type of road found in England and other European countries, that many have at once concluded that similar roads were contemplated for this country. The proposition, viewed from this standpoint, naturally invited antagonism, as the expense consequent upon so radical a change would necessarily entail burdensome taxation.

"Better Roads" rather than "Good Roads" is perhaps more expressive of the true meaning of the present agitation for reform. The roads of to-day have reached their present state by a process of evolution and development, and it is only by a similar process for the future that the desired end can be reached in an economical manner. The difficulty, however, has arisen from the fact that the methods, up to within the last couple of years, have been nearly identical with those pursued by the pioneer settlers of the country a century ago. The same system, statute labor, has been almost universally employed and the same imperfect and erroneous ideas with regard to actual construction have prevailed. It is in the correction of these that the greatest hope for the future lies.

"Better Roads" are everywhere being built in the rural districts. This is to be seen largely in the drainage which heretofore has been so much neglected. A large amount in the aggregate has been spent on modern road machinery, chiefly graders and stone crushers, whereby a much greater extent of improvement has been made. In addition to better drainage and grading and the purchase of machinery, councils are appreciating the need of procuring better gravel and stone, more durable culverts, and are in many different ways evincing their greater interest in improving the roads.

All the cities of Ontario, nearly all of the towns and many of the villages, now use artificial stone (cement-concrete) sidewalks. In some of the smaller towns, such as Ingersoll, Tilsonburg, Aylmer, Ridgetown, nearly all the sidewalks have been relaid with this material, and most of it has been down for five or six years. Collingwood, Owen Sound, Kincardine, Walkerton, Listowel, Mt. Forest, Fergus, Guelph, Barrie, Stratford, St. Mary's, Woodstock, Strathroy, Chatham, Norwich, Galt, Berlin, St. Catharines, St. Thomas, Niagara Falls, Drummondville, Brockville, Cornwall, Kemptville, and others, have all laid large quantities during the past year.

As much interest is being taken in the proper construction of streets as of sidewalks, and where once the municipality provides a proper equipment of machinery and commences to work on right lines, the same progress will be made as in sidewalks. Within the last few years, Chatham, St. Catharines, St. Thomas, Stratford, Galt, Berlin, Woodstock, Brantford, Ingersoll, Niagara Falls, Barrie have all supplied themselves with complete outfits of road-making machinery, including rock crusher and steam roller, and are doing excellent work.

It is needless to say that the larger cities, Toronto, London, Ottawa, Kingston and Hamilton also possess these implements for the construction of macadam roads. In addition reference may be made to Peterborough, Brampton, Renfrew, Brantford, and Cornwall, where equally advanced steps are being taken. In Cornwall a by-law to raise \$35,000 to be spent this year for road machinery and street improvement, was submitted to popular vote and passed. The spirit in which this work is advancing may be instanced by the following communication from G. H. Ferguson, a member of the council of Kemptville:

KEMPTVILLE, Ont. Sept. 27, 1898.

A. W. CAMPBELL, Esq., Prov. Instructor in Road-Making,
TORONTO, Ont. Can.

DEAR SIR,—Your letter of the 16th inst. I duly received. As you will perhaps remember we had about \$6,000 to spend on street improvement this year, the balance of our loan being to take up outstanding debentures. We have built about 20,000 square feet of granolithic sidewalk at a cost of twelve and one half cents per square foot. We also invested in a champion rock crusher and roller at a cost of \$1,000. We have built some macadam road much on the prin-

ciple laid down by you, and apparently it is an excellent piece of work. Of course time will tell what the quality is. The balance of our funds we propose expending also in macadam roads. The sample we put down here in the village has had the effect of starting an agitation in the surrounding municipalities on the good roads question, which no doubt will bear fruit in the near future.

Yours truly,

(Sgd.) G. H. FERGUSON.

ROAD REFORM

The road reform movement has, in Ontario, passed through a number of interesting stages. In spite of the fact that the first organizers were, in the main, farmers, business, and professional men, and journalists, it was first attacked on the score of being a fad of the bicyclists. A little later as the bicycle became more and more generally recognized, not as the toy of a sportsman, but as a vehicle of practical utility even among farmers, this ground became untenable. The opposition then settled into a defence of statute labor in its entirety, while road reform was described as an effort to impose a burdensome tax upon the farmer for the construction of expensively built stone roads. All this has gone by, and we find councils and people accepting more and more the principles of the movement, and everywhere efforts are being put forth to carry them into practical execution. There is everywhere a reasonable desire to give statute labor due credit for the great amount of good it has done; to accept it still where the local circumstances make it advisable as a means of securing better roads; but where the condition of the people has so advanced, that statute labor is no longer a just, equitable and serviceable system of taxation, it should be abolished as any other unjust tax. That is the feeling which is rapidly developing throughout Ontario to-day, one which because of its reasonableness, it is useless to oppose or curb—we can only mould and direct it.

A century has gone by since the passing of the first statute labor law in Ontario. It was an Act of the first parliament of Upper Canada, convened by Lord Simcoe, the first Lieutenant-Governor, at Newark now Niagara-on-the-Lake. This was in 1796. Less than ten years later England and Scotland repealed their statute labor law because of the excessively bad condition of the roads, substituting a system of better management and expenditure which under the methods of Macadam, Telford and McNeil, have amply justified the change. Ontario still retains the system of statute labor, and has learned few of the actual principles of road building from the men who worked so great a change in the roads of the Motherland.

The improvements made to the roads of Ontario are by no means all due wholly to statute labor. As a part of the first settlement duties, road allowances

were cleared of trees, and of stumps sufficiently to form a wagon track. Through the Huron Tract, the Canada Company spent large sums in opening up roads. The Governor's Road, from London to Toronto, the Kingston Road along Lake Ontario from Toronto to Kingston, the old stage road from Kingston to Montreal; the London and Hamilton Gravel road; the Sunnidale Road; the Hurontario Road, Yonge Street, the road from Hamilton to Niagara; these and many other roads in different parts of the Province are examples of the effect which Government aid, toll road companies, and individual efforts apart from statute labor, have had on the roads. In addition to this have been large annual expenditures of money by county and township councils.

Statute labor will do a certain amount toward the making of roads. In the brave pioneer days of this country, it did much toward taking out stumps, grading, corduroying; and has since done considerable toward gravelling or covering with broken stone in certain sections. But past a certain point statute labor will not improve the roads. When the roads have reached this point a backward trend is distinctly noticeable. They commence to deteriorate rather than improve. The incentive which actuates the pioneer who feels, desperately almost, the need of better communication with the market, the mill, his neighbor, has been lost. At this point statute labor commences to be performed in a perfunctory, shiftless, idle manner, becomes the farce that is rapidly condemning the system in all the older sections of the Province.

It is at this juncture that we turn to other countries for information as to the best course to pursue. In none of them do we find a system wholly applicable to Ontario, but our policy in this as in other matters in which we hope to progress should be to adopt that which is suitable to our conditions and surroundings, discarding that which is unsuitable.

A review of the systems used in countries which afford us a model for road construction indicates to us one great defect in our present methods. This is the inefficiency of oversight. In this respect our statute labor has become, not a system, so much as an entire absence of system. With any system rightly so called, there must be centralized control, directly responsible to the people. In France certain of the roads are maintained by the Government of the republic as national roads. The charge of these roads is deputed by the Government to the Chief Engineer who in turn allots different departments to the different members of his staff. Other roads are managed very much as county roads; others, of least importance, would correspond much to our township roads. In all these there is the one general principal—that of centralized control. On the roads of least importance the ratepayers may, if they so desire, work out their special road tax; but this work is done under the direction of a competent supervisor in whose charge the care of the roads is placed, and who is directly responsible for their condition and for the proper outlay of money and labor upon them.

In England, national roads do not exist, but in the county roads, and parish roads, the principle of centralized control again is displayed. So it also is in Germany.

In Massachusetts, on this continent, we find the best models of road construction, built by the state. They are built and maintained by centralized authority; large sums are being expended by the state, and it is expected that ultimately one-tenth of the road mileage will be built in this way. This system it is expected, will serve two purposes. One is the construction, in a permanent and durable manner, of the leading highways of the State; the other that of educating the people of the State as to the necessity of having good roads and as to the best manner of building them, so that the roads of lesser importance, still under the control of township councils, will be built in a durable and economical manner.

As has been emphasized in a preceding paragraph, the lesson to be learned from such examples as this, is not that we should adopt the system of France or England, or Germany or Massachusetts—but that we should adopt the principle of centralized control.

It is useless to say that we need not look to more experienced countries for models, by asserting that we cannot afford to build such roads as France, England, Germany and Massachusetts possess. There is no part of the Ontario Good Roads movement which advocates for this Province such expensive stone roads. But, we, if our resources are less, must husband them more carefully, apply them more judiciously and skilfully—and for this we must seek the advice of older and more experienced countries. If the various municipalities are spending as much as they consistently can on their highways, then the only means of improving the roads is to improve the methods of building and maintaining them.

There is no desire on the part of the road reformer to adopt revolutionary plans. It is rather by a process of development and growth that reform should be accomplished. How then, to apply the principle of centralized control to our present system is the first part of the problem to present itself. The trend of thought as we find it in many parts of Ontario affords a solution. County councils are taking over leading roads to build and maintain and over these competent supervisors and foremen are being placed. The feeling adverse to statute labor is rapidly increasing, township councils will find it necessary to abolish it, and over the township roads will be placed competent foremen and supervisors. However radical the ultimate outcome may seem to those who still have faith in statute labor, the process of reaching it will be one of development and growth, a gradual change. Where the system is retained, statute labor will be less and less faithfully performed, the annual appropriations of township councils will be compelled to increase, and statute labor will shrink to less and less until extinct. It is merely

a process of evolution—an instance of the survival of the fittest. It vanished from England and Scotland nearly a century ago. In the older portions of Ontario it is slowly but surely passing away.

Taxes always have been, probably always will be unpopular. One cause for this is that the reason for taxation is not asked; money paid in taxes is considered a direct loss, almost highway robbery, from which no return is received. The real spirit of a tax, however, is a united effort of citizens in obtaining some common necessity. Take for example, the money spent on roads. What would be the result if this tax were utterly abolished, no work done on the roads, no roads provided, and each man left to build a road to the market-town for himself alone? The supposition is so impracticable that it is absurd. Under such circumstances we would quickly find farmers clubbing together their money and work as they found themselves mutually interested, until we would have practically a township corporation again formed and a tax established. It is the same with schools, the same with the administration of justice, the construction of drains, etc. Taxes are a measure of economy, not extravagance. They are an unalterable example of the old adage—"in union is strength." While the principle of taxation thus commented on may seem to many too simple for consideration, it is an axiom few stop to consider, it is frequently lost sight of, and taxation becomes unreasonably unpopular.

It is this unreasonable unpopularity that so frequently lends strength to the question "What will good roads cost us?" Does the same man refuse to build fences, a stable, a house, because of the cost? What would it cost him if he did not fence his farm? What would it cost if he allowed his horses and cattle to remain unprotected all winter instead of providing warm stabling? What would it cost if he lived in a log cabin instead of a comfortable house? What are bad roads costing him?

When we commence to consider what bad roads are really costing us, the sum of it all becomes very difficult to comprehend. The cost of good roads can be stated in dollars and cents, but it is not so with bad roads. Bad roads are costing some citizens of Ontario half their lifetime in driving over them; to drive one mile occupies the time that it would take to drive five if the roads were good. They are costing many a farmer much of the self respect he feels when driving along a good road at a good pace with clean buggy, clean harness and clean horse. They are costing our farmers small sums which soon aggregate large sums, in repairs to wagons, buggies, harness and whips. They are costing the privilege of attending church and public meetings at many seasons of the year. They are costing the young people of the farms a considerable part of their education; costing them the privileges of society which every healthy mind demands. They are costing the rural districts a large share of their population which, in disgust at the stagnancy of farm life, drifts to the city, overflowing the profes-

sions and trades, and lessening the value of farm property. This and much more is the cost of bad roads. To doubt that the loss is much more is to forget the loss incurred by the business aspect of farming, by trade and commerce, by the railway and transportation system, by the manufacturing and dairying industries, all of which are both direct and indirect losers.

If then we centralize control by placing properly qualified supervision over our roads; and if, in providing proper means to carry out the work under this supervision it is found necessary to substitute for the cumbersome statute labor, a special road tax, what are some of the plain, practical improvements which would be made on the roads? A plain, practical question, requiring, however, a volume to fully answer.

Carried out as it should be, and would be, for the citizens would demand it—the road machinery would be handled to better effect, work would be done at the most suitable season of the year or whenever repairs were demanded; improvements would be made by men accustomed to road-work and therefore skilled; materials for roads, gravel, timber, etc., could be chosen and prepared under one man to better advantage than as now; the work would be done with a view to permanency, and the principles of roadmaking more closely followed.

The first principle applicable to all the roads in this Province, is good drainage. "Take the water out and keep the water out" is a rule which must be followed in the making of every road. It is not enough to dig trenches at the side of the road and call them drains. These trenches must be real drains; that is they must have a good fall to a clear outlet, and be in every respect capable of carrying the water away. The road must be rounded up in the centre so as to shed the water quickly to the side drains. There must be a track in the centre of the road along which travel can pass. This track must be hard and smooth so as to afford the least friction to travel, prevent ruts and at the same time permit water to run off the sides quickly. Ruts and holes must not be permitted to form but must be filled up as soon as they appear. The drainage must not be abandoned with taking care of the surface water, but in many instances must extend below the surface in the form of tile underdrains, so as to maintain a firm foundation and lessen the destructive action of frost—injurious only to wet and water soaked roads. In addition to the simple work on roads, better methods with regard to bridge and culvert construction are greatly needed. The principles above enumerated with respect to roads require an infinity of special methods and treatments in arriving at the best results according as the land is high lying or low, is clay, sand, vegetable-mould, etc.; is difficult or easy of drainage; according to the extent of travel, the quality of road metal obtainable—these and many other varying conditions. That roadmaking and highway work of all kinds is considered a matter in which every man is proficient is probably one of the evils underlying the bad condition of our roads to-day.

In the application of a few simple, fundamental principles lies the skill required in roadmaking as in other arts and sciences which are regarded as difficult. Roadmaking is an art, requiring much experience and skill.

Good roads bring prosperity, and prosperity brings good roads; they go hand in hand, each assisting the other. A recent writer has said that Spain would not have lost Cuba if she had joined the good roads movement. Far-fetched as the assertion may seem, it contains the element of truth. In spite of a long Spanish sovereignty, the common highways of Cuba, most fertile island of the Atlantic, are still of the most wretched description. Barbaric Spain has not yet learned what Rome knew one thousand years ago, what England well knows to-day, that roads are the arteries of commerce along which flows the life stream of a nation, bearing success, civilization and contentment to the inhabitants. The value of good roads to a community, as to a nation, cannot be overestimated.

SHOULD THE TOWNS ASSIST ?

A difficulty in the way of obtaining better roads in Ontario, is that, under the present system of township management, the entire cost of road building falls upon the farmers. The people of the villages, towns and cities, to whom country roads are as necessary as to the farmers, and who compose nearly one-half of the population, pay nothing towards their construction or maintenance.

While it is necessary that the farmer should have roads to haul his produce to the centres of population, he also uses the roads to draw back to the farm the supplies purchased in the towns. It merely happens, as a matter of convenience easily understood, that the farmer draws his produce to the town and his purchases back to the farm instead of the merchant hauling his merchandise to the farmer, and the produce of the farm back to the town.

It is only a century or so ago since the active settlement of Ontario commenced. It was at that time the statute labor system was established. In addition to this, the Provincial Government spent the greater part of its revenue on the construction of roads and bridges. Since that time the distribution of the population has materially changed; the statute labor system remains for the construction of country roads, applying only to the rural districts, while the provincial aid has been withdrawn, thus wholly relieving the people of the towns and cities from their share in the cost of road building.

The change of conditions has taken place so gradually that the evident disparity resulting has remained unnoticed. The system of road control and taxation has not grown and developed with the growth and development of the country. The result, so far as road control is concerned, is similar to a full grown man still wearing the clothes in which he went to school.

There does not appear to be any reason why the farmer and rural population should pay the entire cost of road building, any more than they should meet the entire cost of railway and canal construction. A system of good country roads is an expensive public work, in every way necessary for the development of a country, and so long as the farmer bears the entire burden, it is manifest that the desired end, good roads, will be difficult if not impossible to reach. In any event, the attempt to do so comprises an injustice.

The towns are asking for good country roads. They are beginning to realize how important for them it is to have free and uninterrupted communication with the surrounding country districts at all seasons of the year, and would no doubt be willing to pay a fair proportion of the cost in order that road improvement would progress more rapidly. This very interest which is being displayed, is the strongest evidence as to the mutual right of townsman and farmer to pay for the construction of a country road system.

The principle, since the inauguration of the good roads movement on this continent, has been recognized in a number of American states, such as Massachusetts, Connecticut, New Jersey, Rhode Island, Vermont and New York, by the establishment of systems of state aid, whereby the state funds, derived from the entire population, urban as well as rural, contribute to the cost of country road building. In all European countries national aid in one form or another is given, thus taxing the city and town population.

In England the county council has entire jurisdiction over the roads, and is aided by a grant from the national treasury. A county engineer is appointed, a salaried officer, whose sole duty is the supervision of road work. The county is divided into districts, and the detail of the road work is attended to by assistant engineers acting under the parishes. The money required for the maintenance of highways is obtained by a precept issued by the county council to the various parish councils demanding the amount expended on the roads of the parish. This amount is collected in the general parish (or township) taxes, levied on the assessment values of property. A system of county management has been extended to all parts of Scotland. The roads of Ireland are under a county engineer and several assistants, each of the latter having his own district, improvements being regulated by a grand jury presentment system.

French roads are national, departmental and communal; corresponding largely to state, county and township roads. The national roads radiate from Paris, extending to all the important cities and departments, and are under a special engineering department (department of bridges and roads) attached to the national government. The second and third classes, departmental and provincial roads, are in a general way under local authorities, but departmental roads are usually entrusted to the care of the national corps of engineers.

Germany has a magnificent system of turnpikes built and maintained by the national government. They are under the general management of a state road commissioner, while he is assisted by an extensive staff of road directors and inspectors. Other roads are known as "country roads," and are built and maintained by the several parishes through which they pass.

The highways of Austria are classified as state or Imperial roads, provincial roads, district roads, and community roads, according to the authority constructing and managing them. The cost of building and maintaining the Imperial roads is derived from the national funds, the cost of provincial roads from the provincial funds, district roads from district funds. A little of the cost of community roads is borne by the several communities interested, aided in certain cases from the district funds. For the Imperial and provincial roads the best of engineering skill is employed, while for immediate repairs, road keepers are at work constantly.

Italian roads are under the supervision of the Minister of Public Works, and are national, provincial, communal, or vicinal, according to the source from which taxes for construction and maintenance are derived.

The more important roads of Denmark are controlled by the county councils but are subject to the annual inspection of a state engineer; the roads of lesser importance are governed by the parish or township councils.

The main roads of Belgium, those routes running from one part of the kingdom to another are controlled and managed by the state; another class, provincial roads, are controlled by the province; a third class, communal roads, are controlled by the communal authorities. The construction of these roads is entrusted to a corps of engineers.

In the Netherlands, a network of roads, providing convenient travel from one part of the country to another, is maintained by the general Government; other roads are at the expense of the various provinces and communities benefited.

The federal government of Switzerland controls a few of the important roads, but in the main they are built and maintained by the cantonal government through whose territory they pass. The construction and repair of roads of lesser importance pertains to the several townships through which they pass.

Spain, decayed and tottering, the vestiges of an ancient magnificence falling from her, has not joined the good roads movement; nor has Turkey, the home of barbarism. Russia, too, has been exceedingly backward in road-building, and as a result her extensive and rich dominions are still practically undeveloped.

The immense benefit conferred by good roads has practical illustration in many parts of Europe, where it is of unusual occurrence to see a load drawn by more than one horse. In these countries so excellent are the roads, that any one owning a team of horses does not consider himself dependent upon the rail-

roads for transportation, as wagon loads of from three to six tons are frequently drawn several hundred miles in competition with railway rates. In these cases the roads are, of course, of the best possible construction, built and maintained under experienced engineering supervision, and are a skilful compromise between ease of grade and directness of route.

PETITION OF THE DUFFERIN COUNTY COUNCIL.

The following petition is intimately associated with the principle just discussed, state aid, and was presented to the Legislative Assembly of Ontario by the County Council of Dufferin.

To His Honor the Lieutenant-Governor in Council, and to the Members of the Legislative Assembly of the Province of Ontario :—

The petition of the Council of the County of Dufferin humbly sheweth :—

That an averaged organized and settled township of the Province of Ontario expends annually about \$2,500 for the opening up, repairing and deviation of highways within its limits. That owing to the want of experience of those entrusted with the work, lack of sufficient funds and the absence of uniformity and system as to method, the public do not realize the benefits it is entitled to expect from so large an expenditure as \$25,000 in ten years, as the condition of our highways in the fall and early spring, and the snow drifts in the winter amply testify. That the obstruction to, and sometimes absolute prohibition of traffic owing to the defective highways entail much annual inconvenience and loss to the farming, town and city communities, by reason among many others of the farmers being unable to team their products at a time when the highest prices obtain, and by reason also of them not being able without much inconvenience and loss, to keep up communication for other purposes, with those with whom they have business and social relations.

That, in the opinion of your petitioners, no more important duty can be undertaken, than that which will ensure speedy, safe and easy communication between the rural districts and the towns and cities and between the farmers themselves at all times of the year. That it is the opinion of your petitioners, that uniformity and system in road construction on modern and enlightened principles, are necessary in order to obtain the maximum of advantage to be derived from so large an expenditure on roads.

That recently the Province of Ontario has established the office of Provincial Road Instructor ; and in the opinion of your petitioners, this uniformity and system in road building may best be obtained by the Instructor's supervision, in connection with county roads commissioners ; provided Provincial aid be obtained in the payment of such work.

That the support of education, the poor, the erection and maintenance of county buildings, roads and bridges, administration of the laws, fire protection, lighting, water services and many other municipal necessities already entail as great burden on the local municipalities as they can at present well bear without incurring other financial liabilities.

That many of the States of the adjoining Union have adopted the principles of state aid to highways, with ever increasing advantage.

Thus the small State of Massachusetts in 1895 expended the large sum of \$600,000 in aid of this work. In New York State a bill recently became law to the effect that, on a petition to the county council, certain roads may be adopted as state roads. The petition is presented to the State Engineer and if it obtains his approval, he presents plans and specifications to the Legislature, and if approved by that body, fifty per cent. of the cost of construction is paid by the State.

In New Jersey the State pays three-tenths of the cost, the county six-tenths and the owners of the lands affected by the improvement, one-tenth.

In Connecticut any township may prepare plans and specifications and let the contract, under the supervision of the state commissioner, in which case one-third of the cost is paid by the state, one-third by the county, and one-third by the township.

In Rhode Island, California, Indiana and Kentucky, somewhat similar provisions exist, the principles running throughout all of them being, that the local municipalities prepare the plans, let the contracts under which the work is done, under the supervision of the state commissioner, on a uniform system so as to secure a permanent highway with the least expenditure, to which the states liberally contribute.

That, in the opinion of your petitioners, some method could be adopted by legislation by which the public may secure uniform and permanent highways, with protection from snow drifts, with the most economical expenditure of money, to be constructed under some general and competent supervision, in order that this Province may secure its full development and increase of population, and that it may not lag behind the States of the adjoining Union.

Your petitioners therefore pray That legislation may be adopted in order to accomplish the objects above indicated.

(Sgd.) JOHN C. REID,
Clerk of County of Dufferin.

(Sgd.) W. H. HUNTER,
Warden of County of Dufferin.

A LESSON FROM SCOTLAND.

Everywhere in romantic, hilly Scotland good roads are to be found, and only those who have travelled over them know the great measure in which they add to the real merits of the rural districts. Not merely in the well populated districts are the roads good, but in the craggy, mountainous parts as well, where sheep-grazing is almost the only possible means of utilizing the lands. When any portion of these roads is out of repair, the traveller may know that one man at least is not doing the work for which he is paid.

In Scotland farmers do not work on the road, but spend their time where they can do so to better advantage—on the farm. The repairing of roads is let by contract and is under the supervision of an engineer or inspector, the inspector advertising for tenders on so many miles of road. It is then the duty of the contractor to see that the gutters are kept open and the water drawn off, particularly in the fall and spring of the year. Broken stone is furnished him and, with a one-horse cart, he draws this wherever needed for patching, filling up hollows and ruts as they form. In this way the roads are kept smooth as a floor, so smooth that very rarely in Scotland is a farm conveyance drawn by more than one horse.

Roughly estimating, the settled area of Ontario approximates the total area of Scotland, about 30,000 square miles, while, the population of Ontario is about one-half that of Scotland. At the time when statute labor was abolished in Scotland—nearly a century ago—the population was less than the present population of Ontario, being then only 1,800,000, while Ontario has now over 2,000,000 inhabitants. Ontario with respect to her highways is one hundred years behind Scotland, but is better able in wealth, population and experience to make rapid strides for their improvement. The seed of a better system and better methods is being scattered over the Province, and less than a quarter of a century should find us on equal footing in every respect with the older lands.

A COUNTY SYSTEM,

That every county municipality should undertake the management of its leading roads is very much to be desired in the interest of better roads. There are a number of good reasons for the introduction of such a system, which has been thoroughly tested for several years in the County of Hastings. During the past year it has been actively discussed in the Counties of Oxford, Wentworth and Victoria, and although decided action has not yet been taken, there is every reason to believe that, when better understood, the prejudice which at present exists in opposition to it, will cease to prevent its adoption.

Under county control a properly organized corps of men can be employed to build and repair roads. As at other employments they become experienced and do better work, and in the matter of repairs are ready to make them as soon as signs of wear appear.

By a county plan, uniformity in system and work will be secured throughout the various municipalities. Whereas under township control a diversity of plans is sure to be adopted.

In a county plan an experienced and properly qualified man could be employed to have constant supervision of the work, whereas under township control, each municipality cannot afford to pay the salary of such a man. Under every good system of government it is necessary to have responsibility centralized and defined, not divided and easily shifted from one to another, as it is now under the statute labor system.

Under county control machinery can be handled to better advantage as an experienced operator can be employed for each implement, and a better and more uniform class of work will be secured.

A township can manage its roads properly only by adopting a plan similar to that outlined under a county system. But by extending it over the county it becomes more cheaply operated.

A properly connected system of leading roads throughout the county will be obtained under a county system; whereas with each township working independently of those around it, this will be lost sight of.

There is no community of interest between the townships. In one township there is a certain leading road much travelled and well made and maintained. The adjoining municipality may for various reasons not consider the continuance of that road through it of so much importance as to warrant them in making an expenditure to benefit largely their neighbors who are obliged to travel over it.

If no greater expenditure is made upon the roads than at the present time, the rate will be reduced because most of the township expenditure is now placed on the leading roads; and the township will be relieved of these by a county

system. Under the county system the funds will be sufficiently concentrated to undertake durable work, and consequently these roads will be properly constructed and afterwards maintained at a less cost than at present.

A county road system equalizes the cost of maintaining leading roads. In every county within a certain radius of a market town, traffic constantly increases as the town is approached. The cost of construction and maintenance increases in proportion to the traffic. It is unfair to charge those living near the town with the cost of keeping the roads to support the traffic from a distance; so unfair as to cause discouragement and often withdraws support.

Property is very largely valued according to distance from the market, and the convenience with which the market can be reached. Property a long distance from the market is affected to a greater extent by the bad condition of the roads than is property very near the market. Good roads are therefore of greater value to townships a long distance from the market town than those in the immediate vicinity.

Under a county system proper road construction would be undertaken and the economic value of this work would be seen and appreciated by the people of the different townships. The well-built roads would stand as object lessons, and would teach the better expenditure of the funds spent by the townships on the roads maintained by them.

STATUTE LABOR.

The demand for better roads has directed public attention to the merits and demerits of statute labor, and a general feeling in opposition to the system is accumulating and gathering strength, promising within a few years to make a radical change in the method of caring for roads.

Previous reports have dealt upon this aspect of the case to a considerable extent, so that it scarcely becomes necessary to again enter upon a detailed discussion.

The system, under certain conditions existing chiefly in pioneer districts, is doubtless one of the best which could have been adopted, and has done exceedingly good work. Provision has been made under the Municipal Act, however, for substituting more suitable methods where these conditions cease, as they have done in many localities.

For well populated districts, where more advanced principles of government should be applied, the system of statute labor is fundamentally weak from two main causes which bring in their train many abuses, and cause failure in the effort to improve the roads in a manner corresponding to increasing requirements. These causes are, first: That control is so scattered that no one can be held responsible for a proper expenditure of the labor and money devoted to roads:

And second: That the payment of a tax in labor is too vague and clumsy to meet modern conditions. Some pay the tax with honest work, while others, from various motives, do not.

From these two sources spring many important results; neglected roads, injustice to the men who do honest work, misappropriation of labor and money, and similar abuses.

Under present conditions, all responsibility is, in the first place, distributed among the members of the council; the council again distributes it among fifty or one hundred pathmasters. A considerable sum in the aggregate is annually spent by most townships in the way of money and statute labor, but so scattered is this, together with responsibility, that the people scarcely realize how large a sum is being spent; and when mis-spent and squandered as it generally is, there is no one whom they can call to account. The councillors can shift responsibility from one to another and then to the pathmasters. The latter have nothing to lose, and so can merely shrug their shoulders when any improper or careless expenditure comes to light. Divided responsibility is never a successful form of government.

There should be a public servant having a position similar to that of a clerk or treasurer, who can be held responsible for the proper disposition of money and labor applied to the roads. It is the only means of securing the most beneficial and economical expenditure. The position must be made a desirable one, both from the fees and honor attached, in order that the best men may be secured, and in order that they will be sufficiently anxious to retain the office to give their best services in planning and supervising the work.

The number of enquiries received by the Roads Branch, respecting methods of replacing statute labor is constantly increasing, and strongly indicates a trend of public feeling which will shortly operate throughout the Province generally in remedying the present weaknesses—weaknesses which exist merely because the attention of the public has not heretofore been directed towards the necessity of reform. A communication received from the Clerk of Minto Township, Wellington County, together with the reply thereto, will afford an illustration of the feeling in this respect, and the reforms recommended.

HARRISTON, Nov. 14th, 1898.

A. W. CAMPBELL, Esq.,

Provincial Instructor in Road-Making, Toronto.

DEAR SIR,—We propose submitting a by-law to be voted on by the rate-payers of the Township of Minto at the next municipal elections for the abolition of statute labor, and as several methods have been talked of to replace it, I take the liberty of asking you to give us your views as to the best methods to be adopted.

The number of days statute labor in our township for the past year was 3,508 and it is proposed to levy a rate sufficient to raise an amount equal to about 50 cents per day, or say, \$1,750, to be expended in road improvements in addition to what is now spent, which is about \$1,100 exclusive of bridges.

What we would like your opinion about is the proper method of expending the money. Should it be expended by the Councillors, each acting as Commissioner for his own division; or would it be better to place it in the hands of one Commissioner, or several, not members of the Council; or should it be apportioned to the pathmasters to be expended in their several divisions?

I may mention that there appears to be the greatest opposition to the by-law on the grounds that the present system is good enough, that by the new system the work would not be so evenly distributed and that too much money would be appropriated for commissions.

From the extensive knowledge you have in all matters relating to road making and from a desire of our Council to put the matter in a practical light before the electors before having it voted on, we considered it highly desirable to obtain your opinion on the matter. Any further suggestions outside of what I have indicated in this letter will be fully appreciated.

Yours respectfully,

[Sgd.] W. D. McLELLAN,
Clerk of Minto.

TORONTO, November 17th, 1898.

W. D. McLELLAN, Esq.,

Clerk of the Township of Minto, Harriston, Ont.

SIR,—I am very glad to see that your council has taken the progressive step of submitting a by-law to the ratepayers, at the next municipal election, for the abolition of statute labor.

I think, wherever this move is undertaken, the same opposition which you mention is made, viz., that the money will not be evenly distributed throughout the township, as the work is at present. The chief weakness of the statute labor system is the very fact that the labor is so distributed or scattered that in no section can durable or substantial work be undertaken for the want of a sufficient amount of labor. If the labor could be concentrated and faithful service performed along a well defined plan, good results would be produced. But this would be impossible. The money can be distributed in the same way by returning to each pathmaster the amount collected in his beat; but this would be only a very slight improvement, as experienced by the township of Winchester, where, upon making the change, this plan was adopted.

To reach successful results, the expenditure must be concentrated. All of your roads cannot be improved in one year nor a few years, but the whole of your roads should be planned, the township divided into sections, and the money in each section spent in a methodical way by commencing on the more important roads at one or more important points in each section, as your council may decide, finishing any work undertaken in accordance with your general plan and extending this work from year to year until the whole is finished. Every man

in the section or ward cannot be benefited by the first year's expenditure, but under a well defined system, each year's expenditure will bring the improvement closer to him, until finally each ratepayer is served. It would be necessary to strive to raise your people above jealousies and prejudice, and, in the interest of the township, look beyond the present to the improvement a few years will produce.

Your proposal to raise the money by levying a rate equal to about fifty cents a day is right. This added to the \$1,100 will make a fund which, if judiciously expended, will in a short time work a wonderful improvement on your roads without incurring an excessive tax. I would suggest dividing the township into four sections of about equal area. In apportioning the expenditure to the different sections, the road mileage, extent and character of the travel, accessibility of the gravel or other road material, and the nature of the land surface, whether hilly or comparatively level, should be taken into account.

The division of the township into four sections will insure the ratepayers a fair distribution of the money to each part of the township. I favor the appointment and continuing in office of one commissioner to have charge of the expenditure on all roads and bridges, as well as the care and operation of all road-making machinery; such commissioner of course being an officer of the corporation and a servant of the council. He should report to the council with any recommendations before any work is done, and should proceed with the work only when authorized to do so by the council. Payment for all work should be made only upon the certificate of the commissioner.

As far as practicable, the council should prepare a plan and specification, classifying the roads according to their importance, and should specify width of grade, amount of crown, width and depth of gravel or broken stone to be placed on each, in order to secure, as far as possible, uniformity in work.

The commissioner should be a man of some municipal experience, familiar with the township, with some practical knowledge of handling men and directing work, as far as possible independent of outside influence, and who has sufficient interest in the township to secure his continuance in the position.

He need devote only such portion of his time to the duties of his office as the council may think necessary, but should be free at any time in the spring, fall or winter to make inspections and have such work done as may be needed during these seasons, or made necessary by snow storms, freshets or other emergency.

Such a man can be found in each township, but great care should be taken in his selection, as the success of the whole system depends on him. He need not be an expensive man, as his employment should be by the day, and as a general thing not more than six or eight weeks per year of his time is required. A fair rate, including horse and buggy, is \$2.00 per day.

In the spring of the year the commissioner and the council should make an examination of the roads proposed to be improved, bridges, culverts, etc., and then decide what work is to be done.

Specifications should then be prepared, and as far as possible all works of construction, such as hauling gravel, digging drains, building bridges and culverts, cutting hills, should be let by public competition. Contracts should be entered into and security given. All grading should be done by a grading machine, which is the cheapest, simplest and most economical way of doing the work properly and expeditiously. This machine should be owned by the township, and operated where directed by the commissioner, the commissioner being given authority to employ whatever teams and men are needed.

The same teams and men, as far as possible, should be retained during the season's work, as they thereby become proficient and produce better results. The time for doing this grading should be in the early part of the season before the roads become baked and hard. There are many minor details of the work which the commissioner will regulate after a little experience, and which will be of great service in improving the roads; and a year's experience will show that the expenditure which you contemplate, made in this way, will do a surprisingly great amount of work, and will indicate to your citizens that system and method are what is required to make good roads in your township in a reasonable space of time, without increasing the present taxation for this purpose.

The chief objection to appointing councillors as commissioners is that a change of council means a change of commissioners and a consequent change of methods and ideas of doing work.

Besides, there is the influence which ratepayers are ever willing to exert upon their representative to secure favors, and which experienced councillors know will seriously tend to cripple the very object which your proposed change is intended to accomplish, viz., concentration of expenditure.

The sphere of a council is administrative merely, and the execution of the work should be through independent agents.

The commissioner will relieve councillors from a good deal of worry and humiliation, consequent upon carrying out such works. In one commissioner for the township you have an official whose continuance in office would be similar to that of a clerk or treasurer whose duties are fixed by by-law, and costing no more than a commissioner in each division.

I have recently received a letter from the clerk of N. Monaghan, in which township a commissioner is appointed in each district, but this plan they do not appear to favor and are now making an effort to place the work in the hands of one commissioner.

In fact I think the letter may be of some information to your council, and have taken the liberty of sending you a copy.

This is a matter which should be clearly placed before your people prior to the day of voting, as unfortunately in many municipalities the change is not made simply because the ratepayers do not understand, before voting, the effect the change will produce. Had this plan been adopted ten years ago, with the liberal expenditure you are making (say \$28,000, or its equivalent, expended on such a system) I think the greatest opponent of reform must admit that the results in your township would be vastly better.

Should you require fuller information on any point connected with the proposed plan I would be very glad to discuss it with you, or if your council think I can be of any assistance in discussing the matter with the ratepayers I would willingly address any meetings you may arrange for that purpose.

Trusting to hear from you again with reference to the matter,

I remain, yours truly,

(Sgd.) A. W. CAMPBELL,
Provincial Instructor in Road-Making.

ROAD REFORM IN OUTLINE.

The following is an outline, in brief, of a system of road control which a great many townships in Ontario could consider with profit.

Do away with the statute labor roll entirely.

To raise the money required, levy a rate on the assessment of the township.

For road purposes, divide the township into a convenient number of divisions, usually four.

Apportion the money equally among the road divisions, keeping in view all circumstances, viz: Importance of roads, works needed on them, benefit resulting to the greatest number of people, amount of traffic, assessment, etc.

Appoint one township road commissioner to advise and consult with, and carry out the direction of the council.

The office of road commissioner should be similar to that of the township clerk or treasurer.

Councillors should not act as commissioners, as they are subject to undue influence from the ratepayers, and their term of office is uncertain.

A general plan for road improvement should be laid down by the council for the commissioner to follow.

This plan should specify the width to be graded, width and depth of road metal, character of drainage, etc., of all roads.

Roads of importance should not be less than twenty-four feet between the inside edges of the open ditches. No road should be of less width than eighteen feet.

Early in the year the council and the commissioner should go over all the roads to consider the work to be undertaken.

Work of construction, such as hauling gravel, ditching, and drainage, building of bridges and culverts should be done by contract, and supervised by the road commissioner.

No account for labor or material should be paid by the treasurer except on the certificate of the road commissioner.

Minor work and repairing should be done by day labor, only the road commissioner being authorized to employ, direct or discharge men or teams.

All roadmaking machines should be in the care of the road commissioner.

Only the road commissioner should employ, direct or discharge the men or teams needed to operate the machinery.

Should the council desire to interfere in any of these matters they can do so through the commissioner.

The same man and teams should be hired to operate the machinery for the entire season, or longer if possible, as they become proficient and do better work. This applies particularly to the operation of a road grader.

The commissioner should keep a pay roll to return quarterly to the council showing who have been paid and the amount paid, the roll to be then filed for auditors.

This roll will act as a check on favouritism on the part of the commissioner. Work should be divided as much as possible among the residents of the township desiring it.

Work should be commenced with a definite end in view and continued systematically, from year to year if necessary, until the entire road mileage has been brought to a proper standard.

NORTH MONAGHAN - STATUTE LABOR ABOLISHED.

Enquiry made of the Clerk of the Township of North Monaghan, Peterborough County, elicited the following reply from Mr. G. W. Bennett, Clerk of the Township.

DEAR SIR,—In reply to yours, 31st August, I beg to state: North Monaghan, is a small township, about 13,871 acres with an assessed value of \$611,750, with between 35 and 40 miles of roads—a large proportion of which are earth roads. Situated as we are, adjacent to the town of Peterborough, a large number of our ratepayers are gardeners, butchers, milkmen, etc. In fact a majority of our ratepayers attend the market more or less three or four times a week all the year around. It can easily be seen how important it is to have the roads in a first-class condition, more especially as prices for produce usually incline upwards during the season of bad roads.

Our system now is commutation of our total labor, about 1,150 days, at 50c. a day. This is levied and collected as ordinary rates.

Our township is divided as equally as circumstances will admit into four districts, in each of which a commissioner appointed by the council has entire supervision.

As soon as possible after Court of Revision, I, as clerk, make a list of all parties in each division liable to pay the tax and amount of the same. Each commissioner is supplied with the list for his division, this for the purpose of making as equitable a distribution of the money as conditions will admit. This money is entirely expended in gravelling, *i.e.*, hauling gravel and metal from pits, some of which are private and belong to the township, and dumping upon such portion of the roads as are graded and water-tabled to receive the same.

The gravel from private pits costs us ten cents per yard, that from the township pits infinitely less, so much so, in fact, that no township should have, if possible, to pay for road metal by the yard. Our commissioners find it a great advantage to use a rooter plough in the pits to facilitate filling, and from township pits they can select such material as best serves their purposes on the different kinds of roads.

In addition to the commutation tax, each year, in our estimates there is an appropriation made for gravel, grading and grants, *i.e.*, to operate a road grader,

one of which we have purchased, and grants on boundary lines where we are required to give an equivalent for maintenance, and for repair and reconstruction of culverts and bridges, of which we have a large number.

All these works are directly under the supervision of the commissioners, as such occur within their respective divisions. All gravel is paid for on the certificate of commissioner who receives blank forms for that purpose.

Our council at an early date each year go over each district in company with the commissioner and suggest and lay out work to be undertaken during the year.

The method of letting jobs by auction tender also, is frequently adopted in cases requiring a larger expenditure than ordinary. This plan has been adopted, I may say, to facilitate the distribution of money as much as possible. In each case the contract is subject to the inspection of the commissioner in whose division the job is let, who reports on the same to the council.

Previous to this year we had a permanent gang of three men and two teams on the road grader who were controlled by the council. This method was discarded this year as it was found too much grading was done for the amount of gravelling, and time, too, was wasted going about from place to place at the dictation of any member of the council.

We this year only employ an operator, who has become an expert by practice, and is under the direction of the commissioner when required by him, who supplies teams and uses the grader when and where he can do so to the greatest advantage, observing to do no more grading than can be well metalled in any one year. I may here say that out of twelve scrapers belonging to the municipality previous to the purchase of a grader we can only get track of three, so little are they called for.

Our Council have a fixed rate of pay for all purposes, as follows:—

Commissioner	\$1 50 per day.
Operator on grader.....	1 25 "
Breaking stone or filling.....	1 00 . "
Man and team for any purpose.....	2 50 "

Each Commissioner is supplied with pay sheets and a time book which he is required to fill up, and as soon as required, the Council provide sufficient money which is paid to the order of each Commissioner who distributes it according to his pay sheet. This pay sheet when properly signed and certified is returned and examined by the Council, then if correct is filed for the Auditors, or for future reference. This examination by the Council is a check upon anything like favoritism on the part of a Commissioner.

This year our appropriations were as follows:—

Commutation of statute labor for gravelling.....	\$550 00
Road grader operator.....	50 00
Bridges and culverts.....	400 00
Grants and miscellaneous.....	100 00
	\$1,100 00

I may say that for 1898 I have not the complete returns from the Commissioners as to amount of work done, but for 1897—which would be about the

same—there was about $8\frac{1}{4}$ miles of highway graded and water-tabled at a cost of about \$34 per mile. A considerable portion of this will require the grader over it again as it was found that during the wet weather in spring and fall the “narrow tire” cut into the grade, making ruts. This year there was much less grading done and also at a reduction in the cost. In 1897 there were 2,083 yards of road metal, gravel, etc., drawn at an average cost, laid on the road, of 40 cents per yard.

There are other matters connected with our efforts to secure better roads, but I find this becoming too lengthy. However, although this may not be the best method of road construction one thing is certain, there is not a man of any importance in this township who would go back to the old statute labor way. There are a few kickers whom the Commissioners gave the go-by as not worth their salt. I find from direct communication with both Council and Commissioners, as well as a thorough knowledge of the condition of our roads, that until 1896 there was little or no work done in this municipality with a view of permanency. Now I find stretches of roads all over the township that will compare favorably with almost any street in the Town of Peterborough, where they have procured modern road-making machinery and approved methods.

Another thing, at a meeting of the Council last week attention was drawn distinctly to the fact that amongst other improvements none have stood out so prominently as that of employing two first-class men at the dumps when graveling, supplied with a shovel and two stone hammers. Another important observation I can make.

The Commissioner having control of the work can and does insist upon a uniform box on each wagon with a result that, almost without exception, all the old boxes that were good enough for statute labor have now been discarded and new dump boxes of uniform depth and length have taken their place. This is important in many ways; not only does the township get the full value for their money in material, but each man draws an equal share, and it facilitates the evenness of the gravel when dumped upon the road.

There is an agitation on foot now in our Council to have *one capable Commissioner* over the entire township, and if it is possible to cut loose from the patronage which these appointments give each Councillor and appoint one good man who can employ foremen when desirable, they think it would be a still greater improvement.

Our Council are divided upon the commutation method. Some hold the view, and I endorse it, that the abolition of statute labor and commutation entirely, levying a general rate sufficient to meet the appropriation as found necessary after the annual inspection—in the spring—would be a better way.

In conclusion I might say we hope to have it within our means to acquire a road roller and crusher, and also to see a “wide tire law” within a few years.

I have the honor to be, Sir.

Yours truly,

[Sgd.]

GEORGE W. BENNETT,

Peterborough, 24th September, 1898.

Clerk of North Monaghan.

PREPARING FOR ROAD WORK.

The first thing the municipalities throughout the State should do is to have the roads already in the season of the year when the work is to be done. The improvements to be made should be planned for and money should be thoroughly raised before the work is started. When the men are at work, the work should be done in such a way that the waste of a great portion of the material is avoided. Every step in the work should be planned for by the pathmaster and if there is a capable surveyor, he should be consulted with whom he can consult. The work should be done in such a way that the more prospect is there that the work will be done in a proper and economical way.

When the work is to be undertaken and constructed the pathmaster should know what kind of drainage is needed; he should know what amount of grading is needed, and in what order the various parts of the work should be undertaken; and he should know how many days of labor, what tools and what tools or machinery are needed in each successive step. All lumber, etc., or other material required should be purchased in advance and should be on the ground when needed.

More than knowing what tools are needed, the pathmaster should take proper steps to see that they will be furnished. Otherwise there will be too many wagons and not a scraper to be had, too many with shovels and not enough with teams, half a dozen plows and no shovels. Any implements owned by the municipality should be looked after early in the season to see that they are in proper repair.

In notifying those by whom statute labor is to be performed, only such a number as can be properly directed by the pathmaster and used to the best advantage on the work should be called out for one time. To call the men out without reference to the needs of the work is either an evidence of mismanagement on the part of the pathmaster, or if it cannot be done, is generally an indication that the statute labor of the district is of very little value.

Gravel pits should be attended to before the material is hauled to the road. They should be stripped of the layer of earth which generally covers them, so as to avoid delay when the teams are in the pit; or if this is neglected, the probability is that the dirt will be permitted to fall into the pit, mix with the gravel, and be placed on the road, producing injury rather than benefit. If the gravel needs screening or crushing before being placed on the road, this should not be

left until the teams are in the pit, but is a matter for which money appropriation should be made, as statute labor can be utilized to best advantage in hauling the gravel.

DIRECTIONS FOR PATHMASTERS.

The following directions contain in brief the elemental principles which should be followed in the construction of a country road. They are adapted to the statute labor system, and to the means and conditions which prevail in townships where more radical reform has not been undertaken. Councils in the majority of cases would obtain better results from their statute labor by distributing freely among the ratepayers, prior to the time of performing road work, copies of such a list of instructions.

1. Every good road has two essential features :

(a) A thoroughly dry foundation.

(b) A smooth, hard, waterproof surface covering.

2. The foundation is the natural sub-soil, "the dirt road," which must be kept dry by good drainage.

3. The surface covering is generally a coating of gravel or broken stone, which should be put on the road in such a way that it will not, in wet weather, be churned up and mixed with the earth beneath. That is, it should form a distinct coating.

4. To accomplish this :

(a) The gravel or stone should contain very little sand or clay—it should be clean.

(b) The road should be crowned or rounded in the centre so as to shed the water to the open drains.

(c) Ruts should not be allowed to form as they prevent water passing to the open drains.

(d) The open drains should have a sufficient fall, and free outlet so that the water will not stand in them but will be carried away immediately.

(e) Tile under drains should be laid wherever the open drains are not sufficient and where the ground has a moist or wet appearance, with a tendency to absorb the gravel and rut readily. By this means the foundation is made dry.

5. Do not leave the gravel or stone just as it drops from the wagon, but spread it so that travel will at once pass over and consolidate it before the fall rains commence.

6. Keep the road metal raked or scraped into the wheel or horse tracks until consolidated.

7. Grade and crown the road before putting on gravel or stone.

8. If a grading machine is available, grade the roads which you intend to gravel before the time of statute labor, and use the statute labor as far as possible in drawing gravel.

9. A fair crown for gravel roads on level ground is one inch of rise to each foot of width from the side to the centre.

10. The road on hills should have a greater crown than on level ground, otherwise the water will follow the wheel tracks and create deep ruts instead of passing to the side drains. One and one-quarter inches to the foot from the side to centre will be sufficient.

11. Repair old gravel roads which have a hard centre but too little crown, and which have high square shoulders, by cutting off the shoulders, turning the material outward and placing new gravel or stone in the centre. Do not cover the old gravel foundation with the mixture of earth, sod and fine gravel of which the shoulders are composed. The shoulders can be most easily cut off by means of a grading machine.

12. A width of twenty-four feet between ditches will meet most conditions, with the central eight feet gravelled.

13. Wherever water stands on the roadway or by the roadside, or wherever the ground remains moist or is swampy in the spring and fall, better drainage is needed.

14. Look over the road under your charge after heavy rains and during spring freshets. The work of a few minutes in freeing drains from obstruction or diverting a current of water into a proper channel may become the work of days if neglected.

15. Surface water should be disposed of in small quantities; great accumulations are hard to handle and are destructive. Obtain outlets into natural water-courses as often as possible.

16. Instead of having deep open ditches to underdrain the road and dry the foundation, use tile.

17. Give culverts a good fall and free outlets so that water will not freeze in them.

18. In taking gravel from the pit, see that precautions are taken to draw only clean material. Do not let the face of the pit be scraped down, mixing clay, sand and turf with good gravel. There is a tendency to draw dirty gravel as it is the easiest to handle.

19. Gravel which retains a perpendicular face in the pit in the spring, and shows no trace of slipping, is generally fit for use on the road without treatment. Dirty gravel should be screened.

20. Plan and lay out the work before calling out the men.

21. When preparing plans keep the work of succeeding years in view.

22. Call out for each day only such a number of men and teams as can be properly directed.

23. In laying out the work estimate on a full day's work from each man and see that it is performed. Specify the number of loads of gravel to constitute a day's work. Every wagon box should hold a quarter of a cord.

24. Make early arrangements for having on the ground when required, and in good repair, all implements and tools to be used in the performance of statute labor.

25. Do all work with a view to permanence and durability.

THE BEST USE OF STATUTE LABOR.

Experience, wherever statute labor has been employed, has shown that it can usually be made most productive of good when used in teaming. Without a great deal of friction, wagon boxes of a fair size can be insisted upon, and honest loads drawn. Appropriations from the general funds should be so directed as to utilize as much of the statute labor as possible in this way; they should provide for the preparation of road metal; that is, the crushing of stone or the screening of gravel, or whatever preliminary treatment is needed to provide a good metal. It is better that a grading machine should be used on the road some little time before the gravel or broken stone is applied, and this is a matter which can be profitably attended to by the council. Draining, or a major part of it, and the construction of culverts are details which should be looked after, independently of statute labor. Townships which can anticipate statute labor by preparing the gravel, grading the roads with improved machinery, and constructing culverts and drains out of the general funds, using the statute labor in loading the wagons, teaming and spreading the metal on the roads, will obtain the most satisfactory results.

GRAVEL.

Councils very commonly give but little consideration to the matter of procuring gravel for use on the roads. Pathmasters are authorized to obtain gravel from a certain pit. It is paid for by the load, and the teamster performing statute labor is left to decide the size of the load and quality of the gravel. In the performance of statute labor horses are treated very mercifully, the smallest and oldest wagon is used, the laborer is "working for the Queen," and the result usually is that the owner of the pit obtains for half a load of boulders and turf, the price of a load of clean gravel. The only fair way to buy gravel is to purchase the pit outright. Test pits and borings can be made, and the quality and quantity of gravel obtainable estimated therefrom. Buying gravel by the

load is like buying water by the bucketful. Cases have occurred in which the money received for the gravel in a pit in the corner of a farm has amounted to enough to pay for the entire farm.

If the waste in this way ended with the extravagant amount paid for the gravel, the amount would be comparatively small. The great difficulty arises from the fact that the money that can be spent for gravel is limited to a certain amount. If only half a load is obtained for every full load paid for, only half the possible extent of road is gravelled and the roads suffer in consequence.

The gravel used should be clean, free from clay and sand, and if these are mixed with it in any quantity the gravel should be screened. If large stones or boulders are contained in the gravel it is a good plan to place a stone crusher in the pit and pass all the material through it. By having a rotary screen attached to the crusher the dirt and sand will be removed. Too much stress cannot be placed on the necessity for clean material. There is usually enough dirt on the road before the gravel is put on without going to the trouble of drawing it from the pit.

Some townships are wasting an enormous amount of gravel, and in a few years will have to face a serious difficulty—that of finding a substitute. There are townships now not far from that difficulty; which have for years been piling gravel on badly drained, badly graded, badly maintained roads. The immense hollows on many farms testify to the large amount of material which has been used, without forming in a single instance, a first class and durable road covering.

Gravel placed on a poorly drained, poorly graded road, the repair of which is not attended to at the proper time, is quickly rutted, and in the wet seasons of spring and fall, is mixed with the soil beneath, its usefulness as a road covering being thereby largely lost. But on the other hand, when the road is well drained, and graded, and the gravel is so laid upon it as to form a distinct coating, and ruts are repaired as rapidly as they form, the life of the gravel is very much extended.

There is a mile of road in a township of Western Ontario, which was first gravelled fifteen years ago, has had very little expended upon it since then in comparison with other roads in the vicinity, and is in splendid condition to-day.

The road is on clay loam having moderately good natural drainage. The roadway was graded and the ordinary open drains provided at the sides. Gravel was placed in the centre, dropped from the wagons in the usual manner without spreading.

An old man who lived on the road, having little else to do, was employed by the pathmaster for a small amount to work on the road. He took an interest in his work, kept the drains open; year by year as a new line of gravel was added,

he spread it so that traffic at once passed over it instead of turning to the roadside, ruts were made in the loose material, but with a rake he brought the gravel back until the wheel tracks were like a line of steel rails, hard and solid.

This work was done fifteen years ago, but the road, as has been stated, is to-day the best in the township in spite of the fact that less gravel has been placed on it. A little extra labor has thus effected a saving of gravel, which if extended to the whole township would double the life of the gravel pits now being rapidly exhausted.

THE ESSENTIAL FEATURE OF A GOOD ROAD—DRAINAGE.

The essential feature of a good road is good drainage. This is true in all climates, but it becomes absolutely imperative in Canada where rains are heavy, at seasons almost continuous, and attended in the winter, autumn and spring seasons with severe frost, snow and slush. Every rule has its exception it is frequently said, but the maker of roads can safely follow the principle under all conditions, that drainage cannot be too perfect: that where drainage is perfect there is a perfect road.

On first sight this statement may seem exaggerated. The impression has so long existed that, if we get a surface of gravel or broken stone piled on a ridge of earth we have built a road, it is difficult to adjust our minds to other principles. No doubt working with this object, many fairly good roads have been produced, but they are, in a measure, the result of accident rather than reasonable and clear-headed design. When roads are built with the fundamental purpose of doing so by securing perfect drainage we will be on the most direct and shortest route towards securing good roads. Much unnecessary labor and money will be saved, as well as much disappointment and dissatisfaction.

Doubtless the dweller in the country of sand will be inclined to think lightly of this advice, for he knows that, in wet weather, not too wet, his roads are at their best. He will be inclined, if he has never lived on clay lands, to think that advocacy of road building with such an object is a false doctrine. Yet even on sand lands the principle of good drainage is the principle of good roads.

An essential part of good drainage is to attend to the shape of the road surface. This must be crowned or rounded up towards the centre. The shape of the road will tend to throw off to the side, the water as it falls in rain, and the water of melting snow.

In addition to being crowned, the surface must be smooth; that is, free from ruts, wheel tracks, holes and hollows. If these exist on the surface of the road, instead of being thrown to the side the water is held back, is absorbed into the roadway which softens and yields readily to the wearing of the wheels. Like the water poured on a grindstone, so the water on a road surface assists the

grinding action of wheels. When the road is wet the holes and ruts rapidly increase in size, wagon after wagon sinks deeper, until finally the road becomes utterly bad and perhaps impassable as we so frequently find Canadian roads in late fall and early spring.

In order to provide for a smooth, round roadway, that will remain so in wet weather, it is necessary that the surface of the road be hardened with some stony material, and for this we use gravel and broken stone. By resisting wear, ruts and hollows do not form readily, or when commenced they do not increase with such great rapidity.

Having crowned the road and made it smooth, having placed on it a covering of metal to harden it, and assist in keeping it so, it is necessary to dispose of the water which flows from the travelled roadway to the open drains at the road side. These open drains must have an outlet to which the water flows readily and freely. Drains which have not a good fall and free outlet, which merely catch the water and hold it until it sinks into the earth, are of little service to the road. The reason for this points to another of the main features of good drainage.

That is, it is not sufficient that we crown the surface, make it hard and smooth, and carry away the surface water in open drains at the side of the road. It is absolutely necessary that the natural earth sub-soil which we crown and coat with stone shall be kept dry; for to keep the sub-soil dry is to keep it firm and strong to support the load. This metal, as we have said, resists wear; the natural earth underneath has to support not merely the load but the surface material as well. It is therefore, as previously pointed out, of prime importance that the water caught by the side drains shall be carried away immediately, before it can sink into, and soften the road foundation.

More than this, not merely must the surface water be carried away quickly, before it can sink into the soil, but underdrainage must frequently be resorted to. In many sandy and gravelly localities and even in clay districts, nature sometimes provides sufficient natural underdrainage, but more frequently we find low, wet sections where the water line must be lowered by means of tile drainage. A tile drain under each open drain is, in nearly every case, the best plan to pursue where underdrainage is needed.

The vast good which can be accomplished by means of tile underdrainage on the roads has been too long overlooked in Canada. Agriculturalists who have used tile underdrainage on farm lands, will be able to better understand its action on the roads. Just as there are lands which are useless for farming without underdrainage, so there are roads which are useless without underdrainage. A dry sub-soil can support any load. But with a wet and consequently weak sub-soil, the road metal is at once forced down and buried in the mud, while the mud

rises to the surface. And so, for want of a firm, dry foundation, the crown of the road is destroyed, the stone is mixed with the mud and the surface becomes soft and rough, easily worn, and surface drainage is thereby interfered with.

Here, then, we have, in brief, the principles of roadmaking, which resolve themselves essentially into a matter of good drainage. A smooth, hard, rounded surface throws the water to the side drains and the side drains carry it quickly away. The metal covering is both a roof and floor; a roof since it sheds the water to the side before being absorbed into the sub-soil beneath; a floor, in so far as it resists wear. The underdrainage of the roadway provides a firm, dry foundation, and so, in our structure, we have provided foundation, floor and roof, all by means of good drainage.

Sand, unless in a low section, seldom requires underdrainage, but with a good road covering of gravel or stone frequently makes the strongest of roads—so also with gravel. Clay is most frequently in need of underdrainage, particularly in low and swampy districts. But in every class of soil, at every season of the year, the water line should be kept three feet below the surface of the road. If nature does not do this, then artificial means, tile underdrainage, should be resorted to.

It is by means of good drainage that we provide against the action of frost on our roads. Frost can only be destructive where there is moisture. The upheaving action of frost on soil is caused by the presence of water. Water expands on freezing and forces the soil upwards; when thawing takes place the ground is left spongy and wet, and the roads "break up." Keep the roads dry and they will not break up. Our energies then, in the making of good roads, must be directed to the essential feature, good drainage.

When the road is kept dry by drainage the only care required is to see that the actual wear from the grinding of wheels and hammering of horses hoofs is repaired as it appears. This wear is but a trifle compared with the injury caused by wheels sinking into a wet and saturated subsoil.

By keeping in view the one principle "drainage," the reason for nearly all the steps in constructing and repairing a road at once becomes apparent, and work can be better adapted to all peculiar and unusual circumstances which arise.

One great lack in the present system of caring for roads is the little attention to repairs. Repairs should be made as soon as the signs of wear appear. Otherwise the defect grows with increasing rapidity; and what at first could have been entirely removed with a few minutes work, results in extensive and permanent damage. The reason is generally to be found under the head of "drainage." The track of a single wagon, scarcely perceptible at first, after a shower stands full of water and softens the road at that spot. A vehicle passing along at

such a time, sinks farther into the softened track. Here is a deeper hole to hold water; soon it becomes a rut, and the rut develops into a pitch-hole. The soil underneath is brought up and mixed with the surface covering. The surface covering is broken and forced down into the soil. Here then is a bad road and a permanent weakness, the result of neglect to repair the road at the proper time, neglect to keep the drainage of the road in perfect condition at all times.

THE ROAD GRADER.

During the past season in a large number of townships of Ontario, grading machines were used in improving the roads. That so many different methods of operating these machines should be employed, points strongly to the necessity of laying down a uniform system for regulating this important part of the work of road-making in order that the best results may be produced. A road grader is the most necessary implement for a township to possess, and in fact, some townships are so convinced of this as to have invested in four machines, placing one in each quarter of the township. The crowning of the roadway can be done very rapidly, cheaply and perfectly, by their use. To depend upon manual work for the first grading of roads, and the repair of others that require re-shaping, is a useless waste of labor and money. But these machines in the hands of inexperienced men, are often made to injure rather than improve the road.

The operator must know how a road should be made, and how the grader should be worked in order to make each road with the least expenditure. A study of the different methods employed, and a talk with the different operators, shows that the ideas are as varied as can be imagined; many of them are so ridiculous as to make one wonder why they should be tolerated by any municipal management.

In one township the machine is purchased by the council, placed at the disposal of any pathmaster who sees fit to take it from the railway people, and of any other pathmaster who cares to take it from one beat to another. These men may have then seen the machine for the first time and know nothing whatever about its mechanism or how it should be operated.

The directions specify the number of teams for ordinary soil. The horses are attached and if they do not provide sufficient power others are added. The blade is plunged into solid clay full tilt as if the whole object was to determine its strength or test the merits of the different teams engaged.

Each pathmaster is permitted to plan the work to suit his own ideas, and these usually differ from those of the pathmaster of the next beat. The width of grade differs from 15 to 45 feet in the same township, and sometimes in the same beat. The amount of crown which should be given is, of course, arbitrary, and extends

from the flat and useless, to the ridiculous and dangerous. Roads are often graded higher where the crown is already sufficient, and occasionally the machine is used for ungrading a piece which is already properly made.

Another plan is for the council to hire the operator, the beats requiring his services to furnish the necessary teams, the work to be done according to the direction and dictation of the pathmaster, thus making the operator simply a part ; the operator to be employed for the season's work only, and in fairness to others, the job must be passed around, so that he cannot expect to be hired for more than one season.

Sometimes the council employs the operator and one team, the beats using the machine to furnish the remaining team or teams. Where the township is divided into four wards or sections, one man and one team is sometimes employed in each ward so as to distribute the honors and patronage, these men to have charge of the machine in their respective wards. In townships more advanced in this work, the operator and two teams are employed by the council, he to appear when called on at any beat in the township even if he has to pass from one side of the township to the other. He is given no plan to follow, no directions, but is supposed to exercise his best judgment in the performance of the work ; except where his judgment may conflict with the idea of the pathmaster, in which case the pathmaster rules.

Under all foregoing plans, no attempt is made at operating the machine during the early part of the season when the ground is in a fit condition, and when the work can be easily and properly performed ; but the machine must remain idle awaiting the statute labor season when, as a general thing, especially in clay sections, the soil is baked so hard that perfect work cannot be done and it is unreasonable to expect satisfactory results. Often this work is commenced just when these roads are becoming serviceable and are ruined for the remainder of the summer. In such cases the investment is unprofitable, the labor is wasted, the roads are injured, farmers are deprived of services which are urgently required at this important season on the farm ; and the benefits which would result from improvements upon the roads, if such labor was expended at a proper season and under suitable plans, is lost to the community.

The chief mistake in the use of a grader, and one which has caused a great deal of harm, is the covering of old gravel roads with the earth from the shoulders, in order to crown the road. This soft material at the edges of the road, clay, turf and washings from the road, is precisely what is not wanted on the travelled track. But this stuff is precisely what inexperienced men insist on putting on the road. These square shoulders are, of course, objectionable and should be cut off, as they interfere with drainage ; but they should be thrown outward, across the ditch if necessary, and then the best gravel or stone obtainable should be used to crown the road and form the track. The reverse of this, rounding up

the roads with the material from the edges of the gravel roads has utterly destroyed many fairly good gravel roads. It appears almost necessary to let people destroy a few roads in order to show them what ought not to be done, and so convince them as to what should be done.

A proper plan to adopt would be as follows :—

1. Prepare a complete plan of the township, showing all road allowances, watercourses, bridges, etc.

2. Classify these roads according to their importance, dividing them into (a) main roads, (b) roads of moderate travel, (c) roads of least travel.

3. Prepare a proper specification fully describing the roads according to the above classification, setting forth the width of roadbed, plan of drainage, amount of crown, kind of material to be used as a covering, whether gravel or broken stone, depth and width to which it should be laid, how the material shall be prepared and applied, etc.

4. The grading machine should be owned by the township, one man should be employed to have charge of and operate the machine, and he should be a fixed resident of the municipality in order that his services may be retained for a number of years. He should be thoroughly practical and have some mechanical knowledge. He should be provided with a copy of the above specifications, and each spring, with a committee of the council, should examine the roads and determine the portion to be graded, as far as possible extending the work of previous years, securing continuity and uniformity in all work.

5. The operator should hire a sufficient number of teams to work the machine, and the same teams should always be employed in order that they may become accustomed to the work.

6. Where roads have already been gravelled, but the surface has become flattened by traffic and other causes, they should be carefully examined by the council, and all needed improvements can be then specified. As far as possible, the old gravel or stone roadbed should be preserved, and, except in cases where it is absolutely necessary to raise the grade, no earth should be placed upon it. But the shoulders should be cut off, turned outward and removed, and a new coating of metal applied to the centre of the road.

ROCK CRUSHERS.

Good roads and streets cannot be made perfectly and economically without the use of proper implements and machinery. In many municipalities gravel is scarce, and what is to be found is of very inferior quality, while stone for macadam can be easily and cheaply procured. In its natural condition the stone is useless

for streets, but properly broken, makes excellent material. For this work a rock crusher is required. Municipalities in Ontario are realizing this, and many have purchased complete outfits and are doing excellent work.

The cost of these crushers depends largely upon the capacity of the machine. The smallest make is usually chosen on account of its cheapness and without regard to the amount of work it will have to perform. The smallest machines are intended for township work, where they have to be constantly moved about, and where the work at each point is not very great. With careful handling and not overcrowding, these machines, with necessary repairs, will answer for a number of years.

For city or town work, where a large quantity of material is required, it is a mistake to purchase a small crusher. The breaking of stones is a very severe test on machinery, owing to the varying character of the material; and ample capacity, so that the work can be done with perfect ease, is necessary. A crusher of ten cubic yards per hour run only at three-quarters its capacity, is the most serviceable and economical machine for most towns and cities. The extra cost incurred will prove a profitable outlay when the expense of maintenance and operation is considered.

There is frequently more than one way of doing a thing, but there is usually only one way which is the best way. To discover that one best way, even in doing so simple a thing as breaking stones for roads, generally requires some thought, knowledge and experience.

For example, some municipalities still cling to the oldtime method of breaking stone for roads, by the use of hammers and worn-out men who are pensioners on the municipality. There may be cases in which such a method is satisfactory, but they are very rare. Good labor is cheaper at any price.

In stone-breaking, there is no better laborer than a well-built, well-designed stone crusher. To operate the stone crusher, power is required however. Some municipalities, for this work, use an expensive steam roller attached to the crusher. The roller is made for one purpose, for compacting the metal in a road. To operate the motive machinery at so high a speed as is required for a stone crusher, to subject it to the jolting strains of a crusher, cannot but be very injurious to the roller. Of greater moment, perhaps, is the fact that the roller is unprotected from the dust created in crushing the stone, thereby causing great



THE POPULAR IDEA OF ROAD-MAKING MACHINERY.

wear. This illtreatment of an expensive roller will tend to destroy it very rapidly, the resulting depreciation being very great, varying, of course, according to the extent to which it is so used.

The cost of operating a steam roller for power (in addition to the deterioration of the roller), will cost about \$2.50 per day for fuel, and \$1.50 per day for an engineer.

Better than this is to purchase a separate engine which can be operated for about the same daily outlay \$4. The value of the engine being much less than the steam roller, the resulting loss from depreciation is not so great. Some municipalities have purchased second hand engines which have been used on threshers, for \$250 and \$300, and which have rendered very efficient service.

The town of Berlin, however, has one of the most economical sources of power in the Province. An electric motor, receiving power from the local electric plant, does the work for \$1.75 per day. The motor which cost \$250, is a small piece of machinery, requiring little care other than oiling and "pushing the button," when power is to be turned on or off. The motor is of very simple design, is protected from dust and weather by a small wooden covering, so that loss from depreciation is very slight.

A number of the American States have in various ways experimented with the employment of convicts in road-building, New York, North Carolina, and California being perhaps the chief. There are objections to having the prisoners employed immediately upon the roads in such work as draining and grading, objections which the majority of Canadians will readily appreciate. There are few who desire to see, in public, anything resembling chain gangs, guarded by armed keepers.

The spectacle is a degrading one, and can result in no good to the prisoners. The law has no part in avenging crime, but in preventing it. To elevate the criminal is the righteous and highest aim of justice. To publicly humiliate can have no such influence.

There is, however, a manner of employing convicts, behind the prison walls, which will at once give them healthful employment, take them away from competition with honest labor, and result in benefit to the roads. This is in the preparation of road material. Milton, Cornwall, and we believe other county gaols, already employ the prisoners in this way, the stone being conveyed to the gaol, and broken with stone hammers. Much more, however, could be accomplished if stone crushers were used, and the broken stone supplied free, if necessary, for use on country, as well as town roads.

Where stone or gravel for road metal is not readily available it can be presently shipped a considerable distance by rail at a reasonable rate. Complete crushing plants are now operated and have been in operation for the past few years, at St. Davids, near Niagara Falls, and at Hagersville. Considerable material is

being supplied to different municipalities by these quarries. In western Ontario gravel and stone in many sections, is scarce, but a large crushing plant has been placed in the quarry at Amherstburg, from which an ample supply for all such sections may be obtained. The railway companies are all disposed to grant reduced rates to municipalities on material required for roadmaking.

It would be advisable for municipalities without road metal to obtain prices, and strive to divert as much as possible of their annual appropriation for the purpose of providing such material, leaving the statute labor or other expenditure for hauling and placing it. Many municipal councils consider this too expensive an undertaking, to even consider, but would be surprised to find, taking all things into consideration, how cheaply roads can be made in this way.

“ A STEAM ROLLER AN ABSOLUTE NECESSITY.”

The usefulness of a steam roller in the construction of broken stone roads, has been fully commented upon in previous reports. An article under the above title appeared recently in the ‘Brantford Expositor,’ fully corroborating from entirely independent sources, and in an entirely independent manner, all that has been urged with respect to steam rollers, and is most instructive. The article indicates, too, the widespread interest taken in street improvement, and is as follows :—

“The ‘Expositor’ makes no pretence of being a scientific expert in the matter of road-making, but is prepared to accept experience both in this, and in other lands, as determining that a steam road roller is an absolute necessity in the economical construction of macadam roads, and in their repair. This is abundantly demonstrated by the fact that in countries, and municipalities, where the best macadam roads are to be found, this appliance is always present.

“In our own Province of Ontario the movement in favor of good roads has been advancing rapidly, and simultaneously there has been a demand for proper machinery. The following cities and towns have purchased steam rollers :—

Toronto	London	Hamilton	Chatham	Stratford
Berlin	Galt	St. Catharines	Niagara Falls	Ottawa
Brockville	Woodstock	Owen Sound	Kingston	

Peterborough council has decided to make the purchase, and in St. Thomas a steam roller is owned by a contractor.

“The very fact that almost every Ontario municipality occupying a similar position to Brantford is in possession of this valuable aid to good roadmaking affords strong evidence that such an appliance is an absolute necessity for this city.

"The purchase of a steam roller was one of the strong recommendations made by Mr. Campbell, the Provincial Road Commissioner, in his report of the 30th October, 1897. on Brantford streets, and the same excellent authority in a letter to the editor of this paper of recent date says:

"SIR,—Your favor of the 11th inst., making enquiry as to the utility of steam rollers, was duly received.

"From the list which I sent you, you will see that Brantford is one of the few cities in Ontario which does not own or use a steam roller in the building and maintaining of streets, and, in fact, very few of the larger towns are without such an implement.

"I am satisfied that every municipality using a steam roller will unhesitatingly say that it is indispensable in the proper construction of streets. No one with any experience in street-building will assert that the work can be done with any degree of satisfaction without a roller.

"In building streets, as in any other class of work, proper implements are necessary to to secure perfect and economical results. Without a roller the practice is to excavate and grade the roadbed, then dump the gravel or broken stone on this soft foundation. The material not being compressed will yield to the wheels, rut, separate, and roll about. To prevent this in a degree, sand, cinders, or other foreign material is mixed with the stone and spread over its surface, to fill the voids and act as a 'binder.' This, however, is only temporary work, and lays the foundation for most expensive maintenance, besides proving a very unsatisfactory roadway.

"The sand, ashes, cinders, etc., placed on the surface grinds into dust in the dry weather and becomes unbearable if not constantly sprinkled, and this is difficult to do without converting it several times a day into a body of mud.

"The extra sprinkling is a considerable item of cost, and the mud created is very annoying to the users of carriages and bicycles, as well as being injurious to these vehicles.

"This foreign material in the body of the roadbed attracts moisture at wet seasons, turns into mud, and the bond between the stones is broken, causing them to yield to traffic and turn into ruts and holes. This mud churns to the surface and yields a very heavy crop every season, which must be scraped up and carried away.

"In a properly constructed road, after the excavation is done, and before any stone is placed upon it, the foundation should be rolled until perfectly solid. Upon this the broken stone is placed. This stone should be graded by the screen attached to the crusher into three different sizes, the coarsest being placed at the bottom of the road and the finest at the surface. In crushing the stone, a large quantity of fine stone dust is created, which is removed by the screen, and only this dust should be used for the binder, a sufficient amount of it to fill the voids being scattered over each layer of stone placed on the road and also on the surface.

"Each layer should be rolled sufficiently to set the stones in place, and after the final coating is put on, the rolling should be continued until every stone is firmly set and the mass perfectly consolidated. Each layer should be sprinkled with water before rolling so as to wash the stone dust into the crevices. In this way a road is made of proportionate thickness and uniform strength on an unyielding foundation, impervious to water, with a smooth, hard surface which will readily shed the water and distribute wear and traffic uniformly.

"It is the only way to construct a macadam road in the cheapest, most efficient and economical manner. It can at once be seen that this cannot be accomplished without the use of a heavy roller. For city work, such a roller should be about fifteen tons in weight.

"In constructing a macadam street, the stones are broken into cubical fragments and placed to such a depth as may be required to withstand the traffic. If this stone is dumped loosely on the road each stone lies separately, there is no proper union of the mass, and in wet weather the water passes through the stone as through a sieve, enters the foundation and converts it into mud. The traffic then settles the stone into the weakened foundation, forcing the mud through the voids to the surface.

"Whereas, where a roller is used, its repeated operation over the surface of the stone causes each cubical block to find its place among the cubes below it, forming a perfect combination, keeping them together and filling the voids. By the aid of sprinkling and pressure, the whole is cemented and forms a perfect sheet of clean material, almost equal to solid stone.

"For the sake of appearance and cleanliness the principal streets in the business part of Brantford should be paved with a more substantial material, such as brick or asphalt, but the standard pavement for Brantford, as well as other cities and towns in Ontario, should be properly

constructed broken stone roads. Your city now owns a crusher, but the usefulness of this valuable machine is largely lost for the want of its complement, the heavy roller. By providing this you will have made an important step towards the improvement of your streets.

"As the use of heavy rollers in street construction in Ontario is of recent date, it will be interesting for you to obtain, as you contemplate, the experience of other towns and cities using them."

"The *Expositor*, as Mr. Campbell indicates in his letter, did take steps to obtain the opinions from cities and towns which already own steam rollers, and in no instance was a reply unfavorable to their use obtained. On the contrary, all the authorities were in agreement that good roads could be obtained in no other way. We append the replies thus far received :

From C. H. Rust, City Engineer of Toronto :

"DEAR SIR,—In reply to your letter of the 11th inst., it is impossible to make a good macadam road without the use of a steam road roller, and, if the city of Brantford is building macadam roads, I know no better investment for their money than a steam road roller."

From E. G. Barrow, City Engineer of Hamilton :

"DEAR SIR,—In answer to the question asked in yours of November 12th, I would say that our steam road roller has given very great satisfaction. It has been in constant use since we purchased it, and our macadam streets have been done far better and much cheaper than heretofore without the roller. In picking up rough macadam streets it has done as much work in a day as twenty-five or thirty men, and its use in rolling the foundations to receive the macadam is very great, making a compact bed. I cannot speak too highly of the work of the roller, and I can say with the utmost confidence that our city never spent money to better advantage."

From Robert Surtees, City Engineer of Ottawa :

"DEAR SIR,—In reply to yours of the 12th asking experience of this city with steam road roller, I beg to state that this city has had a fifteen-ton steam road roller for the past ten years, and consider it one of the best investments they have made. In my opinion it is impossible to make good macadam roads without using a ten or fifteen-ton steam roller, and prefer fifteen-ton."

From A. Ormsby Graydon, City Engineer of London :

"DEAR SIR,—In reply to your enquiry as to the benefit or otherwise to the city of having a steam roller for assisting to make good roads, I beg to say that it is not possible to make a good macadam road without one, and for ordinary gravel roads the roller is of immense value. In my opinion and experience no corporation making gravel or macadam roads can afford to do without one.

"I have had one in use for about three or four years and have found it of great advantage, in fact, for a portion of the year I would be glad to have two of them. The season is short in which you can use it, about six months, and we can hardly roll all the streets required to be rolled in that time, as there are 118½ miles of streets in the city."

From Jas. A. Bell, City Engineer of St. Thomas :

"DEAR SIR,—In answer to yours of the 12th, I beg to say that this city has no steam roller and has only used one under contract for laying brick pavement. All our other streets have been made with a five-ton horse roller, which is owned by the city. I may say that from experience much better work can be done at less cost by using a steam roller."

From W. Bolger, City Engineer of Kingston :

"DEAR SIR,—In reply to your letter of the 12th inst., I beg to state our steam road roller is considered the most convenient and useful article the city owns.

"The one we have has been in use for twelve years, and is now as good as ever. Your city would make no mistake in purchasing a road roller if you have much macadam streets. We would not be without one for any consideration."

From W. F. Van Buskirk, City Engineer of Stratford :

"DEAR SIR,—I have the honor to acknowledge receipt of your letter of the 12th inst., re experience with steam road roller.

"You will find a report on the matter written by me for Mr. Campbell, Provincial Instructor in Road Making, in his report for 1897.

"I am of the opinion that good macadam, gravel or earth roads cannot be made without a roller, but it should be borne in mind that a roller itself will not make good roads. It is like any other machine, requires skilled labor to operate properly, and intelligent management to secure the best results.

"The work done here on old gravel roads has been satisfactory, and has been done at a low cost. Old gravel roads have been torn up, regraded and thoroughly consolidated, making as good roads as could be made for four times the cost without the roller.

"A roller is necessary for the repair of roads, as new material can be so easily incorporated with that already in place.

"I was greatly amused at the ideas of the laborers at the time the roller was purchased here. They were of the opinion that the machine would not do the work that they had been doing, and opposed purchase for that reason.

"I consider a steam roller a necessity for any town doing road work by day labor. The question of weight necessary should be left to the engineer and not to the agents of the firms wishing to sell machines. The engineer will know what weight is best for the class of soil and material available, and as he will be held responsible for work done, let him choose the machine he wishes. This is important."

From George A. McMullen, Town Clerk of Brockville :

"DEAR SIR,—Replying to yours of 12th inst., would say that we have a fifteen-ton steam roller and would not be without it under any consideration. We now have street beds which are giving satisfaction in every way. I may also state that outside places such as Cornwall and Township of Elizabethtown have rented our roller with the result that they have decided on having rollers of their own.

"We purchased ours from Kelly Bros., Springfield, Ohio, U.S. (firm name the O. S. Kelly Co.). I understand, however, that steam road rollers can now be procured from Canadian agents at a less price than ours cost in 1894.

"Any further information you may require will be cheerfully given."

From J. M. Hood, Clerk of the Town of Galt :

"DEAR SIR,—Replying to your letter of 12th, the Town of Galt has a Buffalo Pitts Steam Road Roller, and it has been in use nearly all summer on our streets. It is the universal opinion that we have more to show for the money spent on our streets this year than for years before on the old tinkering plan. We have a stone crusher and all the other appliances for making good roads.

"We are especially well pleased with the roller. It is a fifteen-ton one, and in every particular fills the bill.

"Any further particulars will be pleased to give."

From John Robinson, Clerk of the Town of Niagara Falls :

"DEAR SIR,—In reply to yours of 12th, I have no hesitation in saying that the steam roller lately purchased by the town of Niagara Falls has materially contributed to the making of good streets. The steam roller was bought in the latter part of 1897 from the Buffalo Pitts Company and has given perfect satisfaction. This year the town bought a stone crusher, and with these two machines has made the road along the river front the best road we ever had in the town. After seeing the work of the roller and crusher I do not think there is a man in town who would consent to do without them."

From H. Aletter, Town Clerk of Berlin :

"In reply to your letter of 12th inst., I beg to say that how to make good permanent macadam roads without a roller is beyond our comprehension. We are just now using our roller on one of our leading streets, which street we constructed in manner following :

- "(1) Lay four inches cobble stones.
- "(2) Lay four inches coarse crushed stone.
- "(3) Lay four inches fine crushed stone.
- "(4) Lay one inch dressing (fine gravel).

"This street is partly finished and we are all of the opinion that we have an A1 street in every respect.

"We have nothing to show for the vast amount of money spent on our streets heretofore without the use of a roller.

"We have also been using our rollers on old worn-out streets by way of picking up and rolling down roadbeds with quite satisfactory results."

"The *Expositor* has withheld no letters it received in reply to its question; it has given every letter precisely as it was received. Such an emphatic endorsement of the use of the steam roller cannot be safely disregarded by any thinking elector, particularly by one who is anxious, as all should be, to see that proper return is secured for the money expended on our streets, and that the city in which we take so much pride shall fully deserve, in all respects, the claim we are continually making for it of being progressive and up-to-date. It is false economy to work with poor tools, and this applies to the making of corporation streets as it does to everything else. A steam roller will pay for itself many times over, and the investment cannot be made too soon.

"We are satisfied that if the ratepayers of Brantford had been seized of all the facts when they voted upon the steam roller last summer, the result would have been vastly different. It would probably have been different as it was, had not two other by-laws, one for a very large amount, been submitted at the same time, and thus created a certain amount of hesitancy lest expenditures were being pushed forward too rapidly. We are convinced at any rate that a mistake was made at that time which cannot be too soon corrected, and that it will pay the Council to borrow a steam roller rather than continue as we have been doing in the past. It is worth considering, indeed, if another by-law should not be submitted to the electors in January next, and we feel satisfied that if this be done the proposal will carry by a good majority."

THE WEIGHT OF THE ROAD ROLLER.

There are different kinds and classes of rollers. The horse roller weighing six or eight tons, will do fairly well if a steam roller cannot be afforded, but the horse roller is not sufficiently heavy and has to be used much longer on a given section than has a steam roller to produce the best results. The feet of the horses, in exerting sufficient strength to move the roller, sink into and disturb the road metal, and thereby injure the shape and quality of the roadway.

There is danger, on the other hand, of having a steam roller which is too heavy. A very heavy roller will sometimes sink into light or loose soil, force it ahead and create a mound over which it cannot pass; this, however, may sometimes be overcome by spreading over the surface of the soil being rolled, a thin

coating of gravel. The same result will sometimes occur with an excessively heavy roller on a layer of loose stone. The heavy roller is more liable, too, to injure underground pipes, catch-basins, culverts, bridges, or disturb sidewalks.

For these reasons a roller exceeding ten or twelve tons in weight, in some localities where the soil is of a loose or sandy nature, is frequently not desirable. In districts where the natural soil is gravelly or of a stiff clay, a heavier roller may be operated successfully, but some municipalities have found it necessary to use a light horse roller in consolidating the sub-soil and first layers of stone.

Nor, if the stone used in the construction of macadam streets is of a soft nature, is a heavy roller say of twenty tons desirable even in the finishing courses, as the crushing effect has been found in some cases to crumble and pulverize the stone rather than merely consolidate it.

For new work, in which the dirt foundation must be rolled, a weight of twelve tons is generally the most serviceable; but for picking up an old roadway and reconsolidating it or for finishing a new work, fifteen tons is better. Where a town is to own only one roller it will be advisable to consider very carefully the work to be done before purchasing a roller of over twelve tons weight.

REPAIRING WITH A ROLLER.

By proper attention and repairs the life of a broken stone pavement can be made continuous. The surface can be frequently rolled, improving it greatly. It can be scraped and swept as are other pavements. When it begins to lose shape the surface can be loosened up by means of teeth attached to the roller, a light coating of new metal applied, and then rolled down as well as when new. It is by such means as these that broken stone roadways can be made more economical and satisfactory than any other for streets generally. This ease of renewal and repair is a property peculiar to macadam, which renders it most suitable for general purposes—that is, for residence streets. While the cost in the first instance may nearly equal that of cedar block, yet at the termination of the period when cedar block is decayed and has to be torn up and renewed, the macadam, if properly treated, is still in good condition. It forms a permanent basis, and its perpetuation is merely a matter of repair to be met by the general funds. Except under excessive wear or where in business sections a high grade pavement is necessary, broken stone pavements, by the aid of a steam roller, are beyond doubt the most serviceable, economical, and give the greatest satisfaction to the taxpayer.

HIGHWAY CULVERTS AND BRIDGES.

The majority of Canadians, when visiting Europe, are impressed with the durability and solidity which characterizes the structures of that country. Private residences are built to withstand the wear of centuries. Cathedrals.

public halls, libraries, and similar civic institutions are constructed, not merely for the present, but for future generations. Among the works marked by this durability are to be classed the public highways with all that pertains to them. Canada in this regard, presents a very unfortunate contrast.

It can justly be argued that Canada is a very young country, and that England is a very old country; that Canada is not a wealthy country, and that England is a very wealthy country. While England is, in a way, a very old country, yet it is not so much older than this country in the arts of civilization, which should teach our citizens and municipal councils the necessity for and the means of wisely spending money in permanent improvements. And while England is a richer country than Canada, that greater degree of wealth has been brought about, to some extent, by the very durability which we have so long avoided. Permanent improvements are the cheapest. Structures which need props and repairs within a year or two after they have been built, seem to be in a chronic state of starvation, with a ravenous appetite for money. Canadians have not yet entirely outgrown the idea that they live in a pioneer land where the needs of the present entirely overwhelm the future.

In nothing is this temporary building more apparent than in our highways; and in no detail of our highways is it more striking than in the matter of bridges and culverts. At the same time there is no portion of the making of a road that offers more scope to the roadmaker than in providing substantial and permanent waterways. Instead of the handsome stone and concrete arches that span so many of the streams intersecting the highways of England, there are to-day in this country scores of wooden boxes and trusses,—flimsy, disjointed, unsafe; the constant source of accident, and the bottomless pit into which councils are annually throwing money in a vain endeavor to keep them in repair.

Location.

Considerable attention is generally paid to the selection of a good site for a bridge, and an effort is made to decide in the interest of economy, usually with a considerable measure of success. There is, however, a tendency to cling to the line of original survey, rather than deviate the road slightly, when by doing so, much would be gained in lessening the dimensions of the bridge, securing firm foundations for piers and abutments, reducing the cuts and fills at the approaches of the bridge; all of which, while they may not decrease materially the first cost, very frequently are of the utmost consequence with regard to maintenance, and may decide for good or bad the usefulness of the entire roadway. The utility of a road with respect to hauling heavy loads, is not governed so much by the condition of the best section as by the worst; not so much by the level portion as by the steepest grade. Bridges, forming as they do a means of crossing valleys, are intimately associated with the problem of judiciously choosing between directness of route, easy gradients and details of construction.

The location of culverts is a matter of very common error. Water should be disposed of in small quantities, along natural watercourses, before it gathers force and headway. Instead of this principle being followed, water is frequently carried long distances by the roadside, past watercourses after watercourse, rather than build a culvert or culverts to carry it away without injury to the road. Where culverts are needed, they should pass directly across the road and carry the water away from it.

The size of bridge or size of culvert involves nice discrimination, in which local circumstances and the class of construction introduce various factors. For the size of waterway, no hard and fast rule can be given. Many existing culverts and bridges were at one time of sufficient size, but the clearing, draining and cultivating of the land now permits the water after a rainfall to reach the watercourse in a shorter time with increased volume, causing submerged roadway and flooded roadsides, while culverts and bridges are swept away. The best guide to a proper size of waterway is an intimate acquaintance with the locality or the evidence of others who are, with respect to maximum rainfall, height of water line, previous experience as to floods, form and inclination of the stream and area to be drained, kind and condition of the soil and similar details. Talbot's Formula, proposed more as a guide to the judgment than as an unalterable rule is at times very useful:

Area of waterway in square feet = $C \sqrt{(\text{Drainage area, in acres})^3}$. C is a variable co-efficient and the values given are:

"For steep and rocky ground, C varies from $\frac{2}{3}$ to 1. For rolling agricultural country subject to floods at times of melting snow, and with the length of valley three or four times its width, C is about $\frac{1}{2}$; and if the stream is longer in proportion to the area, decrease C. In districts not affected by accumulated snow and where the length of the valley is several times the width, $\frac{1}{3}$ or $\frac{1}{4}$, or even less, may be used. C should be increased for steep side slopes, especially if the upper part of the valley has a much greater fall than the channel at the culvert."

Waterways should be neither needlessly large, nor of too small dimensions, involving on the one hand unnecessary expense for first construction, and on the other hand, injury to the road, washouts, expensive repairs and delay to traffic.

Materials.

The materials available for culvert construction in addition to timber, are sewer pipe, concrete pipe, iron pipe, brick, stone and concrete. Culverts are sometimes made of one of these materials alone, or of two or more in combination. When the dimensions of a bridge are reached, concrete and stone abutments and piers, with iron or steel superstructure; or stone, brick or concrete, alone or in combination, are the materials gaining favor.

Sewer Pipe.

For the small culverts, sewer pipe is very economical and durable if well laid. To render them secure against the test of a Canadian climate, they should be laid with a good grade, and the ends protected with concrete, stone or brick headwalls having deep aprons. The joints should be made water tight with cement. These precautions will provide against the action of frost and will prevent the culvert being undermined by water passing along the outside of the pipe, either from the ends or through the joints. Care should be taken to excavate a concave bed for the pipe to rest in, always laying the spigot ends up grade.

The pipe at the outlet should be set flush with the surface of the ground. If set higher than the surface the fall of water will wash out a depression and will in time undermine the end of the culvert. A too rapid grade will cause the same result. It is frequently well to cobble-pave the outlet, where this undermining action is likely to occur.

Cement-Concrete Pipe.

Excellent culvert pipe of concrete can be manufactured cheaply in any gravel pit under the immediate direction of the municipal engineer. The pipes are from two to four inches in thickness according to diameter; which latter may safely and conveniently reach three feet, in lengths of two and one-half feet.

The implements required are of the simplest kind. The most important are two steel, spring cylinders, one to set inside the other, leaving a space between the two equal to the thickness of the finished concrete pipe. By "spring-cylinder" it may be explained, is meant such a cylinder as would be formed by rolling an iron plate into a tube without sealing the joint. With the smaller of these cylinders the edges overlap or coil slightly; but are so manufactured that the edges may be forced back and set into a perfect cylinder.

These two cylinders with joints flush are set on end, the one centrally inside the other and on a firm board bottom. The concrete, made of first-class cement and well screened gravel in the proportion of one of cement to three of gravel, is then tamped firmly but lightly into the space or mould between the two cylinders. The tamping-iron used to press the concrete into place is so shaped as to fit closely to the cylinders.

The concrete is allowed to stand in the mould for a few hours, when the cylinders are removed; the outer and larger cylinder by inserting an iron wedge into the joint, and forcing the edges apart; the inner cylinder, by inserting the wedge into the joint and turning the edges so as to allow them to again overlap, returning to the shape of a coil. The outer cylinder having thus been made larger and the inner one smaller, they can readily be taken away, and the concrete pipe is then left until thoroughly hardened.

Just such a number of pipe as are actually required for the season's work need be manufactured; the implements required are inexpensive, and the pipe

may be made by the municipality for actual cost, which, after a little experience, can be reduced to a very small amount. Culverts of concrete pipe are laid in a manner similar to those of sewer pipe.

Arch Culverts of Stone and Concrete.

There is no departure which would more enrich the highways than the general use of stone and concrete for the construction of bridges and culverts. They cost more in the first instance, but the longer life, the fewer repairs needed, the greater convenience, the lesser liability to accident, render them in every way desirable.

Concrete and stone are the only materials with which really permanent work of this nature can be constructed. Bridges and culverts of rubble masonry have existed in Scotland and Ireland with scarcely any repairs for more than a century, since before the time of Telford and Macadam. Concrete bridges and roadbeds built by the Romans nearly 2,000 years ago are still in use in spite of efforts to destroy them in military operations. The cost of this class of work is constantly decreasing through the cheapening and improving of cement, through the lessened expense of procuring stone and crushing it, and through growing experience in the use of cement. In Scotland it is common for farmers to contract for rubble concrete bridges, provide the stone, and hire masons to do the work. In this way the entire expenditure is kept in the locality, among the people who pay the taxes, and is therefore, in spite of a slightly greater cost, not unpopular. Up to forty foot spans, this construction is not difficult.

In the construction of a stone arch the first consideration is the foundation. The depth to which the excavation must be made will depend chiefly upon the span of the arch, and the nature of the natural soil on which it will rest. The chief object is that it shall be secure. If bed rock comes to the surface it may be safe to rest the base of the arch upon it without any further excavation. A firm hardpan may exist a short distance below the surface of the ground. But a quicksand, or other insecure footing, may necessitate the sinking of piles, or the placing of a wide, and perhaps deep, concrete base. But the foundation must be sufficient to provide that the washing of water cannot undermine it, that the lateral thrusts of the embankments cannot move it, nor that the weight of loads cannot cause it to sink. No more definite rule can safely be given than to make the most of local circumstances, with always a fair margin for safety.

Full-centre arches, that is, entire semicircles, are easily formed, possess great strength, and have little lateral thrust, but with wide spans, they necessarily rise to a correspondingly great height, and cannot always be employed. A segmental or flat arch will lessen the rise, but has a considerable lateral thrust which necessitates very strong abutments. A compound arch, made up of a number of different circles, when rightly proportioned, combines the advantages of the two, reducing the height, and at the same time having an excellent appearance. The

thickness of the arch and abutments depends upon a number of details, the chief of which are: the form and size of the arch, the quality of the material composing it, and the character of the workmanship. The haunches or shoulders should be built from the spring of the arch half way to the top.

With regard to the masonry, first-class hydraulic cement should be used. The arch stones should be full-bedded in cement, and each course afterwards thoroughly grouted. Each stone should be cleaned and dampened before being placed in the arch. Improperly dressed stones should be re-cut, as no hammering should be allowed after the stones are set. The ring-stones should be dressed into a wedge shape, so that they will radiate truly from the centre of the circle, and should be so dressed that the joints need not exceed three-eighths of an inch in width. The ring-stone should be of such thickness as to expose ten inches on the inside or face of the arch. The exterior of the arch should be flushed with a one inch coat of cement and the surface then smoothed off.

Arch-culverts and bridges of cement-concrete can be more cheaply constructed than can masonry arches, and if careful workmanship is employed, are quite as serviceable. They are formed by constructing a curbing and thoroughly ramming the concrete into it in successive layers. The manner of mixing the concrete depends upon the character of the cement used, some cements being slow setting, others quick setting; some will set in water, while others will not; some will allow a considerable portion of water to be used in forming the mortar, while other cements should be but slightly moistened.

One feature in connection with concrete culvert work is that, with the curbing and centres in place, an intelligent workman can, by following the instructions of the engineer, lay the concrete. Manufacturers complain that masons, in the great majority of cases, entirely disregard the instructions given them with respect to the mixing of cement, and follow their own methods of mixing common mortar; while a man totally unaccustomed to work of this description will obey instructions carefully and minutely. Concrete cannot be mixed and put in place like common mortar, and by overlooking this fact, much concrete work has failed and has brought the material into disrepute in some localities.

Bridge Abutments.

The most substantial substructures of bridges are of either stone or concrete. In their construction sufficient excavation must at first be made to properly contain the abutments, and this earth may be refilled again so as to form approaches to the bridge.

The excavation completed, when concrete is used in whole or in part, the portion thus constructed must be boxed and curbed in a substantial manner the exact size and shape required. After the concrete has set, this boxing is removed and earth filled in solidly around the face of the abutments. Hammer dressed stone should crown the concrete to form a bridge seat.

Concrete should be composed of first-class cement ; a clean sharp silicious sand, entirely free from earthy particles and coarse enough to pass through a twenty mesh and be retained on a thirty mesh seive ; clean screened gravel the largest not to be more than two and one-half inches in diameter ; or in place of gravel, broken stone that will pass through a two inch ring. These materials should be mixed in the proportion of one of concrete, two of sand, and three of gravel or broken stone, with just sufficient water to form a plastic mass. The sand and cement should first be thoroughly mixed when dry, then water added to make a thick paste, and this thoroughly mixed again. This mortar is then spread out and the stone or gravel added, when the whole is mixed together until every stone is thoroughly coated with mortar. When this is done the concrete may be put in place and should be spread out and pounded until the excessive moisture appears on the surface.

Masonry abutments should be of rock-faced ashlar, first-class in every respect. The projection of the rock face should not be more than three inches from the line of pitch. The stone used should be approved quarried stone laid on their natural beds, and all beds of stone dressed parallel and true, the bed to be always as large as the stones will admit. Vertical joints should be dressed not less than twelve inches in from the face, and as much more as the stone will admit, and particular care must be taken to have them well filled with mortar. Joints should in no cases exceed one-half of an inch in thickness. The courses ordinarily should not be less than eight inches in thickness. Each course should be dressed before laying, and not to be moved after being laid, or if moved, should be taken up cleaned and re-laid again in fresh mortar. The stones and work should be kept free from all dirt that will interfere with the adhesion of the mortar. Stones ought to be sprinkled with water before being placed in the work. Every stone must be laid with a full bed of mortar and beaten solid. Spaces in the vertical joints back from the face have to be built up, thoroughly grouted and each course finished off so as to be perfectly solid. Stretchers should be two and one-half feet in length, with a depth of one and one-half times their height.

Headers should be built in each course at least every four feet apart, and so arranged with the adjoining courses as to leave them equally distributed over the face of the structure. They should have a length in the face of the work of at least two feet and a depth of at least twice their length, unless the wall will not admit of this proportion, in which case, they will pass through from side to side of the wall. The backing or filling ought to be of good sized stones, and of such shape and so arranged that they will break joints and thoroughly bond the wall in all directions, and leave no space of more than six inches in diameter. All spaces must be filled in with small stones and spawls laid in mortar and thoroughly grouted.

The coping stones should be of the necessary sizes and shapes, well bedded and closely jointed. The upper surface should be bush hammered and the face

and corners brought to a true line. A tail wall, if built upon each abutment, may be of rubble stone work.

All mortar used in the masonry should be composed of clean, sharp sand, and an approved brand of cement. It should be of the best quality and freshly ground. The cement and sand for the mortar should be mixed in the proportion of three parts of sand by measure and one part of cement, the mortar to be made in a water-tight box or on a floor, and in no case on the ground. The ingredients should be mixed thoroughly in a dry state, and the proper amount of water added afterwards, and again thoroughly mixed. It must be used directly after mixing, or if not used within one hour after mixing should be discarded.

Design of Iron Bridges.

The design of iron or steel bridges commonly erected may be classified under: The plain beam or girder; the beam truss; the suspension truss; and the bowstring or arch truss. The first of these is well understood; the second comprises those trusses in which both bottom and top chords are essential; the third includes those in which the upper chord only is necessary; the fourth is not properly a truss, but an arch in which the horizontal tie takes the place of fixed abutments. The style chosen should be governed by circumstances and economy; but apart from this any design is good so long as it can be accurately analyzed as to the character and amount of strain in all its parts. On the other hand any design which cannot be so analyzed should not for a moment receive consideration.

The course pursued by some, indeed most municipalities, in erecting iron bridges is likely, however, to result disastrously, and throw iron and steel into disrepute. A council advertises for tenders. The companies responding supply their own plans and specifications. Thus far the procedure is entirely satisfactory. The difficulty arises when the councils accept the lowest tender without obtaining the advice of an experienced builder of iron bridges as to the plans and specifications submitted. Cases have occurred in which a difference of five dollars has influenced a council to accept a tender for a bridge which manifestly, to a man of experience, was worth less than the other by several hundred dollars; and which was indeed unsafe, offering every likelihood of failure with attendant loss of life and great expense for reconstruction. It is difficult to understand the action of councillors, shrewd in other matters, in the construction of bridges and other public works proceeding with such apparent disregard for the true interests of those whom they represent. A small sum in securing reliable advice is as much a matter of economy in public as in private affairs.

Painting Iron Bridges.

The prevention of rust is a matter of first importance in the care of iron and steel bridge work. The first principle in so doing is the exclusion of air and moisture. Galvanizing and painting are the two most commonly adopted means

for accomplishing this end. If the coating could in all cases be made continuous the result would be reached, but it is impossible to prevent slight breaks no larger than pin points perhaps, but which admit air and moisture. With these as a point of commencement, corrosion goes on beneath the coating, and is all the more dangerous because concealed. The durability of painted iron work depends largely upon the surface of the metal being properly cleaned and prepared to receive the paint or zinc coating. An ideal paint should have a toughness that does not depend upon a perishable ingredient; its elasticity should not be diminished by cold; it should not soften, but rather hardened by heat; and it should contain nothing which would act as a carrier of oxygen to the metal. Most engineers have a pet compound, that recommended by the chief engineer of the Bay of Quinte Railway being composed of one pound of lamp black, eight pounds of red lead, one gallon of raw linseed oil, the color being a rich chocolate brown.

Painting Wooden Bridges.

The painting of timber in bridges is a matter of some importance. The effect of paint on wet timber is to retain moisture and cause the so called "dry rot." If the timber is dry and well seasoned, the better practice is to apply a coating of paint at once. It is, however, a mistake to paint unseasoned timber in a bridge until it has stood a year; and the work should be done at the end of the summer when the wood is thoroughly dry.

Because of the tendency of paint to retain moisture, there has been some dispute as to the advisability of painting very large timbers in a bridge, the tendency of the paint being to increase the "dry rot." Nevertheless, the timber reaches a certain stage, after sufficient seasoning, when to exclude the atmosphere by painting tends to its preservation.

It is not considered good practice, however, in wooden bridge construction, to use any timber more than six inches in thickness. Where greater strength is required, in caps, beams, cords, braces, etc., a sufficient number of pieces of this or less thickness should be built together by keys, packing pieces and bolts, thus allowing a free circulation of air. Over members thus composed should be placed a covering of galvanized iron, extending a half or three quarters of an inch down on the timber, and secured by galvanized tacks, preventing the rain from entering the spaces or resting in or around the packed points.

The bridge thus built should be given one thorough coat of paint the summer following its construction, and a second one should be applied the third year. The ends of all timbers, all seats, joints and bearings should be well coated with white lead at the time of construction.

Economy of Iron Bridges.

The economy of iron and steel bridges for replacing wooden structures is a matter in which definite statistics are not available. The steel superstructure should last half a century, while the masonry piers and abutments, with first class

material and workmanship, and careful attention to repairs should be good for at least twice that period. The life of a wooden bridge may be placed at fifteen years; and during that period expensive repairs will be required. If the cost of the iron and masonry bridge is twice that of the wooden structure there is still a wide margin for profit. The price of concrete, masonry and steel is constantly decreasing; that of timber is becoming greater, while the quality available is degenerating; and it is not difficult to fortell that, in future, culverts and bridges will necessarily be built of permanent and durable materials.

THE INFLUENCE OF PAVEMENTS ON PUBLIC HEALTH.*

In presenting to this association a paper on the sanitary aspect of pavements, I have been actuated by a desire to obtain information rather than to impart it. Ontario has so recently developed from a wilderness to the home of civilization and culture; our villages have grown so quickly into towns, our towns into cities, and the advance of the various sciences has been so rapid, that our people scarcely realize the changed circumstances and the need of carefully directing their energies in meeting the demands of the times. In my visits to different parts of the Province I am constantly met with evidences of the good wrought by this association. I find that in very small villages even, inefficient drainage, cesspools, piggeries, slaughter houses, and impure water supplies are not now tolerated as they were once, and that this is due to the work of your association. It is with considerable hope, therefore, that I have undertaken to briefly lay before you the subject of pavements and public health, confident that you will lend your assistance in adding to our knowledge of this as of other matters pertaining to perfect sanitation; and where reform is needed your aid will be afforded.

There is no one paving material which possesses every quality desired in a pavement to meet all conditions and uses. The ideal pavement remains to be discovered; but the features which should belong to such an ideal pavement are so numerous and of such varying character as to render the search apparently a hopeless one. The ideal pavement: 1, should be cheap, and economical of maintenance; 2, should be durable; 3, should suit all classes of traffic; 4, should offer little resistance to traction; 5, should give a good foothold to horses; 6, should be adapted to all grades; 7, should have a good appearance; 8, should not be muddy or pervious to water; 9, should be sanitary, that is, non-absorbent, not subject to decay, easily cleaned, not dusty, not noisy.

It is apparent then that, notwithstanding the importance of the sanitary aspect of a pavement, there are other features which must be considered. The primary intention of a pavement is to accommodate travel, and to provide one which will do this satisfactorily, which will be durable, cheap, of good appearance, healthful, and possess in the highest degree the other qualities enumerated,

*A paper read before the Association of Medical Health Officers of Ontario.

in view of the location, nature and extent of traffic, is the problem which presents itself to the paving engineer. Just as no absolutely perfect paving for every time and place has been discovered, it is doubtful if any paving material now used should be utterly condemned. Each has its place in which, until the ideal, universal pavement is found it will be more satisfactory than any other which could be used under that particular set of circumstances of soil, climate, traffic, etc. The purpose of this paper, however, is to treat of the healthfulness of paving in general, of the sanitary aspect of commonly used paving materials, that is, asphalt, stone blocks, vitrified brick, cedar block, and broken stone, (macadam) with respect to absorption, decay, ease of cleaning, dustiness and noise.

Of all these, cedar block has received the greatest censure on the score of unhealthfulness. The late Dr. O. W. Wright, Health Officer of Detroit, is quoted as saying: "On sanitary grounds I must earnestly protest against the use of wooden block pavements. Such blocks, laid endwise, not only absorb water which dissolves out the albuminoid matter that acts as a putrefactive leaven, but also absorbs an infusion of horse manure and a great quantity of horse urine dropped on the street. The lower end of the blocks, resting on boards, clay or sand, soon becomes covered with a fungoid growth thoroughly saturated with albuminous extract and the excreta of animals in a liquid putrescible form. These wooden pavements undergo a decomposition in the warm season, and add to the unwholesomeness of the city. The street in fact, might as well be covered a foot deep with rotting barnyard manure so far as unwholesomeness is concerned. Moreover, the interstices between the blocks and the perforations of decay allow the foul liquids of the surface to flow through, supersaturating the earth beneath and constantly adding to the putrefying mass." Cedar block has been condemned in similar terms by many others. On the other hand, Col. Heywood, Engineer of the City of London, England, has said: "It has been said that wood pavements at all times smell offensively and may be unhealthful; but although some city streets have been paved with wood for thirty years, no complaints that I am aware of have been made to the commission on this head, and the inhabitants at all times have not only expressed great anxiety lest the wood should be replaced by other materials, but have subscribed towards the cost of its renewals. . . . I have at times noticed offensive emanations from it near cab-stands, but am unable to find further evidence of its unhealthfulness. These remarks must be held to apply only to public streets open to the sun and air, and traffic; in confined places and under some conditions, wood might be objectionable. I have seen it decaying in confined places without traffic."

The one statement by the Medical Health Officer of Detroit refers directly to the cedar block pavement as we understand it in this country. The other opinion, that of Col. Heywood of London, is expressed as regarding the wooden pavements as laid in European countries. Between these two pavements there is a vast difference. Under European practice, many of the pavements are of the

Karri and Jarrah woods of Australia, which are thoroughly saturated with resins, are very hard, and are not subject to decay. They are sawn into brick-like blocks and laid on concrete. Where soft woods are used, they are also cut into regular oblong blocks and laid on concrete, but are saturated with creosote or treated with some other preservative process.

Wooden pavements of America, however, represented by cedar blocks, are of a very different order. The round blocks of irregular diameter, are merely the untreated wood, still carrying the bark. These, placed on a bed of sand, are under the most favorable conditions possible for decay, being constantly exposed to moisture, air and warmth. With no preservative treatment they are enabled to absorb to the fullest extent all forms of liquid street-filth, which in the process of putrefaction, feeds on the organic matter of the wood. The surface, which quickly becomes uneven, retains a large quantity of loose matter subject to decay, the whole giving rise at times to noxious odors. The effect, were sufficient of such paving used, would subject us to the conditions favorable to marsh fever. From a sanitary standpoint, the cedar block pavement of this country would indicate a serious menace to health. At the same time, while we are justified as a matter of theory in arriving at this result, there do not appear to be any statistics to prove the conclusion to be a correct one. The death rate of cities most largely paved with cedar blocks does not bear any ratio to the extent of such pavement; nor does a change from cedar block to another less absorbent pavement produce a noticeable effect on the death rate.

In European practice, wood, more suited to a business street than macadam affording a better foot-hold for horses than asphalt, less noisy than granite setts, is exceeding popular in spite of its less sanitary character. In this country, however, there is an unwillingness to renew the wooden pavement when decay has rendered it unfit for further use, and this, coupled with the less careful method of laying, is the cause of the complete disrepute into which it has fallen.

Experiments have been made recently by a Polish scientist with regard to cedar blocks. The bacteriological examinations showed that, in specimens taken from blocks which had been in use for four years, and from a depth of one centimetre and two centimeters below the surface, there were at the end of five days 650,000, 220,000 and 12,100 bacteria per gramme of wood. A later examination showed 1,200,000 colonies per gramme in the surface of the wood, and 8,600 colonies at two centimes below the surface. An estimate, in terms of its nitrogen, was made of the organic matter absorbed by the wood, and indicated that the surface layer of wood contains more nitrogen than the most polluted soil. A comparative estimate of the pollution of the atmosphere was made by placing a definite quantity of sulphuric acid under a glass bell, on the surface of wooden and asphalt pavements, the result, as indicated by the quantity of ammonia absorbed by the acid, being much in favor of asphalt.

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The observations show that while a wooden pavement gives absolute protection to the soil and subsoil water, there was a considerable atmospheric contamination. The experiments were made on blocks of pine, preserved by impregnation with copper sulphate. Such being the case, with a wooden pavement laid under European practice, there can be little doubt of the unwholesome effect of cedar block upon the atmosphere. Further experiments of this description, conducted by members of your association, would doubtless prove instructive and profitable.

Broken stone or macadam would next arouse suspicion with regard to its absorptive qualities. There is this great difference between the two, however, that whereas a wooden pavement itself decays, and affords food for the decay of other organic matter falling on it, the macadam does not in itself decay. With under-drainage, such as a well-built macadam road, possesses it should be little more than a good sewage disposal bed for the comparatively small amount of sewage which falls upon it. A macadam pavement can be scraped, and swept, it is not too noisy, dust can be subdued by sprinkling, and on sanitary grounds appears to be an excellent pavement for residential streets where traffic is not excessive. For business streets, or for heavily travelled thoroughfares of cities, a harder surface is advisable. With regard to absorption, there can be no objection to asphalt, vitrified bricks, nor stone blocks. Asphalt is impervious to water; while the joints of brick or stone pavements are practically perfect so far as absorption is concerned.

A pavement, to be sanitary, should not be dusty. The dust of a pavement is not only an irritant, but carries with it the bacteria of disease, which from various sources are a part of street filth. To prevent dust, the pavement must be so perfectly cleaned that a practically harmless amount is taken up by the wind; or if perfect cleanliness is not possible, dust must be subdued by sprinkling. Unless perfectly cleaned, much better cleaned than is usually the case in this country, an asphalt pavement is apt to be a disagreeably dusty pavement on a windy day in summer.

This, indeed, is one of the greatest faults from a sanitary standpoint. Toronto has the reputation of being a clean city, with a well-organized street department; yet under these favorable conditions, a walk or drive down Yonge St. on a warm, windy day is a very trying experience. The smooth, hot surface quickly dries any matter falling upon it, a wheel passing over this dry substance grinds it to powder, and the result is that clouds of dust find their way into the eyes, nose, mouth, throat and lungs of pedestrians. Business men in their offices are not safe from attack, as it drifts in through the open windows. The dust imbeds itself in clothing, fastens itself on articles of food exposed in the shops, to be eaten finally by the purchaser. One case came to my notice in which a consumptive patient was ordered by his physician to leave Jarvis Street, one of the best resi-

dence streets of Toronto, because of the dust which came from the asphalted roadway. These streets are swept by machines, and are handswept by a corps of city employees, but are not, to my knowledge, flushed as are similar pavements in London and Paris. Flushing is the only method whereby asphalt can be freed from this unsanitary dustiness, but in addition to being expensive and hurtful to the asphalt, such a proposal will meet with the disapprobation of the engineer in charge of sewers. The dust, however, is not a defect of the pavement so much as it is a fault in the method of cleaning.

Asphalt has, nevertheless, the disadvantage of being a very hot pavement. Its smooth surface, reflecting back the heat and light, is productive at times of sun-stroke, and the glare is frequently painful to the eyes. This is most noticeable in closely built business sections where there is least circulation of air, where the sun beats down between high brick walls; and is not so objectionable on a shady residential street with houses well apart. Vitrified brick and stone block pavements are neither so dusty nor so hot as asphalt, since the surfaces are less smooth and assist in retaining in the joints the finer particles of dust. Sprinkling too, is in a greater measure effective in subduing dust on brick or stone block than on asphalt, from the hot, smooth surface of which moisture evaporates rapidly. A macadam pavement is dusty if not properly treated, but if scraped and swept as are other pavements, the dust can be largely subdued by sprinkling.

Noisiness, if excessive, is another unsanitary feature. A noisy pavement is jarring to the nerves, grating upon the sensibilities, and for either a heavily travelled business street, or a residential quarter, a quiet pavement is much to be desired. Noise itself is not always unhealthful. It is doubtful if the workman in a boiler factory, or a railroad engineer, or other employee, is much influenced by the noise incidental to his occupation. Both are muscular of body, constantly taking vigorous exercise. But to the more sedentary man of business, whether at high nervous tension in his office, or resting in the quiet of his home, a din, constant or intermittent, is a source of annoyance, and as such is wearing to the nervous system. The most objectionable in this regard is granite or other stone block pavement. Vitrified brick is apt, unless great precautions are taken, to create a disagreeable rumbling. Asphalt, wood and macadam are the least objectionable with respect to noise.

While we have this to say of the comparative healthfulness of different varieties of pavements, there is another condition of matters common to too many towns and cities, in which the streets, in fall and spring, form a wilderness of mud and stagnant pools, and in the summer are shapeless beds of dust. Many of them are made receptacles for the refuse from private property, which is left to disfigure the street, forming rivers of filth and cesspools of disease. Such streets have been regarded as a zero quantity, doing no particular harm, doing no particular good. Streets, however, which do no good, should do good, and therein

lies the harm. A good street is a well drained street, a well cleaned street, and is a source of healthfulness to the members of the community. Streets should be the public parks, pleasing to the cultivated taste, adding to the culture and refinement of the people, and enticing them to breathe health and vigor, whether walking, bicycling, riding or driving. Passing along the city street we reach the country highway, which as a means of permitting the people of the city to leave the congested portions and to reside in the less thickly populated suburbs, forms an important factor in securing public health

TOWN STREETS.

General descriptions of plans for street improvement have been issued in previous publications. Every town and city, however, possesses local circumstances peculiar to itself, requiring special consideration and adoption. The following report to the City Council of Windsor, Ontario, while containing, briefly, an outline of the means to be employed, and capable of universal application, contains references of a local nature, having much, nevertheless, of useful suggestion to other Councils:

To His Worship the Mayor and Members of the Council of the City of Windsor, Ontario.

GENTLEMEN.—In accordance with the request of your Council forwarded me by your city Clerk, I visited Windsor on the 2nd inst., examined the streets, and conferred with the Mayor, your Board of Works, Street Superintendent and City Clerk; and herewith submit the following report as to the present condition of the streets, the system of control, and the methods which should be adopted for their improvement:

Present Condition of Streets.

A number of streets, Sandwich, Pitt, Ouellette, Glengarry, Windsor, Church, London, Chatham, Dougall and others have been paved with cedar block. The majority of these pavements have survived their usefulness, and are now exceedingly rough from decay and wear. A few sections, such as Sandwich Street, from the G.T.R. bridge to Church Street, have been recently laid, and are still in serviceable condition.

The streets not paved with cedar block are dirt roadways which, during the summer, are kept nicely crowned, and the gutters well shaped and cleaned by the persistent and skillful use of a grading machine.

Broken stone has, in a few instances, been placed on top of these dirt roadways, but broken stone used in this way is of but little service in improving town streets; these are, by the general public, understood as "macadam" roads, but are totally devoid of the principles advocated by McAdam.

Expenditure.

The construction of cedar block pavement has cost \$172,331, during the past twelve years. On the repair of streets about \$10,000 is spent annually, which represents for the past twelve years, \$120,000.

The total amount then, spent by Windsor in the past twelve years, on pavements and sidewalks represents nearly \$300,000—over a quarter of a million dollars. This is a very large expenditure, one which, without better returns, no city of the size of Windsor can afford. The expenditure indicates on the part of the citizens, a most praiseworthy desire to advance the true interests of the city. It indicates that the importance of well paved streets has been fully appreciated by the taxpayer; it indicates a willingness to pay for well-paved streets. It is in fairness to be pointed out that a considerable proportion of this has been spent on work from which permanent results could not be expected. There were annual repairs, street cleaning, removal of snow and similar details, from which only temporary benefit could be had. At the same time, it must be apparent to your council that permanent improvements are very few in comparison with the large expenditure. And notwithstanding its apparent simplicity, it is evident that the paving problem has not been solved in Windsor. Good streets cost a large amount of money, but bad streets cost vastly more.

Temporary Work.

While cedar block may at one time have been, in the opinion of some, a serviceable and economical pavement for business streets in the immediate business section, it was not an economical pavement on residential streets subjected to but little traffic, where failure must result from decay rather than from wear. For the residential streets of Windsor, much has been lost by overlooking the claims of broken stone roadways in place of cedar block. The mistakes which have arisen, and which by no means have been confined to Windsor alone, are the result of insufficient acquaintance with paving methods and materials, and a consequent misuse of them.

The dirt roadways which constantly require to be re-graded, which are good in the summer season only, and which in spring and fall become shapeless, muddy and of little use for travel, are a constant source of expense for repairs. In this case the error, in a sense, is the reverse of that arising from the use of cedar blocks on residential streets, the roadway being too weak to support the traffic, and the consequent outlay for repairs very great. Here again the use of macadam has been neglected. The expense of grading these streets year after year is very great. To obtain good results from the least expenditure, a larger amount should be spent in the first construction of permanent broken stone pavements, and a very much less amount in the repair of temporary dirt roads.

In sidewalks, no use has been made as yet of cement-concrete in place of plank, and the consequence is a large outlay for repairs.

Throughout the whole, there is an evidence of temporary construction which, cheapest in first cost, after a term of years when the cost of repairs is included, is the dearest. The waterworks system has been constructed at a cost of \$304,000. Since 1884 the sum of \$302,000 has been spent on sewers in addition to a considerable expenditure previous to that time for main sewers, the exact amount being difficult to ascertain, as it was raised in connection with funds for school purposes.

On the sewerage system, waterworks, public buildings, churches, places of business, and private residences, permanency has been sought. But in street work, sidewalks and pavements, it is apparent that a different policy has been pursued. Everything connected with streets has been built in the most temporary manner, little remaining to represent the expenditure but rotting blocks which will soon have to be removed; clay streets which dissolve after a shower; decaying sidewalks which are constantly being renewed.

Frontage Tax.

The Frontage Tax system has its advantages and, where the by-law has been carefully prepared, should give satisfaction. But very great care should be taken in framing the by-law so as to adjust it to local circumstances as far as possible. It is a simple means of raising the money, and of levying the tax, and when an equitable mode of assessment is provided, regulating corner, triangular, or irregularly-shaped lots, side-hill and similiar property, as well as street intersections, levying a just proportion against the general funds, it invariably stimulates street improvement.

Unfortunately for Windsor, the system has not operated equitably in all cases. The residents of certain streets, with commendable loyalty to the best interests of the city have petitioned and paid for expensive pavements, in the expectation that other streets, equally able to bear the tax, would follow in due time. The result, however, has been that a number of important streets are still unpaved, and an undue amount of travel has been thrown on the improved streets; while the unimproved streets, by the use of the grader, and because of the little travel which they receive, are in as good condition.

The injustice does not appear to lie in the fact that streets other than those improved did not follow with improvements but that, unfortunately, the wrong materials were used, and the property-owners who made the improvement, are now called upon to repeat the expenditure. The fault does not lie in the frontage system but in the character of the work performed under it. Viewing it in this light, it may have been well for you citizens that no more work was undertaken with this class of material. Because in nearly every instance where cedar block was laid on residence streets, first-class macadam should have been used. Had this been done, with due attention and repair those streets would be in as good condition to-day as when first constructed.

Now that the subject is better understood, and councils and citizens are taking so much interest in the question, certain amendments to the Act would render the frontage system more just and workable, and these, no doubt, could be secured if proper representation were made. It would, for example, be better, in cases where the council considers it necessary to take the initiative, to require an adverse petition of two-thirds of the property owners; after which if the council still considers the work necessary they may proceed on a two-thirds vote of their own number.

Where the improvement of leading entrances to the city is undertaken and the property on either side of the street in the outlying sections is not built up or improved, or for any reason is unfit for building purposes, and the Council does not deem it equitable to assess such property in the same proportion as other property, then the Council should have the power, in all such cases, to determine the proportion.

And further, it would be well to provide by by-law for the payment of say one-third of the cost of all street improvements out of the general funds. When the Act was framed little interest was taken in the question of street improvement, and the intention undoubtedly was to permit the residents on certain streets, desiring their improvement, to have it made and the payments extended over a term of years; the principle being to allow them to get what they wanted and pay for it. This was taken advantage of by the more progressive citizens, but very seldom in a general way. With the changes of time and conditions, the demand to-day for improved streets is general, and no system should be adopted which will not to a reasonable extent provide for this.

An injustice to the outlying sections, less inclined to realize the advantages of improved streets, would be apparent at the first sight of this proposition, but

upon examination it will be seen that it would be found just and equitable. The opinion of the property holders in such sections generally is that streets demanding any improvement beyond the safe condition of the roadway, should pay for such improvement, believing that the benefits are purely local.

This, however, is erroneous, as no improvement can be made upon the streets of a city without benefitting the whole place. The nature, and consequently the cost of these improvements should be proportionate to the service which they render, and should not be wholly regulated according to the desire of the owners. Before any work is undertaken a comprehensive plan of the city outlining the character of the improvements proposed for the different streets, width of roadway, location of sidewalks, estimated cost, should be prepared, and followed as nearly as possible. The highest assessed property, where the most expensive pavement is required, pays a proportionately great share of the cost of that pavement as well as of the least expensive pavement.

Street intersections should be charged to the general funds, otherwise a street first undertaking improvement would be charged with all intersections, and other streets crossing this and subsequently undertaking improvements would be relieved from the cost of these street intersections.

In assessing the cost, the engineer should make an allowance on corner, triangular, or other irregularly shaped pieces of land situated at the intersection of streets. It is difficult to specify what this allowance should be, as regard must be taken to situation, superficial area, etc. A plan much favored is to charge two-thirds of the frontage on the side of the lot to be assessed.

The frontage tax system is a simple means of raising the money. No matter how small the work, the money can be easily provided. It is a check on careless management, because contracts must be separate, are not large, are understood by each property owner who feels directly interested, and studies the contract, watching closely its execution.

Whereas, under a plan of general assessment, money must be raised by by-law voted by the freeholders generally; consequently in large sums which would not likely be expended as judiciously or the work performed as economically. Where an effort is made to raise as large a sum of money as would be required to perform even the most important work, considerable opposition would be offered by sections not directly benefitted, and subsequent by-laws would meet much opposition from those previously served. The frequent submission of by-laws for any purpose is found objectionable.

Some expression was given to the wisdom of returning to a system of general taxation for street improvements. Under this system the main streets and entrances to the city might be more quickly paved; nevertheless it would form a total denial of the principle embodied in the frontage tax system, that the property directly benefitted should bear the greater proportion of the cost; and at the same time, as pointed out, there are difficulties to be encountered in passing the necessary by-laws raising large amounts of money, created by sectional jealousies and other causes which cannot definitely be foreseen.

The Ward System.

Good streets and the ward system are incompatible. Under any circumstances, the ward system of representation is not favorable to the best city government, and with a system of general taxation for street improvement, its abolition (for which recent legislation provides) becomes imperative. Every year, in

the larger municipalities, several thousands of dollars are divided among the wards; and this again is subdivided by the representative, or under his influence, and is scattered over the street area. Frequently work is attempted which is never completed, and not infrequently, work is done which would be better undone. It is in direct violation of the more economic management of streets whereby the expenditure would be concentrated in work of a substantial and finished character, and afterwards systematically maintained. Instead of this, under the ward system the effort is to build streets by an extensive series of disconnected patches, and to repair streets which were never constructed.

A Standard Pavement.

A standard pavement in Windsor for streets other than those in the immediate business section should be macadam. A well-kept macadam driveway is in keeping with well-kept boulevards, lawns and shade trees, the characteristics of a residential street; it has a cool appearance, the dust can readily be kept down by sprinkling, and for light driving is the favorite among horsemen. Bicyclists, now an important section of the community, usually favor macadam, in preference to the more costly classes of pavement. A comparison of macadam with asphalt or vitrified brick, in point of utility and beauty, will not result unfavorably to the former, for use on residential streets. It is not to be inferred, however, that broken stone roadways are recommended for streets in the immediate business section, where a harder and, in a sense, a cleaner surface is desirable.

By proper attention to repairs, the life of this class of pavement can be made continuous. The surface can be frequently rolled, improving it greatly. It can be scraped and swept as are other pavements. When it begins to lose shape the surface can be loosened up by means of teeth attached to the roller, a light coating of new metal applied, and then rolled down as well as when new. It is by such means as these that broken stone roadways can be made much more economical and satisfactory than any other for streets generally. This ease of renewal, and repair is a property peculiar to macadam, which renders it most satisfactory for general purposes. While the cost in the first instance may nearly equal that of cedar block, yet at the termination of the period when cedar block is decayed and has to be torn up or renewed, the macadam, if properly treated, is still in a good condition. It forms a permanent basis, and its perpetuation is merely a matter of repair to be met by the general funds.

Except under excessive wear, or where in business sections a high-grade pavement is necessary, broken stone pavements, by the aid of a steam road roller, are beyond doubt the most serviceable and economical, and give greatest satisfaction to the taxpayer.

Cedar Block.

The real life of cedar block paving will average only about seven years. Up to that time the surface is moderately smooth. Many of these pavements are allowed to stand, however, for ten or twelve years, at the end of which time they have become almost impassable, for the last half of their existence having received constant repairs, the cost of which is very great, a fact which your council must be experiencing in the effort to keep your cedar pavement in repair. Except for a few years after construction, cedar block makes a very rough road surface, is temporary, dirty, unsanitary, and the appearance is unsatisfactory. This is the result of Canadian experience with cedar block, except under favorable conditions which do not exist in Windsor.

Asphalt and Brick.

For a pavement such as is required in the business sections of your city, the essential qualities are:

- (1) A secure and pleasant footing for horses.
- (2) Smooth, so as to render travelling and traction agreeable, easy and noiseless.
- (3) It should be sanitary; the form and material such that it will be impervious, liquids will not have permanent lodgement, and dust will not be easily produced.
- (4) The durability and service rendered will be commensurate with the cost of construction and maintenance; that is, it must be economical.
- (5) It must be easy of removal, replacement and repair at reasonable cost, and with the appliances and materials within the control of the corporation.

In view of the above it will be apparent, I believe, that brick and asphalt are the two competing materials. With regard to the first quality, the foothold afforded to horses, brick must have the preference. It is one of the objectionable features of asphalt that it is exceedingly slippery when wet, and even when dry it is not always safe.

As to the second quality, asphalt must take first place since it is in a slight degree smoother and less noisy than brick. It is doubtful, however, if traction is any easier, owing to the insecure footing afforded to horses.

There is little if any difference between the sanitary status of the two pavements. Both are, of course, impermeable and offer little resistance to the flow of liquids; the joints of the brick are just sufficient to retain moisture and subdue dust. With the smooth asphalt surface no amount of sprinkling will keep the surface moist in hot weather.

As to cost, asphalt is from one-half to one-third more than brick, and experience has not proven its life to be any greater. As to durability, there must always be the proviso that good material and proper plans and methods of construction are used in any case.

The laying or repairing of brick does not require skilled labor as does asphalt, and this difficulty in connection with the latter is felt more particularly in places where only a short section of asphalt is used. In large cities the inconvenience is not so great.

While asphalt is extensively used in the United States and Canada, and is unquestionably a good paving material for certain streets of large cities, I believe brick to be more suited to the requirements of your City.

The quality of a brick pavement is not to be gauged by the best brick used in its construction, but by the poorest. For this reason it is necessary, in deciding on the kind to use, to see that it comes up to the standard of scientific tests. More than this, while the building of the pavement is in progress there should be careful inspection to see that no brick of an inferior quality is used.

There is a tendency also, to reduce the cost of a pavement by having a cheap foundation. Foundations of gravel, sand and macadam have proven successful in a number of cases, but only where the natural sub-soil is of a loose and porous nature. In this climate where we are subjected to alternatives of frost and slush, the experiment is a dangerous one. A foundation of six inches of concrete should be used. On this, place a one-inch cushion of sand, and fill the joints of the brick with a cement of tar and sand. The earth sub soil should, of course, have been previously graded and thoroughly consolidated with a heavy roller.

Cost of Pavements.

A suitable asphalt pavement on a six inch concrete foundation, with a five year guarantee, in Windsor would cost \$2.50 per square yard; with maintenance guarantee for fifteen years, the cost would be about \$3.00 per square yard. A vitrified brick pavement laid on a similar foundation would cost approximately, \$1.75 per square yard and should last for fifteen years. A first-class macadam pavement would cost about 75 cents per square yard, the life of which would be continuous by means of proper repair. On a 24-foot roadway where the curbing is now set, the cost to an average lot having fifty feet of frontage, for such a macadam road would be \$4.21 per annum for ten years, providing that street intersections and one-third of the cost were charged to the general funds.

Classification of Streets.

To place any system on a satisfactory and equitable basis, the improvements must be made in a business-like manner according to a carefully prepared plan which will adjust the right class of pavement to each of the streets, according to traffic and requirements.

In preparing such a plan the streets must be classified. It may emphasize the effect of classification by pointing out that a light macadam roadway costing 25 cents per square yard for a little travelled street, and an asphalt pavement, costing \$2.50 per square yard for a business block, would each be economical and satisfactory, yet they could not be interchanged except at a loss. A light macadam on a business street is quite as unwise an investment as asphalt would be on one of the least important streets. The main streets in the business sections, naturally fall into a class by themselves, requiring the most expensive form of pavement. These in Windsor, are Sandwich Street from the C.P.R. Bridge to Church Street; Ouellette Street from Sandwich Street to London Street; Pitt Street from Windsor Avenue to Ferry Street; and Goyeau Street, from Pitt to Sandwich Street. On these sections, as has been pointed out, vitrified brick is the most suitable pavement (except, for the present, that portion of Sandwich Street from the C.P.R. bridge to Church Street, which has recently been paved with cedar block).

There are other streets heavily travelled, but which do not require so expensive a form of pavement as those just enumerated, and for which a strong macadam is most suitable. Such streets are: Sandwich Street from the C.P.R. bridge westerly; Sandwich Street from Glengarry Avenue to Walkerville; Wyandatte Street; Howard Avenue the main entrance to the City from the south; Glengarry Avenue, Aylmer, and London Streets the main entrance to the City from the west.

A lighter form of macadam may be used on a third class, comprising such residential streets as Goyeau Street, Ouellette Avenue, Dougall Avenue, Chatham, Church and Pellissier Streets (the last mentioned newly block-paved).

On streets of less importance a still lighter macadam may be used; constructed however, in all respects according to the best principles of road-making.

In laying down a system of waterworks, the first step to be taken is to prepare a plan of the whole municipality and determine the size of the pipes to be laid on each street to meet the requirements. The sizes of pipes are classified, main, circuit and lateral, and range possibly in size from a 24-inch main to a 4-inch lateral. The 4-inch pipe is as efficient as the 24-inch, considering the service it has to perform. Interchanged they would be useless. Much economy is practised in the careful designing of such a plant. So also should it be with

sewers, and before any work is undertaken, a plan should be prepared, not only providing for present, but as far as may be, for future requirements in capacity and disposal. As with waterworks and sewers so it is with streets. Before anything is done a similar plan should be prepared which will adjust the right class of pavement to each street according to requirements.

Construction of Broken Stone Roadways.

Macadam or broken stone roadways have not yet been understood in your city. The practice has been to pile broken stone in the centre of the roadway on top of the natural soil, or to bring the earth from the gutters to the centre of the roadway, placing the stone on top of this. This is the plan usually followed in grading township roads, but is unsuited to city street construction.

The roadbed should be excavated to the required width to receive the broken stone, and the excavated earth may be used in making boulevards or filling in the low lots to bring them to the grade of the street. The side of the street should be levelled to conform to the surface of the roadway. The general plan is much the same as for cedar blocks, except that a broken stone roadbed is used instead of cedar blocks. From 24 to 26 feet between curbs is a sufficient width of pavement, except where the street railway is laid, when extra provision should be made, for a single or double track, as the case may be.

The stone should be broken by means of a rock crushing machine which should be owned and operated by the city. In order to lessen the initial cost, small crushers are often purchased with the intention of operating them to their full capacity. Owing to the great variation in material, and the severe trials to which these machines are subjected, it is advisable to provide a machine which will not have to be operated to its full capacity. One of about two cords per hour of which only two-thirds the capacity should be used, would be suitable. The extra cost would prove a profitable investment.

By having a rotary screen attached to the crusher, the stone as it passes through is graded according to size. The grading of the metal is the most important part of the work which the crusher has to perform. When stone is placed on a road without being graded, large stones surrounded by smaller, the latter wear more quickly, and the surface becomes uneven. The large stones also, do not rest firmly on the surface, but are more easily disturbed and are apt to roll loosely under the wheels of vehicles and the feet of horses. Material of uniform size, placed in regular layers of coarseness, properly consolidated, will form a smooth surface, distributing traffic and wear uniformly. The grades of stone frequently are:

- 1st. Such as will pass through a $2\frac{1}{2}$ inch ring.
- 2nd. Such as will pass through a $1\frac{1}{2}$ inch ring.
- 3rd. Such as will pass through a 1 inch ring.
- 4th. Chips and dust screenings.

The stone should be placed on the roadbed in layers of not more than four inches in thickness, the total thickness of the covering to range from eight to twelve inches according to the strength of road required to support the travel. With each layer should be mixed a quantity of the chips and stone dust to assist in consolidation; and the surface should be finished off with a thin coating of screenings.

More stone should be placed at the centre, than at the sides. Where the thickness in the center is twelve inches, that at the curb should be about eight

inches. The thickness of the different courses from the bottom upwards, would be eight inches of two and one-half inch stone; three inches of one and one-half inch stone; and one inch of inch stone. At the curb, the thickness of each layer will be proportionately less.

In the process of laying, the material should be thoroughly rolled, and consolidation assisted by a liberal application of water. Excellent samples of this class of road may be conveniently seen by your Council in London, St. Thomas, Ingersoll, Stratford, Galt and numerous other points.

After the excavation and boulevarding has been completed, the curbing set, the foundation underdrained and sub-grade thoroughly rolled, an excellent plan where the material can be procured, is to place a layer of quarry flake stone from two to four inches in thickness, in the bottom of the roadway, and then to place on top of this about nine inches of hard, crushed stone coursed and in layers as previously specified. Quarry stone for the foundation, or for the first layers of the roadway, could be obtained from Amherstburg at a cost of \$3.00 per cord f. o. b. Amherstburg. The material can also be obtained in large quantities at Pelee Island. Field stone for finishing the road can be obtained in the vicinity of Kingsville, Harrow and Ruthven, and costs f. o. b. at those places about \$2.75 per cord—to which must be added the cost of freight and crushing. In all this would amount to about \$6.50 per cord crushed at Windsor.

Of course the harder and tougher the stone used for surfacing the street, the more durable will be the roadway. Trap rock is undoubtedly the best material. Its scarcity, however, limits its use. It can be had in large quantities from the north shore of Lake Superior, the city of Cleveland importing largely from that district by boat. Being located on the route, you should be in a position to obtain this material at reasonable cost.

Rolling.

For economical, durable and serviceable roadmaking, a heavy roller is indispensable. A road should be sufficiently smooth and compact to shed the water readily to the side gutters. If the gravel or other metal is dropped from the wagon loosely on a soft earth foundation, as is your practice, water passes into the sub-soil as into a sieve. Wheels passing over the road when in such a condition at once sink into and rut not only the gravel, but the earth beneath. Water is held in the ruts and each succeeding vehicle renders their condition worse. The road is less durable since, the gravel being mixed with the dirt beneath, it obtains, when fully consolidated, a dusty, easily-worn surface.

The weight of roller must depend upon various circumstances—the amount of work it will be required to do, the quantity of road metal used, the strength of the bridges and culverts over which it must pass. A steam roller costs more than a horse roller, but does so much better and faster work that it is more economical. A nominal 12½ ton steam roller is commonly used, and would be most suitable for your city.

Curbing.

The streets which are block paved have been provided with stone curbing which is a very important detail of a well-designed street. It is the one part of the work in Windsor which, in first construction, was made of durable material and does not need to be renewed. This curbing defines the roadway, protects the boulevards and gutters and keeps the paving material in place; and on all other the same plan should be continued.

Drainage.

From the lay of the land, the quality of the soil and the climate, the necessity of perfect drainage is possibly greater in Windsor than in any other city of Ontario. The land is flat, the soil retentive of moisture, and the changes of freezing and thawing are frequent. Without properly draining the foundation it would be useless to attempt to construct and maintain good macadam or other pavement, but with a thorough system of drainage no better foundation could be desired.

Underdrainage is one of the first points to consider. In making streets it is the native soil which must really support the weight of the traffic, no matter what paving material is used to surface it. Gravel, stone, brick or asphalt are not sufficiently strong to bridge over a wet and yielding subsoil. But if this natural soil is kept in a dry state it can support any weight, and to this end underdrainage is necessary. Underdrainage of common field tile, four to six inches in diameter, should be placed on each side of the carriageway underneath the gutters, and below frost. This "lowers the water line" and secures a good foundation.

There must be surface drainage, and for this the roadway should be crowned or rounded up, covered with suitable surface material, and open gutters provided to carry away the surface water. The surface metal, of course, resists wear, but on streets which are lightly travelled the main object is to provide a cover which will prevent the water penetrating the natural soil underneath, making it unfit to support traffic. By crowning the surface of the road, water is shed at once to the side, where provision should be made to carry it away immediately in open gutters.

The amount of crown varies with the width of the roadway and character of the surfacing. For a macadam roadway, the crown ranges from one inch to one-half an inch per foot from the curb to the center of the road, the crown decreasing as the width of roadway increases.

Gutters or underdrains are useless unless outlets are provided, and care should be taken to see that these do not become obstructed. Surface drains and underdrains should have outlets into catch-basins, leading into the sewers, if capacity for storm water is provided.

Springs underneath roadways should be tapped with blind drains and the water carried diagonally to the underdrains at the side of the streets.

Main Entrances to the City.

There are portions of main entrances to the city, such as Sandwich street West, Howard avenue, Gt-ngarry avenue and London street West, in the outlying sections of the city, where it is not advisable to pave the full width of the roadway, nor to curb it. With the foundation underdrained, and gutters made to carry away the surface water, the roadway should then be graded, leaving an excavation in the central portion a width of twelve feet and a depth of twelve inches, to receive the road metal. In this excavation should first be placed a layer about four inches thick of quarry flake stones laid closely together. On this should be placed eight inches of stone broken and coursed according to size as before specified, and thoroughly rolled. The underdrainage should be obtained by placing a row of common field tile on each side of the roadway underneath the open gutters; and diagonal drains across the roadbed at wet points. From four to six-inch tile is usually sufficient.

Sidewalks.

Sidewalks will be required in Windsor as long as the city exists and their construction in the most permanent manner possible is a part of the same principle that directs the erection of the places of business of a durable material. Planks used in sidewalks are subjected to an exceedingly severe test. They are lying close to the ground, always absorbing moisture on the underside, and are exposed to repeated changes of wet and dry. The average life of plank in this work is not more than five years until decay commences and repairs are demanded. These repairs increase annually, and walks may be and are carried for ten or twelve years, by which time the cost of repairs has almost equalled that of renewal. Every city in Ontario has laid and is laying large quantities of granolithic (cement-concrete), while nearly all towns and in fact many villages have made use of this more durable material.

In cement-concrete, sand finish, the surface coat is composed of sand and cement; while with cement-concrete granite finish (granolithic), granite, cement and sand are used. The sand finish is useful on residential streets, or other streets subjected to moderate traffic; while the granolithic forms a harder wearing surface for much-travelled streets.

The life of this class of walk is indefinite. The first walks of this material in Toronto were laid in one of the most heavily travelled sections about thirteen years ago, and are to all appearance as good as when new. Other cities have used sidewalks of cement-concrete for twenty years, which are still in excellent condition. There appears to be no reason why perfectly built walks of this material should not last half a century.

Artificial stone (cement-concrete) makes a durable, economical and satisfactory walk and this material should replace plank as rapidly as possible in your city. Of course, as with any work, care must be exercised to select the best material, and provide the best workmanship. A point frequently overlooked in concrete sidewalk construction is the necessity for perfect drainage, a matter of importance in all paving work in this country where the action of frost is so severe. Should your council desire it, I will furnish a standard specification for the construction of these walks.

Recommendations.

Summarizing the preceeding discussion, the main recommendations which I would lay before your council are as follows:—

1. That a plan of the city be prepared, classifying the streets according to traffic and requirements, adjusting the right class of pavement to each street, fixing grades, width of roadway, boulevards and location of sidewalks.
2. That the frontage tax system for street improvement be retained but so amended as to require the adverse petition of two-thirds of the property owners to annul the initiative of the council; after which the further power is given the council to perform the work on a two-thirds vote of the council. The system should be amended also so as to assess one-third the cost of all work against the general funds of the city if favorable legislation can be obtained.
3. That the ward system be abolished.
4. That less be expended in the temporary grading of dirt roadways, and more diverted to permanent work each year.
5. That broken stone roadways be laid on residential streets, the general plan to be as specified herein.

6. That vitrified brick be used on business streets in the immediate business sections.

7. That a nominal 12½-ton steam roller, and a rock crusher of at least two cords per hour capacity, be purchased.

8. That plank sidewalks be replaced with cement-concrete as rapidly as possible—granolithic finish in business sections, sand finish on residential streets.

SPECIFICATIONS FOR CEMENT-CONCRETE WALKS.

The following form of specification includes the most necessary requirements for laying artificial stone walks by contract. They should, however, be varied as circumstances may render expedient :—

1. The location and approximate extent of artificial stone side walks to be laid under these specifications are as follows : Location and extent of walks.
2. The corporation shall remove the old plank, stone, brick, and other sidewalk from the street before the construction of the new walk shall be commenced by the contractor, all such material being the property of the town, to be disposed of as the engineer may direct. Removal of old sidewalks.
3. All excavated earth, stones, posts, stumps, other obstacles or rubbish shall remain the property of the town, to be removed by the contractor to such point or points as the engineer may direct ; if not hauled for a distance exceeding one half mile, such removal to be without extra charge. Removal of excavated earth and rubbish.
4. The walk shall be laid to the lines and levels given by the engineer. No stakes or bench-marks placed for this purpose by the engineer shall be moved or effaced by the contractor without the permission of the engineer so to do. Levels, stakes and bench-marks.
5. The space over which the walk is to be laid shall be excavated to a depth of twelve inches below the elevation of the finished walk in accordance with the plans and profiles, on file at the office of the engineer. Perishable or objectional material shall be removed to a further depth, to secure a firm foundation if so required by the engineer. Such excavation in excess of twelve inches shall be filled with gravel, or other material approved by the engineer, and the bottom of the sub-grade thus obtained shall be then made thoroughly firm and solid by pounding or rolling. For all excavation or filling ordered by the engineer in excess of twelve inches below the grade of the finished walk, the contractor shall be entitled to the sum of 35 cents per cubic yard. Preliminary excavation and earth-work.
6. A porous tile drain shall be laid centrally beneath the walk, to the depth, grade, of such diameter, and carried to such outlets as are specified upon the plan and profile on file at the office of the engineer ; and tile drains for carrying surface and other water through or under the walk shall be laid as indicated upon the aforesaid plan and profile. All tile used shall be of the best quality of clay, manufactured expressly for drainage purposes, in lengths not less than one foot, and of uniform diameter throughout. All earth excavated in the laying of these drains shall be returned to the trench, being thoroughly rammed and pounded in layers not exceeding one foot in thickness, and rendered perfectly firm and solid, to the satisfaction of the engineer. When sewer pipe is required in place of common tile, such pipe shall be furnished to the contractor by the engineer ; and shall be laid in all respects to the satisfaction of the engineer. Tile drainage
7. Upon the sub-grade thus excavated, drained and consolidated, shall be spread a layer of clean gravel or broken stone to be thoroughly wetted, rolled or pounded, and brought to an even surface. The layer of gravel or stone so placed shall have a thickness of seven inches ; and shall be uniformly not less than five inches below the elevation of the surface of the finished walk, having preference thereto. A layer of gravel or broken stone seven inches in thickness

A layer of concrete, four inches in thickness.

8. Upon the foundation thus prepared, a layer of concrete shall be laid in the following manner : It shall be composed of one part by measure of fresh cement, of a quality approved by the engineer, and in accordance with the specifications for such elsewhere herein described ; three parts by measure of clean, sharp sand ; and five parts by measure of broken stone of such a size as will pass through a two-inch ring. The concrete shall be mixed in a water-tight box placed close to the work, by first spreading evenly a layer of sand ; upon this shall be evenly spread the proportionate quantity of cement and the two thoroughly mixed while dry. To this water shall be added and the whole thoroughly mixed and brought to the consistency of mortar. The proportionate amount of stone shall then be spread evenly over this mortar and thoroughly intermixed therewith. The concrete when mixed as aforesaid, shall be immediately put in place and thoroughly pounded until it has an even surface, is perfectly and uniformly solid, and is four inches in depth over the foundation and within one inch of the finished surface of the walk. Slab or flag divisions shall then be marked off, sixteen feet in area, the joints to be filled with clean sand or other approved separating material.

A layer of concrete one inch in thickness.

9. Before the aforesaid layer of concrete has set, and while it is still adhesive, there shall be laid upon it a wearing surface one inch in thickness. It shall be composed of one part by measure of Portland cement, and two parts by measure of clean sharp sand. The cement and sand shall be mixed dry, water then added to moisten sufficiently, the whole again thoroughly manipulated and mixed in a water-tight box or floor and immediately put in place. The layer shall then be thoroughly pounded, and worked to a true and even surface. Over this shall be sifted a layer of Portland cement, the whole to be neatly levelled to a perfectly smooth surface, and rolled with a tooth roller to make a surface that will not be slippery. This surface layer shall be cut into sections, the joints to correspond exactly with those of the first described layer of concrete, the edges of the walk to be rounded, and the whole finished in a neat and workmanlike manner.

Temporary curbs to be supplied by contractor.

10. Before any concrete is placed in the walk, temporary curbs, of 2 x 6 pine, with edges dressed so as to be perfectly straight, shall be firmly and accurately placed along the outer edges of the walk, to be removed after the walk has hardened ; these curbs to be furnished by, and remain the property of the contractor. When the curb is removed, the vacant space must be filled with good soil, and any sodding disturbed in so doing must be carefully restored.

Total thickness and slope of walk.

11. The total thickness of the walk, including foundation layer, concrete layer, and the wearing surface, shall be uniform throughout, and shall have a slope towards the roadway of $\frac{1}{4}$ -inch to the foot, unless otherwise required by the engineer.

Cement.

12. All cement used in the work must be of some well and favorably known brand, and shall be approved by the engineer. It shall be delivered in barrels or equally tight receptacles, and must be protected from the weather by storing in a tight building or by suitable covering, the packages to be placed on boards or flooring raised above the ground. All cement rejected by the engineer shall be conspicuously marked "Condemned," and shall be immediately removed from the site of the work. Should any cement so rejected be thereafter used in the walk, such sections as may be required by the engineer, shall be immediately torn up by the contractor, and replaced with cement of proper quality, without extra compensation. The supply of cement must be so gauged that a sufficient quantity will be kept on hand to allow ample time for testing and examination by the engineer, without delay to the work of construction ; the cement to conform to the following tests, and such others as the engineer may require :—

(a) At least 90 per cent. shall pass through a sieve having 10,000 holes to the square inch.

(b) Pats made of neat cement, with thin edges, on pieces of glass, covered with a damp cloth, and allowed to set in air ; then placed in boiling water 48 hours, must not show expansion cracks, distortion, nor curling of the thin edges.

(c) Samples of cement shall be made into the consistency of a stiff mortar, pressed firmly into moulds, and covered with a damp cloth, then allowed to set in air 24 hours, then in water three, and seven days. When in water three days, six samples shall show an average tensile strength of 300 pounds per square inch ; when in water seven days, six samples shall show an average tensile strength of 450 pounds per square inch.

13. The stone and sand shall at all times be subject to the approval of the engineer, the sand to be clean, sharp and silicious.

Sand and stone.

14. At street crossings, lanes and private driveways, the walk shall be so rounded, placed, and at such an elevation as to give a convenient passage for vehicles, to the satisfaction of the engineer. The surface layer of concrete shall be 1½ inches thick, composed of equal parts by measure of cement and sand, and marked into diamond shaped blocks by lines crossing the walk diagonally six inches apart. The edges shall be rounded, and faced to the bottom of the concrete with the cement mortar used for surfacing the walk. At all street crossings, and elsewhere if so desired by the engineer, the edges of the walk shall be protected by a curbing of 4 x 6 cedar, placed in a permanent and durable manner, flush with the surface of the walk.

Street crossings, lanes and private driveways.

15. The contractor, in doing the work, shall excavate or fill in around trees in a careful manner so as not to injure the said trees; and all gratings, areas, tree-spaces, or other interruptions to the walk shall be regarded as continuous in the payment of the walk. The repairing or building up of area walls or other supports for gratings shall be performed and the material supplied by the contractor, as the engineer shall direct.

Trees, tree-spaces, gratings, areas, etc.

16. No concrete shall be laid in wet or freezing weather.

Laying concrete in wet or freezing weather.

17. Care must at all times be taken to prevent injury to waterworks stopcock-boxes, down pipes, door sills, steps, areas, gratings, or other appliances which may be under, project into, or pass through the walk, and the pavement shall be carefully and neatly filled around such appliances. When required by the engineer, all gratings or covers furnished the contractor shall be properly fitted into and conform to the surface of the walk.

Prevention of injury to waterworks and other appliances.

18. The contractor shall be bound to maintain the walks and crossings in perfect repair, free from all cracks and defects, for the term of five years from the date of completion thereof, and should the contractor fail to do so at any time during the said term, the engineer may cause the necessary repairs to be made, retaining the cost from moneys due, or becoming due to the said contractor on this or any other contract between the town and the contractor, or may recover the same from the contractor, or his sureties in this contract, as money paid at their request. The certificate of the engineer is to be final as to the necessity of repairs and amount expended upon them.

Contractor to maintain walk in perfect repair for five years from date of completion.

19. Care shall be taken at all times not to interfere with business or travel more than is absolutely necessary for the faithful performance of the work. The contractor shall make suitable and adequate provision for the safe and free passage of persons by or over the work, as may in the opinion of the engineer be necessary, and must confine himself to that half of the street on which the sidewalk is being put down, leaving the other half for the regular traffic.

Interference with traffic.

20. At all times during the progress of the work care must be taken not to unnecessarily injure or destroy private lawns, nor boulevards adjacent to the walk. On the completion of the work all surplus or refuse material must be immediately removed from the street by the contractor. If not removed within forty-eight hours after notice in writing so to do from the engineer, it shall be removed by the engineer at the contractor's expense.

Care of private lawns, boulevards, and removal of surplus material.

21. The contractor shall, during the progress of the work, use all proper precautions by good and sufficient barriers, red lights, or watchmen, for the prevention of accident, and he will indemnify and save the corporation of the town of from all suits and actions, and all costs and damages occasioned by the negligence or carelessness of the contractor or his agents, or employees.

Liability in case of accident.

22. The contractor or his duly authorized agent or foreman shall at all times while work is in progress, be on the ground, and instruction given by the engineer to such agent or foreman shall be of the same effect as if given to the contractor.

Contractor or his agent to be on the work while in progress.

23. The word engineer, where and whenever used herein, refers to the engineer of the town of or his authorized assistants, or other person appointed by the council of the town of to have charge and oversight of the work.

Engineer defined.

24. The decision of the engineer shall be final in case of ambiguity of expression in the specifications, or doubt as to the correct interpretation thereof.

Interpretation of specification.

Disorderly or incompetent employees.

25. Any disorderly or incompetent person or persons who may be employed on the work shall be removed when required by the engineer, and no person so removed shall thereafter be employed upon any portion of the work.

Defective work.

26. Any defective work that may be discovered by the engineer before the final acceptance of the work or before final payment shall be made, shall be removed and replaced by work and materials which shall conform to the spirit of the specifications; failure or neglect on the part of the engineer to condemn or reject bad or inferior work or materials, shall not be construed to imply an acceptance of such work or materials.

Payment of contractor.

27. The contractor is entitled to receive 90 per cent. of the value of any portion of the work completed under these specifications, at the end of each fortnight, such completion being certified by the engineer, and by the chairman of the board of works. At the expiration of sixty days after the acceptance of the work, the whole of the moneys accruing to the contractor under these specifications, excepting such sum or sums of money as may be retained under any of the provisions herein contained, and such sums as may have been paid in the form of partial payments upon the fortnightly estimates of the engineer.

Necessary notices.

28. All necessary notices to waterworks, gas, electric light, telephone, or telegraph officials, owners or occupants of property, or other interested parties shall be given by the contractor.

Payment of workmen.

29. The contractor shall punctually pay the workmen who shall be employed on the work comprised in these specifications, in cash current, and not what is denominated as "store" pay. And final payment for the work shall not to be made until satisfactory vouchers are furnished the engineer by the contractor showing all wages to have been paid.

Unforeseen obstruction delay or hindrance.

30. All loss arising from unforeseen obstructions or difficulties encountered in the performance of the work under these specifications, or from delay or hindrance from any cause during the prosecution of the same, shall be sustained by the contractor.

Commencing the work.

31. The work to be done under these specifications shall be commenced on such day and at such place or places as the engineer may direct. Failure so to commence without good and valid reason therefore, will be authority for the engineer to declare the contract forfeited. Nor shall the contractor commence work on any street without the order of the engineer in writing so to do.

Forfeiture of contract.

32. The Board of Works reserves the right to declare the contract forfeited at any time it should appear to the engineer that the work, or any part thereof, is being unnecessarily delayed by the contractor, or that the contractor is wilfully violating any of the conditions of the contract, or is executing the same in bad faith.

Removal of rejected work or material.

33. In case the contractor should refuse to remove or replace any rejected work or material within forty-eight hours after written notice, such work or material shall be removed by order of the engineer at the contractor's expense.

Tender to be accompanied by a cheque for \$100.

34. Each tender must be accompanied by a certified cheque for the sum of \$100, as a guarantee of good faith on the part of the person tendering, all such cheques to be retained in the possession of the town treasurer until the contract and bond for the performance of the work are signed and filed with the engineer.

Bond for \$1,000 required.

35. Before the contract shall be signed, or the work commenced, the contractor shall furnish a bond for the sum of \$1,000 for the satisfactory completion of the work, signed by two responsible sureties, and approved by the chairman of the board of works.

Right to reject tenders.

36. The right to reject any or all tenders is reserved by the town of and the lowest or other tender is not necessarily accepted.

Form of tender.

37. Tenders for the work under these specifications must be made on the form for this purpose, which may be had on application to the engineer.

Receiving tenders.

38. Sealed tenders, endorsed "Tender for Artificial Stone Sidewalk," will be received by the engineer up to noon, the day of next.



THE NEW
PUBLIC LIBRARY
ASTOR, LENOX AND
TILDEN FOUNDATIONS

A STREET IN BARRIE.

A broken stone (macadam) pavement well laid and properly cared for.



IN MERRITT.
Street monumental in 1909.

THE NEW YORK
PUBLIC LIBRARY



IN MERRITON.
Street macadamized in 1898.



DRAIN THE ROADS !



DRAIN THE ROADS !



DRAIN THE ROADS !

A knowledge of how to drain the roads most perfectly is the basis of successful roadmaking.



ARCH CULVERT OF CONCRETE.



ARCH CULVERT OF CONCRETE.



THE LAKE SHORE ROAD.

One of the oldest roads in Ontario, once an Indian trail, but now improved (?) by statute labor.



IMPROVED ROADS LEAD TO IMPROVED²FARMS.



WELL NAMED.
"Canal Street," Ottawa.



ON THE OLD TALBOT ROAD.

FOURTH ANNUAL REPORT

OF THE



PROVINCIAL INSTRUCTOR IN ROAD-MAKING ONTARIO

1899.

(PUBLISHED BY THE ONTARIO DEPARTMENT OF AGRICULTURE.)

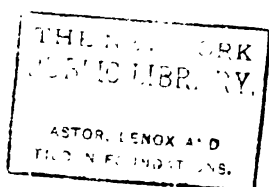
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1900.







**A Street in Renfrew
Macadamized in 1899.**



**Grove Street, Perth,
Macadamized in 1899.**



FOURTH ANNUAL REPORT

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ONTARIO

1899.

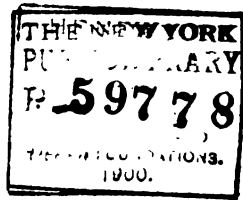
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TORONTO:

WARWICK BRO'S & RUTTER, PRINTERS, &c., 68 & 70 FRONT STREET WEST.
1900.



To the Honorable JOHN DRYDEN,
Minister of Agriculture.

SIR,—I have herewith the honor to submit to you my fourth annual report for the year 1899, on Road and Street Improvement in Ontario.

I have the honor to be,

Sir,

Your obedient servant,

A. W. CAMPBELL,
Provincial Instructor in Road-making.

PARLIAMENT BUILDINGS, TORONTO,
ONTARIO, 7th February, 1900.

CONTENTS.

	PAGE
LETTER OF TRANSMISSION	4
 ROAD REFORM :	
Amount Involved—Underlying Defects—Details—Result of the Agitation.....	7
 THE RELATION OF TOWNS AND CITIES TO COUNTRY ROADS :	
Equal Economy Demanded—Cost of Streets and Roads compared—A Matter of Self Interest for Cities to assist in Road-making—Comparison of Benefits.....	12
 A COUNTY SYSTEM :	
The Main Roads—A Larger Area for the Distribution of Revenue—County Systems now Operated in Ontario—Advantages of a County System.....	15
ROAD EXPENDITURE AND STATUTE LABOR,—Township Reports	20
ENDORISING THE ABOLITION AND COMMUTATION OF STATUTE LABOR.....	26
THE TOWNSHIP SYSTEM	30
ROAD-MAKING OUTLINED.....	31
IMPROVING SAND ROADS	33
ROAD MACHINERY	34
TRANSPORTATION OF ROAD MATERIAL BY RAILWAYS.....	35
PERMANENT ROAD WORK	36
 HIGHWAY CULVERTS :	
Township Reports.....	38
 CULVERT SPECIFICATIONS :	
A Stone Arch Culvert—A Cement Concrete Arch Culvert.....	41
CAUSES OF DEFECTIVE SIDEWALKS.....	43
A BY-LAW COMMUTING STATUTE LABOR IN TORONTO GORE.....	46
A BY-LAW TO COMMUTE STATUTE LABOR IN NIAGARA TOWNSHIP	47

FOURTH ANNUAL REPORT

OF THE

PROVINCIAL INSTRUCTOR IN ROAD-MAKING

ROAD REFORM.

The value of good country roads as a part of Ontario's transportation system is far from being fully understood and appreciated. The Province has not spared its energies in the building of railways and canals, largely for the benefit of "our undeveloped resources."



There are resources existing in Ontario, not usually regarded as "undeveloped" because they exist in the older and more thickly populated districts. Nevertheless agriculture, the most important industry of the Province has by no means assumed its proper proportions, largely for the reason that the country roads, the veins and arteries reaching the farmer's lands, have been mismanaged and neglected.

A remarkable example of this has recently been given by a number of Hamilton and Toronto merchants and manufacturers who have established a freight wagon service between the two cities, a distance of forty miles, in competition with railway rates, resulting in a saving of seven and eight cents per hundred on the freight carried. That this can be accomplished on the highway traversed, which is no better than the average bad road, would seem to be proof conclusive that the common roads offer advantages which have been overlooked.

AFTER THREE YEARS TRAFFIC.—IN NORTH MONAGHAN.

In France we find teamsters competing with railways in drawing goods two and three hundred miles over country roads. In Belgium there are instances, as between Liege and Brussels, or Antwerp and Brussels, in which teamsters commonly haul their loads sixty and seventy miles in competition with the railways. Similar examples could also be found in Germany and England.

That traffic over the common roads can compete with railway rates is regarded by many as an anomaly, a condemnation of freight rates, and a return to the primitive conditions of our grandfathers. Nevertheless for limited hauls, as great as three hundred miles in France, the common roads provide a means of competing with the railways which we cannot afford to disregard. That such competition is successful on the very inferior roads between Hamilton and Toronto is an index to what the farmer should be able to procure for himself. With good roads, farmers would equip themselves with better stock, journeys could be made more quickly, and double the present loads could be carried. There are no better means of regulating freight rates than to render ourselves largely independent of railways, by improving the common roads, a matter which European experience tells us, is not beyond our power to accomplish.

The agitation in favor of good roads, carried on by the Government during the past few years, has not only advanced public opinion in favor of better roads, but has revealed to the general public that the system by which our roads are at present being made and maintained has been by no means conducive to the best results. The effect of the movement has been and no doubt will be to increase the expenditure upon roads. The popular idea, which at first prevailed, that this was the chief object has, however, been largely dispelled and it has become generally recognized that the aim of the reform already in part accomplished in Ontario has been to procure a proper administration of the money now being spent upon the highways. That there is urgent need of this there is everywhere abundant evidence, and is universally accepted. This applies not merely to country roads, but to town and city streets, in all of which a great annual waste is going on without producing the desired results, thereby entailing both a direct and reactive loss.

The Large Amount Involved.

The actual money spent upon roads, streets and bridges is of very much larger dimensions than is commonly supposed. During the ten years, 1887 to 1896, it amounted in cities to \$14,483,643, in towns to \$6,759,626, in townships and counties to \$9,380,689, a total of \$30,623,958. If to this we add about 11,000,000 days of statute labor expended in the townships, we have a money and labor value of \$41,623,958, an average of over \$4,000,000 per year. This is more than one-third of the taxes levied on the people in municipal and school rates.

It forms the greatest outlay of the municipalities on any one public service, and is one which will constantly increase. It is estimated that the canal system

of Canada, chiefly in Ontario, has cost since its inception about \$80,000,000. The cost of the railway system within the Province has been about \$150,000,000. In ten years, as has been shown, the expenditure on roads in money and labor has amounted to \$41,623,958. While an exact estimate cannot be made, it is apparent that the common roads of the country have in all probably cost in the aggregate, fully as much as both railway and canal systems put together.

This large sum has been spent on the roads in comparatively small amounts throughout the Province, the ratepayers see only these small sums, and in consequence do not realize the enormous total. The expenditure is made too, with little or no attempt at careful supervision, with the result that a large proportion is actually wasted and much of the remainder turned to poor account. Much of the expenditure is undoubtedly devoted to a very good purpose, but a careful examination of the methods pursued in townships, villages, towns, and as a general thing in cities, will disclose that owing to faulty administration, careless and inexperienced supervision, and the improper influences which are brought to bear upon those in charge, our system of road-making is incompetent, utterly unjust and extravagant, and is almost solely responsible for their poor condition.

That this vast expenditure should be made upon such an important public work, in so slipshod a manner, without rule or design, seems inexplicable. That, as a general rule, work is done without plan or specification, without system, organization or proper supervision, seems startling in view of the fact that wherever any other public work is done systematically it is watched with the closest interest by the ratepayers, and the expenditure criticized to the most trifling item.

When the people of a municipality are shown what the roads are costing them, it is a revelation, and for the first time are they now commencing to realize the importance of the work and the magnitude of the expenditure. They see that better roads, and even good roads, do not necessarily mean an increased outlay, but a better administration of the present expenditure. Any department of government which can even successfully criticize such an outlay would be performing a valuable service. And where the services of such a department can be extended so as to reduce the work to a system, and insure such consultation and assistance as will bring about the most profitable expenditure of this money, it will confer a lasting benefit upon the country.

Such a heavy expenditure made upon a public service so important, certainly warrants the attention of the Government, which can be directed in various ways. In addition to educating and instructing the people in the proper methods of road-making; educating the people by a criticism of the work done, and methods of expenditure; construction of sample sections as object lessons to elevate their ideals of what a road should be; and circulation of proper literature, there should be legislation so revised as to be in keeping with advancing conditions, and constantly changing requirements; municipalities might be furnished with the

means required at the lowest rate of interest on easy terms of re-payment; the co-operation of railway companies could be secured in the cheaper carriage of road-making material for districts without a local supply, the railways and people thereby obtaining mutual benefit.

Ratepayers generally, are possessed of the idea that their roads are now built and maintained almost solely by statute labor, and that if this work is not performed in the most faithful manner, no serious crime is committed. They fail to realize, however, that owing to the very imperfect manner of doing the work, municipal councils are obliged to impose a direct tax. The amount of this tax, the indifference displayed in expending it, and the inferior results produced, comprise the serious phase of the whole question.

Underlying Defects.

For well populated districts, where more advanced principles of government should be applied, the present system is fundamentally weak from two main causes which bring in their train many abuses, and cause failure in the effort to improve the roads in a manner corresponding to increasing requirements. These causes are: First—the control is so scattered that no one can be held responsible for a proper expenditure of the labor and money devoted to the roads; and second—the payment of a tax in labor is too vague and clumsy to meet modern conditions.

From these two sources spring many important results: neglected roads, injustice to the men who do honest work, misappropriation of labor and money, and similar abuses. Some pay the tax with honest work, while others, from various motives, do not.

Under present conditions, all responsibility is, in the first place, distributed among members of the council; the council again distributes it among fifty or one hundred pathmasters. The large amount annually spent by most townships in the way of money and statute labor is also so scattered, together with responsibility that the people do not realize how large a sum is being spent; but when mis-spent and innocently squandered as it generally is, there is no one whom they can call to account. The councillors can shift responsibility from one to another, and then to the pathmasters. The latter have nothing to lose and so can merely shrug their shoulders when any improper or careless expenditure comes to light. Divided responsibility is never successful even in township government.

That the money expenditure is so largely scattered in patchwork, in temporary and perishable work, is due largely to the fact that all roadwork is viewed from the standpoint of statute labor. The money appropriations, while large in the aggregate, are made in small amounts with the object of supplementing statute labor. If this were reversed, a system of money expenditure used as a basis, and the money concentrated in durable and permanent work.

from year to year, statute labor being the supplementary force, it would greatly tend to correct present weaknesses. It is found, however, to be almost impossible to combine the two systems in this latter way. Immediately the greater economy of a concentrated money expenditure is realized, with its greater flexibility, and adaptation for securing better work, the awkwardness and wastefulness of statute labor becomes apparent, and it is found to be, in many ways, a hindrance to its more efficient rival.

The Details.

In pursuing a study of the details we find that statute labor is slighted in many well known ways by the taxpayers individually and collectively. Some do their duty and others try to do it. Some will not do their duty, and try to keep others from doing it. Some would do their duty, but seeing the carelessness of others, feel the injustice of the system, and so are as careless as the rest. Others neglect the work in order to have greater claim for money appropriations.

Even if there were honest endeavour to do the required amount of labor, the system would still be unproductive of the best results for many reasons. It is not a system which provides for the needs of the roads, so much as for the wishes of the men who do the work. This applies to the kind of work done, whether grading, ditching or gravelling; to the time it is done, and to the locality where it is done.

No matter how much grading may be needed, no matter how much ditching is required, if it is more convenient for the farmer to draw gravel he does so, and the other work is neglected. It is the same under other conditions, and grading and ditching are very often done to the neglect of gravelling if the former involve less bother. And so it goes on, the man who is doing the work, not the work itself, being the first consideration.

In the care of roads, there is a proper time for doing the work, and no other time will do so well. No other time will be so satisfactory. No other time will keep the roads in good condition at the minimum of expense. But this, under the statute labor system, is an impossibility. This labor is available only at one time of the year, that, a very unfavorable time for road-making, and it is not then devoted to a proper scheme of repairs. Repairs are made only when the road becomes entirely worn out. Once placed in a good condition, they are neglected and allowed to become almost impassable before repairs are made.

The work under statute labor is not done at the points where it is most needed, but it is an unwritten law that the pathmaster for the year will look after his own interests, even to the extent of digging a ditch, which will benefit only his own farm, without benefit to the road, possibly to its detriment.

The Result of the Agitation.

All these are things which the average ratepayer will now admit. Three years ago the statute labor question was one of exceeding delicacy to handle, but

by a continuous campaign, pointing out the small amount of good produced by statute labor, its unfairness and injustice, the large amount of money spent in addition and the poor results from the united expenditure; and by pointing out what could be produced if proper methods were resorted to, the devotion of the people to statute labor has been largely removed. A number of townships have abolished statute labor, and others are preparing to do so. The feeling is growing that statute labor should be abolished by the Provincial Legislature. In the meantime, as a result of the agitation, townships are putting forth greater effort to obtain better results. They are organizing their forces more efficiently, purchasing modern road machinery, and are learning and applying in a better manner the principles of roadmaking.

This however, in the main, is but educative. The efforts they are putting forth are but teaching them the need of a more efficient system. The Township of North Monaghan is probably the most advanced in this respect. The municipality has been divided into four divisions and a road commissioner placed in charge of each. All taxes are paid in money, and work is done by contract or day labor as its importance dictates. The details of the system are being gradually perfected, and it is giving entire satisfaction to the ratepayers and the roads are rapidly improving.

The demand for information has been general, and meetings for the purpose of discussing the question have been held in every part of the Province. The question is now practically in the hands of councils or other executive bodies, the minds of the people being pretty well prepared for action. Consultation with municipal councils will always be necessary, and much detailed information required on methods of taxation, principles of construction, supervision, material, the use of modern machinery, etc. This would be accomplished largely by correspondence and the issuing of bulletins.

This is but one phase of the question of road reform which the people are gradually working out, and problems of a different nature are presenting themselves.

THE RELATION OF TOWNS AND CITIES TO COUNTRY ROADS.

A map of the main roads through Ontario as they existed half a century ago presents a striking similarity to a map of the railroads of to-day. The construction of railways was, in effect, but the improvement of the former trunk lines of common highways, providing them with firm and well graded roadbeds, smooth, hard tracks of steel, and discarding the less efficient horse power for mechanical traction. We are most readily impressed with the larger and newer enterprises, so it is that we are inclined to regard the railways as the chief factor in transportation. It is not the massive root branches of the tree which abstract nourishment from the soil, but the diminutive, thread-like off shoots from them. It is not the steam railway which is the active agent of a nation's growth; but

like the root branches of the tree, the railway is secondary, subservient to the lesser avenues, to the net-work of which is deputed the task of first gathering the means of subsistence. In proportion to the excellence of the network of common roads will the country be occupied and productive. As the roads are good, the country occupied and therefore productive, so will the railways have employment. The activity of the railways is a certain index to the magnitude of the flow of commerce. And commerce is the life stream of national prosperity.

Equal Economy Demanded by Country Roads and City Streets.

The city is the product of the country; the country is not the product of the city. The first roads on the continent were country roads, not city streets. As country roads were first in origin, so they are first in importance. There is too great a tendency among townspeople to overlook the wider application of the question of roads in general. With city paving, there is certainly a greater demand for engineering skill, and the engineering difficulties appear to be farther from solution than is the case with country roads. At the same time, the simplicity of country road construction is not always so real as it appears, and the difficulties are greatly increased by the deficiency of funds with which to overcome them. The same obstacle, it is true, is met with in city paving, and the question becomes in each case, one of obtaining the best results with a minimum or within a limited cost. In view of the strict economy demanded, the construction of country roads, in selection of materials, location, drainage, grading, bridge and culvert building, and the various details, becomes a matter in which the greatest skill is not wasted.

Cost of Streets and Roads Compared.

The people in the cities are very apt to urge that, because their pavement cost so much per foot frontage more, because the farmer receives the reciprocal use of the city street in return for the city man's use of the country road, they have, therefore discharged their obligation with regard to roads. Contrasting an eighty-acre farm, however, with a fifty-foot town lot, and a farm road at \$1,000 with a city road at \$5,280, we find the cost to the individual farmer is \$125, and to the city property owner, \$25.

Many of the streets, it may be said, cost much more than the amount named. The same is true of the country roads, almost the minimum has been named in each case, and if we double the cost in the one instance we must do so in the other, so that the proportions remain about constant. Then too, a fifty-foot city lot is double the frontage occupied by the majority of city houses. An eighty-acre farm is not uncommon in the country, and in levying it with one-eighth of a mile, there has not been included its proportion of flankage existing in every block.

The work of country road building is one of considerable expense, as compared with the number and wealth of those upon whom it now commonly rests. Wherever it is left solely to the farmer it will be years before the condition of the roads will be adequate to the complete development of the resources of any country.

A Matter of Self-Interest for the Cities to Assist in Roadmaking.

The sum of the matter is that, whether or not the towns and cities discharge their strict duty in the construction of streets within their limits, their prosperity is dependent upon the prosperity of the country districts, and it is but a matter of self-interest, of profitable investment, to assist the farmers in road building.

There is, first of all, the broader aspect of the question which regards the welfare of the nation as a whole. In this aspect of the question, we are led to regard the relations which exist between the great metropolitan centres which have as territory upon which they depend for support, the nation as a whole; which, whether for agricultural, mineral or forest wealth, are ultimately dependent upon the rural highways for the materials of manufacture and consumption.

There are again the towns and cities of lesser magnitude which draw their support largely from the agriculture of the immediate vicinity. From these latter there comes a more urgent demand for good roads, a demand which all urban communities have made of late years, for it is these lesser cities which would be more directly benefited by the improvement of roads in the immediate district. The benefits being more direct, the value of good roads becomes more apparent.

The larger cities, less directly benefited, but benefited to no less, indeed to a much greater degree, appreciate less perfectly their value because, being less direct, the benefit is less apparent. They have usually many manufacturing industries, and therefore seem more self-supporting than do the towns more directly dependent upon the agriculture of the district. The more apparent independence is, however, deceptive. If the town is a manufacturing centre, it must have country roads over which to draw the material for manufacture; to a much greater degree, however, must it have a wealthy territory surrounding it to purchase its articles of manufacture. It is to the agricultural country that good roads are most beneficial, and no agricultural country can become wealthy and to the highest degree prosperous, without good roads. From greatest to smallest, towns and cities are dependent upon good rural roads.

Comparison of Benefits.

Many towns, it has been said, have learned that it is most important to have free and uninterrupted communication with the surrounding farm districts at all seasons of the year. If the farmer must come over the roads to the centres of population and the railway station, to dispose of his farm produce, it is equally necessary to the townsman that he should use the roads to draw the merchant's

goods back to the farm. It merely happens as a matter of convenience for obvious reasons that the farmer draws his produce to the town and his purchases back to the farm, instead of the merchant hauling his merchandise to the farmer and the produce of the farm back to the town. The country roads are nearly, if not quite, as much benefit to the townsman as to the farmer.

Without the means of access, a country is valueless for production purposes. A farm of highest fertility within fifty miles of Toronto, if there were not roads by which it might be reached, would be as valueless as if situated in the heart of Africa. Distance is not measured by miles, but by rapidity and ease of travel and transportation. It naturally follows that, with the opening of the first wagon track leading to it, the value of a farm commences, and as the road improves the value of the farm advances, other conditions remaining constant. It is true that the more the country districts become filled with population the more rapidly the improvement of roads will advance, but it is equally true that the more rapidly the roads are improved the more rapidly will population advance. The construction of the Canadian Pacific Railway has been the means of enticing a population to the Canadian North-west. The improvement of roads will have the same influence upon the less distant territory, and there is every need for greater attention to the home field for missionary effort in this respect. As population increases, the wealth of our cities will increase, and it therefore points forcibly to the conclusion that one of the potent means of improving and lengthening town and city streets is to provide at the distant end of the chain of transportation, good country roads.

A COUNTY SYSTEM.

Throughout Ontario, in the great majority of municipalities, the country roads, in addition to being constructed entirely by the farmers, are kept up by the statute labor system, supplemented by money grants from the county and township councils, chiefly the latter. This applies to all the roads however important they are. The result is that there are inadequate and imperfect methods employed. No experienced skill, an absolute necessity for good road making, is employed, so that the roads are, for the greater part of the year, in an exceedingly deplorable condition, materially interfering with all commercial and social functions within the Province.

There are distinct classes of roads throughout the country districts. One of these classes is composed of the main roads, the through roads, the leading roads. They serve large areas, an entire township, or several townships, and usually lead to the chief market towns in the district or county. They are the trunk roads from which branch another class of less importance which serve only the two adjacent concessions, or perhaps only a small neighborhood, perhaps only a few farms.

The Main Roads.

The construction and maintenance of these main roads is much more expensive and exacting than the roads of lesser importance. They already absorb most of the expenditure and energy of the townships. Upon them the greater part of the township money appropriations is spent, and the greater part of the statute labor, to the neglect of the little-travelled roads. Heavily-travelled, they quickly get out of repair, and, under the system which townships can adopt, they cannot be constructed so durably nor maintained so thoroughly and economically as under a county system, in which latter the most workmanlike system, together with the best and most suitable machinery, can be employed.

Passing, as these main roads do, through an entire township, and continuously through several townships, it is necessary that they should be maintained in a uniform manner throughout their entire length, if they are to serve their best purpose. At the present time, so far are they from being maintained uniformly, even through one township, that they vary from statute labor beat to statute labor beat; one beat being in good condition and well kept, the next, perhaps, almost impassable, according to the relative skill of the various path-masters, the opportunity for drainage, the convenience and quality of the road material. These main trunk roads should not be subject to such caprice, but should be maintained after a uniform plan, under all circumstances, from township to township, and through the greatest area possible.

A Larger Area for the Distribution of Revenue.

The necessity for providing that towns and cities contribute to the maintenance of roads has been explained. We find, however, that in some townships there are none of these urban municipalities, while in others there are several, and perhaps very large cities. The duty and interests of these towns do not remain, except in rare instances, with so small an area as a township, and indeed, in some cases, the entire county is too small to spread their contribution over. The tax thus raised from urban municipalities cannot be equitably distributed on a less scale than a county system.

A county road system does not necessarily mean that a greater expenditure will be made upon the roads of the county, but means rather that the townships will spend less on the roads and the county council will spend more.

At the present time, under a county system, all towns and villages may be taxed for roads at the county rates. Cities within the county, however, are not so taxed, and this it might be advisable to rectify.

County Systems Now Operated In Ontario.

The only counties in Ontario which maintain county roads are Wentworth, Lincoln, Wellington, Frontenac and Hastings. Wentworth County has for some time been considering the advisability of extending its system, and during the past year a commission was at work in the County of Oxford. Victoria,

Grey and York Counties have also canvassed the question to some extent. For the best information on the subject in Ontario, however, the County of Hastings will most repay consideration. The County of Peterborough appointed a committee to look into the Hastings system two years ago, and this committee fully endorsed the system as they found it.

The Hastings system commenced about twenty years ago with the purchase of the toll roads throughout the county. These have been added to until at present about 370 miles are maintained as county roads. These roads are among the finest in the Province of Ontario, and in many parts are undoubtedly the best. The County Council appropriates annually about \$18,000 for the county system, including the expenditure on county bridges which are expensively built of iron, concrete and stone. The management of these roads is deputed to a committee of the council, by whom a superintendent of roads is employed. The roads are divided into four sections, and a gang of men under a competent foreman is placed over each. Whenever a section of roadway requires re-construction or repairs it is inspected by the superintendent, from whom the foreman with his gang receive proper instructions and are sent to the place to carry out the work.

The plan was at first vigorously opposed, but after a year's trial it proved so satisfactory and economical that it was continued, with the result that the added experience enables them to do the work at about one-half the first cost, and the roads are becoming better every year, owing to the careful supervision exercised and the ability to make repairs as soon as they are needed on the road.

The expenditure on the county roads proper in Hastings county, exclusive of bridges, is estimated at \$35 per mile only. A careful study of the present expenditure on township roads including statute labor will show a comparison very favorable to county roads in view of the better results obtained. At present the main roads absorb the greater part of the energies of the townships so that the side roads are apt to be neglected, whereas by relieving the townships of the main roads they are enabled to very much improve the roads of less importance under their particular jurisdiction.

Advantages of a County System.

Under a county system, a portion of the cost of road building is levied, in the county rate, against the towns and villages within the municipality for road purposes. In order to extend this tax to the cities, it would be necessary to amend the statutes. At the present time under township systems, the farmers bear the entire cost. All the expenditure placed on roads would be spent in the county, and thus returned in a great measure, to those who contributed it in the first place.

Under county control a properly organized corps of men can be employed to build and repair roads. As at other employments they become experienced and

do better work, and in the matter of repairs are ready to make them as soon as signs of wear appear.

By a county plan, uniformity of work and system will be secured throughout the various municipalities. Whereas under township control a diversity of plans is sure to be adopted.

In a county plan an experienced and properly qualified man could be employed to have constant supervision of the work, whereas under township control, each municipality cannot afford to pay the salary of such a man. Under every good system it is necessary to have responsibility centralized and defined, not divided and easily shifted from one to another, as it now is under the statute labor system.

Under county control modern machinery too expensive for individual townships can be purchased and handled to advantage, an experienced operator can be employed for each implement, and a better and more uniform class of work will be secured.

A township can manage its roads properly only by adopting a plan similar to that outlined under a county system, but by extending it over a county it becomes more cheaply operated.

There is not community of interest between the townships. In one township there is a certain leading road, much travelled and well made and maintained. The adjoining municipality may, for various reasons, not consider it of so much importance as to warrant them in making an expenditure to benefit largely their neighbors who are obliged to travel over it. A properly connected system of leading roads throughout the county will be obtained under a county system; whereas with each township, and even each statute labor beat, working independently of those around it, this will be lost sight of.

During the winter months, when snow is plentiful, the necessity for use of the roads is as great as at any other season. But, unfortunately, often in one night snow drifts may render them almost impassable, especially for heavy teaming; or at least, owing to drifts and pitchholes, they are made very unserviceable. The opening of the roads would be part of the work of county maintenance, and the necessity would supply implements for doing the work promptly and cheaply.

If no greater expenditure is made upon the roads than at present, the rate will be reduced, because most of the township's expenditure is now placed on the leading roads; and the township will be relieved of these by a county system. Under the county system the funds will be sufficiently concentrated to undertake durable work, and consequently these roads will be properly constructed and afterwards maintained at a less cost than at present.

A county road system equalizes the cost of maintaining leading roads. In every county within a certain radius of a market town, traffic constantly increases as the town is approached. The cost of construction and maintenance increases in proportion to the traffic. It is unfair to charge those living near the town with

the cost of building the roads to accommodate traffic from a distance ; so unfair as to cause discouragement, and often withdraws support.

Property is very largely valued according to distance from the market and the convenience with which the market can be reached. Property a long distance from the market is affected to a greater extent by the bad condition of the roads than is property very near the market. Good roads are therefore of greater value to townships a long distance from the market town than those in the immediate vicinity.

Under a county system durable road construction would be undertaken. The economic value of this work would be seen and appreciated by the people of the different townships. The well-built roads would stand as object lessons, and would teach the better expenditure of funds spent by the townships on the roads maintained by them.

While there may be some feeling averse to townships parting with any control of their roads, it is nevertheless impossible, under a township system, to levy taxation equitably, or employ the most economical and at the same time serviceable system. The trend of opinion has turned towards collecting the most important roads of each county, placing them under the management of the county council. It has been shown that by such a means road-making can be placed on a more business-like basis, and consequently greater efficiency is secured. It provides for a more equitable system of levying the cost, for a better use of modern machinery and for a higher grade of oversight and workmanship. At the present time township councils are unable to maintain the roads by statute labor, and are in consequence compelled to make annual appropriations of money from the general tax. This money is, in the main, spent on the roads which would comprise a county system, but owing to the contracted character of the system, township councils cannot expect to apply this expenditure to the greatest advantage.

Despite the advantages of a county system of roads, township councillors are frequently unwilling to give up any of their present powers, and the influence and patronage which accompany them. Their sessions are mainly occupied with the distribution of money over the roads, the building of culverts and bridges, and the settlement of road accounts. This distribution, as a general thing, is made at the request of a community of favor-seeking electors, and not in accordance with the real requirements of the roads, as would be the case if reported upon by a responsible official under a county system.

ROAD EXPENDITURE AND STATUTE LABOR.

Statute labor should be worth one dollar a day. This is a moderate estimate for a day's work, and there are few men who would be willing to sell their time for so small an amount in the open market. And yet there are some communities



A GRAVEL ROAD IN N. MONAGHAN.—FINISHED IN 1896, AND NOT TOUCHED SINCE.

which protest against the commutation of statute labor, or its abolition, for a tax amounting to half this rate, or fifty cents a day. Indeed, where it has been proposed to commute for twenty-five cents a day, there have been those who oppose the change, preferring to work on the road for a day in order to retain so small an amount.

The farmer whose work on the farm is not worth one dollar a day is not likely to be worth that amount in statute labor, and opposition to commutation of statute labor at that

rate is, to a great extent, one of the best reasons for adopting a commutation system, indicating, as it does, that the roads are not receiving the amount of honest labor which it was intended should be placed upon them. Roads are of much too great importance to be slighted in any degree.

With the purpose of comparing the present annual expenditure upon roads in the Province with the amount needed to keep roads in good repair, the money and labor spent by townships and counties has been collected under county totals. With these it is interesting to compare the expenditure upon the Hastings county roads of 370 miles, kept under a management admirably suited to the needs of every county. These Hastings roads cost in 1896 an amount equal to \$49.82 per mile; or, deducting the expenditure on bridges which is included in this sum, the roads proper cost, it is estimated, about \$35 per mile.

An analysis shows that there are two counties, York and Wentworth, which show an expenditure exceeding \$60 per mile; the former, to be more exact, \$69.87, and the latter \$61.34. Three counties show an expenditure of between \$50 and \$60; Elgin \$51.25, Middlesex \$51.20, Oxford \$56.23. Seven counties show an expenditure of between \$40 and \$50; Essex \$45.32, Lambton \$40.52

Perth \$47.84, Wellington \$41.17, Waterloo \$41.57, Lincoln \$45.07, Hastings \$40. Fifteen counties spend between \$30 and \$40 ; eleven spend between \$20 and \$30. Only three, Frontenac, Renfrew and Haliburton, fall below \$20.

Out of thirty-eight municipalities thirty show an average expenditure per county of \$39.46, while of these the expenditure of five exceeds that of the county of Hastings on their county road system. The roads of Hastings, including township and county roads, it has been stated, cost \$40 per mile, an amount which certainly compares most favorably with other sections of the Province, in view of the condition of the roads.

These estimates, it may be pointed out, consist of all expenditure upon roads within the several counties ; that is, the township money appropriations, county money appropriations, and statute labor at the value provided by law, one dollar per day. These statistics are deserving of careful consideration, and point out that a straightforward, business-like policy with regard to roads is no more costly than the present unsystematic methods in common use ; that good roads are no more expensive than bad roads in actual monetary outlay if right principles are adopted for their management ; and, in addition, bad roads mean the many losses which bad roads necessarily entail, and the absence of the benefits conferred by good roads.

Township Reports.

Returns received from township clerks in reply to a circular dated October 30th, 1899, indicate that the reform of statute labor is making rapid headway and point to a great change of feeling from that exhibited by the returns included in the report of 1897. Extracts from the returns of 1899 respecting statute labor are as follow :

NELSON TOWNSHIP.—At the first council meeting in 1900 a by-law was passed commuting statute labor at fifty cents a day.

PELEE ISLAND.—Statute labor is abolished by by-law.

BLANSHARD.—Statute labor is abolished.

BURFORD.—A by-law is in force whereby a majority of any road division may petition to commute their labor at 50 cents per day. Eight divisions commuted this year.

SALT FLEET.—Statute labor is commuted at 35 cents per day. Three commissioners are appointed by the council there being three divisions in the township.

BINBROOK.—The statute labor system is not used in this township. We have one road commissioner appointed in each school section, whose duty it is to keep the bridges in proper repair. The system seems to be working very well, our roads are certainly improving every year. The plan is to get over all the roads with the grader in three years ; we have nearly 60 miles.

EAST ZORRA.—Some divisions commute statute labor.

EUPHEMIA.—Commutation tax was reduced from 75 cents to 25 cents per day. Ratepayers were given their choice either to labor or avail themselves of the use of the road machine, the council paying for same and charging the several road divisions accordingly. More than one half of the divisions availed themselves of the use of the grader. The other divisions did no work, all were charged at the rate of 25 cents per day.

FRONT OF YONGE AND ESCOIT.—Statute labor is commuted at 75 cents per day and collected with other taxes.

LOUTH.—Half the statute labor was commuted for this year to pay for the road machine work. In 1898 the whole of the statute labor was commuted and double the work performed that year over all years, but this year a few old fogies wanted to work it out, and when the machine came over their beats the most of them were in their harvest or other work and could not attend to the statute labor and about one third was thrown away.

SARAWAK.—The statute labor system is still used in two or three wards only. The statute labor is commuted at 60 cents per day in the largest half of the municipality, and to be paid with the taxes and the plan works well. More work is done with 60 cents than for one days labor in the old way. The other two divisions are agitating for commutation at 60 cents per day, and believe it will be in force before two years. It is four or five years since the change in first two wards and they would not go back to statute labor.

EAST OXFORD.—This year we have commuted our statute labor at 60 cents per day and I think we have got one third more gravel on the roads. I think all municipalities should commute their statute labor and the roads will be better made under one or two men to build them.

ANCASTER.—Statute labor is commuted at 50 cents per day and an account opened called the statute labor account and a column on collector's roll used for same and collected as other taxes. Commutation has been adopted for three years, but the system and appointment of overseers is not complete.

TORONTO GORE.—The statute labor system is not used in this township, a change was made last spring. Have been spending from \$400 to \$500 in gravelling leading roads, but are working at a disadvantage as there is no supply except creek gravel.

WHITCHURCH.—The statute labor system is still used except on Yonge St. and some village lots north of Stouffville, being commuted at above places at 75 cents per day. A member of council is commissioner for Yonge St.

NORTH GRIMSBY.—Statute labor is commuted at 50 cents per day. Since commuting the statute labor our roads are in very much better condition. We raise a trifle over \$1000 for road purposes which amount is expended by the road commissioners.

SARNIA.—Statute labor is commuted in one road division only, the amount of such commutation is expended in said division by the commissioner.

MALDEN.—Statute labor is abolished.

NORTH CROSBY.—Statute labor is commuted in one road division only.

NIAGARA.—Statute labor is abolished.

NORTH MONAGHAN.—Statute labor is commuted at rate of 75 cents per day.

TILBURY NORTH.—In two road divisions the statute labor is commuted at 50 cents per day. These divisions bought a road making machine.

REACH.—The statute labor system is still used with the exception of four divisions in which the statute labor has been commuted at 50c. per day instead of 75c. In the four divisions three commissioners were appointed whose duty was to collect the commutation tax, and supervise the expenditure when most needed. On the whole the change is working satisfactorily with a few exceptions (there are always some grumblers as you doubtless know). I may say that there is a growing feeling in the township for a better system of roadmaking and think it time that the old worn out statute labor system was entirely abolished.

BARTON.—Statute labor is commuted at so much per day according to outlay rate struck at the time of other rates. We find the commutation of statute labor a benefit, works well in the township.

WINCHESTER.—The statute labor system is abolished and road work is done by commissioners of whom we have eight in the township.

AMELIASBURG.—Statute labor is commuted in one road division, and it is proposed to commute one-half the statute labor throughout the entire township in 1900.

STAMFORD.—The statute labor system of road improvement is still used in about one-third of the township. In 1896 a by-law was passed commuting the statute labor in two polling divisions, and we hope to see the whole township commuted in a short period.

CLINTON.—Statute labor is commuted at 40 cents per day.

BASTARD AND BURGESS.—A number of road divisions have within the past two years commuted statute labor at 50 cents per day; money is expended by overseer appointed by council—improved roads the result. The township owns a stone crusher; breaks free of charge all at we got out in any road division, whether statute labor is commuted or not. The tendency is to do away with statute labor, spend money in getting out stone and spreading it on road, townships paying expense of breaking.

BECKWITH.—Statute labor is commuted in some cases.

YORK.—The statute labor system is used only in a portion (about one-half, according to assent). In commuted portion of the township the money is all expended under the supervision of the township engineers. The feeling is growing stronger to entirely abolish statute labor, if the law would permit of rate being levied in each division (4) according to its requirements. The two northern divisions of township being rural and the southern portion urban, the requirements of the latter for sidewalks are greater than the north.

The holdings in the north are large as compared with the south, therefore they prefer commutation to abolition as the law stands at present. With the amendment in the law as suggested, I think abolition would follow and a rate in the dollar (the only equitable mode of levy) would be levied for "public works" in each division (not more than four), and spent within the division. The provision as to division of the township could be left subject to by-law of the municipality.

SCOTT.—Statute labor is a wretched failure. The 2,000 days of statute labor and half that number of dollars fail to keep our roads good.

WHITBY.—The great hindrance to good roads is the lack of system occasioned by appointment of yearly overseers and commissioners, resulting in patchwork. This year's (1900) council proposes to abolish statute labor.

SOUTH SHEERBROOKE.—I believe if statute labor was done away with altogether and money raised it would be a great benefit to the roads. I consider a great part of the statute labor work is miserably laid out.

ALBION.—We work on the same old plan, but it is not a good one.

TOWNSEND.—The statute labor system is still used in this township, but there is a strong feeling to do away with it.

OPS.—Some few are in favor of abolishing statute labor, but at present at least 90 per cent. would rather work than pay any reasonable equivalent. In some beats statute labor is done systematically and well. Fifty cents per day is accepted in lieu of work if paid before the month of June. After that a commutation of one dollar per day is exacted for any work appearing undone on pathmaster's returns—generally about \$60 on collector's roll annually. An effort was made last year to place the leading roads under the control of the County Council and borrow a heavy loan on thirty years' debentures for macadamizing, but the councils of the municipalities declined to assent to such change, for the present at least.

MADOC.—Statute labor system not satisfactory. I think ere long it will be abolished and a tax of so much per day of statute labor added to ratepayers' taxes, and all the roads in the township put under the supervision of township road surveyor.

RAMA.—The township officials are talking of doing away with statute labor.

GRANTHAM.—There has been considerable discussion about commuting the statute labor, but the change has not yet been tried.

SOUTHWOLD.—The statute labor system is still used, and a great humbug it is. Statute labor is not abolished, but it is about time the unprofitable system was abolished.

EGREMONT.—Statute labor system is still used in this township, but many regard it as a relic and are longing for a change. Our road improvements have not kept pace with the other improvements, and it is owing entirely to the system. The sooner it is abolished the better as we cannot have a worse.

McMURRICH.—The present system of statute labor gives very poor satisfaction, the general impression is that the sooner a change is made the better.

BRANT.—It is probable that statute labor will be commuted in the near future.

MOORE.—The statute labor system is still adhered to, although many advocate a change.

ERAMOSA.—There is a strong growing feeling that the system of statute labor has outlived its usefulness, and should be abolished and a competent road engineer appointed, instead of the council overseeing the roads.

CARDEN.—It is time the statute labor system was abolished.

SOUTH FREDERICKSBURG.—I am in favor of the abolition of the statute labor system and in lieu thereof impose a tax of 50 or 75c. per day, and think if it were left to a vote the tax system would carry.

AMHERST ISLAND.—I am of the opinion it would be very much more advantageous to do away with the old system of statute labor, substitute tiles for culverts and do away with wooden coverings.

WOOLWICH.—I and a few others are in favor of having statute labor commuted, but our people do not seem to be ready for it yet.

ORFORD.—I have advocated for years a payment of 50c. to commute instead of work, and eventually it will come to that.

MARA.—I believe that if statute labor was commuted at 50c. per day we would have more and better work done than under the present system.

ENFISKILL.—Would like statute labor abolished by Act of Parliament.

SOUTH EASTHOPE.—The same old way used with some awakening that this old way does not give best results.

BRANTFORD.—The statute labor system is still used but it should be abolished.

CLARKE.—There is considerable dissatisfaction with system of statute labor and a growing desire for some better way.

SULLIVAN.—The statute labor system is getting into disfavor with most of the progressive farmers in the municipality, and I think the time is near when there will be a change—and it cannot come too soon.

FRONT OF LEEDS AND LANSDOWNE.—The plan adopted in this municipality for improving the roads is to advertise for stone in September, to be delivered by the 1st of June of the following year, about 25 cords in each place in twin piles (near the highway and convenient for crusher) on all the leading roads. The crusher here is owned by a private party who crushes and puts the stone on the roads with special carts for the purpose after the road has been graded with statute labor, at a rate per cord. This plan is giving good satisfaction. But it is thought that statute labor should be abolished.

PORTLAND.—The statute labor done in this township is of very little use. It should be made compulsory to commute all statute labor in the township.

GLENELG.—While the statute labor system is still in use, it is giving very poor satisfaction. Our councils are trying their best to make the statute labor more successful but with no apparent result. This is the result of: 1st—the ignorance of the pathmasters; 2nd—the indifference of the pathmasters. A change would seem to be necessary but the ratepayers as a rule are not in favor of commutation.

MARMORA AND LAKE.—We have still the old blundering way of doing or pretending to do statute labor. Although there are a few conscientious pathmasters. A good many people pay 75c. a day for commutation. In the village of Marmora, unincorporated and included in the municipality of Marmora and Lake a little more than one-half pay a commutation of their own free will without being asked to do so. It is very beneficial to the village. Statute labor as performed in the municipality is with a few exceptions a farce, all sorts of dodges and schemes are put in use to get out of it. The grants made by the Council are not all put to good use; in some cases the money might as well be put down a rat hole. The system of work is uncertain and needs a thorough reforming, and compulsory at that.

GLOUCESTER.—The statute labor system is still used in this township but a great deal worse than it was fifty years ago. It is commuted to the extent of reducing it to 50c. per day and pay \$1.00 per day for men to do the work, and as a further reduction the statutory scale is reduced two-thirds, as you will observe by the instructions enclosed herewith. The statute labor system as practiced in this township is virtually a failure—as the reduction made in the statutory scale, and the payment per day in lieu of performing the work and the price per day for performing the work and the chronic determination of the men to do just as little work as they can—has resolved itself into a sort of play day for the neighborhood to gossip and smoke away the time.

SHEFFIELD.—The statute labor system is better worked than formerly, but it is generally acknowledged to be unsatisfactory yet the ratepayers are not ready to give it up.

MELANCTHON.—The statute labor system is a fraud. It would be much better to charge 50c. per day and expend the money under a commissioner. Years ago we had a great amount of non-resident lands and the statute labor system yielded a large sum. We have very little non-resident lands now.

SOUTH MARYSBURG.—We have very bad roads during the spring and fall, every one does just as little road work as he can. The old system is a failure, we want improved legislation in this direction.

SOUTH NORWICH.—I have received instructions from the Council to submit to the ratepayers in January the question of doing away with the statute labor system. It is commuted at 50c. per day in the Police Village of Otterville and a portion of the township and it works wonders. The village is nearly all cement walks.

HUMBERSTONE.—The Government should pass an Act abolishing statute labor system.

HILLIER.—To me the statute labor system is very unjust.

OSNABRUCK.—Think we will be able to abolish the statute labor system inside of two years.

SYDENHAM.—There has been some talk about doing away with statute labor, but no action has been taken in the matter as yet. The Council of this year (1900) proposes to raise \$40,000 for road improvement.

FARADAY.—The system is very faulty as carried out here. Too many pathmasters, some only have their own work to oversee and others only have one or two to oversee.

NASSAGAWEYA.—The statute labor system is still used and many are sick of it, believing that a commutation tax of 50c. per day would give much better results.

HUNGERFORD.—In many sections of the township the road work is not half done.

MARCH.—I am of the opinion that if the Legislature would abolish the old fashioned system of statute labor, division of the township could be left subject to by-law of the municipality.

WOLFE ISLAND.—There is very little interest taken in roadmaking, and I do not look for much improvement until our people are educated to the benefit of better roads, and until statute labor is commuted instead of performed.

USBORNE.—There is some agitation to have the statute labor system abolished.

PELHAM.—Statute labor is commuted only in two small unincorporated villages, and giving good satisfaction there.

CRAMAHE.—Statute labor is not commuted in our township, but I think it would give better satisfaction if it was; a portion of the work each year is not done, and the council is asked each year to do so much repairing I think it will have to be done in time. The only objection I see in commuting statute labor is the laboring class who would far rather work for 50c. per day than pay it in their taxes.

SAULT STE. MARIE.—Statute labor system is still used in the township, I am sorry to say, as I consider that the better way would be to commute it.

MONCK.—In some instances statute labor is commuted but not abolished, nor otherwise modified. It should be abolished by statute, as it is not only detrimental to progress but a source of municipal corruption. I have had 30 years of experience to convince me.

DAWN.—We are now in Dawn township just upon the limit or divide between the old system and a new one ; the unsettled or new parts, however, will hold to the old methods yet for some years.

SHERBROOKE.—Our township makes poor progress ; one man will grade and gravel and another will come along and cover it up with earth.

CARADOC.—The abolition of statute labor and a better system of road improvement is very slowly gaining ground. How to make it practicable is the question.

ALDBOROUGH.—There is quite a feeling that the present statute labor system should be abolished.

MIDDLETON.—Our Council is in favor of doing away with statute labor, but there is a strong feeling among the ratepayers against it, and they do not like to make a move in the matter.

KALADAR, ANGLESEA AND EFFINGHAM.—This municipality sticks to the old system, and getting no aid from county and being a rough country, results—bad roads.

CARLOW.—The statute labor system is still used and a very poor system it is.

WILBERFORCE.—The statute labor system is still used but many express the desirability of abolishing it.

MCKELLAR.—The statute labor system of road destruction is still used in this township.

NEPEAN.—Yes, I am sorry to say, statute labor is still in use in this township.

SOUTH WALSHINGHAM.—Our statute labor is a perfect fraud.

ALBEMARLE.—There is an agitation in favor of abolishing the statute labor system, but whether it will be successful in accomplishing the same is yet uncertain.

PUSLINCH.—The statute labor system is persistently adhered to with its defective and unsatisfactory results.

YARMOUTH.—The present Reeve, after two years' experience, is in favor of the abolition of statute labor. The councillors are of the same opinion, but the approaching elections have an influence and the matter has been laid over.

BERTIE.—During the five years that the road grader has been in use more actual improvement has been wrought than for a decade before. Some permanent work has been accomplished through use of rock crusher, but the amount done in any year will be necessarily small so long as the statute labor system is in practice. Ready funds is the desideratum.

CHRISTIE.—Many of the more intelligent ratepayers of this township would be pleased to see statute labor abolished, but we still have too many who like their annual holidays.

WOODHOUSE.—There seems to be a growing opinion that the present system of bridging, building culverts and road-making needs revising.

PITTSBURG.—I am of the opinion that if statute labor were commuted better results would follow if commutation money were better laid out.

TUDOR & CASHEL.—Our roads are in a very bad state and will be as long as the old statute labor system is used.

ARTEMESIA.—There is considerable talk of abolishing statute labor.

NORTH EASTHOPE.—We spend a great deal of time and money upon the roads, and they are gradually improving, but we can never—in the opinion of the writer—have really good roads until the whole statute labor system is abolished and a new uniform system is adopted.

STORRINGTON.—A by-law was passed in March abolishing statute labor, but such a storm was raised that the council repealed the law before it was acted upon.

DERBY.—The idea of abolishing statute labor and introducing up-to-date methods is being freely discussed and growing in favor.

EAST GARAFRAXA.—There was a vote of the electors taken on a by-law to commute statute labor two years ago, but it failed.

SANDWICH EAST.—The council have the idea to abolish statute labor system which seems to be a failure in our municipality.

PERRY.—The statute labor system is still used in this township, I am sorry to say.

ASPHODEL.—The statute labor system is still used in the township and gives very poor results.

CHARLOTTEVILLE.—Same old style of pathmaster work.

ORO.—A great many of the ratepayers in Oro would like to see the statute labor system abolished, while a good many would oppose it.

FLOS.—Since the commutation of statute labor in Elmvale a very great improvement in the streets is noticeable.

WEST MISSOURI.—Some of our people are in favor of having statute labor abolished and others think taxation is as high as they can stand. If we could enforce the performance of the work the system is all right.

RYDE.—The idea is gaining ground in this municipality that the statute labor system is not the best if we are to have and maintain good roads.

TUCKERSMITH.—The statute labor system is still used in this township but its days are numbered, at least in its present form. The trouble about abolishing statute labor system is to get some good feasible scheme to replace it with. Are considering the advisability of reducing the pathmasters from 86 to 4, giving each one a ward of his own under council's control and allow in rate a ers to draw the gravel as at present.

ENDORISING THE ABOLITION AND COMMUTATION OF STATUTE LABOR.



ALMOST IMPASSABLE UNTIL STATUTE LABOR WAS ABOLISHED.—NOW IN FAIR CONDITION.

A by-law to commute statute labor at fifty cents a day, was submitted to the ratepayers of Orillia Township at the municipal elections in January, 1900, a majority favoring commutation. In discussing the proposed reform the *Orillia Packet* of December 28th, 1899, published the following article and letters which are self explanatory:

"Experience, it has been said, is the best teacher, but the wise man benefits as far as possible by the experience of others. With a view to ascertaining how the system of roadmaking advocated by Mr. Campbell worked in the municipalities where it had been put into operation, the *Packet* wrote to the Clerks of ten of these townships, asking them to give the ratepayers of North and South Orillia the benefit of their practical experience. Eight have complied with the request, and their replies, given below, make interesting reading. It will be noticed that while the details differ widely, the verdict is unanimous that commutation gives much better results than the old statute labor system. This unanimity of opinion where the commutation has been given a trial should be proof enough to the ratepayers of Orillia Township that they will make no mistake in voting for the change. The question has been asked, how it is possible that fifty cents in cash should build more road than a day's statute labor, which represents \$1. Whatever the reason, it is perfectly plain that such is the case, as is exemplified plainly in the case of every municipality heard from. The most plausible explanation would seem to be that the performance of statute labor is, as has often been said, a farce, and that there is truth in the statement that much of the labor is never performed. The farmer who believes in good roads and works hard to get them is thus made to suffer in passing over beats where the call to statute labor is not looked on seriously, but rather as an invitation to a picnic. It is, however, of little use discussing the cause. The fact is plain that commutation at fifty cents a day does produce better results than the statute labor system.

"There are one or two other points which evidently need to be cleared up. It was a mistake to refer to the proposed change as the abolition of statute labor system. It would be better described as a reduction of the commutation from \$1 to fifty cents a day. The road tax will still be reckoned on the same basis as at present, and those who wish will be given an opportunity to work out their statute labor, provided they are willing to work faithfully and to earn the money. Most of those who do work will doubtless get much more than enough to pay their road tax, and will be paid the extra sum out of the amount received from those who prefer to commute at the low rate. It should also be distinctly understood that, according to the scheme outlined by Mr. Campbell, and to the intention of the Councillors, the money received in commutation from each beat will be expended upon the improvement and up-keep of the road on that beat, so long as may be necessary to render it perfect. The only difference in this respect will be that the work will be done on a cash instead of on a picnic basis."

From J. F. Kelly, Mallorytown, Clerk Front of Yonge & Escott :

"SIR,—Yours of 18th to hand and noted. In reply would say that we have only been commuting our statute labor for two years and am hardly able to tell how it will work yet, as they have not decided on any regular system ; but think it is the proper thing to do and will work all right when properly arranged."

From H. Bryant, Barton, Clerk of Barton :

"SIR,—In reply to yours of 18th inst. The Township Council of Barton has commuted the statute labor and, after a trial of six years, we find it very satisfactory. Our people, I am quite sure, would not go back to the old rut. I am very busy at present, as we are on the eve of a lively municipal contest. But if we can give you any assistance in the new year by sending copies of by-laws, rates, etc., I shall be pleased to do so."

From N. Harrison, Castlemore, Clerk of Toronto Gore :

"SIR,—Your letter of 18th inst. to hand, and has come at a very busy time. Therefore I cannot answer so fully as I would wish. I enclose you a copy of by-law abolishing statute labor. At the time the by-law was passed, about one-half the ratepayers raised a strong protest against it, but are now coming round to look at the question in a different light, and I think after another year's trial all objections will have disappeared. I claim that more work has been performed and a more uniform quality this year for about \$600 than was done in previous years by over one thousand days under the statute labor system."

From W. H. Nelles, Grimsby, Clerk of North Grimsby :

"SIR,—Yours of 18th inst. regarding commutation of statute labor duly received. We have commuted our statute labor for the last four or five years, with very satisfactory results. Our roads are now in first-class condition. We employ two commissioners to look after the work. We use a stone crusher and a road grader. The latter is handled by a party owning two good teams. We pay him \$6.25 per day. He will do as much work and do it far better than fifty men would do after the old style. At first we commuted at sixty cents a day, which gave us something over \$1,000. This year we have lowered it to fifty cents, not requiring so much to keep our roads up. The ratepayers are fully satisfied with the system, and we would never think of returning to the old way."

From G. W. Tinline, Beamsville, Clerk of Clinton :

"SIR,—I received a letter from you asking about the abolition of statute labor. About four years ago we bought a road machine and commuted half the statute labor at sixty cents per day. We tried this plan for two years and found it was too expensive, as we had to appoint pathmasters for the half of the time, and some pathmasters will charge a day for coming after their list and time for warning men to work, and some pathmasters would return their list and mark it all worked when probably they have not worked at all. So two years ago we commuted the whole of the time at forty cents a day. I think we have better roads than we had on the old statute labor system. The first year that we commuted the whole time we hired two teams and men to work road machine and would get some farmer who was living on the beat where the machine was working to do the ploughing. We had a great deal of trouble in this way, as a great many men do not know how to plough for a road machine, and in busy time we had trouble to get them when we wanted them. So this year we hired three teams, one team to go along with the plough and two teams on the road machine ; and when the team was not ploughing they would put the three teams on road machine. We have road commissioners to look after road machine, to mend culverts and bridges. I might say that the first year that we commuted half the time and bought a road machine there was a lot of fault found. But before the season was over the most of the ratepayers were in favor of it and would not like to go back to the old system."

From F. A. Hutt, Southend, Clerk of Stamford :

"SIR,—I have your letter of 18th inst. In the beginning of the year 1896 our Council passed a by-law amending our old statute labor by-law by commuting the statute labor in two of the largest electoral districts in the township. These two divisions have been worked on that system ever since. Whenever any road divisions in the two remaining uncommuted districts petition to be added to the divisions that are now commuted, the Council has granted the petition. We have had to commute a road division in the uncommuted district every year since, and I do not think the time is far distant when it will be worked wholly on the commuted system."

"The statute labor in the commuted district is commuted at 50 cents per day, and in the uncommuted district at 75 cents where unperformed. The roads have been greatly improved by the adoption of the system, by the purchase of two road grading machines and by using vitrified sewer pipe for waterways. It is very probable that if the whole township were worked on the commuted plan better roads could be made for a tax of 25 or 30 cents per day than are now obtained under the old system in the uncommuted districts. Besides, the roads are worked under improved methods and uniformity in making is obtained. Where the path-

masters work the roads there is diversity and difference, and not being skilled in handling the road machine the roads are not constructed in the best possible way, but are sometimes left in a worse condition than found. But they have performed their statute labor and the travelling public have to put up with the inconvenience. I am quite certain that if a vote was taken of the ratepayers of this township that the statute labor system would be abolished. Kindly let me know the result."

From W. B. Switzer, Binbrook, Clerk of Binbrook.

"SIR: I received your letter some days ago and would have answered it sooner, but as you are aware we have a very busy time at this season of the year closing up our township matters for the year. So you will have to forgive me for not being more prompt. We did away with the old system in 1896. So we have had four years' trial of the new system, and we are all very well satisfied with the change. Of course there are a few who are not satisfied, but I find they are mostly those who in years gone by used to dump their scrapers full on the roads and leave them for time and travel to level them, or those whose time is of no value on the farm and who would like to work for the Queen and do nothing.

"But the almost universal opinion is that our roads are a great improvement on those of some years ago. I have heard travellers say that when they entered Binbrook they could immediately see a difference in the roads. For general work we have a road machine and our calculations are to go over the road once in three years. We have between sixty and seventy miles of roads. We take about one third of this and put it in good shape each year, and if there is a bad piece anywhere in the township we try to reach that too, and we find a great improvement in our roads. We appoint an overseer of roads in each school section, whose duty it is to see that the roads are levelled as early in the spring as possible, to repair all culverts and washouts, and report to municipal council when any large bridges are in need of repairs. They are responsible also for keeping the roads open in winter, having power to call out any number they may require. The municipality pays for all work performed at the rate of 10 cents an hour. I might say this seems to be the most difficult part of the whole system, as there is a disposition on the part of some persons to take all the advantage of the municipality they can. But of course you would not have any of them in Orillia. We began the system of commutation of statute labor at 60 cents per day the first year. The second year 50 cents per day. Third year 35 cents per day. There was where we made our mistake, as we had to pay out hundreds of dollars for shovelling snow in December of last year, leaving us short for the present year. We have now gone back to the 50 cents per day basis which for a few years will probably be the amount of commutation. Our rate is any assessment up to \$800 two days, and for every additional \$800 or fractional part over \$400 one day, giving us about \$850 in our township from this rating. I do think in large townships the appointment of a superintendent of roads would be a good thing. But our township is too small for that expense."

From J. Honor, Amherstburg, Clerk of Malden.

"SIR,—Your letter of December 18th to hand, and in reply I beg to state: Malden Township, County Essex, contains twenty-one thousand acres with an assessed value of \$730,000, and about sixty-five miles of road, about one-half of which are earth roads, the rest gravelled, with two exceptions of about four miles of crushed stone. The land is very flat except a few hundred acres lying close to the town of Amherstburg. About twelve years ago we passed a by-law to commute statute labor, and this continued in force and seemed to work well for about five years. Then a vote was taken at the time of the municipal election. The necessary number of ballots were printed as follows: 'For the abolition of statute labor,' 'Against the abolition of statute labor.' And statute labor was wiped out. I notice that you say that in North Monaghan statute labor is abolished. I beg to differ from you. They have only commuted it. We have abolished it. Our council divided our township into four divisions for their own convenience, each councillor looking after his part. But on large contracts they consult each other. There is no commissioner employed. In the month of April they make a tour of inspection of the township and inspect each job of work that is most necessary to be done, and put an estimate on it. Later on they advertise that on a certain day they will sell jobs of work around the township, and each job is sold to the lowest bidder. There are a certain number of men who make a business of ditching for a living, and they become very expert at the work, so that all work is done in a thorough, workmanlike manner. We have two road graders. We let the contract of operating them to the lowest bidder by the day, he to furnish two teams and two men. One half of the township costs us about \$4 a day and the other half \$4.50. What I mean by grading is going over the roads in the spring when they are beginning to dry up, and levelling them. Our township usually expends about \$2,000 a year on our roads and bridges. And I think that I can say without fear of honest contradiction that there is not another municipality in the Province that gives its ratepayers better value for their money than we do. Our council receives the handsome salary of \$32 a year each for their services. All money required is raised in the general township rate. The rich land owner pays the bulk of the taxes, and the poor man who is assessed for only \$100, who under statute labor had to do either his two days work or pay fifty cents a day, now pays only about twenty cents taxes. In

conclusion you will notice that our council control the whole work. No commissioner is employed. Their salary is saved to the township. All work is let by contract to the lowest bidder, and is received by the council. So that if a councillor does not prove himself to be a good road-maker and ditcher he knows what his reward will be. We advertise for tenders for cutting thistles and all noxious weeds from growing upon the roads, each tenderer to state the price for the job and which of the four divisions he wishes to work in. We prohibit all stock from running at large, and employ a bailiff to enforce the by-law. There is no corruption or mismanagement. In our township everything is done openly and above board. I herewith send you one of our financial statements for the year."

From G. W. Bennett, Peterboro', Clerk of N. Monaghan.

"SIR.—Yours of December 18th reached me at a very busy time for township clerks. I can scarcely do justice in general terms to this question (i.e., Good Roads) as we have found it in North Monaghan, other than the plan outlined in the letter of 1898 to which you refer and which is still in the main the principal upon which we work. I may say that any alteration to this plan adopted since has been only of such a nature as to widen the breach between our present system and the hydra-headed statute labor. In the beginning of the year 1899 our council increased the commutation rate to seventy-five cents per day, reduced the road districts by one, purchased another gravel pit at so much per acre, and also supplied each commissioner with a number of short-handled shovels for filling gravel. These changes together with a few minor ones in the matter of detail were adopted after practical experience, and a favorable comment throughout our township. Our system seems complete in every detail, and at present is working excellently. Of course varying conditions will necessitate corresponding changes from time to time. What is of vast importance, our commissioners are, through experience and the progressive literature supplied from time to time, becoming expert and more self-reliant. I mean by the latter that they are becoming independent of the patronage that is frequently bestowed by councillors in view of a coming election, and which in no case can be applied beneficially to the roads. At our nomination meeting to-day our Reeve and Treasurer presented a financial statement which differs from its predecessors in this township inasmuch as it shows all our taxes for the year paid on December 15th, all our debts paid, and a substantial balance on hand to begin next year, and this without either charging interest or allowing discount. This is largely due to the fact that the people are recognizing the benefits of well-directed efforts and well-spent money upon the roads, and signify their approval by their readiness to pay their taxes for these purposes. If there is any specific information or any detail that I can furnish you I will be pleased to do so, but as I said before, I cannot cover the ground in this general way to the extent I would like, and I do not know the circumstances under which you are working. In 1896 we voted upon the question of the abolition of statute labor. This was carried against by two to one. In closing I might say that the objective in reform of this kind is success. One or two good live men can carry the thing through with a little push, and very soon you will hear the last of road beats and pathmasters. By all means procure a road grader, and when procured, do not over-do the amount of grading. Make haste slowly. Crown and thoroughly build the work for the first year so as to avoid adverse criticism and enlist sympathy in the beginning. See that your pay lists are kept accurately and publish them with your auditor's statements, for no doubt you understand the peculiar desire some of your ratepayers may have to know 'where the money goes,' 'or who gets it.' I strongly advocate and advisedly so, the abolition of statute labor. There can be no doubt about the result. Any of the methods open to you by the 'Act' is better and more business like, and as such encourages better men to come forward as councillors. I wish you every success."

THE TOWNSHIP SYSTEM.

The system which townships are advised to adopt in place of statute labor, is included in the summary which follows. This has not been uniformly adopted, several townships making slight alterations such as they deem more suited to local conditions, instances of which are shown in the by-laws of two townships included in this report:

Do away with the statute labor roll entirely. If this plan is not favored the labor may be commuted at a rate per day. This ranges from 25 cents to 75 cents in the townships now commuting.

If statute labor is entirely abolished, levy a rate on the assessment of the township to raise the money required.

For road purposes, divide the township into a convenient number of divisions, usually four.

Apportion the money among the road divisions, keeping in view all circumstances, viz. Importance of roads, works needed on them, benefit resulting to the greatest number of people, amount of traffic, assessment, etc.

Appoint one township road commissioner, or if preferred one commissioner for each road division, to advise, consult with, and carry out the direction of the council.

The office of road commissioner should be similar to that of the township clerk or treasurer.

Councillors should not act as commissioners, as they are subject to undue influence from the ratepayers, and their term of office is uncertain.

A general plan for road improvement should be laid down by the council for the commissioner or commissioners to follow.

This plan should specify the width to be graded, width and depth of road metal, character of drainage, etc., of all roads.

Roads of importance should not be less than twenty-four feet between the inside edges of the open ditches. No road should be of less width than eighteen feet.

Early in the year the commissioner should go over the roads to consider the work to be undertaken, and report to the council.

Work of construction, such as hauling gravel, ditching and drainage, building bridges and culverts should be done by contract, and supervised by the road commissioner.

No account for labor or material should be paid by the treasurer except on the certificate of the road commissioner.

Minor work and repairing should be done by day labor, only the road commissioner being authorized to employ, direct or discharge the men and teams needed to operate the machinery.

If a member of the council desires to interfere in any of these matters he should do so through the commissioner.

The same man and teams should be hired to operate the machinery for the entire season, or longer if possible, as they become proficient and do better work. This applies particularly to the operation of a road grader; or for the grader a traction engine may be hired.



JUST GRADED AND GRAVELLED—A ROLLER WOULD COMPLETE IT.

The commissioner should keep a pay roll to return quarterly to the council, showing who have been paid and the amount paid, the roll to be then filed for auditors.

This roll will act as a check on favoritism on the part of the commissioner. Work should be divided as much as possible among the residents of the township desiring it.

Work should be commenced with a definite end in view and continued systematically, from year to year if necessary, until the entire road mileage has been brought to a proper standard.

ROADMAKING OUTLINED.

Previous reports of this Department have dealt as fully as seems advisable with the practical work of roadmaking. It therefore does not appear necessary to again enlarge upon it at the present time, other than to briefly outline the subject. This can best be done by quoting from a previous report, a list of instructions for pathmasters, which contain briefly the elementary principles of roadmaking, as applicable to the roads of Ontario, and the systems under which they are maintained :

1. Every good road has two essential features :
 - (a) A thoroughly dry foundation.
 - (b) A smooth, hard, waterproof surface covering.
2. The foundation is the natural sub-soil, the "dirt road," which must be kept dry by good drainage.
3. The surface covering is generally a coating of gravel or broken stone, which should be put on the road in such a way that it will not, in wet weather, be churned up and mixed with the earth beneath. That is, it should form a distinct coating.
4. To accomplish this :
 - (a) The gravel or broken stone should contain very little sand or clay—it should be clean.
 - (b) The roads should be crowned or rounded in the centre so as to shed the water to the open drains.
 - (c) Ruts should not be allowed to form as they prevent water from passing to the open drains.
 - (d) The open drains should have a sufficient fall, and free outlet so that the water will not stand in them but will be carried away immediately.
 - (e) Tile under-drains should be laid wherever the open drains are not sufficient and where the ground has a moist or wet appearance, with a tendency to absorb the gravel and rut readily. By this means the foundation is made dry.
5. Do not leave the gravel or stone just as it drops from the wagon, but spread it so that travel will at once pass over and consolidate it before the fall rains commence.
6. Keep the road metal raked or scraped into the wheel or horse tracks until consolidated.
7. Grade and crown the road before putting on gravel or stone.

8. If a grading machine is available, grade the roads which you intend to gravel before the time of statute labor, and use the statute labor as far as possible in drawing gravel.

9. A fair crown for gravel roads on level ground is one inch of rise to each foot of width from the side to the centre.

10. The road on hills should have a greater crown than on level ground, otherwise the water will follow the wheel tracks and create deep ruts instead of passing to the side drains. One and one-quarter inches to the foot from the side to centre will be sufficient.

11. Repair old gravel roads which have a hard centre but too little crown, and which have high, square shoulders, by cutting off the shoulders, turning the material outward and placing new gravel or stone in the centre. Do not cover the old gravel foundation with the mixture of earth, sod, and fine gravel of which the shoulders are composed. The shoulders can be most easily cut off by means of a grading machine.

12. A width of twenty-four feet between ditches will meet most conditions, with the central eight feet gravelled or metalled with broken stone.

13. Wherever water stands on the roadway or by the roadside, or wherever the ground remains moist or is swampy in the spring and fall, better drainage is needed.

14. Look over the road under your charge after heavy rains and during spring freshets. The work of a few minutes in freeing drains from obstruction or diverting a current of water into a proper channel may become the work of days if neglected.

15. Surface water should be disposed of in small quantities; great accumulations are hard to handle and are destructive. Obtain outlets into natural watercourses as often as possible.

16. Instead of having deep open ditches to underdrain the road and dry the foundation, use tile.

17. Give culverts a good fall and free outlet so that water will not freeze in them.

18. In taking gravel from the pit, see that precautions are taken to draw only clean material. Do not let the face of the pit be scraped down, mixing clay, sand and turf with good gravel.

19. Gravel which retains a perpendicular face in the pit in the spring, and shows no trace of slipping is generally fit for use on the road without treatment. Dirty gravel should be screened.

20. Plan and lay out the work before calling out the men.

21. When preparing plans keep the work of succeeding years in view.

22. Call out for each day only such number of men and teams as can be properly directed.

23. In laying out the work estimate on a full day's work from each man and see that it is performed. Specify the number of loads of gravel to constitute a day's work. Every wagon box should hold a quarter of a cord.

24. Make early arrangements for having on the road when required, and in good repair, all implements and tools to be used in the performance of statute labor.

25. Do all work with a view to permanence and durability.

A study of the foregoing will point to three main faults commonly to be found in the roads. These are, bad drainage, poor gravel, and improper methods

of placing the metal, (gravel or broken stone) on the roads. It is doubtful if any of these evils can be fully remedied under the statute labor system. To overcome bad drainage it is essential that there should be a constant system of repairs keeping the road well crowned, free from ruts, the gravel or broken stone raked into place, and the side drains and culverts open and without stop-pages. The use of poor road metal, (gravel or stone) is likely to continue until there is someone, who, by experience, is able to select the best material available, and is provided with proper implements to screen and crush it when necessary, Nor can statute labor ever provide the means necessary for putting metal on the roads in the right way, first preparing the road by the use of machinery, then properly spreading the metal and rolling it.

IMPROVING SAND ROADS.

A degree of moisture is necessary in the summer season in keeping sand roads or roads over sandy ground in their best condition. Drains are necessary but they should not be deeper than will provide suitable drainage in spring and fall. Under-drains, if they are needed at all, should be shallow.

One of the most lasting and beneficial improvements to sand roads, is the planting of rows of trees on each side of the road, and close enough together to provide continuous shade. They will prevent, in part, the drying effect of the winds, as well as intercept the rays of the sun. For this purpose the white elm with its arching branches is most serviceable, and will add many times to the appearance of the country.

ROAD MACHINERY.

The importance of using modern machinery in making good roads, has become pretty generally recognized since the commencement of the good roads movement in Ontario. A road grader has become a standard implement, employed by a great majority of townships, and wherever it is properly used is a great saving of labor, and does the work more efficiently than can be done by hand. Municipal clerks report their use in 215 townships, a considerable number of these having two. Several, among which are Minto, Trafalgar, Tecumseh, Toronto and Plympton own three and four.

Several townships and numerous towns have supplied themselves with rock crushers. Among the townships owning these implements are:—Nottawasaga, Ameliasburg, Front of Leeds and Landsdowne, Mara, Winchester, Bastard and Burgess, Bathurst and Emsley, Bertie, Rear Yonge and Escott, Smith, Williamsburg, Charlottenburg (crusher and traction engine), and North Grimsby. A number of townships of which are Keppel, Elmsley South, Beckwith, Pittsburg, Elizabethtown and Ramsay, hire a crusher or contract for broken stone. The county of Hastings uses a complete outfit of roadmaking machinery, roller, crusher, grader and traction engine. Victoria also has an outfit of roadmaking machinery for the use of the townships, and Wentworth rents its crusher to the townships within the county.

Broken stone roads are, when properly made, very much more serviceable than gravel roads, and wherever there is a plentiful supply of good stone, particularly if gravel is scarce or of poor quality, a crusher with screen attachment should be used. It is profitable also to put certain qualities of gravel through a stone crusher, such gravel as contains a quantity of large stone. If there is much clay or earthy material in the gravel it will be well to remove it by using a revolving screen attached to the crusher.

Townships should, wherever possible, use a road roller, particularly if broken stone is used. A roller will at once consolidate the metal, leaving the road in the best condition for travel, at the same time resulting in a great saving of gravel or stone. Without the use of a roller, the road reaches a condition fit for travel only after a long period, during which time the loose material causes much inconvenience, and is mixed with the soil beneath, largely destroying its usefulness. Townships should own a horse roller, or better, should rent a steam roller from a town or city in the vicinity possessing one.

If a steam roller is available, its use may be further extended to picking up the road and loosening the old material before placing the new material on the road; after which the road may be rolled down solidly, the new material and the old road surface having a perfect bond.

Road graders are commonly operated by two teams of horses at a cost of \$6 per day. A traction engine, such as is used for threshing machines, is, however, a very much better power, and is sometimes hired for about \$3 a day, or

half the cost of using horses. The engine gives a steady draught, and does not need to stop to rest, two advantages over the use of horses. It is customary to go two or three miles with a traction engine before turning and by making two or three circuits, up one side of the road and down the other, it is generally possible to cut off the square edges which have formed, restoring the crown and preparing it for a new coating of gravel or stone.

TRANSPORTATION OF ROAD MATERIAL BY RAILWAYS.

In several counties, such as Essex, Kent, Lambton, Norfolk, Haldimand, Monck, Welland, there is a total absence of roadmaking material, and in these counties the improvement of streets of towns and cities, as well as the rural roads, appears to be a hopeless task, and almost outside of their consideration. The ratepayers suffer through bad roads, are anxious for good roads, but helplessly complain of having no material, and have almost given up hope of any improvement. With an earnest endeavor to better their condition, they are lavishing their money and labor in raking and scraping the mud, which work is almost completely destroyed every spring and fall.

These counties are completely traversed by the Michigan Central, Grand Trunk, Canadian Pacific, Lake Erie and Detroit River, Erie and Huron Railways. Along these lines are several stone quarries, at Amherstburg, Hagersville, St. Davids and Fort Erie, in each of which have been recently placed stone crushing plants, capable of turning out first-class road material in large quantities. In addition to this, at Komoka, Blenheim and many points along these lines, this material can be cheaply loaded by machinery and scraped off the cars at any road crossing convenient to the work, reducing the haul to a minimum. Railway companies have in every way signified a desire to assist in the building of roads, especially those leading the traffic to their own lines, and in some instances have delivered the material free, in others not charging more than half freight rates. When they are permitted to carry the material at slack seasons, they have expressed a willingness to do the work at actual cost of hauling, which would place these municipalities in a position to procure first-class material at less cost than others where material is considered plentiful. In this way, too, many towns and cities now using gravel of an inferior quality would be enabled to use a first-class quality of durable material.

In many townships, by a liberal expenditure of money and labor, the roads have been improved, as in the township of Ekfrid, where few miles remain ungravelled; but in doing this the supply has been exhausted, and now these roads, built of an inferior quality of gravel, are suffering for want of repairs. Unless some method is immediately adopted this work will suffer, and the expenditure of labor and money already made will be lost. At the last municipal

nomination the chief question in that township was as to how an additional supply of gravel or broken stone could be best secured.

Railway haulage of broken stone is now being looked upon as impossible by most municipalities, but through the influence of the Government it may be possible to arrange a general plan of delivering this material at a low freight rate, resulting in an immense advantage to the municipalities and solving their difficulty.

PERMANENT ROADWORK.

With merely sufficient exceptions to prove the rule, permanency and durability are entirely lacking in road construction throughout the Province. This is characteristic of all new countries where immediate needs are pressing and



where the money and material for substantial work are lacking. This we find exemplified not only in Ontario, but in the whole of Canada, the United

States and the entire American continent. Not only is this temporary character of construction characteristic of roads (and the bridges which are a necessary part of the roads), but it is to be seen in the quality of work placed in other works public or private as well.

As a country emerges from the days of pioneer settlement, a different feeling should, and does, prevail. Temporary work, while it may be the best investment under the conditions surrounding early settlement, becomes very expensive and unsatisfactory under circumstances attached to a denser population and its attendant commercial and social requirements.

Permanent work, with its fewer repairs, its less interruption to traffic when undergoing these needed repairs, the better service at all times, renders durable construction, particularly in highway building, a measure of economy, in spite of the fact that the first cost of such durable work is more than that of temporary construction.

An effort should be made to get away from the spirit which prompts the building of flimsy wooden culverts and bridges that render but poor service at their best, and in a few years are twisted and bent out of all resemblance to a bridge or a culvert. The expenditure for the repair of such structures will generally, at the end of a term of years, make up much more than the cost of a permanent work.

The only really permanent bridge or culvert is the arch of masonry or concrete, which, up to spans of forty feet, can be very cheaply built for ordinary highway traffic. Such arches have been in use in Scotland and Ireland for more than a century, and, with scarcely any repair, are still in good condition. It is common there for a farmer living in the vicinity of the work to contract for it, supply stone and other material, hiring one or two masons to do the work. In this way the money paid out for the work is all kept in the locality, and the method has proven very popular.

Iron bridges, with masonry or concrete abutments or piers, come next, in order of durability and economy. The iron superstructure, with proper repairs, should last for fifty years, while the masonry should be good for at least a century, if rightly built and attended to when signs of wear appear.

During the life of the superstructure three or more wooden bridges would be needed, even with close attention to repairs, while the service rendered would not be at all equal to that of the stronger and more durable work.

For small culverts, which at present are the cause of so much annoyance on our highways, sewer pipe, iron pipe, or concrete work should replace the wooden boxes now so much used. Care should be taken to provide a level roadway at these points, so as to avoid the disagreeable bumping and jolting which so many of these boxes now create.

For the roadway itself, attention should be given first to a permanent system of drainage in all its parts, involving tile under-drainage where necessary. The grade of the road should be brought to its permanent level, before gravel or stone are placed on it. Hills should be permanently graded, and low sections permanently filled. Ontario has emerged from the days of temporary work, and the time has come when we should avoid the past policy which, by scattering money and labor in patch-work, has left our roads in a most unsatisfactory condition.

HIGHWAY CULVERTS.

The construction of culverts is a detail of road improvement which demands, in many townships, a large annual expenditure. Until very recently timber, principally cedar, was the only material employed, with the exception of an occasional rubble-stone structure. Timber is a very temporary material for this class of work. It quickly decays, warps and is displaced by the frost, so that the life of wooden culverts is frequently not more than eight, and rarely reaches fifteen years. During this period it is very apt to be in an unsafe condition and is generally an impediment to traffic. In a term of years the cost of repairs, frequently necessitating a man's labor for half a day if only to put in a new plank, is considerable.

The price of timber, too, is constantly increasing while the quality obtainable is not equal to that of former years. The returns received from municipal clerks with regard to last year's work, show a decided tendency on the part of municipal councils to follow the suggestion of this Department in adopting, in place of timber, sewer pipe, concrete pipe, and stone, brick and concrete arches. There will be failures, undoubtedly, of these new materials. Inexperience in their use, carelessness in workmanship, and an inferior quality of material, will almost of necessity be found in some cases, but with care and growing experience they will be found a decided advantage both in point of economy and service. Experience, too, in the use of these new materials will, in nearly every case, enable councils to largely lessen their cost.

Returns from municipal clerks show that 176 townships in the Province are using cement-concrete and sewer tile, cement, concrete, stone or brick arches. Of these, only eight report unfavorably to their use. The common expressions are: "Growing in favor," "Good satisfaction," "Not in use long enough to express an opinion," "Full satisfaction," "Are doing well," "Give general satisfaction," "Excellent satisfaction," "So far have proven satisfactory," "Good results," "Very satisfactory," "The best satisfaction," "No complaints," "First-class," "Fair satisfaction," "Entire satisfaction," "Very satisfactory results."

A selection from these reports is appended, such as are most instructive, and including the eight unfavorable replies to which reference has been made. These replies indicate that tile are most satisfactory for small sizes of culverts; that care must be taken to put a sufficient covering of earth over them; that they must have a free outlet so that water will not freeze in them; and that for larger sizes, arch culverts of concrete, stone or brick should be used. It may be well to recommend here also, that the earth covering over and around these culverts should be of clay or other heavy soil, such as is not readily washed out by freshets, and this covering should be firmly tramped into place.

Township Reports.

The comments of a number of township clerks regarding culverts are as follows :

WEST OXFORD.—The council are using cement concrete pipe for culverts up to 3 ft. 6 in. with good satisfaction.

WINDHAM.—The tile gives very good satisfaction in all cases where there is fall enough to take the water from the lower end of culvert.

KING.—Have been using sewer pipe for 8 years. They give good satisfaction for culverts and are much cheaper than cedar.

TORONTO.—Tile gives very good satisfaction when you can get them well covered, so that heavy loads do not break through them.

LOBO.—Cement concrete tile is used for culverts very extensively and gives good satisfaction.

WEST FLAMBOUR.—We have one cement concrete culvert. Is giving good satisfaction. We have several vitrified or sewer pipe culverts. They are all right where there is a good fall from the discharge so that the water will not stand in them and freeze.

WATERLOO.—Cement concrete is used extensively with good satisfaction.

BROOKE.—Concrete tile for deep culverts is coming into use, and where properly placed are satisfactory. A large concrete culvert has just been completed.

EAST HAWKESBURY.—We purchased a car load of vitrified pipes for culverts and have used them with good results. They are still good and sound after three years usage.

McKILLOP.—Built three cement abutments for steel bridges this summer.

OAKLAND.—We have been using vitrified tile for some time and find the results very satisfactory. Have not had to renew any as yet except where they were not large enough.

CARRICK.—We put in one tile culvert in 1898 and it gave good satisfaction. We put in three this year. The tiles we use are cement concrete.

BOSANQUET.—Using the small sizes of tile with satisfaction.

BLANSHARD.—We have been using all concrete tile for culverts and have built one cement arch. They have given the best of satisfaction.

ADELAIDE.—Cement-concrete is used in all culverts where they are of sufficient size to carry water. We put in a three foot one this summer, which gives good satisfaction.

NORTH DORCHESTER.—Cement-concrete is used for culverts and gives good satisfaction.

TRAFALGAR.—Cement-concrete and vitrified tile are used very extensively and are most satisfactory ; have not had any broken by frost although frozen solid when over three-quarters full.

WEST ZORRA.—Concrete tile is used almost entirely, with good success.

ROCHESTER.—Tile are now used generally for culverts.

BATHURST AND ELMSLEY.—Tile for culverts has been used to some extent but it is said that where tiles fill with water in the fall that the freezing destroys them. Where there is plenty of fall to keep the water away and no danger of it freezing, they have given good satisfaction.

WILLOUGHBY.—Vitrified tile are in quite common use in this township and are giving good satisfaction.

MARKHAM.—Cement-concrete and vitrified tile have been used with very good satisfaction.

THURLOW.—The county has used vitrified tile in a couple of places in the township to my knowledge with satisfaction. I do not think the township has used any.

ELMA.—Vitrified tile has been used in some culverts this year for the first time, but as they have not stood a winter yet cannot tell the result but if they give good satisfaction will be entirely used.

HAY.—Vitrified tile was used 10 years ago and they give the best satisfaction. But since then nothing but timber has been used, with very unsatisfactory results.

WEST WILLIAMS.—About eight years ago we bought a carload of vitrified tile. They have stood all right. Since then we have used cement-concrete tile, the contractor manufacturing them at a convenient place in the township. They give perfect satisfaction where properly put in, but much care must be taken to pack them well around to prevent them washing out. We built, four years ago, a 51 foot long culvert of 3 feet cement concrete tile with masonry at each

end ; there is 17 feet of earth above it, giving entire satisfaction. We built this year a 56 foot arched stone culvert at a cost of \$1,000.

KINCARDINE.—A few tile have been used this year and are giving good satisfaction being much cheaper than timber.

BINBROOK.—We have two cement-concrete culverts in the township. It was merely for a test and it would appear as if they were going to be very satisfactory. We have no vitrified tile.

EAST ZORRA.—Cement concrete mostly is used. Very few wooden culverts built in last ten years. Very good when properly put in, otherwise not.

YARMOUTH.—Cement concrete and vitrified tile are both used and found to be satisfactory when properly put in. Cull pipe are usually supplied where vitrified tile is used and these do not fit, causing leaky joints and washouts.

BERTIE.—Vitrified tile is used for culverts and sluiceways with excellent results where properly put in.

CARADOC.—Concrete tile for culverts has taken the place of wood altogether. Good satisfaction. Some glazed tile is used but not much.

NEPEAN.—Vitrified tile is used occasionally, giving fairly good satisfaction where the place is suitable.

SOUTH WALSLINGHAM.—Built three culverts with cement and stone in 1898 and they are all right ; have used vitrified tile a number of years ; in deep gullies use iron pipe, both giving good satisfaction.

WHITBY.—Vitrified sewer pipe has been used for culverts for a few years, almost entirely this year. Give good satisfaction at less cost than timber when properly placed.

TOWNSEND.—Tile is used somewhat, but not with satisfaction. Stone culverts are being used in place.

WOOLWICH.—Vitrified tile used for culverts, etc., and have given good satisfaction during the past ten years.

THOROLD.—Tile have been used for about ten years with perfect satisfaction.

ANCASTER.—Vitrified tile is used in some cases, also plank. The plank gives better satisfaction in shallow culverts.

NASSAGAWEYA.—Cement concrete and vitrified tile have been used with very good satisfaction, as no injury has resulted from frost or from any other cause.

WOODHOUSE.—More tile is used every year and, where suitable, gives good satisfaction. No concrete cement has been used.

OPS.—Ops has one large culvert, or sluice, arch-shaped, built of brick and cement. It cost about \$300 and, being filled overhead, it serves as a bridge, and is supposed to outlive any bridge, but no tile culverts have yet been used.

USBORNE.—Cement concrete tile very largely used during the past three years. Also arched culverts of the same, 3 to 6 feet wide. So far they seem to be perfectly satisfactory.

CHARLOTTEVILLE.—Vitrified tile used altogether for the last eight or ten years ; good satisfaction so far ; no complaint.

WEST NISSOURI.—Vitrified tile have given the greatest satisfaction where the flow of water is not too large. Cement concrete is used.

TUCKERSMITH.—We use cement concrete tile wherever possible, also cement arches for small culverts. Good satisfaction if put in large enough and carefully.

GRANTHAM.—Tile as large as two feet in diameter have been used with good result for a number of years.

SOUTHWOLD.—About all the culverts constructed with vitrified tile and concrete, principally vitrified. I believe the cement concrete will give the best satisfaction.

BARTON.—Tile is used where practicable, otherwise plank. Our soil in many places is too shallow for tile ; we are near the rock in the south portion of the township.

BURFORD.—We have been using cement tile for culverts for the last five years and they are giving good satisfaction.

NELSON.—Vitrified sewer pipe is used to a limited extent and, where properly put in, is giving good satisfaction.

HULLETT.—Cement concrete is very generally used in our township both for culverts and bridge abutments, but so far is not quite satisfactory for culverts.

GLANFORD.—The use of cement concrete has been proposed but not yet tried. Sewer pipe has been used a good deal for culverts, but is, I think, not so much thought of as formerly, especially the large sizes. The medium and smaller sizes are fairly satisfactory.

MIDDLETON.—Tile is used in a few instances, but not very satisfactory.

SOUTH NORWICH.—A few years ago I bought a car-load of drain tile a foot through. They were not given a good show and were not used after that lot. I think they would be all right if properly put in.

WALPOLE.—We have been using cement concrete for culverts this year, but cannot give results as yet. We have not found tile very satisfactory.

DAWN.—We have used concrete some, but not satisfactory. We have been using vitrified tile for about five years, and satisfaction is fairly good.

DUNWICH.—Tile used, but have not proved very satisfactory.

CULVERT SPECIFICATIONS.

The following specifications for highway culverts include only those clauses necessary to the actual work of construction, and are intended more as a basis for a complete specification, rather than to be followed in detail in every case.

If the work is done by a contractor, there are a number of clauses relating to excavation, protecting the public from accident while the work is in progress, manner of payment and other agreements which it will be necessary to include.

The details of construction should be altered and amplified to suit local conditions, such as the quality of stone and sand obtainable. The paving of the stone culvert requires the use of flag-stone, whereas this may not be obtainable locally, and a concrete or rubble stone paving may be more suitable. Local conditions may suggest various alterations.

A culvert of cement concrete will generally be found most economical but there are circumstances, no doubt, in which municipal councils will wish to avail themselves of a plentiful supply of suitable stone, within easy reach, for a stone masonry waterway. The specification for a stone arch culvert will indicate the character of masonry most suitable for coursed rubble stone or broken range work, and is adapted to highway culverts up to a considerable span.

A Stone Arch Culvert.

Plans and drawings. 1. The culvert shall be built in accordance with the dimensions indicated upon the plans and drawings hereunto attached and forming part of these specifications.

Masonry. 2. The masonry shall consist of coursed rubble or other approved stone laid in cement mortar. No stone shall be less than six (6) inches in thickness and not less than twelve (12) inches in its horizontal dimensions. At least one-fourth of the stone face shall be headers, evenly distributed throughout the wall. The stones shall be roughly squared on joints, bed and faces, laid so as to break joints and in full mortar beds. The face shall be "rock face" with edges pitched to line, with no face projections exceeding two inches. All vertical spaces shall be flushed with good cement mortar and then be packed full with spalls. No spalls will be allowed in the beds. Selected stones shall be used at all angles, and shall be neatly pitched to true lines and laid on hammer dressed bed; draft lines may be required at the more prominent angles. The foundation shall be of large sound stones, roughly squared, no stone to measure less than two cubic feet.

Coping and step stones. 3. The top of the parapet wall shall be capped with stones extending entirely across the wall, and having a front and end projection of not less than six (6) inches. The steps of wing walls shall be capped with stone, covering the entire step, and extending at least six (6) inches into the wall. Coping and step stones are to be roughly hammer dressed on top, their outer faces pitched to true lines, and shall not be less than six (6) inches in thickness.

Details of masonry work. 4. Care should be taken that all stones are laid on their natural bed; that they are brought to an even bearing; that there shall be no vertical openings between the stones. Mortar shall be used throughout the structure and the practice of using mortar only on the face and back of the walls shall not be permitted. The inside of the wing and side walls, that side on which the earth rests, shall have a frost batter of one (1) inch to the foot.

Arch stones. 5. Arch stones must invariably extend through the entire thickness of the arch and have a minimum thickness of five (5) inches on the soffit. Each stone is to be well and closely fitted so as to give half-inch joints and to break joints with its fellow 9 to 7 inches. The whole must be laid in cement mortars, and each course well grouted immediately after being laid.

6. The face stones of the arch are to be as nearly uniform in depth as possible, of large size, and neatly incorporated with the perpendicular face of the masonry. The key-stones are to be 10 or 12 inches on the soffit, to have a chisel draught around their edges, and to project beyond the face of the wall 2 or 3 inches.
- Face and key stones.**
7. The extrados of the arch shall be flushed with cement mortar two (2) inches thick, levelled up and rounded to a moderately even and smooth surface.
- The extrados.**
8. All outside joints shall be raked out to a depth of one (1) inch and neatly pointed with a mortar made of one part Portland cement and one part sand.
- Pointing the masonry.**
9. Each course of masonry as laid shall be grouted with a mixture of one part of Portland cement to two parts of sand, no more water being used than that necessary to give the required plasticity.
- Each course to be grouted.**
10. The waterway of the culvert between the side walls, and to such distance between the wing walls at both ends as the engineer may direct, shall be paved with stones not less than three feet long, eight inches wide and four inches thick. The stones shall be cut and squared so as to form close joints with each other and with the walls of the culvert. The stones shall be laid on a bed of gravel two (2) inches thick, the joints to be filled with cement mortar.
- Waterway to be paved.**
11. All cement furnished must be of a well and favorably known brand of Portland cement, and shall be approved by the engineer or superintendent in charge of the work. It shall be delivered in barrels or equally tight receptacles, and after delivery must be protected from the weather by storing in a tight building or by suitable covering. The packages shall not be laid directly on the ground, but shall be placed on boards raised a few inches from it.
- Cement.**
12. The water employed for mortar shall be fresh and clean, free from mud or other objectionable matter.
- Water.**
13. Mortar shall be composed of two parts of sand and one part of cement, mixed thoroughly dry, and tempered to the required consistency. It shall be used as soon as made, and any mortar that may have "set" while unused shall be wasted. No variations from the above proportions shall be permitted, unless to make the mortar richer when required in special cases. The thorough mixing and incorporation of the materials will be insisted upon. The dry cement and sand shall be turned over and mixed with shovels by skilled workmen not less than ten (10) times before the water is added; after the water is added the paste shall be again turned over and mixed by skilled workmen not less than six (6) times before it is used. Mortar shall be mixed on a flooring to be provided by the contractor.
- Mortar.**
14. The centring must be well formed, an exact semi-circle, of ample strength, securely placed in position, and in every respect conform to the requirements of the superintendent. The ribs must not be placed farther apart than three (3) feet. The lagging shall be three (3) inches thick; the supports of centres shall be substantial and well constructed. Centres shall not be struck without permission from the superintendent.
- Centring.**

A Cement-Concrete Arch Culvert.

- (1) The culvert shall be built in accordance with the dimensions indicated upon the plans and drawings hereunto attached, and forming part of these specifications.
- Plans and drawings.**
- (2) Concrete referred to in this specification shall be known as "fine concrete" and "rubble concrete." Unless rubble concrete is definitely specified, fine concrete shall be used.
- "Fine" and "rubble" concrete.**
- (3) The abutments are to be erected within a framework of dressed lumber, closely boarded up against the work as it proceeds. The centring for the arch must be well formed, an exact semi-circle. The ribs must not be placed farther apart than three (3) feet, and the lagging shall be three (3) inches thick, dressed to the intrados of the arch. The framework, centring and their supports shall be substantial and well constructed and shall not be removed less than fourteen days from the completion of the work.
- Framework and centring.**
- (4) Fine concrete shall be composed of one part by measure of Portland cement, two parts by measure of sand, and four parts by measure of broken stone. The concrete shall be mixed in a water-tight box placed close to the work, by first spreading evenly a layer of sand; upon this shall be evenly spread the proportionate quantity of cement and the two thoroughly mixed in a dry state. To this water shall be added and the whole thoroughly mixed and brought to the consistency of a stiff mortar. The proportionate amount of stone shall then be spread evenly over the mortar, and thoroughly intermixed therewith. The concrete, when mixed as described, shall be immediately put in place and thoroughly pounded and rammed until it is perfectly and uniformly solid, moisture appearing on the surface.
- Fine concrete.**

(5) Within the body of the abutments of culverts, of not less than four foot span, but not nearer than six inches to the surface in any direction, large stones may be placed by hand in layers. These stones shall be in "rack and pinion" order, and not less than two inches apart. Concrete shall be carefully inserted between the stones thus placed and thoroughly packed and rammed so as to fill all voids. Concrete shall cover each layer of stones to a thickness of half the depth of the stones, when another layer of stones may be placed. A facing of concrete is at all times to be kept at least six inches higher than the rubble concrete; and shall be united with the rubble concrete so as to form a continuous and solid mass. This outer rim of concrete shall precede the placing of the rubble work within, and shall be placed around the interior of the casing to a height of nine inches and a thickness of six inches. It is to be thoroughly pounded so that no cavities shall remain when the outside casing is removed. In no instance is the rubble concrete to extend higher than one foot below the top of the abutment, which top of the abutment shall be finished with fine concrete. The rubble stone is not under any circumstance to extend into the arch.

(6) All cement employed in the work must be of a favorably known brand of Portland Cement, cement, and approved by the superintendent in charge of the work. It shall be delivered in barrels or equally tight receptacles, and after delivery must be protected from the weather by storing in a tight building or by suitable covering. The packages shall not be laid directly on the ground, but shall be placed on boards raised a few inches from it.

(7) The stone used shall be granite, quartzite, fine-grained limestone or other equally strong and durable stone, care being taken to exclude soft limestone, friable sandstone, and stone affected by the atmosphere. It shall be broken into varying sizes, the largest to pass, any way, through a two-inch ring. The sand used shall be clean, sharp, silicious and of varying sized grain. The water used shall be clean and care shall be taken not to use an excessive amount, the concrete when mixed and ready for the work to have the consistency of freshly dug earth.

8. When gravel is used instead of broken stone, it shall be screened to remove all sand and earthy material. If excessively dirty it shall be well flushed to remove loamy matter, the dirty water being allowed to run off; nor shall it contain stones, any diameter of which exceeds two inches.

9. Care should be taken to make the extrados of the arch a smooth, regular surface such that moisture will not find lodgment. All framework and centring shall be of dressed, well-fitted lumber, and the concrete shall be perfectly rammed into place so that all surfaces shall be smooth, without cavities, when the casing is removed.

10. While the work is in progress, it shall be so arranged that a steady supply of mixed concrete shall pass from the mixing box to the point where it is to be placed. At any time when the work is interrupted before its completion, or at the end of the day, a wet covering shall be placed over the last layer of concrete; before the work of depositing the concrete is resumed, this surface shall be thoroughly flushed with water to remove any foreign material which may have gathered thereon. No concrete shall be laid in wet or freezing weather.

CAUSES OF DEFECTIVE SIDEWALKS.

The advisability of replacing old plank sidewalks in very many cases with the more permanent concrete has been frequently urged, and a specification was included in the report of this Department for 1898. In so doing, however, it has not been advocated that an inferior quality of concrete would be preferable to a good quality of plank walk, the comparison being, in every case between good quality of both plank and concrete.

The first cost of plank walks is about five cents a square foot. The life may be extended to about fifteen years, but for the last ten years repairs will have been many. Each repair requires that a man be sent with material so that the patch-

ing process is very expensive in proportion to the first cost. The result is to about double the original cost of the work in a term of fifteen years, making the cost ten cents a square foot.

Cement concrete walks are being laid for eleven and twelve cents a square foot, although the circumstances of some cases, the difficulty of obtaining broken stone, gravel, or sand, may require a slightly increased outlay. The life is indefinite for we really do not know how long concrete work will endure, but we do know that concrete structures of the Romans, build 2,000 years ago, are still in existence. It is not too much to expect that the concrete walks now being laid will do service for fifty years if laid as they should be. The cost after a term of fifteen years is very little more than that of plank walks, so that their life beyond a term of fifteen years is a practical saving to the municipality.

The use of concrete is far from being properly understood by those who have charge of sidewalk construction in some municipalities, and the result has been in many instances, that most unsatisfactory work has been obtained.

Cement concrete in walks, culverts and other structures of a like nature, is lasting and serviceable, and when properly mixed and rightly put in place, is in the best sense economical. But defective work shows itself quickly. Wherever there has been careless or "scamped" work, it is very likely to show itself after exposure to a winter's frost and dampness. Work which stands for two years without signs of cracking, shaling or crumbling is likely to be permanent, its life conditional chiefly upon the actual wear which it receives.

Among the most frequent causes of poor concrete is the use of an inferior brand of cement. A cheap sidewalk obtained by the use of cheap cement will, in the end, prove exceedingly costly. Cement is a material of widely varying degrees of excellence, and experiments should not be tried by the smaller municipalities. There are good and reliable brands of Canadian Portland cement which have proven their qualities in large Government works, and in other extensive contracts, and to go beyond these is unnecessary and in many cases entirely unsafe.

Just as to how cement can be of poor quality is not understood by many, and it may be pointed out, as one example, that the presence of free lime causes what is known as "blowing" in the finished concrete, and rapidly results in decomposition of the sidewalk.

Again among causes of inferior sidewalks is the use of gravel and cement to form the entire work, in place of a proper mixture of sand, cement, and stone. In the composition of concrete we first prepare a mortar of sand and cement with just sufficient water to make a plastic mass; this is mixed with a quantity of stone in such proportion that the mortar will completely fill the voids between the stones, with a proper excess of mortar for safety. There are certain definite proportions of cement, sand and stone which must be uniformly mixed in producing a good quality of artificial stone.

In using gravel only, the difficulty arises from the fact that the cement, sand and stone are not and cannot be expected to be mixed uniformly in these proportions. Gravel usually contains sand, but not in uniform quantities, nor in any definite proportions. Some pockets of so-called gravel will be almost completely sand, while adjoining it there is scarcely any sand, perhaps almost clean stone. To mix such material with cement to produce concrete, while in many cases it may be successful, is always hazardous. The only safe way, where gravel is the material to be used, is to separate by screening, the sand and stone which compose the gravel, then to mix the sand and cement in proper proportions to form the mortar, to which is then added the stone, uniformly, and in its proper proportion.

A cause of an inferior sidewalk is that the materials composing it have been carelessly mixed. The cement and sand should first be mixed dry. This should be turned over and mixed with shovels not less than ten times before the water is added. Water is then added in just sufficient quantity to slightly dampen the mixture, and the paste should be again turned over and mixed not less than six times. If this work is properly done, each grain of sand will be surrounded by a sufficient coating of cement. This mortar having been spread out, the stone should be added and the whole turned over and mixed not less than ten times before being used, the last mixing ensuring that each stone is completely surrounded by a coating of mortar.

The use of too much water in mixing will produce, when set, a concrete of spongy texture. Concrete when ready to be put in the work should have the consistency of freshly dug loam, and should be rammed into place until moisture appears on the surface.

Neglect to keep a newly laid walk damp, and protected from the rays of the sun, will permit the surface to set too rapidly. Minute, hair-like cracks will appear on the surface. These fill with water, which freezes in cold weather, and the expansion destroys the surface of the walk, causing a thin layer to shale off in patches. The same shaling will result from laying concrete in frosty weather; nor should it be mixed nor laid in wet weather.

Another cause of the failure of sidewalks is that the concrete has not been properly and completely separated into flags to allow for expansion, with the result that in hot weather, when expansion takes place, the sidewalk is thrown up at points of least resistance. The expansion of concrete is about the same as that of steel, and no railway company would contemplate for an instant the laying of a continuous steel track in which there are not joints at proper intervals to allow for this expansion. This is true to the same extent with a concrete sidewalk, and every care must be taken to first cut through the foundation layer of concrete, then through the surface layer directly in line with the joint underneath. The entire joint should then be filled with sand to ensure complete separation.

The surface layer of the sidewalk must be laid upon the foundation layer while the latter layer is still damp and before it is set, otherwise there is not a proper union of the two, and shaling will result.

The effect of frost on a wet sub-soil must also be guarded against by the use of under-drains, otherwise the upheaval of the ground, under the expanding influence of frost, is very apt to crack the walk. The separation of the sidewalk into flag divisions, however, is also of service in this respect by giving the separate divisions an opportunity to rise and subside with the soil underneath. No dependence, however, should be placed on this, as a saturated sub-soil will eventually cause a very uneven surface, since the separate flags when once upheaved seldom return exactly to their original position.

Walks are very commonly laid by contractors who furnish their own specifications and agreement. Where such is the case, the agreement should contain a clause specifying in the most stringent terms that the walk is to be kept in perfect condition, order and repair, so that at the end of the term of guarantee, usually five years, the walks shall have given satisfactory evidence of their proper construction and durability. The guarantee is generally so worded that it can be interpreted to mean that the contractor is simply to keep the walk in repair for five years, and entailing no responsibility upon the contractor even if the walk at the end of that period it is ready to crumble to pieces, demanding almost immediate reconstruction. A town, however, should not depend upon the contractor's guarantee, but should employ an experienced and reliable inspector to see that the specifications are faithfully carried out.

BY-LAW COMMUTING STATUTE LABOR IN TORONTO GORE.

By-Law No. 347.

To make provisions for Making, Grading, Graveling, and otherwise Improving and Keeping in Repair the Highways in the Township and under the Control of the Council of the Corporation of Toronto Gore, and to Repeal Certain By-Laws relating to Statute Labor and the Duties of Overseers of Highways.

Whereas, by chapter 223, section 561, sub-section 6, Revised Statutes of Ontario, 1897, power is given to the councils of township municipalities to pass by-laws for reducing the amount of statute labor to be performed by ratepayers and others within the municipality, or for entirely abolishing such statute labor ;

Therefore, the Corporation of Toronto Gore by the council thereof, enacts as follows :—

First.—That from and after the passing of this by-law no statute labor shall be levied or assessed against any person, resident or assessed, in this municipality.

Second.—That the council shall in each and every year levy an amount equal to a commutation of the prescribed statute labor at 65 cents per day over and above the amount heretofore raised for roads and bridges, and that said sum shall be expended in grading the highways and graveling or otherwise permanently improving and paving the leading roads of the township.

Third.—That the council shall annually appoint a commissioner or commissioners or an engineer to carry out the provisions of this by-law, whose duty shall be to go over the whole township as early as convenient in the spring, and present report for adoption to the council at:

the April meeting, giving plans and estimates of the work required to be performed for the year.

Fourth.—The said engineer, commissioner or commissioners, as the case may be, so appointed shall have full supervision of all work performed within the township and shall superintend all work ordered by them, but may appoint some competent person to operate and take charge of the grader, with power to employ men and teams, establish route and to have full supervision of the grading for the portion of the township in which he has been appointed to act.

Fifth.—That the council shall appoint an experienced and trustworthy person overseer in each statute labor division whose duty shall be :—

(1) In the event of any breach or sudden giving away of any bank or bridge, or any other work required to be done for the convenience or safety of travellers, they shall immediately report the same to the engineer or nearest commissioner.

(2) They shall see that thistles and other noxious weeds detrimental to husbandry are prevented from going to seed, and they shall employ men to cut such weeds and thistles on the highway within their respective divisions.

(3) They shall employ men and teams where considered necessary for the purpose of making and keeping open the township roads within their road divisions during the season of sleighing in each year, and shall report from time to time, stating the work performed, to the council.

Sixth.—The before-mentioned overseers shall remain in office from year to year until removed by by-law of the council.

Seventh.—That the pay for all work under this by-law shall be regulated by the following scale :

Commissioner or Engineer 20 cents per hour.

Operators of grader while at work on road, 17½ cents per hour.

Man and team for any purpose, 25 cents per hour.

Man filling gravel or other work, 10 cents per hour.

Eighth.—That all by-laws or parts of by-laws contrary or inconsistent with the provisions of this by-law be and the same are hereby repealed.

Passed this 10th day of April, A.D. 1899.

BY-LAW TO COMMUTE STATUTE LABOR IN NIAGARA TOWNSHIP.

Whereas, by vote of the ratepayers of the township of Niagara, taken at the same time as the Municipal Elections held January 2nd, 1899, a majority of the said ratepayers voted in favor of abolishing the statute labor system, and in paying a commutation fee in lieu thereof, and as by section 561 of chap. 223 of the Revised Statutes of Ontario for 1897, every township council is authorized and empowered to pass by-laws for that purpose.

Therefore, be it enacted, and is hereby enacted by the municipal council of the

Township of Niagara.

1st.—That from and after the passing of this by-law all statute labor in this township shall be abolished, and in lieu thereof all ratepayers and others liable for statute labor shall pay a commutation in money, at the rate of twenty-five cents per day for every day's statute labor for which they are liable, in accordance with section 102 of chap. 224 of the Revised Statutes of Ontario for 1897; and the clerk when making out the collector's roll shall charge to the parties therein named the amount for which they may be liable, and the same shall be collected at the same time and in the same manner as any other taxes.

2nd. That the township be divided into four divisions as may be agreed upon by the council and one councillor appointed a committee for each division; each committeeman to engage a foreman who shall do all small jobs under the direction of the committee, and if any extra help or teams are required, the committee in charge of the division shall employ those nearest the work, in case they can be got at reasonable rates; and when required the foreman shall be paid at the rate of one dollar and twenty-five cents per day, and shall keep an exact account of the time he is engaged in every piece of work, and also of the time put in by any extra help required and shall render an account of the same to the committee by whom he is employed and the committee shall furnish a copy of the same to the township council for their approval.

3rd. That the Committees for the north and south portions of the township shall in each and every year, and as early in the spring as the roads and weather will permit, travel over and inspect the roads and bridges in their respective divisions, and make a detailed statement of all work required, naming the different roads where the road machine can be used to advantage, and the approximate length of road required to be graded; the bridges to be re-built, style and size of stringers, width of opening and amount of plank required; also the different bridges that need repairing and materials required for said repairs. They shall also state where tile can be used to advantage in replacing wooden bridges, and the number and size of tile required for that purpose; and keep an account of good plank in old bridges that can be utilized for repairing, and dispose of all others to the best advantage for the township.

4th. That when any piece of work is required to be done, whether in ditching, grading, road-building or repairing bridges, removing obstructions on the road, on account of snow in winter, or any other cause whatsoever, it shall be the duty of the Committee for the division in which the work is to be done to give written instructions in regard to the same to the foreman, and the foreman shall do such work in accordance with said instructions; and in case at any time the said foreman shall not work faithfully, or shall neglect or refuse to perform any of the duties required of him, it shall be lawful for the Committee having charge of the division, and he is hereby authorized to discharge such foreman, and to hire some other competent person to take his place.

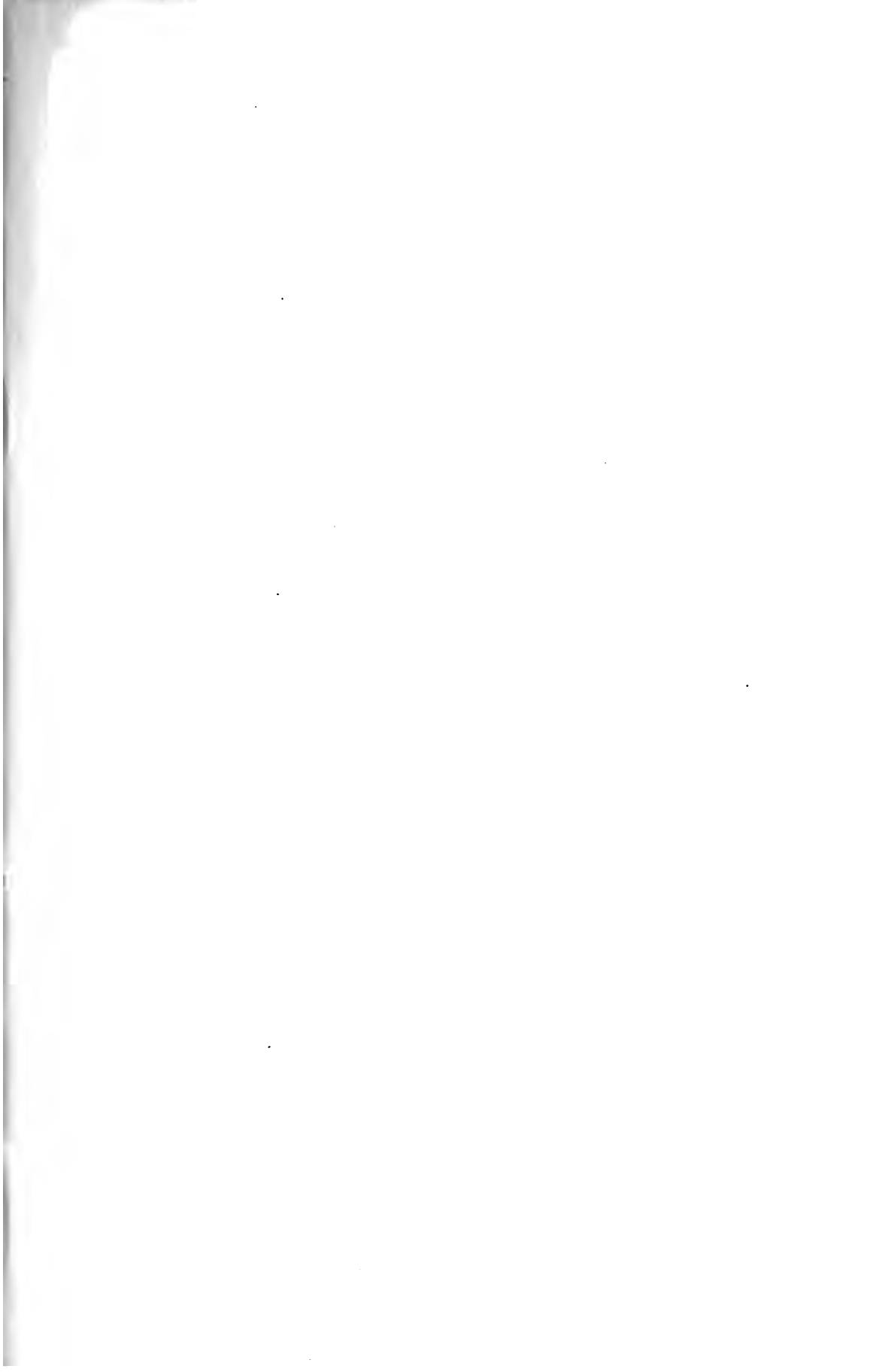
5th. That the Clerk is hereby instructed to furnish each Committeeman, also each foreman, with a memorandum book, in which they shall keep the necessary instructions and accounts of the different divisions.

6th. That by-laws numbered 159, 204, 214, 271, 273, 290 and 324, and all other by-laws contrary to the provisions of this by-law, be, and the same are hereby repealed.

Passed in Council this 20th day of February, A.D., 1899.







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